In[5061]:=

```
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Remove["Global`*"]

(*Parameters of the model*)

T = 294.2; (*room temperature [K]*)
kb = 13.81; (*Boltzmann Constant [nm*fN *K-1]*)
kbTd = kb * T(*Boltzmann constant times temperature [fN.nm]*)
Ncd = 32;(*number of channels*)
kPd = 1872; (*pivot stiffness [fN.nm-1]*)
lPd = 0.0369535;(* pivot damping /100 for noise comparison*)
kGSd = 468; (*gatging spring. stiffness [fN.nm-1]*)
Delta = 700; (*10-90% width of activation curve [nm]*)
tstartFd = 0;(*initial time when force is applied [s]*)


Dvd = 32; (*gating swing 1[nm]*)
(*Dvd=4.4*kbTd*Ncd/(kGSd*Delta);*) (*gating swing 2[nm]*)

Pinfrest = 0.17; (*resting open probabability 1*)
(*Pinfrest=0.5;  (*resting open probabability 2*)*)

fmult = 180(*amplitude force factor*);

alphad = 0.5;(*Fractional transition-state distance[nm*)
Fzd = 1000; (* modulus of the force 1000[fN]*)
wstd = 5000; (*frequency of stimulus 5000 [Hz]*)
ytlrest = 155.7 ; (*resting  tip link length [nm]*)
R = 230;(* radius row 2 stereocilium [nm]*)

Yrestd = -427; (*resting position of IHB [nm]*)


Deltangsd = 20;
(*difference in resting position with and without gating springs [nm]*)
taoval1d = 10^(-5); (*known gating time scale for resting current of 0.1 [s]*)
Pval1d = 0.1; (*resting current corresponding to gating timescale of 10^-5*)

(*find gating timescale for resting current of 0.17*)
ssp = Solve[{taorestd1 == taoval1d * (Pinfrest / Pval1d)^(1 - alphad) *
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```
      ((1 - Pinfrest) / (1 - Pval1d)) ^ alphad}, taorestd1]
taorest = taorestd1 /. First[ssp]
(*gating timescale for resting current of 0.17*)

(*definition of quasistatic current function*)
Pinf2[Yh_, kGS_, Dv_, kbT_, Nc_] := (1 + Exp[- (Y[t] - Yh) * kGS * Dv / (Nc * kbT)]) ^ (-1)

(*definition of equations of motion*)
eqa[lP_, kP_, kGS_, Yug_, Yup_, Dv_, wst_, Fz_, tstartF_] :=
  lP * Y'[t] == -kP * (Y[t] - Yup) - kGS (Y[t] - Yug - Dv * P[t]) +
    Fz * Sin[2 * Pi * wst * (t - tstartF)] * HeavisideTheta[(t - tstartF)];
taofun[alpha_, Yh_, kGS_, Dv_, kbT_, taoval1_, Pval1_, Nc_] :=
 taoval1 * (Pinf2[Yh, kGS, Dv, kbT, Nc] / Pval1) ^ (1 - alpha) *
  ((1 - Pinf2[Yh, kGS, Dv, kbT, Nc]) / (1 - Pval1)) ^ alpha
eqb[alpha_, Yh_, kGS_, Dv_, kbT_, taoval1_, Pval1_, Nc_] :=
  taofun[alpha, Yh, kGS, Dv, kbT, taoval1, Pval1, Nc] * P'[t] ==
   -P[t] + Pinf2[Yh, kGS, Dv, kbT, Nc];

(*derivation of parameters Yup, Yug, Yh and Yrestngs*)
(*define resting equations*)
eqarest = eqa[lPd, kPd, kGSd, Yug, Yup, Dvd, wstd, 0, tstartFd] /.
  {Y'[t] → 0, Y[t] → Yrestd, P[t] → Pinfrest}
eqbrest = Pinfrest == Pinf2[Yh, kGSd, Dvd, kbTd, Ncd] /. {Y[t] → Yrestd}
eqcrest = eqa[lPd, kPd, 0, Yug, Yup, Dvd, wstd, 0, tstartFd] /.
  {Y'[t] → 0, Y[t] → Yrestngs, P[t] → Pinfrest}
eqdrest = (Yrestngs - Yrest == Deltangs) /. {Yrest → Yrestd, Deltangs → Deltangsd}
ss = NSolve[{eqarest, eqbrest, eqcrest, eqdrest}, {Yh, Yug, Yup, Yrestngs}]
Yhd = Yh /. First[ss];(* deflection for which resting current is 0.5 [nm]*)
Yupd = Yup /. First[ss]; (* unloaded state of the pivots [nm]*)
Yugd = Yug /. First[ss]; (* unloaded state of gating spring [nm]*)
Yrestngsd = Yrestngs /. First[ss];
(* resting position of IHB without gating spring [nm]*)
```

Out[5064]=

```
4062.9
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Out[5083]=

```
{{taorestd1 → 0.0000125211}}
```

Out[5084]=

```
0.0000125211
```

Out[5089]=

$$0. == -468\ (-432.44 - Yug) - 1872\ (-427 - Yup)$$

Out[5090]=

$$0.17 == \frac{1}{1 + e^{0.115189\ (427+Yh)}}$$

Out[5091]=

$$0. == -1872\ (Yrestngs - Yup)$$

Out[5092]=

$$427 + Yrestngs == 20$$

NSolve : Inverse functions are being used by NSolve, so some solutions may not be found; use Reduce for complete solution information.

Out[5093]=

$$\{\{Yh \to -413.235, Yug \to -512.44, Yup \to -407., Yrestngs \to -407.\}\}$$

In[5143]:=

```
(*plot settings*)
Remove[XTicks]
Remove[YTicks]
TickLength = 0.02;
Xmin = 0;
Xmax = 10 - 1;
Ymin = 0;
Ymax = 20;
deltaY = 2;
plotrangepsd = {{10^(2), 10^6}, {0.5 * 10^(-12), 10^(-2)}};
plotrangecoher = {{10^(2), 10^6}, {-1.05, 1.05}};
framelabels =
  {{"power spectrum (\!\(\*SuperscriptBox[\(nm\), \(2\)]\)/Hz)", None},
   {"noise frequency (Hz)", None}};
TickLength1 = 0.03;
Yticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, -24, 14, 2}];
Yticks2 = Table[{10^k, Null, {0, TickLength1}}, {k, -23, 14, 2}];
TickLength2 = 0.02;
aa = Table[0, {38}, {4}];
For[k = 1, k ≤ 38, k++,
aa[[k]] = Table[{l * 10^(-k + 10), , {0, TickLength2}}, {l, 2, 8, 2}];
]
YTicks = Union[Yticks1, Yticks2, aa[[1]], aa[[2]], aa[[3]], aa[[4]], aa[[5]],
    aa[[6]], aa[[7]], aa[[8]], aa[[9]], aa[[10]], aa[[11]], aa[[12]], aa[[13]], aa[[14]],
    aa[[15]], aa[[16]], aa[[17]], aa[[18]], aa[[19]], aa[[20]], aa[[21]], aa[[22]], aa[[23]],
    aa[[24]], aa[[25]], aa[[26]], aa[[27]], aa[[28]], aa[[29]], aa[[30]], aa[[31]], aa[[32]]];

Xticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, 3, 8, 1}];
```

```
aa2 = Table[0, {20}, {4}];
For[k = 1, k ≤ 12, k++,
aa2[[k]] = Table[{l * 10^k, , {0, TickLength2}}, {l, 2, 8, 2}];
]

Xticks2 = {{100, 0, {0, 0.03}}};

XTicks = Union[Xticks1, Xticks2, aa2[[2]], aa2[[3]], aa2[[4]], aa2[[5]], aa2[[6]],
   aa2[[7]], aa2[[8]], aa2[[9]], aa2[[10]], aa2[[11]], aa2[[12]], aa2[[13]], aa2[[14]],
   aa2[[15]], aa2[[16]], aa2[[17]], aa2[[18]], aa2[[19]], aa2[[20]], Xticks2];


pady = 120;
padx = 70;
pad2 = 30;
fntsize = 18;

frameticksy = {{{1 / 100 000, Superscript[10, -5], {0, 0.03}},
    {1 / 10 000, Superscript[10, -4], {0, 0.03}}, {1 / 1000, Superscript[10, -3],
     {0, 0.03}}, {1 / 100, Superscript[10, -2], {0, 0.03}}}, None};
frameticksyENLAR =
  {{{1 / 100 000, Superscript[10, -5], {0, 0.03}}, {1 / 10 000, Superscript[10, -4],
     {0, 0.03}}, {1 / 1000, Superscript[10, -3], {0, 0.03}},
    {2 * 10^-5, , {0, 0.02}}, {4 * 10^-5, , {0, 0.02}}, {6 * 10^-5, , {0, 0.02}},
    {8 * 10^-5, , {0, 0.02}}, {2 * 10^-4, , {0, 0.02}}, {4 * 10^-4, , {0, 0.02}},
    {6 * 10^-4, , {0, 0.02}}, {8 * 10^-4, , {0, 0.02}}}, None};
frameticksx =
  {{{1000, Superscript[10, 3], {0, 0.03}}, {10 000, Superscript[10, 4], {0, 0.03}},
    {100 000, Superscript[10, 5], {0, 0.03}}, {2 * 10^3, , {0, 0.02}},
    {4 * 10^3, , {0, 0.02}}, {6 * 10^3, , {0, 0.02}}, {8 * 10^3, , {0, 0.02}},
    {2 * 10^4, , {0, 0.02}}, {4 * 10^4, , {0, 0.02}}, {6 * 10
       ^4, , {0, 0.02}}, {8 * 10^4, , {0, 0.02}}}, None};
frameticksycoher = {{{0, 0, {0, 0.03}}, {0.2, 0.2, {0, 0.03}}, {0.4, 0.4, {0, 0.03}},
    {0.6, 0.6, {0, 0.03}}, {0.8, 0.8, {0, 0.03}}, {1.0, "1.0", {0, 0.03}}}, None};
frameticksycoherneg =
  {{{0, 0, {0, 0.03}}, {0.25, , {0, 0.02}}, {0.5, 0.5, {0, 0.03}}, {0.75, , {0, 0.02}},
    {1.0, "1.0", {0, 0.03}}, {-0.25, , {0, 0.02}}, {-0.5, -0.5, {0, 0.03}}, {-0.75, ,
     {0, 0.02}}, {-1.0, "-1.0", {0, 0.03}}, {-1.5, "-1.5", {0, 0.03}}}, None};

fntsize = 18;

xmin = 2.5 * 10^2; (*physiological frequency range*)
xmax = 7 * 10^3; (*physiological frequency range*)
ymin = 10^(-10);
```

```
ymax = 10^(10);
ymincoher = -2;
ymaxcoher = 2;
dl1 = {{xmin, ymin}, {xmax, ymin}};
dl2 = {{xmin, ymax}, {xmax, ymax}};
dl1coher = {{xmin, ymincoher}, {xmax, ymincoher}};
dl2coher = {{xmin, ymaxcoher}, {xmax, ymaxcoher}};

plotstylecoh = {{Thickness → 0.01, RGBColor[0 * 251 / 255, 0 * 41 / 255, 0 * 67 / 255]},
   {Thickness → 0.01, RGBColor[106 / 255, 90 / 255, 205 / 255]},
   {Thickness → 0.01, RGBColor[225 / 255, 156 / 255, 36 / 255]}};
plotstylepsd = {{Thickness → 0.01, ■}, {Thickness → 0.01, ■},
   {Thickness → 0.01, ■}, {Thickness → 0.01, ■}, {Thickness → 0.01, ■}};
```

In[ ]:=

In[2992]:=

In[5188]:=

```
SeedRandom[47 741] (* set seed for reproducibility of results *)

gx =  2 * kbTd * lPd;
y0in = Yrestd;

t1 = AbsoluteTime[];

(*define Ito process for simulations of deflection and current
 Fzd multiplied by 180 for stimulated psd, 0 for unstimulated psd*)
proc = ItoProcess[
  lPd * ⅆY[t] == (-kPd) * (Y[t] - Yupd) * ⅆt - kGSd * (Y[t] - Yugd - Dvd * Po[t]) * ⅆt +
     Sqrt[gx] * ⅆn1[t]  + Fzd * fmult * Sin[2 * Pi * wstd * (t - tstartFd)] * ⅆt &&
   taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd] * ⅆPo[t] ==
    (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] - Po[t]) ⅆt +
     Sqrt[taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd]]
      Sqrt[(Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] (1 - Clip[Po[t], {0.0, 1.0}]) + (1 - Pinf2[
               Yhd, kGSd, Dvd, kbTd, Ncd]) Clip[Po[t], {0.0, 1.0}]) / Ncd] ⅆn2[t],
  {Y[t], Po[t]}, {{Y, Po}, {y0in, Pinfrest}}, t,
{n1 ≈ WienerProcess[], n2 ≈ WienerProcess[]}]

totallength = 1.014;(*length of simulation*)
factot = 4;
deltatsima = 0.000001 / factot;(*time-step of simulation*)
ap = RandomFunction[proc, {0, totallength, deltatsima}];
(*simulation of deflection and curent*)

t2 = AbsoluteTime[];
t2 - t1
```

Out[5188]=

```
RandomGeneratorState[ Method: ExtendedCA
                      State hash: 2 152 022 931 536 407 856 ]
```

Out[5192]=

$$\text{ItoProcess}\Big[\Big\{\Big\{0. + 27.061\,(180\,000.\,\text{Sin}[31\,415.9\,t] - 1872.\,(407. + Y[t]) - 468.\,(512.44 - 32.\,\text{Po}[t] + Y[t])),\ \Big(0.\,\sqrt{\Big(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}}\Big)^{0.5}\Big(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\Big)^{0.5}}\ \sqrt{\Big(\frac{1. - 1.\,\text{Clip}[\text{Po}[t], \{0., 1.\}]}{1. + e^{0.115189\,(-413.235 - Y[t])}} + \Big(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\Big)\,\text{Clip}[\text{Po}[t], \{0., 1.\}]\Big)\Big/\Big(\Big(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}}\Big)^{0.5}\Big(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\Big)^{0.5}\Big) + \frac{30\,000.\,\Big(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}} - 1.\,\text{Po}[t]\Big)}{\Big(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}}\Big)^{0.5}\Big(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\Big)^{0.5}}\Big\},$$

$$\Big\{\{468.927, 0.\}, \Big\{\Big(0.\,\sqrt{\Big(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}}\Big)^{0.5}\Big(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\Big)^{0.5}}\ \sqrt{\Big(\frac{1. - 1.\,\text{Clip}[\text{Po}[t], \{0., 1.\}]}{1. + e^{0.115189\,(-413.235 - Y[t])}} + \Big(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\Big)\,\text{Clip}[\text{Po}[t], \{0., 1.\}]\Big)\Big/\Big(\Big(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}}\Big)^{0.5}\Big(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\Big)^{0.5}\Big),$$

$$\Big(30.6186\,\sqrt{\Big(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}}\Big)^{0.5}\Big(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\Big)^{0.5}}\ \sqrt{\Big(\frac{1. - 1.\,\text{Clip}[\text{Po}[t], \{0., 1.\}]}{1. + e^{0.115189\,(-413.235 - Y[t])}} + \Big(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\Big)\,\text{Clip}[\text{Po}[t], \{0., 1.\}]\Big)\Big/\Big(\Big(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}}\Big)^{0.5}\Big(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\Big)^{0.5}\Big)\Big\}\Big\},$$

$$\{Y[t], \text{Po}[t]\}\Big\}, \{\{Y, \text{Po}\}, \{-427, 0.17\}\}, \{t, 0\}\Big]$$

Out[5198]=

10.242228

In[ ]:=

In[5199]:=

```
(*Calculation of power spectra from simulated deflection and current*)

(*Yrestd=-412.681; (*resting position of IHB with channels blocked [nm]*)
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Yrestd=-407;(*resting position with gating spring cut [nm]*)*)

ap0 = ap["PathComponent", 1] - Yrestd;
(* subtract resting position from deflection time series*)

ap1 = ap["PathComponent", 2] - Pinfrest;
(*subtract resting current from current time series*)

resampfactor = factot;(*resempling factor for time series*)

randtime100 = 0.005 * 10; (*starting time to ensure that values
 of time series are independent of initial value of simulation*)

(*round to multiple of time-step of simulation to avoid interpolation*)
randtimenm = Round[randtime100, deltatsima];
stpoint = IntegerPart[randtime100 / deltatsima];

deltatsampl = Round[deltatsima * resampfactor, deltatsima];
(*time-step of resampled time series*)
tsampl = 0.005; (* length of a single window of the splitted time series*)
endpoint = IntegerPart[Length[ap["PathComponent", 1]["Values"]] -
    tsampl / deltatsima - stpoint];(*end point of time series*)
tsamplnm = Round[tsampl, deltatsima];
(*round to fraction of delta of time series to avoid interpolation*)
(*determine length of sampled time series*)
b1 = Table[f[i], {i, randtimenm, tsamplnm + randtimenm, deltatsampl}];
lengthares = Length[b1];

ncopiesa = (totallength - randtimenm) / (tsamplnm + deltatsampl);
(*number of non-overlapping windows*)
ncopies = IntegerPart[ncopiesa];
(*round number of overlapping windows to nearest integer *)



ares0 = Table[0, {ncopies}];
(*table of non-overlapping blocks of deflection time series*)
ares1 = Table[0, {ncopies}];
(*table of non-overlapping blocks of deflection time series*)
sizea = lengthares;(*length of a single block*)
fousampled0 = Table[0, {ncopies}, {lengthares}];
(*table of deflection power spectrum of each block*)
fousampled1 = Table[0, {ncopies}, {lengthares}];
(*table of current power spectrum of each block*)
```

```
cfousampled = Table[0, {ncopies}, {lengthares}];
(*table of cross-power spectrum of each block*)

(*derivation of power spectra and cross-power spectra*)
Do[

  ares0[[ll]] = ap0["Values"][[(ll - 1) * tsampl / deltatsima + stpoint
     ;; IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];
    ares1[[ll]] = ap1["Values"][[(ll - 1) * tsampl / deltatsima + stpoint
     ;; IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];




    fousampled0[[ll]] = ResourceFunction["WelchSpectralEstimate"][ares0[[ll]],
    sizea / tsampl, "Window" → None, "Detrend" → None, "OverlapOffset" → None,
    "SegmentSize" → sizea]; (*deflection power spectrum of single block*)
    cfousampled[[ll]] = ResourceFunction["WelchSpectralEstimate"][
    ares1[[ll]], ares0[[ll]], sizea / tsampl, "Window" → None,
    "Detrend" → None, "OverlapOffset" → None, "SegmentSize" → sizea];
  (*cross-power spectrum of single block*)
    fousampled1[[ll]] =
   ResourceFunction["WelchSpectralEstimate"][ares1[[ll]], sizea / tsampl,
    "Window" → None, "Detrend" → None, "OverlapOffset" → None, "SegmentSize" → sizea];
  (*current power spectrum of single block*)
      ,
{ll, ncopies}];

powspmeansampledt0 = Drop[Mean[fousampled0], -1];
(*mean of deflection power spectra of all blocks*)

powspmeansampledt1 = Drop[Mean[fousampled1], -1];
(*mean of current power spectra of all blocks*)

cpowspmeansampledt = Drop[Re[Mean[cfousampled]], -1];
(*mean of cross-power spectra of all blocks*)

lengthpsd = Length[powspmeansampledt0]
(*tables of values of simulated psd*)
meandataplot0 = Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt0[[i]][[2]]},
   {i, 1, lengthpsd}]; (*deflection psd*)
meandataplot1 = Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt1[[i]][[2]]},
   {i, 1, lengthpsd}]; (*current psd*)
meandataplot1cpsd = Table[{cpowspmeansampledt[[i]][[1]], cpowspmeansampledt[[i]][[2]]},
```

```
    {i, 1, lengthpsd}]; (*cross-psd*)
coher01 = Table[{meandataplot0[[i]][[1]],
    meandataplot1cpsd[[i]][[2]] / Sqrt[(meandataplot0[[i]][[2]] * meandataplot1[[i]][[2]])]},
    {i, 1, lengthpsd}]; (*coherency*)

fc = 610; (*corner frequency of the hair-cell membrane [Hz]*)
filtersq1 =
  Table[{meandataplot0[[i]][[1]], Abs[1 / (1 + I * meandataplot1[[i]][[1]] / fc)]^2},
   {i, 1, Length[meandataplot0]}];(*table of values of the filter*)
meandataplot1filt =
  Table[{meandataplot0[[i]][[1]], filtersq1[[i]][[2]] * meandataplot1[[i]][[2]]},
   {i, 1, Length[filtersq1]}];(*table of values of the voltage psd*)
```

Out[5223]=

```
2501
```

In[776]:=

In[779]:=

In[606]:=

In[ ]:=

In[ ]:=

```
(*derivation of analytical power spectra*)


(*for channels blocked*)
(*PYY=4 kbT lP / ((kGS+kP)^2+lP^2w^2);*)


PYY = (4 kbT (kG^2 lP + lG (kGS^2 + lG lP w^2))) / ((kGS^2 - kG (kGS + kP))^2 +
      ((kGS + kP)^2 lG^2 + 2 kGS^2 lG lP + kG^2 lP^2) w^2 + lG^2 lP^2 w^4);
(*analytical expression of deflection psd*)
PYG = (4 kbT kGS ((kGS + kP) lG + kG lP)) /
    ((kGS^2 - kG (kGS + kP))^2 +
      ((kGS + kP)^2 lG^2 + 2 kGS^2 lG lP + kG^2 lP^2) w^2 + lG^2 lP^2 w^4);
(*analytical expression of cross-psd*)
PGG = (4 kbT (2 kGS kP lG + kGS^2 (lG + lP) + lG (kP^2 + lP^2 w^2))) /
    ((kGS^2 - kG (kGS + kP))^2 +
      ((kGS + kP)^2 lG^2 + 2 kGS^2 lG lP + kG^2 lP^2) w^2 + lG^2 lP^2 w^4);
(*analytical expression of gating-distance psd*)
```

```
(*tables of values of analytical psd*)
theorPSDtaplota0 =
  Table[{meandataplot0[[i]][[1]], PYY /. {w → (2 * Pi * meandataplot0[[i]][[1]]),
      lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
      kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2), kbT → kbTd,
      lP → lPd, kGS → kGSd, kP → kPd}}, {i, 1, lengthpsd}];
theorPSDtaplota1 =
  Table[{meandataplot0[[i]][[1]], (PGG / (Dvd^2)) /. {w → (2 * Pi * meandataplot0[[i]][[1]]),
      lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
      kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2), kbT → kbTd,
      lP → lPd, kGS → kGSd, kP → kPd}}, {i, 1, lengthpsd}];
theorCPSDtaplota =
  Table[{meandataplot0[[i]][[1]], (PYG / (Dvd)) /. {w → (2 * Pi * meandataplot0[[i]][[1]]),
      lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
      kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2), kbT → kbTd,
      lP → lPd, kGS → kGSd, kP → kPd}}, {i, 1, lengthpsd}];




(*derivation of aliased analytical power spectra*)

aliased0 = Table[0, lengthpsd];
aliased1 = Table[0, lengthpsd];
aliased01 = Table[0, lengthpsd];
Paliased0 = Table[0, lengthpsd];
Paliased1 = Table[0, lengthpsd];
CPSDaliased = Table[0, lengthpsd];

nser = 15; (*number of terms for the aliasing expansion*)
fs = 1 / (deltatsampl);(*sampling frequency*)

(*aliased expression for deflection psd*)
aliased0 = Table[
   {meandataplot0[[i]][[1]], (∑_{j=1}^{nser} (PYY /. {w → (2 * Pi * (meandataplot0[[i]][[1]] + j * fs)),
          lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
          kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
          kbT → kbTd, lP → lPd, kGS → kGSd, kP → kPd}) +
      ∑_{j=1}^{nser} (PYY /. {w → (2 * Pi * (meandataplot0[[i]][[1]] - j * fs)),
          lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
          kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2), kbT → kbTd,
          lP → lPd, kGS → kGSd, kP → kPd})) }, {i, 1, lengthpsd}];
```

```
In[•]:= Paliased0 = Table[{meandataplot0[[i]][[1]],
          theorPSDtaplota0[[i]][[2]] + aliased0[[i]][[2]]}, {i, 1, lengthpsd}];

In[•]:= (*aliased expression for current psd*)
        aliased1 = Table[{meandataplot0[[i]][[1]],
            (Sum[((PGG / (Dvd^2)) /. {w → (2 * Pi * (meandataplot0[[i]][[1]] + j * fs)),
                  lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
                  kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
                  kbT → kbTd, lP → lPd, kGS → kGSd, kP → kPd}), {j, 1, nser}] +
              Sum[((PGG / (Dvd^2)) /. {w → (2 * Pi * (meandataplot0[[i]][[1]] - j * fs)),
                  lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
                  kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2), kbT → kbTd,
                  lP → lPd, kGS → kGSd, kP → kPd}), {j, 1, nser}])}, {i, 1, lengthpsd}];

In[•]:= Paliased1 = Table[{meandataplot0[[i]][[1]],
          theorPSDtaplota1[[i]][[2]] + aliased1[[i]][[2]]}, {i, 1, lengthpsd}];

In[•]:= (*aliased expression for cross-psd*)
        aliased01 = Table[{meandataplot0[[i]][[1]],
            (Sum[((PYG / (Dvd)) /. {w → (2 * Pi * (meandataplot0[[i]][[1]] + j * fs)),
                  lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
                  kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
                  kbT → kbTd, lP → lPd, kGS → kGSd, kP → kPd}), {j, 1, nser}] +
              Sum[((PYG / (Dvd)) /. {w → (2 * Pi * (meandataplot0[[i]][[1]] - j * fs)),
                  lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
                  kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2), kbT → kbTd,
                  lP → lPd, kGS → kGSd, kP → kPd}), {j, 1, nser}])}, {i, 1, lengthpsd}];

In[•]:= CPSDaliased = Table[{meandataplot0[[i]][[1]],
          theorCPSDtaplota[[i]][[2]] + aliased01[[i]][[2]]}, {i, 1, lengthpsd}];

        (*Fig 4 plots of stimulated power spectra and coherency vs noise frequency. Set D=
         32 and fmult=180 in parameters *)

        plotstyleY = {Darker[■]};
```

```
plotstyleP = {Darker[■]};
plotstyleV = {Darker[■]};

meandataplot0c = {100, meandataplot0[[1]][[2]]};
meandataplot1c = {100, meandataplot1[[1]][[2]]};
meandataplot1v = {100, meandataplot1filt[[1]][[2]]};
meandataplot1cpsdc = {100, meandataplot1cpsd[[1]][[2]]};
filtersq1[[1]][[1]] = 100;


commondirgridlines = {Dashed, Thickness → 0.01 / 2};
yLineStyle = Directive[Darker[■], Dashed, Thickness → 0.01 / 2];

plotrangeY = {{80, 6 * 10^5}, {0.4 * 10^(-7), 4 * 10^(1)}};

ListLogLogPlot[{meandataplot0},
  PlotStyle → plotstyleY, Joined → True, PlotRange → plotrangeY,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"Fourier frequency (Hz)", "deflection psd \n (nm^2/Hz)"},
  GridLines → {{5000}, {}},
  GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLogPlot[{meandataplot0c}, PlotStyle → plotstyleY]]

plotrangeP = {{80, 6 * 10^5}, {10^(-10), 2 * 10^(-3)}};

ListLogLogPlot[{meandataplot1},
  PlotStyle → plotstyleP, Joined → True, PlotRange → plotrangeP,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"Fuorer frequency (Hz)", "current psd \n (Hz^-1)"}, GridLines →
   {{5000}, {}}, GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLogPlot[{meandataplot1c}, PlotStyle → plotstyleP]]



coher01c = {meandataplot1c[[1]],
   meandataplot1cpsd[[1]][[2]] / Sqrt[(meandataplot0[[1]][[2]] * meandataplot1[[1]][[2]])]};
plotrangeco = {{80, 6 * 10^5}, {-0.55, 1.05}};
```

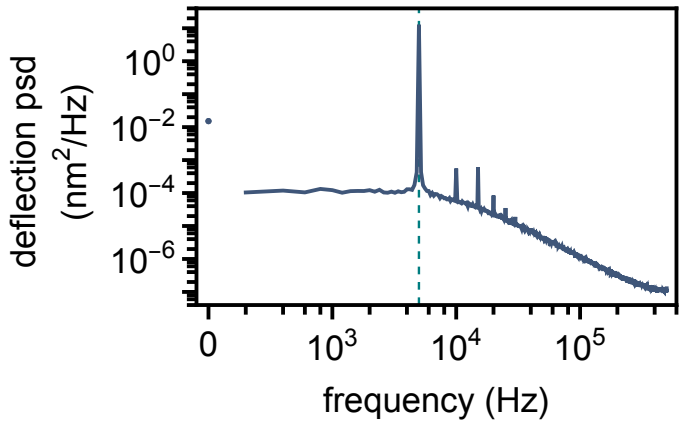
```
ListLogLinearPlot[{coher01}, PlotStyle → plotstyleY,
  Joined → True, Frame → True, PlotRange → plotrangeco,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {frameticksycoherneg, {XTicks, None}},
  FrameLabel → {"frequency (Hz)", "coherency"}, GridLines → {{5000}, {}},
  GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLinearPlot[{coher01c}, PlotStyle → plotstyleY]]
```

```
plotrangeV = {{80, 6 * 10^5}, {10^(-16), 2 * 10^(-3)}};
```

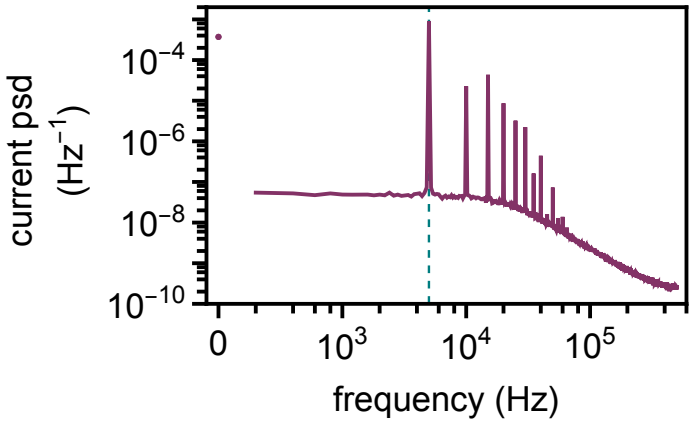
```
ListLogLogPlot[{meandataplot1filt},
  PlotStyle → plotstyleV, Joined → True, PlotRange → plotrangeV,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"Fourier frequency (Hz)", "voltage psd \n (Hz^-1)"}, GridLines →
   {{5000}, {}}, GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLogPlot[{meandataplot1v}, PlotStyle → plotstyleP]]
```

Set : Symbol filtersq1c in part assignment does not have an immediate value.

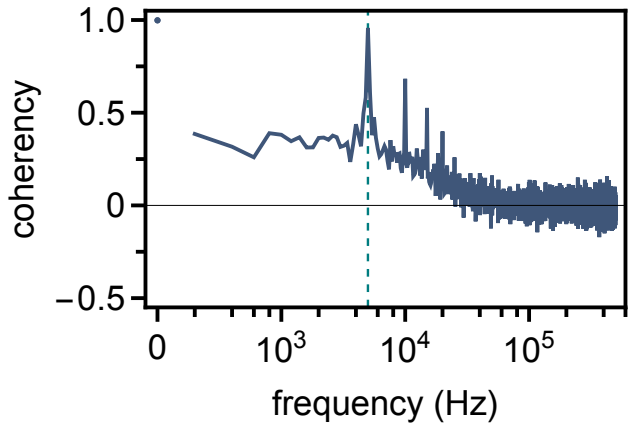
Out[ ]=



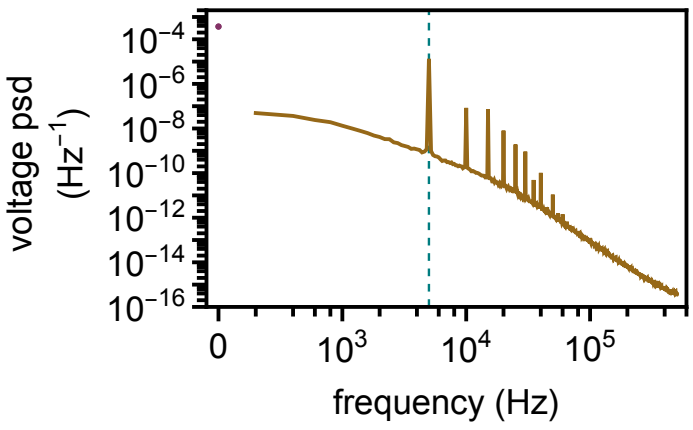
Out[ ]=



Out[ ]=



Out[ ]=




```
(*Fig 4 plots of unstimulated power spectra and coherency vs noise frequency.  Set D=
 32 and fmult=0 in parameters *)
```

```
plotstyleY = {Lighter[■]};
plotstyleP = {Lighter[■]};
plotstyleV = {Lighter[■]};

meandataplot0c = {100, meandataplot0[[1]][[2]]};
meandataplot1c = {100, meandataplot1[[1]][[2]]};
meandataplot1v = {100, meandataplot1filt[[1]][[2]]};
meandataplot1cpsdc = {100, meandataplot1cpsd[[1]][[2]]};
filtersq1[[1]][[1]] = 100;


commondirgridlines = {Dashed, Thickness → 0.01 / 2};
yLineStyle = Directive[Darker[■], Dashed, Thickness → 0.01 / 2];

plotrangeY = {{80, 6 * 10^5}, {0.4 * 10^(-7), 4 * 10^(1)}};

ListLogLogPlot[{meandataplot0},
  PlotStyle → plotstyleY, Joined → True, PlotRange → plotrangeY,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"frequency (Hz)", "deflection psd \n (nm^2/Hz)"}, GridLines →
   {{5000}, {}}, GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLogPlot[{meandataplot0c}, PlotStyle → plotstyleY]]

plotrangeP = {{80, 6 * 10^5}, {10^(-10), 2 * 10^(-3)}};

ListLogLogPlot[{meandataplot1},
  PlotStyle → plotstyleP, Joined → True, PlotRange → plotrangeP,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"Fourier frequency (Hz)", "current psd \n (Hz^-1)"}, GridLines →
   {{5000}, {}}, GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLogPlot[{meandataplot1c}, PlotStyle → plotstyleP]]



coher01c = {meandataplot1c[[1]],
   meandataplot1cpsd[[1]][[2]] / Sqrt[(meandataplot0[[1]][[2]] * meandataplot1[[1]][[2]])]};
```

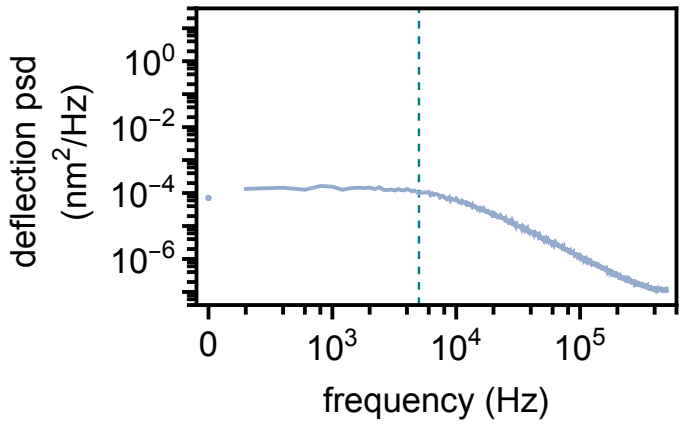
```
plotrangeco = {{80, 6 * 10^5}, {-0.55, 1.05}};
ListLogLinearPlot[{coher01}, PlotStyle → plotstyleY,
  Joined → True, Frame → True, PlotRange → plotrangeco,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {frameticksycoherneg, {XTicks, None}},
  FrameLabel → {"Fourier frequency (Hz)", "coherency"}, GridLines → {{5000}, {}},
  GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLinearPlot[{coher01c}, PlotStyle → plotstyleY]]


plotrangeV = {{80, 6 * 10^5}, {10^(-16), 2 * 10^(-3)}};

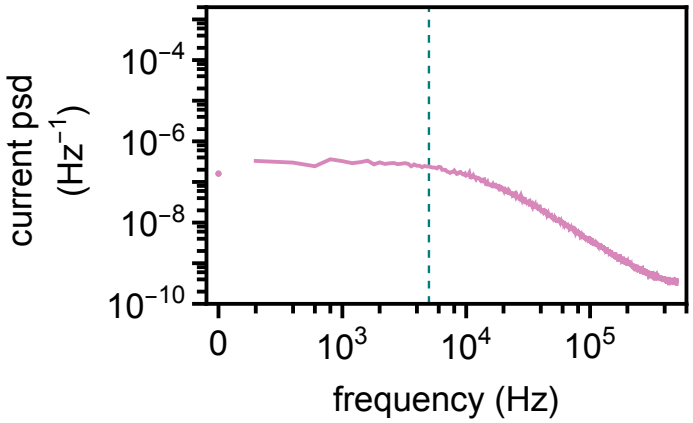
ListLogLogPlot[{meandataplot1filt},
  PlotStyle → plotstyleV, Joined → True, PlotRange → plotrangeV,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"Fourier frequency (Hz)", "voltage psd \n (Hz^-1)"}, GridLines →
   {{5000}, {}}, GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLogPlot[{meandataplot1v}, PlotStyle → plotstyleP]]
```

Set : Symbol filtersq1c in part assignment does not have an immediate value.

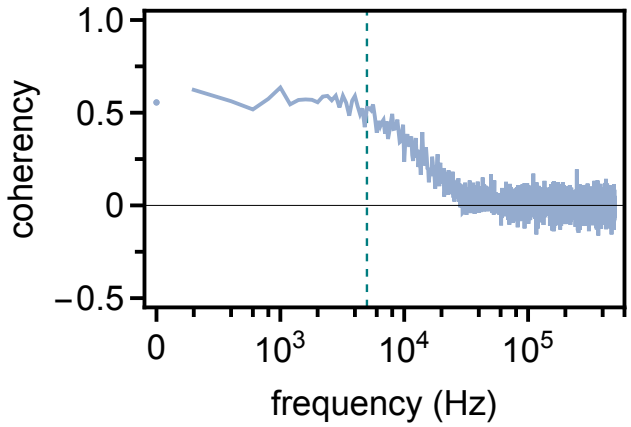
Out[ ]=



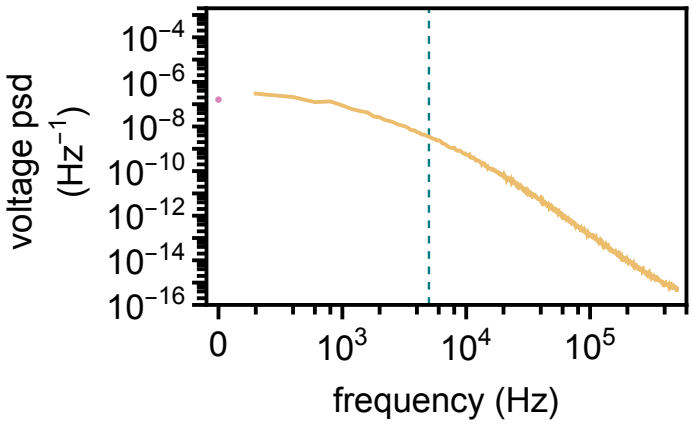
Out[ ]=



Out[ ]=



Out[ ]=




```
(*Fig 6  & S9 plots of stimulated power spectra and coherency vs noise frequency
  for a gating swing of 1.75 nm.  Set D=1.75 and fmult=180 in parameters *)
```

```
plotstyleY = {Darker[■]};
plotstyleP = {Darker[■]};
plotstyleV = {Darker[■]};

meandataplot0c = {100, meandataplot0[[1]][[2]]};
meandataplot1c = {100, meandataplot1[[1]][[2]]};
meandataplot1v = {100, meandataplot1filt[[1]][[2]]};
meandataplot1cpsdc = {100, meandataplot1cpsd[[1]][[2]]};
filtersq1[[1]][[1]] = 100;


commondirgridlines = {Dashed, Thickness → 0.01 / 2};
yLineStyle = Directive[Darker[■], Dashed, Thickness → 0.01 / 2];

plotrangeY = {{80, 6 * 10^5}, {0.4 * 10^(-7), 4 * 10^(1)}};

ListLogLogPlot[{meandataplot0},
  PlotStyle → plotstyleY, Joined → True, PlotRange → plotrangeY,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"Fourier frequency (Hz)", "deflection psd \n (nm^2/Hz)"},
  GridLines → {{5000}, {}},
  GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLogPlot[{meandataplot0c}, PlotStyle → plotstyleY]]

plotrangeP = {{80, 6 * 10^5}, {10^(-10), 2 * 10^(-3)}};

ListLogLogPlot[{meandataplot1},
  PlotStyle → plotstyleP, Joined → True, PlotRange → plotrangeP,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"Fourier frequency (Hz)", "current psd \n (Hz^-1)"}, GridLines →
   {{5000}, {}}, GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLogPlot[{meandataplot1c}, PlotStyle → plotstyleP]]



coher01c = {meandataplot1c[[1]],
```

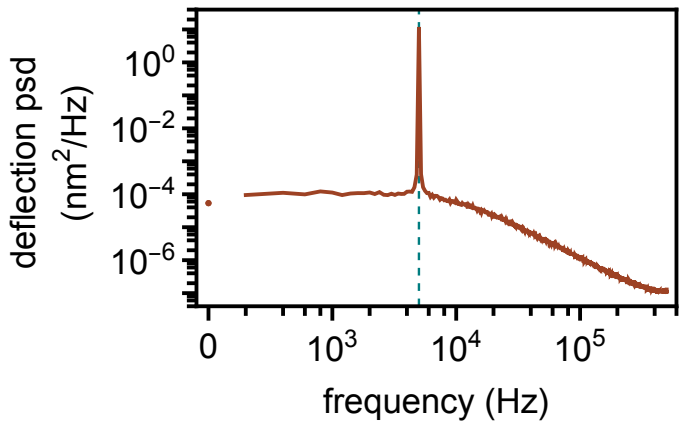
```
    meandataplot1cpsd[[1]][[2]] / Sqrt[(meandataplot0[[1]][[2]] * meandataplot1[[1]][[2]])]};
plotrangeco = {{80, 6 * 10^5}, {-0.55, 1.05}};
ListLogLinearPlot[{coher01}, PlotStyle → plotstyleY,
  Joined → True, Frame → True, PlotRange → plotrangeco,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {frameticksycoherneg, {XTicks, None}},
  FrameLabel → {"frequency (Hz)", "coherency"}, GridLines → {{5000}, {}},
  GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLinearPlot[{coher01c}, PlotStyle → plotstyleY]]


plotrangeV = {{80, 6 * 10^5}, {10^(-16), 2 * 10^(-3)}};

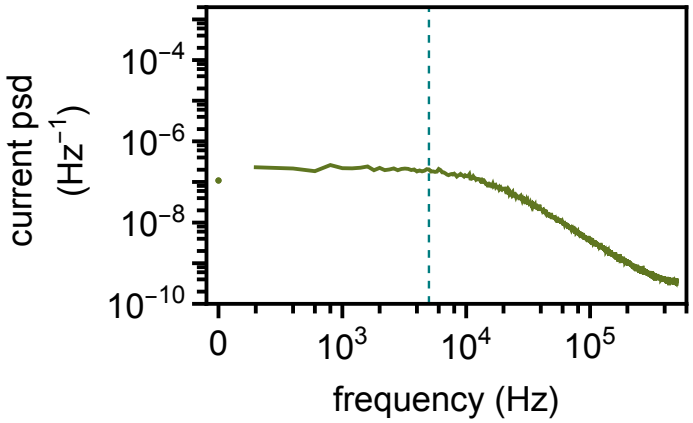
ListLogLogPlot[{meandataplot1filt},
  PlotStyle → plotstyleV, Joined → True, PlotRange → plotrangeV,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"Fourier frequency (Hz)", "voltage psd \n (Hz^-1)"}, GridLines →
   {{5000}, {}}, GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLogPlot[{meandataplot1v}, PlotStyle → plotstyleP]]
```

••• Set : Symbol filtersq1c in part assignment does not have an immediate value.

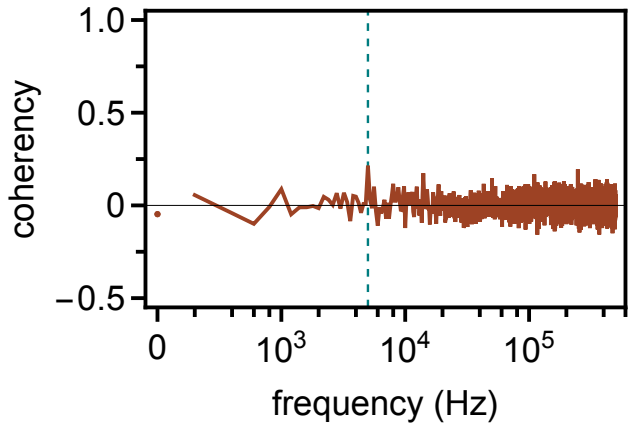
Out[ ]=



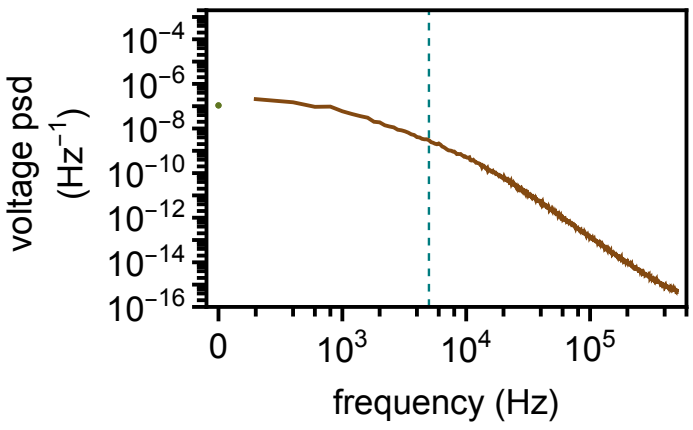
Out[ ]=



Out[ ]=



Out[ ]=




```
(*Fig 6 & S9 plots of unstimulated power spectra and coherency vs noise frequency
  for a gating swing of 1.75 nm.  Set D=32 and fmult=0 in parameters *)
```

```
plotstyleY = {Lighter[■]};
plotstyleP = {Lighter[■]};
plotstyleV = {Lighter[■]};

meandataplot0c = {100, meandataplot0[[1]][[2]]};
meandataplot1c = {100, meandataplot1[[1]][[2]]};
meandataplot1v = {100, meandataplot1filt[[1]][[2]]};
meandataplot1cpsdc = {100, meandataplot1cpsd[[1]][[2]]};
filtersq1[[1]][[1]] = 100;


commondirgridlines = {Dashed, Thickness → 0.01 / 2};
yLineStyle = Directive[Darker[■], Dashed, Thickness → 0.01 / 2];

plotrangeY = {{80, 6 * 10^5}, {0.4 * 10^(-7), 4 * 10^(1)}};

ListLogLogPlot[{meandataplot0},
  PlotStyle → plotstyleY, Joined → True, PlotRange → plotrangeY,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"Fourier frequency (Hz)", "deflection psd \n (nm^2/Hz)"},
  GridLines → {{5000}, {}},
  GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLogPlot[{meandataplot0c}, PlotStyle → plotstyleY]]

plotrangeP = {{80, 6 * 10^5}, {10^(-10), 2 * 10^(-3)}};

ListLogLogPlot[{meandataplot1},
  PlotStyle → plotstyleP, Joined → True, PlotRange → plotrangeP,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"frequency (Hz)", "current psd \n (Hz^-1)"}, GridLines →
   {{5000}, {}}, GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLogPlot[{meandataplot1c}, PlotStyle → plotstyleP]]



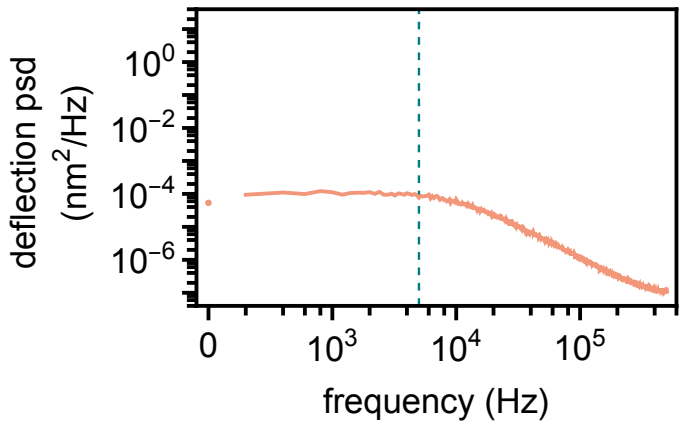
coher01c = {meandataplot1c[[1]],
```

```
    meandataplot1cpsd[[1]][[2]] / Sqrt[(meandataplot0[[1]][[2]] * meandataplot1[[1]][[2]])]};
plotrangeco = {{80, 6 * 10^5}, {-0.55, 1.05}};
ListLogLinearPlot[{coher01}, PlotStyle → plotstyleY,
  Joined → True, Frame → True, PlotRange → plotrangeco,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {frameticksycoherneg, {XTicks, None}},
  FrameLabel → {"Fourier frequency (Hz)", "coherency"}, GridLines → {{5000}, {}},
  GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLinearPlot[{coher01c}, PlotStyle → plotstyleY]]

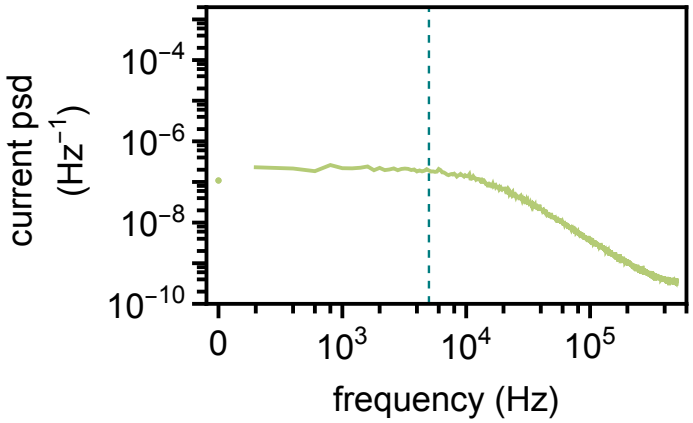

plotrangeV = {{80, 6 * 10^5}, {10^(-16), 2 * 10^(-3)}};

ListLogLogPlot[{meandataplot1filt},
  PlotStyle → plotstyleV, Joined → True, PlotRange → plotrangeV,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"Fourier frequency (Hz)", "voltage psd \n (Hz^-1)"}, GridLines →
   {{5000}, {}}, GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLogPlot[{meandataplot1v}, PlotStyle → plotstyleP]]
```

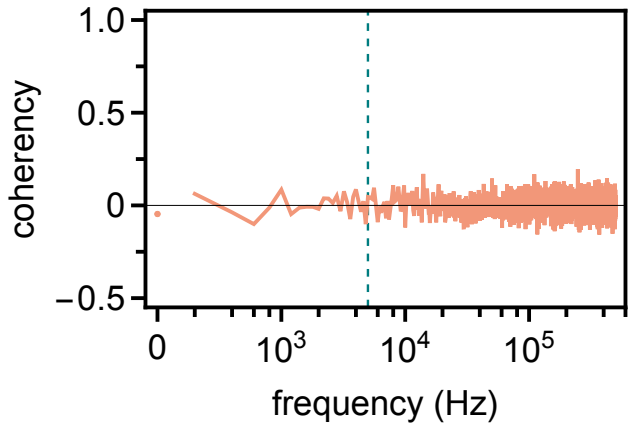
Out[ ]=



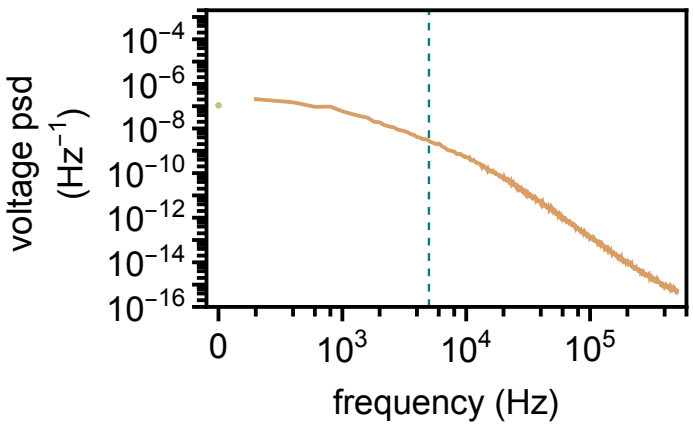
Out[ ]=



Out[ ]=



Out[ ]=




```
(*Fig 7 plots of stimulated power spectra and coherency vs noise
  frequency for a resting current of 0.5 and a gating swing of 32 nm.  Set D=
```

```
 32 and fmult=180 and Prest=0.5 in parameters. *)


plotstyleY = {Darker[■]};
plotstyleP = {Darker[■]};
plotstyleV = {Darker[■]};

meandataplot0c = {100, meandataplot0[[1]][[2]]};
meandataplot1c = {100, meandataplot1[[1]][[2]]};
meandataplot1v = {100, meandataplot1filt[[1]][[2]]};
meandataplot1cpsdc = {100, meandataplot1cpsd[[1]][[2]]};
filtersq1[[1]][[1]] = 100;


commondirgridlines = {Dashed, Thickness → 0.01 / 2};
yLineStyle = Directive[Darker[■], Dashed, Thickness → 0.01 / 2];

plotrangeY = {{80, 6 * 10^5}, {0.4 * 10^(-7), 4 * 10^(1)}};

ListLogLogPlot[{meandataplot0},
  PlotStyle → plotstyleY, Joined → True, PlotRange → plotrangeY,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"frequency (Hz)", "deflection psd \n (nm^2/Hz)"}, GridLines →
   {{5000}, {}}, GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLogPlot[{meandataplot0c}, PlotStyle → plotstyleY]]

plotrangeP = {{80, 6 * 10^5}, {10^(-10), 2 * 10^(-3)}};

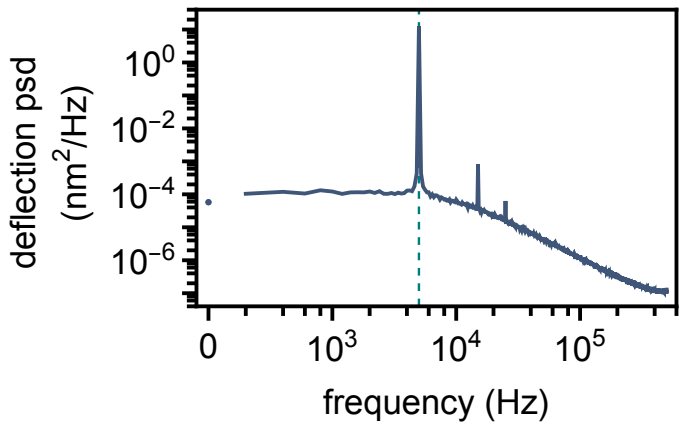
ListLogLogPlot[{meandataplot1},
  PlotStyle → plotstyleP, Joined → True, PlotRange → plotrangeP,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"Fourier frequency (Hz)", "current psd \n (Hz^-1)"}, GridLines →
   {{5000}, {}}, GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLogPlot[{meandataplot1c}, PlotStyle → plotstyleP]]
```

```
coher01c = {meandataplot1c[[1]],
    meandataplot1cpsd[[1]][[2]] / Sqrt[(meandataplot0[[1]][[2]] * meandataplot1[[1]][[2]])]};
plotrangeco = {{80, 6 * 10^5}, {-0.55, 1.05}};
ListLogLinearPlot[{coher01}, PlotStyle → plotstyleY,
  Joined → True, Frame → True, PlotRange → plotrangeco,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {frameticksycoherneg, {XTicks, None}},
  FrameLabel → {"Fourier frequency (Hz)", "coherency"}, GridLines → {{5000}, {}},
  GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLinearPlot[{coher01c}, PlotStyle → plotstyleY]]

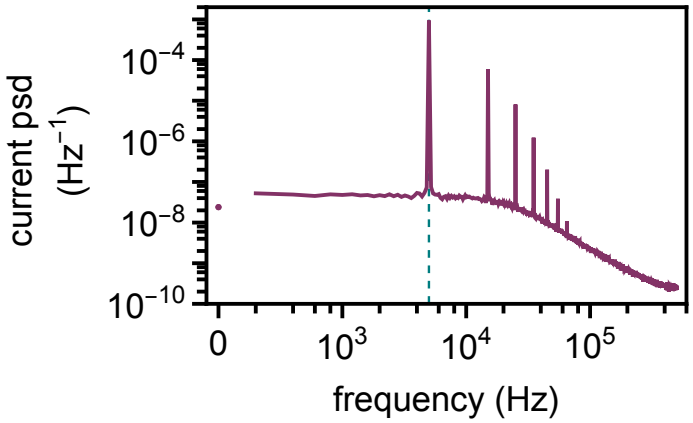

plotrangeV = {{80, 6 * 10^5}, {10^(-16), 2 * 10^(-3)}};

ListLogLogPlot[{meandataplot1filt},
  PlotStyle → plotstyleV, Joined → True, PlotRange → plotrangeV,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"Fourier frequency (Hz)", "voltage psd \n (Hz^-1)"}, GridLines →
   {{5000}, {}}, GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLogPlot[{meandataplot1v}, PlotStyle → plotstyleP]]
```

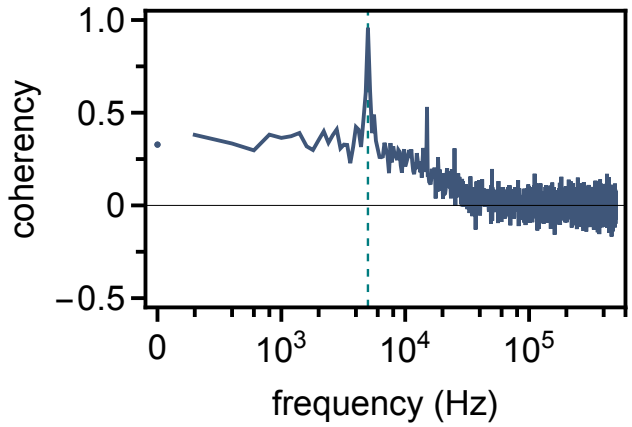
Out[ ]=



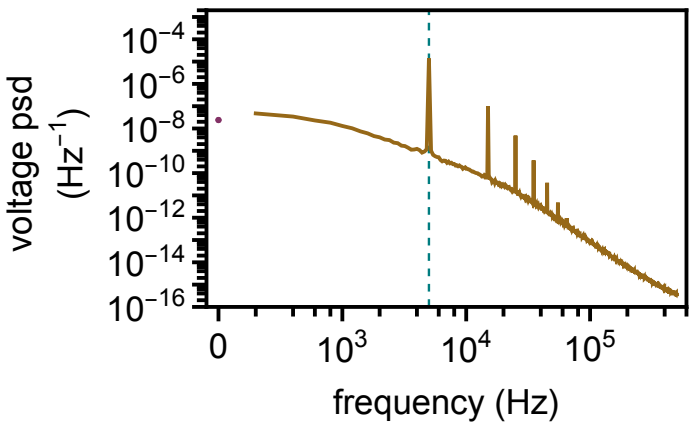
Out[ ]=



Out[ ]=



Out[ ]=




```
(*Fig 7 plots of unstimulated power spectra and coherency vs noise
  frequency for a resting current of 0.5 and a gating swing of 32 nm. Set D=
```

```
 32 and fmult=0 and Prest=0.5 in parameters. *)


plotstyleY = {Lighter[■]};
plotstyleP = {Lighter[■]};
plotstyleV = {Lighter[■]};

meandataplot0c = {100, meandataplot0[[1]][[2]]};
meandataplot1c = {100, meandataplot1[[1]][[2]]};
meandataplot1v = {100, meandataplot1filt[[1]][[2]]};
meandataplot1cpsdc = {100, meandataplot1cpsd[[1]][[2]]};
filtersq1[[1]][[1]] = 100;


commondirgridlines = {Dashed, Thickness → 0.01 / 2};
yLineStyle = Directive[Darker[■], Dashed, Thickness → 0.01 / 2];

plotrangeY = {{80, 6 * 10^5}, {0.4 * 10^(-7), 4 * 10^(1)}};

ListLogLogPlot[{meandataplot0},
  PlotStyle → plotstyleY, Joined → True, PlotRange → plotrangeY,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"frequency (Hz)", "deflection psd \n (nm^2/Hz)"}, GridLines →
   {{5000}, {}}, GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLogPlot[{meandataplot0c}, PlotStyle → plotstyleY]]

plotrangeP = {{80, 6 * 10^5}, {10^(-10), 2 * 10^(-3)}};

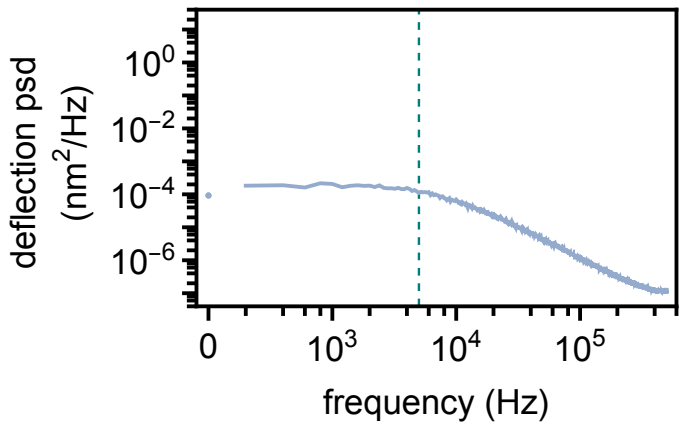
ListLogLogPlot[{meandataplot1},
  PlotStyle → plotstyleP, Joined → True, PlotRange → plotrangeP,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"Fourier frequency (Hz)", "current psd \n (Hz^-1)"}, GridLines →
   {{5000}, {}}, GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLogPlot[{meandataplot1c}, PlotStyle → plotstyleP]]
```

```
coher01c = {meandataplot1c[[1]],
    meandataplot1cpsd[[1]][[2]] / Sqrt[(meandataplot0[[1]][[2]] * meandataplot1[[1]][[2]])]};
plotrangeco = {{80, 6 * 10^5}, {-0.55, 1.05}};
ListLogLinearPlot[{coher01}, PlotStyle → plotstyleY,
  Joined → True, Frame → True, PlotRange → plotrangeco,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {frameticksycoherneg, {XTicks, None}},
  FrameLabel → {"Fourier frequency (Hz)", "coherency"}, GridLines → {{5000}, {}},
  GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLinearPlot[{coher01c}, PlotStyle → plotstyleY]]

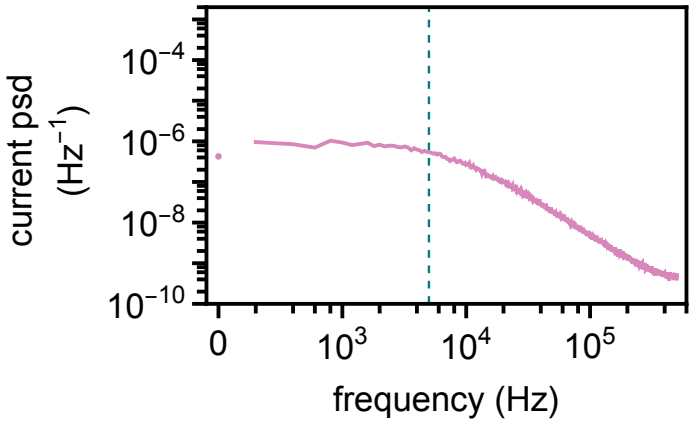

plotrangeV = {{80, 6 * 10^5}, {10^(-16), 2 * 10^(-3)}};

ListLogLogPlot[{meandataplot1filt},
  PlotStyle → plotstyleV, Joined → True, PlotRange → plotrangeV,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"Fourier frequency (Hz)", "voltage psd \n (Hz^-1)"}, GridLines →
   {{5000}, {}}, GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLogPlot[{meandataplot1v}, PlotStyle → plotstyleP]]
```

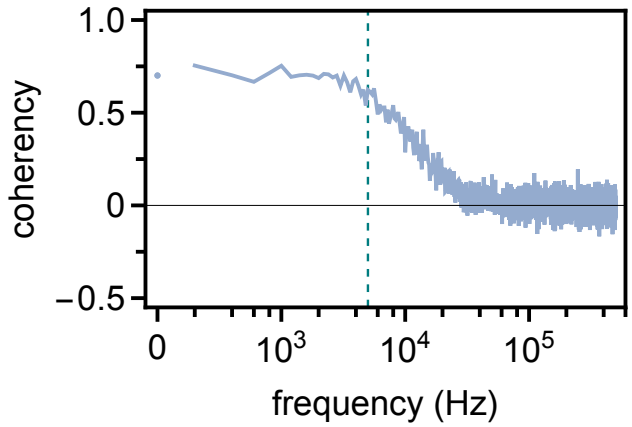
Out[ ]=



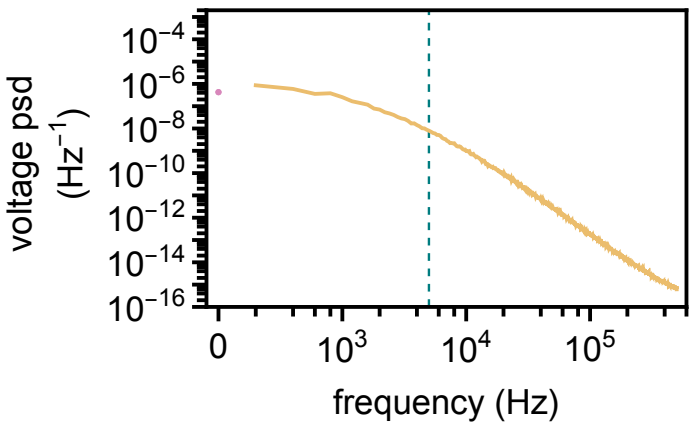
Out[ ]=


current psd
(Hz^{-1})
10^{-4}
10^{-6}
10^{-8}
10^{-10}
0
10^3
10^4
10^5
frequency (Hz)

Out[ ]=


coherency
1.0
0.5
0
−0.5
0
10^3
10^4
10^5
frequency (Hz)

Out[ ]=


voltage psd
(Hz^{-1})
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
10^{-16}
0
10^3
10^4
10^5
frequency (Hz)

```
In[ ]:= (*Calculation of residuals*)
(*residuals are defined as the fractional difference between simulated
 power spectra and their analytical expression including aliases*)
exppsd = Table[(powspmeansampledt0[[i]][[2]] - Paliased0[[i]][[2]]) * 100 / Paliased0[[i]][[2]],
   {i, 1, lengthpsd}];
exppsd2 = Table[{powspmeansampledt0[[i]][[1]] / 1000, exppsd[[i]]}, {i, 1, lengthpsd}];
exppsdP = Table[(powspmeansampledt1[[i]][[2]] - Paliased1[[i]][[2]]) * 100 / Paliased1[[i]][[2]],
   {i, 1, lengthpsd}];
exppsdP2 = Table[{powspmeansampledt1[[i]][[1]] / 1000, exppsdP[[i]]}, {i, 1, lengthpsd}];

In[ ]:=
(*plot settings*)
(*FigS2 C,D deflection adn current residuals*)
PlotrangereY = {{0, 500}, {-100, 100}}

plotstyleY = {Darker[■]};
plotstyleP = {Darker[■]};
plotstyleV = {Darker[■]};

pady = 100;
padx = 70;
pad2 = 30;
fntsize = 16;


Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 500, 100}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 500, 25}];
XTicks = Union[Xticks1, Xticks3];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, -100, 100, 50}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, -100, 100, 10}];
YTicks = Union[Yticks1, Yticks3];

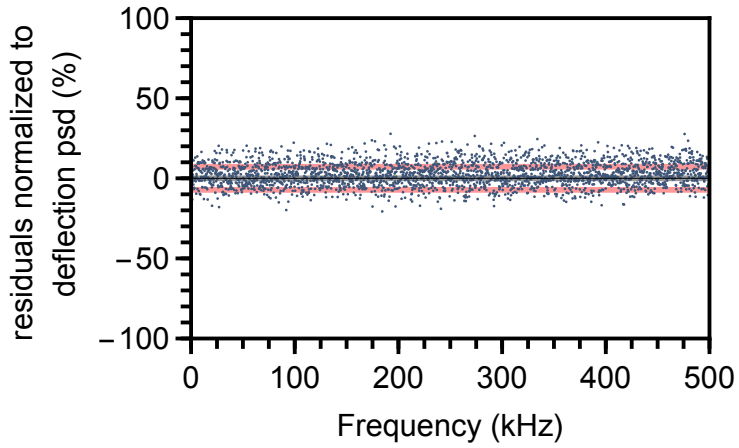
(*FigS2 plots of residuals for a nonlinear IHB with state dependent noise*)
resalpsd =
 ListPlot[exppsd2, PlotRange → {{0, 1 / (1000 * 2 * deltatsampl)}, {-100, 100}},
  PlotStyle → plotstyleY, ImagePadding → {{pady, pad2}, {padx, pad2}},
  ImageSize → 400, LabelStyle → {FontSize → fntsize},
  GridLines → {{}, {{100 / Sqrt[ncopies], Directive[Red, AbsoluteThickness[3]]},
     {-100 / Sqrt[ncopies], Directive[Red, AbsoluteThickness[3]]},
     {0, Directive[Black, AbsoluteThickness[3]]}}}, Frame → True,
  Background → White, FrameStyle → Directive[Black, Thick], FrameLabel →
   {"Frequency (kHz)", "residuals normalized to \n deflection psd (%)"},
  FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

```
resalpsdP =
 ListPlot[exppsdP2, PlotRange → {{0, 1 / (1000 * 2 * deltatsampl)}, {-100, 100}},
  PlotStyle → plotstyleP, ImagePadding → {{pady, pad2}, {padx, pad2}},
  ImageSize → 400, LabelStyle → {FontSize → fntsize},
  GridLines → {{}, {{100 / Sqrt[ncopies], Directive[Green, AbsoluteThickness[3]]},
     {-100 / Sqrt[ncopies], Directive[Green, AbsoluteThickness[3]]},
     {0, Directive[Black, AbsoluteThickness[3]]}}},
  Frame → True, Background → White, FrameStyle → Directive[Black, Thick],
  FrameLabel → {"Frequency (kHz)", "residuals normalized to \ncurrent psd (%)"},
  FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

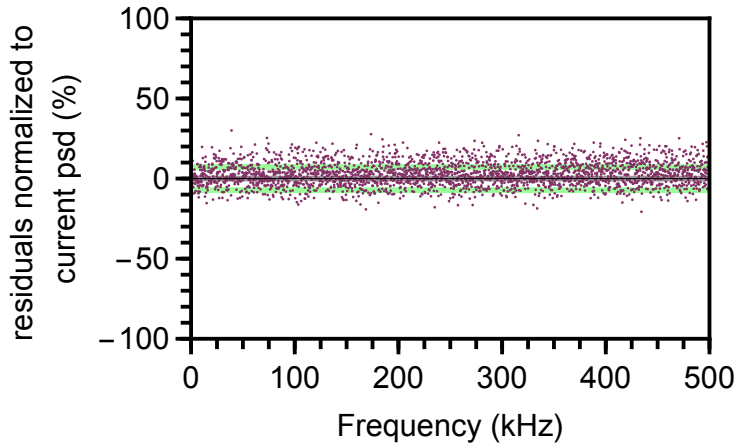
Out[ ]=

```
{{0, 500}, {-100, 100}}
```

Out[ ]=



Out[ ]=



In[ ]:= (*Calculation of power spectra for linear IHB with state-independent noise*)

```
t1 = AbsoluteTime[];

SeedRandom[47 741]

Ncv = 32;

t1 = AbsoluteTime[];

SeedRandom[47 741]
gx =  2 * kbTd * lPd;
kGd = Ncv * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2);
lGd = Ncv * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2);
gy =  2 * kbTd * lGd;
y0in = Yrestd;
G0in = Dvd * Pinfrest;

proc =
 ItoProcess[lPd * dy[t] == (-kPd) * (y[t] - Yupd) * dt - kGSd * (y[t] - Yugd - g[t]) * dt +
     Sqrt[gx] * dn1[t] + 0 * Fzd * 180 * Sin[2 * Pi * wstd * (t - tstartFd)] * dt &&
   lGd * dg[t] ==
    -kGd * (g[t] - Dvd * Pinfrest) * dt + kGSd * (y[t] - Yrestd) * dt + Sqrt[gy] * dn2[t],
  {y[t], g[t]}, {{y, g}, {y0in, G0in}}, t,
{n1 ≈ WienerProcess[], n2 ≈ WienerProcess[]}]
totallength = 2 * 0.507;
factot = 4;
deltatsima = 0.000001 / factot;
ap = RandomFunction[proc, {0, totallength, deltatsima}];

ap0 = ap["PathComponent", 1] - Yrestd;

ap1 = ap["PathComponent", 2] - Dvd * Pinfrest;

resampfactor = factot;

randtime100 = 0.005 * 10;
randtimenm = Round[randtime100, deltatsima];
stpoint = IntegerPart[randtime100 / deltatsima];

deltatsampl = Round[deltatsima * resampfactor, deltatsima];

tsampl = 0.005;
endpoint = IntegerPart[
   Length[ap["PathComponent", 1]["Values"]] - tsampl / deltatsima - stpoint];
tsamplnm = Round[tsampl, deltatsima];
```

```
b1 = Table[f[i], {i, randtimenm, tsamplnm + randtimenm, deltatsampl}];
lengthares = Length[b1];
Clear[ares];
totallength2 = totallength;
ncopiesa = (totallength2 - randtimenm) / (tsamplnm + deltatsampl);
ncopies = IntegerPart[ncopiesa];

ares0 = Table[0, {ncopies}];
ares1 = Table[0, {ncopies}];
sizea = lengthares;
fousampled0 = Table[0, {ncopies}, {lengthares}];
fousampled1 = Table[0, {ncopies}, {lengthares}];
cfousampled = Table[0, {ncopies}, {lengthares}];

Do[

  ares0[[ll]] = ap0["Values"][[(ll - 1) * tsampl / deltatsima + stpoint
     ;; IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];
    ares1[[ll]] = ap1["Values"][[(ll - 1) * tsampl / deltatsima + stpoint ;;
      IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]] / Dvd;

    fousampled0[[ll]] = ResourceFunction["WelchSpectralEstimate"][
    ares0[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
    "OverlapOffset" → None, "SegmentSize" → sizea];
    cfousampled[[ll]] = ResourceFunction["WelchSpectralEstimate"][
    ares1[[ll]], ares0[[ll]], sizea / tsampl, "Window" → None,
    "Detrend" → None, "OverlapOffset" → None, "SegmentSize" → sizea];
    fousampled1[[ll]] = ResourceFunction["WelchSpectralEstimate"][
    ares1[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
    "OverlapOffset" → None, "SegmentSize" → sizea];

    ,
{ll, ncopies}];
powspmeansampledt0 = Drop[Mean[fousampled0], -1];

powspmeansampledt1 = Drop[Mean[fousampled1], -1];

cpowspmeansampledt = Drop[Re[Mean[cfousampled]], -1];

lengthpsd = Length[powspmeansampledt0]
meandataplot0 =
  Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt0[[i]][[2]]}, {i, 1, lengthpsd}];
meandataplot1 =
  Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt1[[i]][[2]]}, {i, 1, lengthpsd}];
```

```
meandataplot1cpsd =
  Table[{cpowspmeansampledt[[i]][[1]], cpowspmeansampledt[[i]][[2]]}, {i, 1, lengthpsd}];

lengthpsd = Length[powspmeansampledt0]
PYY = (4 kbT (kG^2 lP + lG (kGS^2 + lG lP w^2))) / ((kGS^2 - kG (kGS + kP))^2 +
      ((kGS + kP)^2 lG^2 + 2 kGS^2 lG lP + kG^2 lP^2) w^2 + lG^2 lP^2 w^4);
PYG = (4 kbT kGS ((kGS + kP) lG + kG lP)) /
    ((kGS^2 - kG (kGS + kP))^2 +
      ((kGS + kP)^2 lG^2 + 2 kGS^2 lG lP + kG^2 lP^2) w^2 + lG^2 lP^2 w^4);
PGG = (4 kbT (2 kGS kP lG + kGS^2 (lG + lP) + lG (kP^2 + lP^2 w^2))) /
    ((kGS^2 - kG (kGS + kP))^2 +
      ((kGS + kP)^2 lG^2 + 2 kGS^2 lG lP + kG^2 lP^2) w^2 + lG^2 lP^2 w^4);

theorPSDtaplota0 =
  Table[{powspmeansampledt0[[i]][[1]], PYY /. {w → (2 * Pi * powspmeansampledt0[[i]][[1]]),
      lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
      kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2), kbT → kbTd,
      lP → lPd, kGS → kGSd, kP → kPd}}, {i, 1, lengthpsd}];
theorPSDtaplota1 = Table[{powspmeansampledt0[[i]][[1]],
    (PGG / (Dvd^2)) /. {w → (2 * Pi * powspmeansampledt0[[i]][[1]]),
      lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
      kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2), kbT → kbTd,
      lP → lPd, kGS → kGSd, kP → kPd}}, {i, 1, lengthpsd}];
theorCPSDtaplota = Table[
   {cpowspmeansampledt[[i]][[1]], (PYG / (Dvd)) /. {w → (2 * Pi * powspmeansampledt0[[i]][[1]]),
      lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
      kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2), kbT → kbTd,
      lP → lPd, kGS → kGSd, kP → kPd}}, {i, 1, lengthpsd}];

meandataplot0 =
  Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt0[[i]][[2]]}, {i, 1, lengthpsd}];
meandataplot1 =
  Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt1[[i]][[2]]}, {i, 1, lengthpsd}];
meandataplot1cpsd =
  Table[{cpowspmeansampledt[[i]][[1]], cpowspmeansampledt[[i]][[2]]}, {i, 1, lengthpsd}];

aliased0 = Table[0, lengthpsd];
aliased1 = Table[0, lengthpsd];
aliased01 = Table[0, lengthpsd];
Paliased0 = Table[0, lengthpsd];
Paliased1 = Table[0, lengthpsd];
CPSDaliased = Table[0, lengthpsd];

nser = 15;
```

```
fs = 1 / (deltatsampl);

aliased0 = Table[{powspmeansampledt0[[i]][[1]],
    (Sum_{j=1}^{nser} (PYY /. {w → (2 * Pi * (powspmeansampledt0[[i]][[1]] + j * fs)),
          lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
          kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
          kbT → kbTd, lP → lPd, kGS → kGSd, kP → kPd}) +
      Sum_{j=1}^{nser} (PYY /. {w → (2 * Pi * (powspmeansampledt0[[i]][[1]] - j * fs)),
          lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
          kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2), kbT → kbTd,
          lP → lPd, kGS → kGSd, kP → kPd}))}, {i, 1, lengthpsd}];
Paliased0 = Table[{powspmeansampledt0[[i]][[1]],
    theorPSDtaplota0[[i]][[2]] + aliased0[[i]][[2]]}, {i, 1, lengthpsd}];

aliased1 = Table[{powspmeansampledt0[[i]][[1]],
    (Sum_{j=1}^{nser} ((PGG / (Dvd^2)) /. {w → (2 * Pi * (powspmeansampledt1[[i]][[1]] + j * fs)),
          lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
          kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
          kbT → kbTd, lP → lPd, kGS → kGSd, kP → kPd}) +
      Sum_{j=1}^{nser} ((PGG / (Dvd^2)) /. {w → (2 * Pi * (powspmeansampledt1[[i]][[1]] - j * fs)),
          lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
          kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2), kbT → kbTd,
          lP → lPd, kGS → kGSd, kP → kPd}))}, {i, 1, lengthpsd}];
Paliased1 = Table[{powspmeansampledt1[[i]][[1]],
    theorPSDtaplota1[[i]][[2]] + aliased1[[i]][[2]]}, {i, 1, lengthpsd}];

aliased01 = Table[{powspmeansampledt0[[i]][[1]],
    (Sum_{j=1}^{nser} ((PYG / (Dvd)) /. {w → (2 * Pi * (cpowspmeansampledt[[i]][[1]] + j * fs)),
          lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
          kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
          kbT → kbTd, lP → lPd, kGS → kGSd, kP → kPd}) +
      Sum_{j=1}^{nser} ((PYG / (Dvd)) /. {w → (2 * Pi * (cpowspmeansampledt[[i]][[1]] - j * fs)),
          lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
          kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2), kbT → kbTd,
          lP → lPd, kGS → kGSd, kP → kPd}))}, {i, 1, lengthpsd}];
CPSDaliased = Table[{powspmeansampledt1[[i]][[1]],
    theorCPSDtaplota[[i]][[2]] + aliased01[[i]][[2]]}, {i, 1, lengthpsd}];
```

Out[ ]=

```
RandomGeneratorState[ Method: ExtendedCA
                      State hash: -8833868631830390932 ]
```

Out[ ]=

```
RandomGeneratorState[ Method: ExtendedCA
                      State hash: -8833868631830390932 ]
```

Out[ ]=

```
ItoProcess[{{0. + 27.061 (-1872. (407. + y[t]) - 468. (512.44 - 1. g[t] + y[t])),
    0. + 88.7561 (-899.828 (-5.44 + g[t]) + 468. (427. + y[t]))},
  {{468.927, 0.}, {0., 849.244}}, {y[t], g[t]}}, {{y, g}, {-427, 5.44}}, {t, 0}]
```

Out[ ]=

```
2501
```

Out[ ]=

```
2501
```

In[ ]:=

```
(*Calculation of residuals for a linear IHB with state-independent noise*)
(*residuals are defined as the fractional difference between simulated
 power spectra and their analytical expression including aliases*)
exppsd = Table[(powspmeansampledt0[[i]][[2]] - Paliased0[[i]][[2]]) * 100 / Paliased0[[i]][[2]],
   {i, 1, lengthpsd}];
exppsd2 = Table[{powspmeansampledt0[[i]][[1]] / 1000, exppsd[[i]]}, {i, 1, lengthpsd}];
exppsdCP = Table[(Abs[meandataplot1cpsd[[i]][[2]]] - Abs[CPSDaliased[[i]][[2]]]) *
    100 / Abs[CPSDaliased[[i]][[2]]], {i, 1, lengthpsd}];
exppsdCP2 = Table[{meandataplot1cpsd[[i]][[1]] / 1000, exppsdCP[[i]]}, {i, 1, lengthpsd}];
```

In[ ]:=

```
(*plot settings*)
(*FigS2 G,H deflection and current residuals*)
PlotrangereY = {{0, 500}, {-100, 100}}

plotstyleY = {Darker[■]};
plotstyleP = {Darker[■]};
plotstyleV = {Darker[■]};

pady = 100;
padx = 70;
pad2 = 30;
fntsize = 16;


Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 500, 100}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 500, 25}];
XTicks = Union[Xticks1, Xticks3];
```

```
Yticks1 = Table[{k, k, {0, 0.03}}, {k, -100, 100, 50}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, -100, 100, 10}];
YTicks = Union[Yticks1, Yticks3];

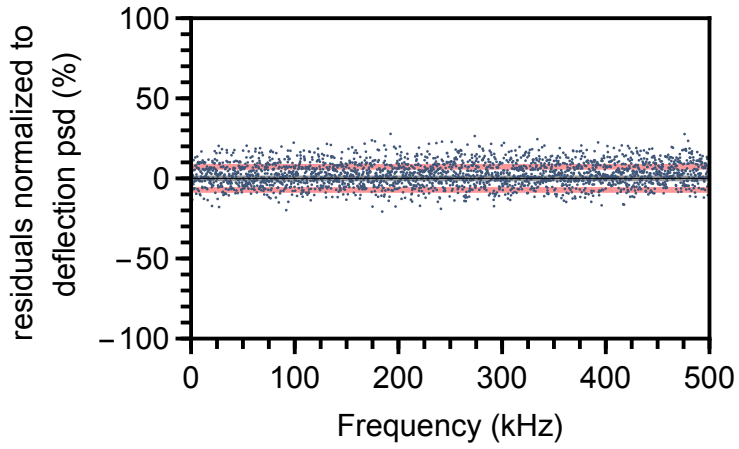
(*FigS2 plots of residuals for a linear IHB with state-independent noise*)
resalpsd =
 ListPlot[exppsd2, PlotRange → {{0, 1 / (1000 * 2 * deltatsampl)}, {-100, 100}},
  PlotStyle → plotstyleY, ImagePadding → {{pady, pad2}, {padx, pad2}},
  ImageSize → 400, LabelStyle → {FontSize → fntsize},
  GridLines → {{}, {{100 / Sqrt[ncopies], Directive[Red, AbsoluteThickness[3]]},
     {-100 / Sqrt[ncopies], Directive[Red, AbsoluteThickness[3]]},
     {0, Directive[Black, AbsoluteThickness[3]]}}}, Frame → True,
  Background → White, FrameStyle → Directive[Black, Thick], FrameLabel →
   {"Frequency (kHz)", "residuals normalized to \n deflection psd (%)"},
  FrameTicks → {{YTicks, None}, {XTicks, None}}]

exppsdP = Table[(powspmeansampledt1[[i]][[2]] - Paliased1[[i]][[2]]) * 100 / Paliased1[[i]][[2]],
   {i, 1, lengthpsd}];
exppsdP2 = Table[{powspmeansampledt1[[i]][[1]] / 1000, exppsdP[[i]]}, {i, 1, lengthpsd}];
resalpsdP =
 ListPlot[exppsdP2, PlotRange → {{0, 1 / (1000 * 2 * deltatsampl)}, {-100, 100}},
  PlotStyle → plotstyleP, ImagePadding → {{pady, pad2}, {padx, pad2}},
  ImageSize → 400, LabelStyle → {FontSize → fntsize},
  GridLines → {{}, {{100 / Sqrt[ncopies], Directive[Green, AbsoluteThickness[3]]},
     {-100 / Sqrt[ncopies], Directive[Green, AbsoluteThickness[3]]},
     {0, Directive[Black, AbsoluteThickness[3]]}}},
  Frame → True, Background → White, FrameStyle → Directive[Black, Thick],
  FrameLabel → {"Frequency (kHz)", "residuals normalized to \ncurrent psd (%)"},
  FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

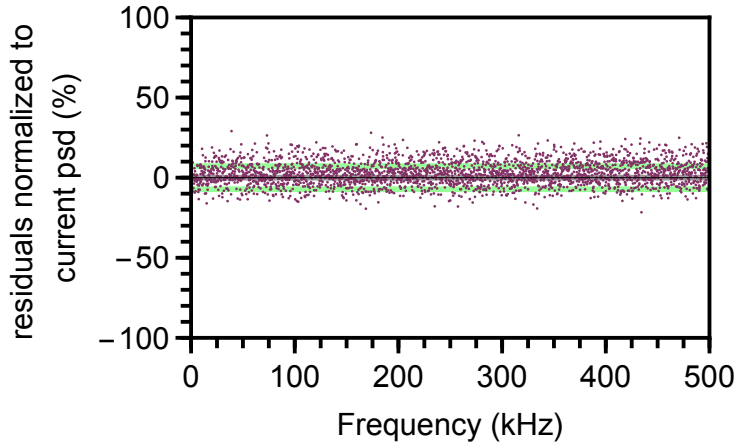
Out[ ]=

```
{{0, 500}, {-100, 100}}
```

Out[ ]=



Out[ ]=



In[ ]:=

```
(*FigS2 E,F. plots of deflectiona and current psd vs
  fourier frequency for a linear IHB with state-independent nosie*)

(*plot settings*)
Remove[XTicks]
Remove[YTicks]
TickLength = 0.02;
Xmin = 0;
Xmax = 10 - 1;
Ymin = 0;
Ymax = 20;
deltaY = 2;
plotrangepsd = {{10^(2), 10^6}, {0.5 * 10^(-12), 10^(-2)}};
plotrangecoher = {{10^(2), 10^6}, {-1.05, 1.05}};
```

```
framelabels =
  {{"power spectrum (\!\(\*SuperscriptBox[\(nm\), \(2\)]\)/Hz)", None},
   {"noise frequency (Hz)", None}};
TickLength1 = 0.03;
Yticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, -24, 14, 2}];
Yticks2 = Table[{10^k, Null, {0, TickLength1}}, {k, -23, 14, 2}];
TickLength2 = 0.02;
aa = Table[0, {38}, {4}];
For[k = 1, k ≤ 38, k++,
aa[[k]] = Table[{l * 10^(-k + 10), , {0, TickLength2}}, {l, 2, 8, 2}];
]
YTicks = Union[Yticks1, Yticks2, aa[[1]], aa[[2]], aa[[3]], aa[[4]], aa[[5]],
   aa[[6]], aa[[7]], aa[[8]], aa[[9]], aa[[10]], aa[[11]], aa[[12]], aa[[13]], aa[[14]],
   aa[[15]], aa[[16]], aa[[17]], aa[[18]], aa[[19]], aa[[20]], aa[[21]], aa[[22]], aa[[23]],
   aa[[24]], aa[[25]], aa[[26]], aa[[27]], aa[[28]], aa[[29]], aa[[30]], aa[[31]], aa[[32]]];

Xticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, 3, 8, 1}];
aa2 = Table[0, {20}, {4}];
For[k = 1, k ≤ 12, k++,
aa2[[k]] = Table[{l * 10^k, , {0, TickLength2}}, {l, 2, 8, 2}];
]

Xticks2 = {{100, 0, {0, 0.03}}};

XTicks = Union[Xticks1, Xticks2, aa2[[2]], aa2[[3]], aa2[[4]], aa2[[5]], aa2[[6]],
   aa2[[7]], aa2[[8]], aa2[[9]], aa2[[10]], aa2[[11]], aa2[[12]], aa2[[13]], aa2[[14]],
   aa2[[15]], aa2[[16]], aa2[[17]], aa2[[18]], aa2[[19]], aa2[[20]], Xticks2];


pady = 120;
padx = 70;
pad2 = 30;
fntsize = 18;


fntsize = 18;


plotstyleY = {Darker[■]};
plotstyleP = {Darker[■]};
plotstyleV = {Darker[■]};

meandataplot0c = {100, meandataplot0[[1]][[2]]};
meandataplot1c = {100, meandataplot1[[1]][[2]]};
```

```
meandataplot1cpsdc = {100, meandataplot1cpsd[[1]][[2]]};


commondirgridlines = {Dashed, Thickness → 0.01 / 2};
yLineStyle = Directive[Darker[■], Dashed, Thickness → 0.01 / 2];

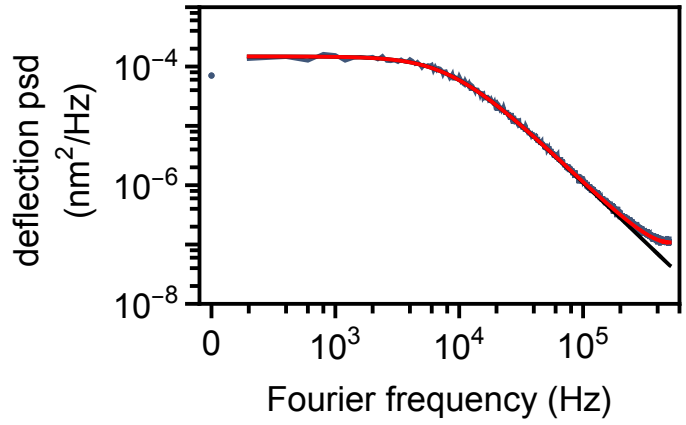
plotrangeY = {{80, 6 * 10^5}, {10^(-8), 10^-3}};

ListLogLogPlot[{meandataplot0},
  PlotStyle → plotstyleY, Joined → True, PlotRange → plotrangeY,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"Fourier frequency (Hz)", "deflection psd \n (nm^2/Hz)"}];
Show[%, ListLogLogPlot[{meandataplot0c}, PlotStyle → plotstyleY]];
Show[%, ListLogLogPlot[{theorPSDtaplota0}, PlotStyle → Black, Joined → True]];
Show[%, ListLogLogPlot[{Paliased0}, PlotStyle → Red, Joined → True]]

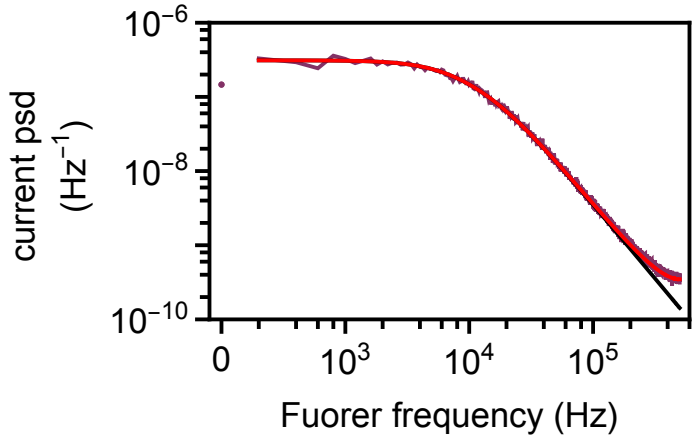
plotrangeP = {{80, 6 * 10^5}, {10^(-10), 10^(-6)}};

ListLogLogPlot[{meandataplot1},
  PlotStyle → plotstyleP, Joined → True, PlotRange → plotrangeP,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"Fuorer frequency (Hz)", "current psd \n (Hz^-1)"}];
Show[%, ListLogLogPlot[{meandataplot1c}, PlotStyle → plotstyleP]];
Show[%, ListLogLogPlot[{theorPSDtaplota1}, PlotStyle → Black, Joined → True]];
Show[%, ListLogLogPlot[{Paliased1}, PlotStyle → Red, Joined → True]]
```

Out[ ]=



Out[ ]=



In[ ]:=
```
(*Calculation of power spectra for linear IHB with state dependent noise*)

t1 = AbsoluteTime[];

SeedRandom[47 741]

Ncv = 32;

t1 = AbsoluteTime[];

SeedRandom[47 741]
gx =  2 * kbTd * lPd;
kGd = Ncv * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2);
lGd = Ncv * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2);
gy =  2 * kbTd * lGd;
y0in = Yrestd;
```

```
G0in = Dvd * Pinfrest;

proc =
 ItoProcess[lPd * dY[t] == (-kPd) * (Y[t] - Yupd) * dt - kGSd * (Y[t] - Yugd - g[t]) * dt +
     Sqrt[gx] * dn1[t] + 0 * Fzd * 180 * Sin[2 * Pi * wstd * (t - tstartFd)] * dt &&
   lGd * dg[t] == -kGd * (g[t] - Dvd * Pinfrest) * dt +
     kGSd * (Y[t] - Yrestd) * dt + Ncd * kbTd * (1 / (Pinfrest * Dvd * (1 - Pinfrest))) *
      Sqrt[taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd]]
      Sqrt[(Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] (1 - Clip[g[t] / Dvd, {0.0, 1.0}]) +
           (1 - Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd]) Clip[g[t] / Dvd, {0.0, 1.0}]) / Ncd]
      dn2[t], {Y[t], g[t]}, {{Y, g}, {y0in, G0in}}, t,
{n1 ≈ WienerProcess[], n2 ≈ WienerProcess[]}]
totallength = 2 * 0.507;
factot = 4;
deltatsima = 0.000001 / factot;
ap = RandomFunction[proc, {0, totallength, deltatsima}];

ap0 = ap["PathComponent", 1] - Yrestd;

ap1 = ap["PathComponent", 2] - Dvd * Pinfrest;

resampfactor = factot;

randtime100 = 0.005 * 10;
randtimenm = Round[randtime100, deltatsima];
stpoint = IntegerPart[randtime100 / deltatsima];

deltatsampl = Round[deltatsima * resampfactor, deltatsima];

tsampl = 0.005;
endpoint = IntegerPart[
   Length[ap["PathComponent", 1]["Values"]] - tsampl / deltatsima - stpoint];
tsamplnm = Round[tsampl, deltatsima];
b1 = Table[f[i], {i, randtimenm, tsamplnm + randtimenm, deltatsampl}];
lengthares = Length[b1];
Clear[ares];
totallength2 = totallength;
ncopiesa = (totallength2 - randtimenm) / (tsamplnm + deltatsampl);
ncopies = IntegerPart[ncopiesa];

ares0 = Table[0, {ncopies}];
ares1 = Table[0, {ncopies}];
sizea = lengthares;
fousampled0 = Table[0, {ncopies}, {lengthares}];
```

```
fousampled1 = Table[0, {ncopies}, {lengthares}];
cfousampled = Table[0, {ncopies}, {lengthares}];

Do[

  ares0[[ll]] = ap0["Values"][[(ll - 1) * tsampl / deltatsima + stpoint
     ;; IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];
    ares1[[ll]] = ap1["Values"][[(ll - 1) * tsampl / deltatsima + stpoint ;;
      IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]] / Dvd;

    fousampled0[[ll]] = ResourceFunction["WelchSpectralEstimate"][
    ares0[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
    "OverlapOffset" → None, "SegmentSize" → sizea];
    cfousampled[[ll]] = ResourceFunction["WelchSpectralEstimate"][
    ares1[[ll]], ares0[[ll]], sizea / tsampl, "Window" → None,
    "Detrend" → None, "OverlapOffset" → None, "SegmentSize" → sizea];
    fousampled1[[ll]] = ResourceFunction["WelchSpectralEstimate"][
    ares1[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
    "OverlapOffset" → None, "SegmentSize" → sizea];

    ,
{ll, ncopies}];
powspmeansampledt0 = Drop[Mean[fousampled0], -1];

powspmeansampledt1 = Drop[Mean[fousampled1], -1];

cpowspmeansampledt = Drop[Re[Mean[cfousampled]], -1];

lengthpsd = Length[powspmeansampledt0]
meandataplot0 =
  Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt0[[i]][[2]]}, {i, 1, lengthpsd}];
meandataplot1 =
  Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt1[[i]][[2]]}, {i, 1, lengthpsd}];
meandataplot1cpsd =
  Table[{cpowspmeansampledt[[i]][[1]], cpowspmeansampledt[[i]][[2]]}, {i, 1, lengthpsd}];

lengthpsd = Length[powspmeansampledt0]
PYY = (4 kbT (kG^2 lP + lG (kGS^2 + lG lP w^2))) / ((kGS^2 - kG (kGS + kP))^2 +
      ((kGS + kP)^2 lG^2 + 2 kGS^2 lG lP + kG^2 lP^2) w^2 + lG^2 lP^2 w^4);
PYG = (4 kbT kGS ((kGS + kP) lG + kG lP)) /
    ((kGS^2 - kG (kGS + kP))^2 +
      ((kGS + kP)^2 lG^2 + 2 kGS^2 lG lP + kG^2 lP^2) w^2 + lG^2 lP^2 w^4);
PGG = (4 kbT (2 kGS kP lG + kGS^2 (lG + lP) + lG (kP^2 + lP^2 w^2))) /
    ((kGS^2 - kG (kGS + kP))^2 +
```

```
      ((kGS + kP)^2 lG^2 + 2 kGS^2 lG lP + kG^2 lP^2) w^2 + lG^2 lP^2 w^4);

theorPSDtaplota0 =
  Table[{powspmeansampledt0[[i]][[1]], PYY /. {w → (2 * Pi * powspmeansampledt0[[i]][[1]]),
      lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
      kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2), kbT → kbTd,
      lP → lPd, kGS → kGSd, kP → kPd}}, {i, 1, lengthpsd}];
theorPSDtaplota1 = Table[{powspmeansampledt0[[i]][[1]],
    (PGG / (Dvd^2)) /. {w → (2 * Pi * powspmeansampledt0[[i]][[1]]),
      lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
      kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2), kbT → kbTd,
      lP → lPd, kGS → kGSd, kP → kPd}}, {i, 1, lengthpsd}];
theorCPSDtaplota = Table[
   {cpowspmeansampledt[[i]][[1]], (PYG / (Dvd)) /. {w → (2 * Pi * powspmeansampledt0[[i]][[1]]),
      lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
      kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2), kbT → kbTd,
      lP → lPd, kGS → kGSd, kP → kPd}}, {i, 1, lengthpsd}];

meandataplot0 =
  Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt0[[i]][[2]]}, {i, 1, lengthpsd}];
meandataplot1 =
  Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt1[[i]][[2]]}, {i, 1, lengthpsd}];
meandataplot1cpsd =
  Table[{cpowspmeansampledt[[i]][[1]], cpowspmeansampledt[[i]][[2]]}, {i, 1, lengthpsd}];

aliased0 = Table[0, lengthpsd];
aliased1 = Table[0, lengthpsd];
aliased01 = Table[0, lengthpsd];
Paliased0 = Table[0, lengthpsd];
Paliased1 = Table[0, lengthpsd];
CPSDaliased = Table[0, lengthpsd];

nser = 15;
fs = 1 / (deltatsampl);

aliased0 = Table[{powspmeansampledt0[[i]][[1]],
    (∑_{j=1}^{nser} (PYY /. {w → (2 * Pi * (powspmeansampledt0[[i]][[1]] + j * fs)),
          lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
          kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
          kbT → kbTd, lP → lPd, kGS → kGSd, kP → kPd}) +
      ∑_{j=1}^{nser} (PYY /. {w → (2 * Pi * (powspmeansampledt0[[i]][[1]] - j * fs)),
          lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
```

```
            kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2), kbT → kbTd,
            lP → lPd, kGS → kGSd, kP → kPd})) }, {i, 1, lengthpsd}];
Paliased0 = Table[{powspmeansampledt0[[i]][[1]],
    theorPSDtaplota0[[i]][[2]] + aliased0[[i]][[2]]}, {i, 1, lengthpsd}];

aliased1 = Table[{powspmeansampledt0[[i]][[1]],
    (∑j=1^nser ((PGG / (Dvd^2)) /. {w → (2 * Pi * (powspmeansampledt1[[i]][[1]] + j * fs)),
            lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
            kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
            kbT → kbTd, lP → lPd, kGS → kGSd, kP → kPd}) +
      ∑j=1^nser ((PGG / (Dvd^2)) /. {w → (2 * Pi * (powspmeansampledt1[[i]][[1]] - j * fs)),
            lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
            kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2), kbT → kbTd,
            lP → lPd, kGS → kGSd, kP → kPd})) }, {i, 1, lengthpsd}];
Paliased1 = Table[{powspmeansampledt1[[i]][[1]],
    theorPSDtaplota1[[i]][[2]] + aliased1[[i]][[2]]}, {i, 1, lengthpsd}];

aliased01 = Table[{powspmeansampledt0[[i]][[1]],
    (∑j=1^nser ((PYG / (Dvd)) /. {w → (2 * Pi * (cpowspmeansampledt[[i]][[1]] + j * fs)),
            lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
            kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
            kbT → kbTd, lP → lPd, kGS → kGSd, kP → kPd}) +
      ∑j=1^nser ((PYG / (Dvd)) /. {w → (2 * Pi * (cpowspmeansampledt[[i]][[1]] - j * fs)),
            lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
            kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2), kbT → kbTd,
            lP → lPd, kGS → kGSd, kP → kPd})) }, {i, 1, lengthpsd}];
CPSDaliased = Table[{powspmeansampledt1[[i]][[1]],
    theorCPSDtaplota[[i]][[2]] + aliased01[[i]][[2]]}, {i, 1, lengthpsd}];
```

Out[ ]=

```
RandomGeneratorState[ Method: ExtendedCA
                      State hash: 4828366434699046382 ]
```

Out[ ]=

```
RandomGeneratorState[ Method: ExtendedCA
                      State hash: 4828366434699046382 ]
```

Out[ ]=

$$\text{ItoProcess}\left[\left\{\{0.+27.061\,(-1872.\,(407.+Y[t])-468.\,(512.44-1.\,g[t]+Y[t])),\right.\right.$$
$$0.+88.7561\,(-899.828\,(-5.44+g[t])+468.\,(427.+Y[t]))\},\left\{\{468.927,0.\},\right.$$
$$\left\{0.,2608.39\sqrt{\left(\frac{1}{1.+e^{0.115189\,(-413.235-Y[t])}}\right)^{0.5}\left(1.-\frac{1.}{1.+e^{0.115189\,(-413.235-Y[t])}}\right)^{0.5}}\right.$$
$$\sqrt{\left(\frac{1.-1.\,\text{Clip}[0.03125\,g[t],\{0.,1.\}]}{1.+e^{0.115189\,(-413.235-Y[t])}}+\right.}$$
$$\left.\left.\left.\left(1.-\frac{1.}{1.+e^{0.115189\,(-413.235-Y[t])}}\right)\text{Clip}[0.03125\,g[t],\{0.,1.\}]\right)\right\}\right\},$$
$$\left.\{Y[t],g[t]\}\right\},\{\{Y,g\},\{-427,5.44\}\},\{t,0\}\right]$$

Out[ ]=

2501

Out[ ]=

2501

In[ ]:=

```
(*Calculation of residuals for a linear IHB with state dependent noise*)
(*residuals are defined as the fractional difference between simulated
 power spectra and their analytical expression including aliases*)
exppsd = Table[(powspmeansampledt0[[i]][[2]] - Paliased0[[i]][[2]]) * 100 / Paliased0[[i]][[2]],
   {i, 1, lengthpsd}];
exppsd2 = Table[{powspmeansampledt0[[i]][[1]] / 1000, exppsd[[i]]}, {i, 1, lengthpsd}];
exppsdCP = Table[(Abs[meandataplot1cpsd[[i]][[2]]] - Abs[CPSDaliased[[i]][[2]]]) *
    100 / Abs[CPSDaliased[[i]][[2]]], {i, 1, lengthpsd}];
exppsdCP2 = Table[{meandataplot1cpsd[[i]][[1]] / 1000, exppsdCP[[i]]}, {i, 1, lengthpsd}];


(*plot settings*)
(*FigS2 K,L deflection and current residuals*)
PlotrangereY = {{0, 500}, {-100, 100}}

plotstyleY = {Darker[■]};
plotstyleP = {Darker[■]};
plotstyleV = {Darker[■]};

pady = 100;
padx = 70;
pad2 = 30;
fntsize = 16;


Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 500, 100}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 500, 25}];
XTicks = Union[Xticks1, Xticks3];
```

```
Yticks1 = Table[{k, k, {0, 0.03}}, {k, -100, 100, 50}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, -100, 100, 10}];
YTicks = Union[Yticks1, Yticks3];

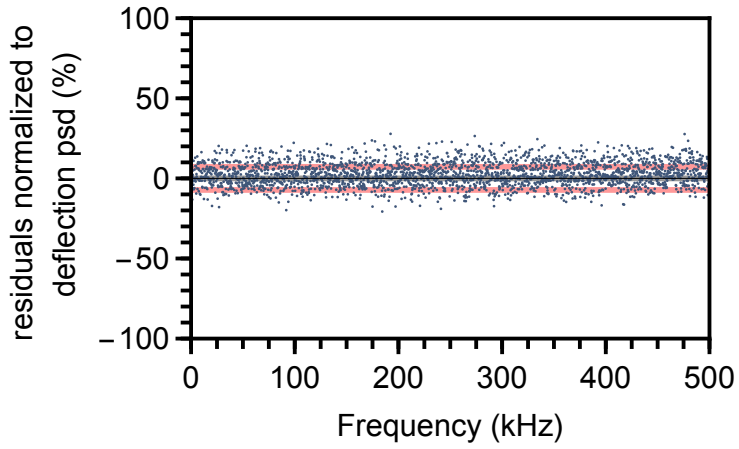
(*FigS2 plots of residuals for a linear IHB with state dependent noise*)
resalpsd =
 ListPlot[exppsd2, PlotRange → {{0, 1 / (1000 * 2 * deltatsampl)}, {-100, 100}},
  PlotStyle → plotstyleY, ImagePadding → {{pady, pad2}, {padx, pad2}},
  ImageSize → 400, LabelStyle → {FontSize → fntsize},
  GridLines → {{}, {{100 / Sqrt[ncopies], Directive[Red, AbsoluteThickness[3]]},
     {-100 / Sqrt[ncopies], Directive[Red, AbsoluteThickness[3]]},
     {0, Directive[Black, AbsoluteThickness[3]]}}}, Frame → True,
  Background → White, FrameStyle → Directive[Black, Thick], FrameLabel →
   {"Frequency (kHz)", "residuals normalized to \n deflection psd (%)"},
  FrameTicks → {{YTicks, None}, {XTicks, None}}]

exppsdP = Table[(powspmeansampledt1[[i]][[2]] - Paliased1[[i]][[2]]) * 100 / Paliased1[[i]][[2]],
   {i, 1, lengthpsd}];
exppsdP2 = Table[{powspmeansampledt1[[i]][[1]] / 1000, exppsdP[[i]]}, {i, 1, lengthpsd}];
resalpsdP =
 ListPlot[exppsdP2, PlotRange → {{0, 1 / (1000 * 2 * deltatsampl)}, {-100, 100}},
  PlotStyle → plotstyleP, ImagePadding → {{pady, pad2}, {padx, pad2}},
  ImageSize → 400, LabelStyle → {FontSize → fntsize},
  GridLines → {{}, {{100 / Sqrt[ncopies], Directive[Green, AbsoluteThickness[3]]},
     {-100 / Sqrt[ncopies], Directive[Green, AbsoluteThickness[3]]},
     {0, Directive[Black, AbsoluteThickness[3]]}}},
  Frame → True, Background → White, FrameStyle → Directive[Black, Thick],
  FrameLabel → {"Frequency (kHz)", "residuals normalized to \ncurrent psd (%)"},
  FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

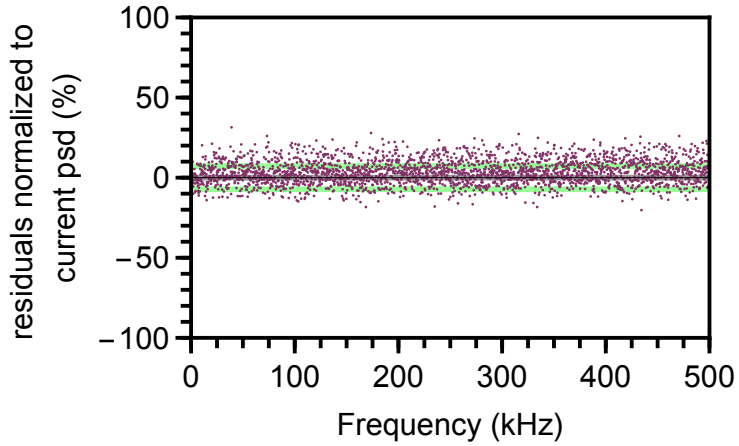
Out[•]=

```
{{0, 500}, {-100, 100}}
```

Out[ ]=



Out[ ]=



In[ ]:=

In[ ]:=

```
(*FigS2 I,J. plots of deflectiona and current psd vs
  fourier frequency for a linear IHB with state-dependent nosie*)

(*plot settings*)
Remove[XTicks]
Remove[YTicks]
TickLength = 0.02;
Xmin = 0;
Xmax = 10 - 1;
Ymin = 0;
Ymax = 20;
deltaY = 2;
```

```
plotrangepsd = {{10^(2), 10^6}, {0.5 * 10^(-12), 10^(-2)}};
plotrangecoher = {{10^(2), 10^6}, {-1.05, 1.05}};
framelabels =
  {{"power spectrum (\!\(\*SuperscriptBox[\(nm\), \(2\)]\)/Hz)", None},
   {"noise frequency (Hz)", None}};
TickLength1 = 0.03;
Yticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, -24, 14, 2}];
Yticks2 = Table[{10^k, Null, {0, TickLength1}}, {k, -23, 14, 2}];
TickLength2 = 0.02;
aa = Table[0, {38}, {4}];
For[k = 1, k ≤ 38, k++,
aa[[k]] = Table[{l * 10^(-k + 10), , {0, TickLength2}}, {l, 2, 8, 2}];
]
YTicks = Union[Yticks1, Yticks2, aa[[1]], aa[[2]], aa[[3]], aa[[4]], aa[[5]],
   aa[[6]], aa[[7]], aa[[8]], aa[[9]], aa[[10]], aa[[11]], aa[[12]], aa[[13]], aa[[14]],
   aa[[15]], aa[[16]], aa[[17]], aa[[18]], aa[[19]], aa[[20]], aa[[21]], aa[[22]], aa[[23]],
   aa[[24]], aa[[25]], aa[[26]], aa[[27]], aa[[28]], aa[[29]], aa[[30]], aa[[31]], aa[[32]]];

Xticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, 3, 8, 1}];
aa2 = Table[0, {20}, {4}];
For[k = 1, k ≤ 12, k++,
aa2[[k]] = Table[{l * 10^k, , {0, TickLength2}}, {l, 2, 8, 2}];
]

Xticks2 = {{100, 0, {0, 0.03}}};

XTicks = Union[Xticks1, Xticks2, aa2[[2]], aa2[[3]], aa2[[4]], aa2[[5]], aa2[[6]],
   aa2[[7]], aa2[[8]], aa2[[9]], aa2[[10]], aa2[[11]], aa2[[12]], aa2[[13]], aa2[[14]],
   aa2[[15]], aa2[[16]], aa2[[17]], aa2[[18]], aa2[[19]], aa2[[20]], Xticks2];


pady = 120;
padx = 70;
pad2 = 30;
fntsize = 18;


fntsize = 18;


plotstyleY = {Darker[■]};
plotstyleP = {Darker[■]};
plotstyleV = {Darker[■]};
```

```
meandataplot0c = {100, meandataplot0[[1]][[2]]};
meandataplot1c = {100, meandataplot1[[1]][[2]]};
meandataplot1cpsdc = {100, meandataplot1cpsd[[1]][[2]]};


commondirgridlines = {Dashed, Thickness → 0.01 / 2};
yLineStyle = Directive[Darker[■], Dashed, Thickness → 0.01 / 2];

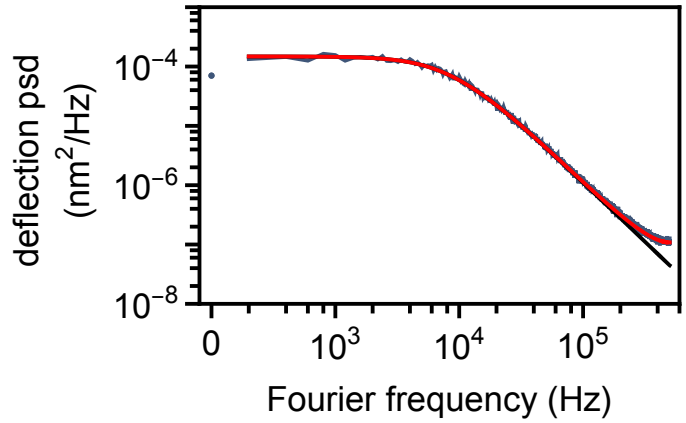
plotrangeY = {{80, 6 * 10^5}, {10^(-8), 10^-3}};

ListLogLogPlot[{meandataplot0},
  PlotStyle → plotstyleY, Joined → True, PlotRange → plotrangeY,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"Fourier frequency (Hz)", "deflection psd \n (nm^2/Hz)"}];
Show[%, ListLogLogPlot[{meandataplot0c}, PlotStyle → plotstyleY]];
Show[%, ListLogLogPlot[{theorPSDtaplota0}, PlotStyle → Black, Joined → True]];
Show[%, ListLogLogPlot[{Paliased0}, PlotStyle → Red, Joined → True]]

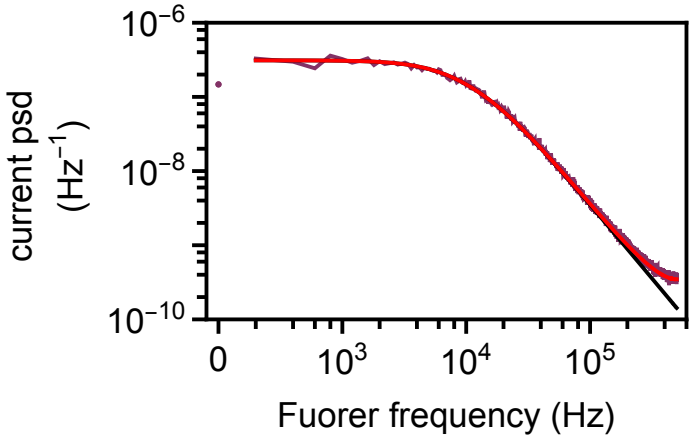
plotrangeP = {{80, 6 * 10^5}, {10^(-10), 10^(-6)}};

ListLogLogPlot[{meandataplot1},
  PlotStyle → plotstyleP, Joined → True, PlotRange → plotrangeP,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"Fuorer frequency (Hz)", "current psd \n (Hz^-1)"}];
Show[%, ListLogLogPlot[{meandataplot1c}, PlotStyle → plotstyleP]];
Show[%, ListLogLogPlot[{theorPSDtaplota1}, PlotStyle → Black, Joined → True]];
Show[%, ListLogLogPlot[{Paliased1}, PlotStyle → Red, Joined → True]]
```

Out[ ]=



Out[ ]=



In[ ]:=
```
meandataplot0
meandataplot1
```

Out[ ]=

{{0., 0.0000702428}, {200., 0.000133489}, {400., 0.000145061}, {600., 0.00012561}, {800., 0.000163825}, {1000., 0.000155966}, {1200., 0.000125434}, ··· 2488 ···, {499 000., 1.04424×10⁻⁷}, {499 200., 1.13081×10⁻⁷}, {499 400., 1.0487×10⁻⁷}, {499 600., 1.1463×10⁻⁷}, {499 800., 1.11046×10⁻⁷}, {500 000., 1.26803×10⁻⁷}}

Full expression not available (original memory size: 41.1 kB)

Out[ ]=

{{0., 0.000136036}, {200., 3.28892×10⁻⁷}, {400., 2.98009×10⁻⁷}, {600., 2.42911×10⁻⁷}, {800., 3.59814×10⁻⁷}, {1000., 3.2372×10⁻⁷}, {1200., 2.88805×10⁻⁷}, ··· 2488 ···, {499 000., 3.81615×10⁻¹⁰}, {499 200., 3.82953×10⁻¹⁰}, {499 400., 3.52623×10⁻¹⁰}, {499 600., 3.26347×10⁻¹⁰}, {499 800., 3.6954×10⁻¹⁰}, {500 000., 3.16363×10⁻¹⁰}}

Full expression not available (original memory size: 41.1 kB)

In[ ]:=
```
(*Calculation of signal-to-noise ratio (SNR)*)
```

```
fmult1 = {0, 1, 5, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110,
  120, 130, 140, 150, 160, 170, 180, 200} (* amplitude force factor*)
(*table of deflection, current and votlage powers*)

powY = Table[0, {Length[fmult1]}];
powP = Table[0, {Length[fmult1]}];
powV = Table[0, {Length[fmult1]}];

SetSharedVariable[powY];
SetSharedVariable[powP];
SetSharedVariable[powV];

t1 = AbsoluteTime[];

ParallelDo[


  SeedRandom[47 741];

gx =  2 * kbTd * lPd;
  y0in = Yrestd;



  proc = ItoProcess[
    lPd * dY[t] == (-kPd) * (Y[t] - Yupd) * dt - kGSd * (Y[t] - Yugd - Dvd * Po[t]) * dt +
       Sqrt[gx] * dn1[t] + Fzd * fmult1[[pp]] * Sin[2 * Pi * wstd * (t - tstartFd)] * dt &&
     taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd] * dPo[t] ==
      (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] - Po[t]) dt +
       Sqrt[taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd]] Sqrt[
         (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] (1 - Clip[Po[t], {0.0, 1.0}]) + (1 - Pinf2[
                 Yhd, kGSd, Dvd, kbTd, Ncd]) Clip[Po[t], {0.0, 1.0}]) / Ncd] dn2[t],
    {Y[t], Po[t]}, {{Y, Po}, {y0in, Pinfrest}}, t,
{n1 ≈ WienerProcess[], n2 ≈ WienerProcess[]}];


  totallength = 2 * 0.507;
  factot = 4;
  deltatsima = 0.000001 / factot;
   ap = RandomFunction[proc, {0, totallength, deltatsima}];


  ap0 = ap["PathComponent", 1] - Yrestd;
```

```
ap1 = ap["PathComponent", 2] - Pinfrest;


resampfactor = 4;

  deltatsampl = Round[deltatsima * resampfactor, deltatsima];

  randtime100 = 0.005 * 10;

  stpoint = IntegerPart[randtime100 / deltatsima];

  tsampl = 0.005;

  randtimenm = Round[randtime100, deltatsima];
  tsamplnm = Round[tsampl, deltatsima];

  b1 = Table[f[i], {i, randtimenm, tsamplnm + randtimenm, deltatsampl}];
lengthares = Length[b1];

  sizea = lengthares;

  (*Yrestd=-421.681;*)(*channels blocked D=32 nm and Po=0.17*)
  (*Yrestd=-407;*)(*gating spring cut D=32 nm and Po=0.17*)
  ap0 = ap["PathComponent", 1] - Yrestd; (*ONLY§ FOR NONLIN CASE*)

 ap1 = ap["PathComponent", 2] - Pinfrest; resampfactor = factot;
randtime100 = 0.005 * 10;
randtimenm = Round[randtime100, deltatsima];
stpoint = IntegerPart[randtime100 / deltatsima];
deltatsampl = Round[deltatsima * resampfactor, deltatsima];
tsampl = 0.005;
endpoint = IntegerPart[
    Length[ap["PathComponent", 1]["Values"]] - tsampl / deltatsima - stpoint];
tsamplnm = Round[tsampl, deltatsima];
b1 = Table[f[i], {i, randtimenm, tsamplnm + randtimenm, deltatsampl}];
lengthares = Length[b1];
Clear[ares];
totallength2 = totallength;
ncopiesa = (totallength2 - randtimenm) / (tsamplnm + deltatsampl);
ncopies = IntegerPart[ncopiesa];
ares0 = Table[0, {ncopies}];
ares1 = Table[0, {ncopies}];
sizea = lengthares;
```

```
fousampled0 = Table[0, {ncopies}, {lengthares}];
fousampled1 = Table[0, {ncopies}, {lengthares}];
cfousampled = Table[0, {ncopies}, {lengthares}];

  Do[

   ares0[[ll]] = ap0["Values"][[(ll - 1) * tsampl / deltatsima + stpoint ;;
      IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];
    ares1[[ll]] = ap1["Values"][[(ll - 1) * tsampl / deltatsima + stpoint ;;
      IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];



       fousampled0[[ll]] = ResourceFunction["WelchSpectralEstimate"][
     ares0[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
     "OverlapOffset" → None, "SegmentSize" → sizea];
    cfousampled[[ll]] = ResourceFunction["WelchSpectralEstimate"][
     ares1[[ll]], ares0[[ll]], sizea / tsampl, "Window" → None,
     "Detrend" → None, "OverlapOffset" → None, "SegmentSize" → sizea];
    fousampled1[[ll]] = ResourceFunction["WelchSpectralEstimate"][
     ares1[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
     "OverlapOffset" → None, "SegmentSize" → sizea];


      ,
{ll, ncopies}];
  powspmeansampledt0 = Drop[Mean[fousampled0], -1];

  powspmeansampledt1 = Drop[Mean[fousampled1], -1];

lengthpsd = Length[powspmeansampledt0];


  meandataplot0 =
   Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt0[[i]][[2]]}, {i, 1, lengthpsd}];
  meandataplot1 =
   Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt1[[i]][[2]]}, {i, 1, lengthpsd}];

  fc = 610;
  filtersq1 = Table[{meandataplot0[[i]][[1]],
     Abs[1 / (1 + I * meandataplot1[[i]][[1]] / fc)]^2}, {i, 1, Length[meandataplot0]}];
  meandataplot1filt = Table[{meandataplot0[[i]][[1]],
     filtersq1[[i]][[2]] * meandataplot1[[i]][[2]]}, {i, 1, Length[filtersq1]}];
```

```
    powY[[pp]] = Integrate[
      Interpolation[meandataplot0, InterpolationOrder → 1][x], {x, 0, 5 * 10^5}];
    powP[[pp]] = Integrate[
      Interpolation[meandataplot1, InterpolationOrder → 1][x], {x, 0, 5 * 10^5}];
    powV[[pp]] = Integrate[
      Interpolation[meandataplot1filt, InterpolationOrder → 1][x], {x, 0, 5 * 10^5}];



    ,
    {pp, 1, Length[fmult1]}];


SNRY = Table[{fmult1[[i]], -1 + powY[[i]] / powY[[1]]}, {i, 2, Length[fmult1]}];
SNRP = Table[{fmult1[[i]], -1 + powP[[i]] / powP[[1]]}, {i, 2, Length[fmult1]}];
SNRV = Table[{fmult1[[i]], -1 + powV[[i]] / powV[[1]]}, {i, 2, Length[fmult1]}];

t2 = AbsoluteTime[];
t2 - t1
```

Out[ ]=

```
{0, 1, 5, 10, 20, 30, 40, 50, 60, 70, 80,
 90, 100, 110, 120, 130, 140, 150, 160, 170, 180, 200}
```

Out[ ]=

```
72.051713
```

In[ ]:=

In[ ]:=

In[ ]:=

In[ ]:=

```
(*plot settings*)

Remove[XTicks]
Remove[YTicks]
TickLength = 0.02;
Xmin = 0;
Xmax = 10 - 1;
Ymin = 0;
Ymax = 20;
deltaY = 2;
plotrangepsd = {{10^(2), 10^6}, {0.5 * 10^(-12), 10^(-2)}};
plotrangecoher = {{10^(2), 10^6}, {-1.05, 1.05}};
framelabels =
```

```
  {{"power spectrum (\!\(\*SuperscriptBox[\(nm\), \(2\)]\)/Hz)", None},
   {"noise frequency (Hz)", None}};
TickLength1 = 0.03;
Yticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, -24, 10, 2}];
Yticks2 = Table[{10^k, Null, {0, TickLength1}}, {k, -23, 10, 2}];
TickLength2 = 0.02;
aa = Table[0, {38}, {4}];
For[k = 1, k ≤ 38, k++,
aa[[k]] = Table[{l * 10^(-k + 10), , {0, TickLength2}}, {l, 2, 8, 2}];
]
YTicks = Union[Yticks1, Yticks2, aa[[1]], aa[[2]], aa[[3]],
   aa[[4]], aa[[5]], aa[[6]], aa[[7]], aa[[8]], aa[[9]], aa[[10]], aa[[11]], aa[[12]],
   aa[[13]], aa[[14]], aa[[15]], aa[[16]], aa[[17]], aa[[18]], aa[[19]], aa[[20]],
   aa[[21]], aa[[22]], aa[[23]], aa[[24]], aa[[25]], aa[[26]], aa[[27]], aa[[28]]];

pady = 100;
padx = 70;
pad2 = 30;
fntsize = 18;

frameticksy = {{{1 / 100 000, Superscript[10, -5], {0, 0.03}},
    {1 / 10 000, Superscript[10, -4], {0, 0.03}}, {1 / 1000, Superscript[10, -3],
     {0, 0.03}}, {1 / 100, Superscript[10, -2], {0, 0.03}}}, None};
frameticksyENLAR =
  {{{1 / 100 000, Superscript[10, -5], {0, 0.03}}, {1 / 10 000, Superscript[10, -4],
     {0, 0.03}}, {1 / 1000, Superscript[10, -3], {0, 0.03}},
    {2 * 10^-5, , {0, 0.02}}, {4 * 10^-5, , {0, 0.02}}, {6 * 10^-5, , {0, 0.02}},
    {8 * 10^-5, , {0, 0.02}}, {2 * 10^-4, , {0, 0.02}}, {4 * 10^-4, , {0, 0.02}},
    {6 * 10^-4, , {0, 0.02}}, {8 * 10^-4, , {0, 0.02}}}, None};
frameticksx =
  {{{1000, Superscript[10, 3], {0, 0.03}}, {10 000, Superscript[10, 4], {0, 0.03}},
    {100 000, Superscript[10, 5], {0, 0.03}}, {2 * 10^3, , {0, 0.02}},
    {4 * 10^3, , {0, 0.02}}, {6 * 10^3, , {0, 0.02}}, {8 * 10^3, , {0, 0.02}},
    {2 * 10^4, , {0, 0.02}}, {4 * 10^4, , {0, 0.02}},
    {6 * 10^4, , {0, 0.02}}, {8 * 10^4, , {0, 0.02}}}, None};
frameticksycoher = {{{0, 0, {0, 0.03}}, {0.2, 0.2, {0, 0.03}}, {0.4, 0.4, {0, 0.03}},
    {0.6, 0.6, {0, 0.03}}, {0.8, 0.8, {0, 0.03}}, {1.0, "1.0", {0, 0.03}}}, None};
frameticksycoherneg =
  {{{0, 0, {0, 0.03}}, {0.5, 0.5, {0, 0.03}}, {1.0, "1.0", {0, 0.03}}, {-0.5, -0.5,
     {0, 0.03}}, {-1.0, "-1.0", {0, 0.03}}, {-1.5, "-1.5", {0, 0.03}}}, None};

fntsize = 18;
```

In[ ]:=

In[ ]:=
```
(*Fig 8 plots of SNR for a gating swing of 32 nm and resting current of 0.17. Set D=
 32 and Prest=0.17 in parameters. *)

plotrangeSNRY = {Full, Full};
plotstyleSNRY = {{Thickness → 0.01, Darker[■]}, {Thickness → 0.01, Darker[■]}};
framelabelsSNRY =
  {{"deflection SNR", None}, {"stimulus-force amplitude (pN)", None}};

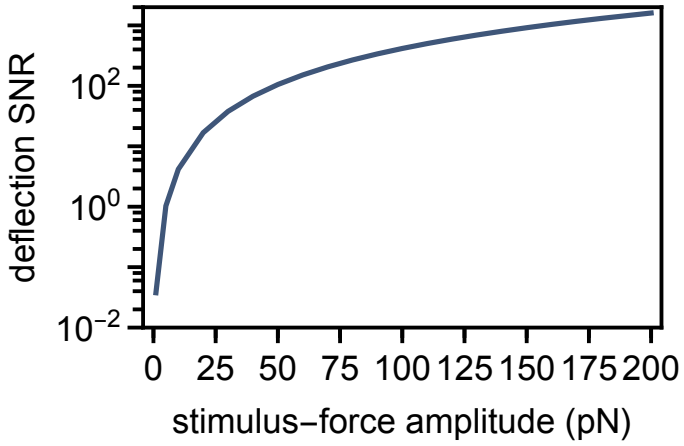

plotrangeSNRP = {{10^-1, 11}, Full};
plotstyleSNRP = {{Thickness → 0.01, Darker[■]}, {Thickness → 0.01, Darker[■]}};
framelabelsSNRP =
  {{"current or voltage SNR", None}, {"stimulus-force amplitude (pN)", None}};


Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1000, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1000, 25}];
XTicks = Union[Xticks1, Xticks3];

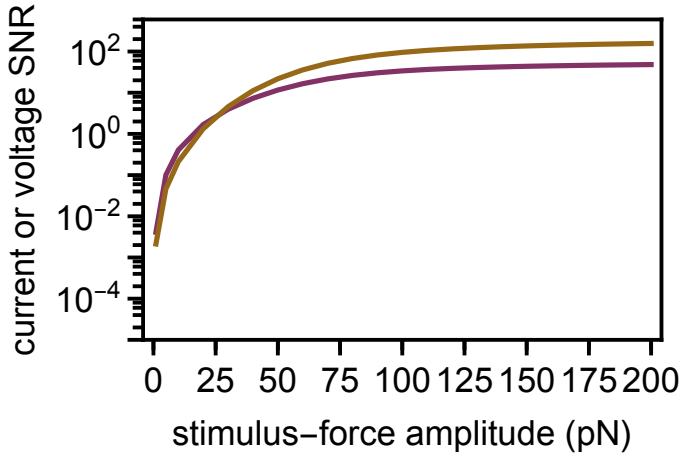

ListLogPlot[{SNRY}, Joined → True, PlotRange → {All, {10^-2, 2 * 10^3}},
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleSNRY, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsSNRY, FrameTicks → {{YTicks, None}, {XTicks, None}}]

ListLogPlot[{SNRP, SNRV}, PlotRange → {All, {10^-5, 6 * 10^2}}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleSNRP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsSNRP,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

Out[ ]=


deflection SNR
10^2
10^0
10^{-2}
0 25 50 75 100 125 150 175 200
stimulus−force amplitude (pN)

Out[ ]=


current or voltage SNR
10^2
10^0
10^{-2}
10^{-4}
0 25 50 75 100 125 150 175 200
stimulus−force amplitude (pN)

```
(*Fig 8 plots of SNR for a gating swing of 32 nm and resting current of 0.5. Set D=
 1.75 and Prest=0.5 in parameters. *)

plotrangeSNRY = {Full, Full};
plotstyleSNRY = {{Thickness → 0.01, Darker[■]}, {Thickness → 0.01, Darker[■]}};
framelabelsSNRY =
  {{"deflection SNR", None}, {"stimulus-force amplitude (pN)", None}};

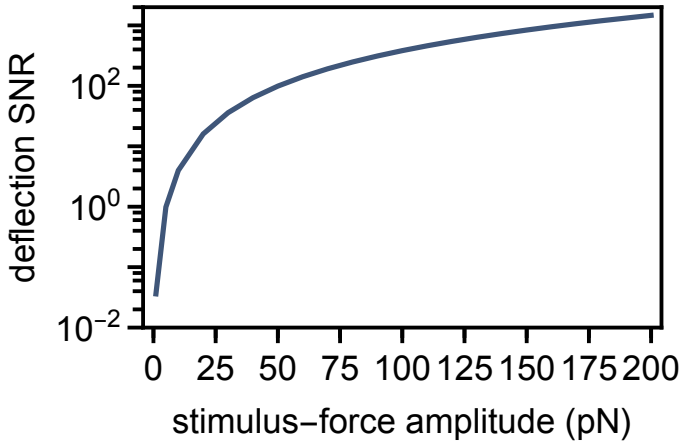

plotrangeSNRP = {{10^-1, 11}, Full};
plotstyleSNRP = {{Thickness → 0.01, Darker[■]}, {Thickness → 0.01, Darker[■]}};
framelabelsSNRP =
  {{"current or voltage SNR", None}, {"stimulus-force amplitude (pN)", None}};


Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1000, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1000, 25}];
XTicks = Union[Xticks1, Xticks3];

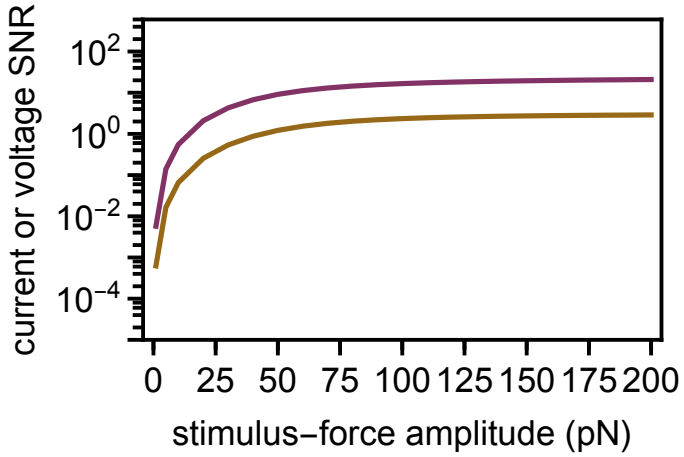

ListLogPlot[{SNRY}, Joined → True, PlotRange → {All, {10^-2, 2 * 10^3}},
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleSNRY, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsSNRY, FrameTicks → {{YTicks, None}, {XTicks, None}}]

ListLogPlot[{SNRP, SNRV}, PlotRange → {All, {10^-5, 6 * 10^2}}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleSNRP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsSNRP,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

Out[ ]=


deflection SNR
10^2
10^0
10^{-2}
0 25 50 75 100 125 150 175 200
stimulus−force amplitude (pN)

Out[ ]=


current or voltage SNR
10^2
10^0
10^{-2}
10^{-4}
0 25 50 75 100 125 150 175 200
stimulus−force amplitude (pN)

In[•]:=
```
(*Fig 8 plots of SNR for a gating swing of 1.75 nm and resting current of 0.17. Set D=
 1.75 and Prest=0.17 in parameters. *)

plotrangeSNRY = {Full, Full};
plotstyleSNRY = {{Thickness → 0.01, Darker[■]}, {Thickness → 0.01, Darker[■]}};
framelabelsSNRY =
  {{"deflection SNR", None}, {"stimulus-force amplitude (pN)", None}};

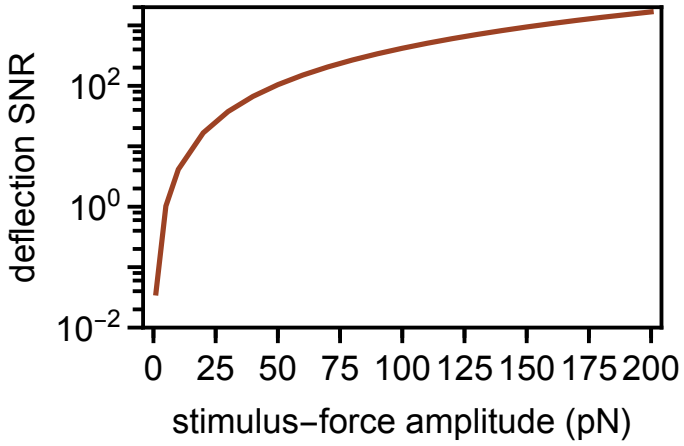
plotrangeSNRP = {{10^-1, 11}, Full};
plotstyleSNRP = {{Thickness → 0.01, Darker[■]}, {Thickness → 0.01, Darker[■]}};
framelabelsSNRP =
  {{"current or voltage SNR", None}, {"stimulus-force amplitude (pN)", None}};


Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1000, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1000, 25}];
XTicks = Union[Xticks1, Xticks3];

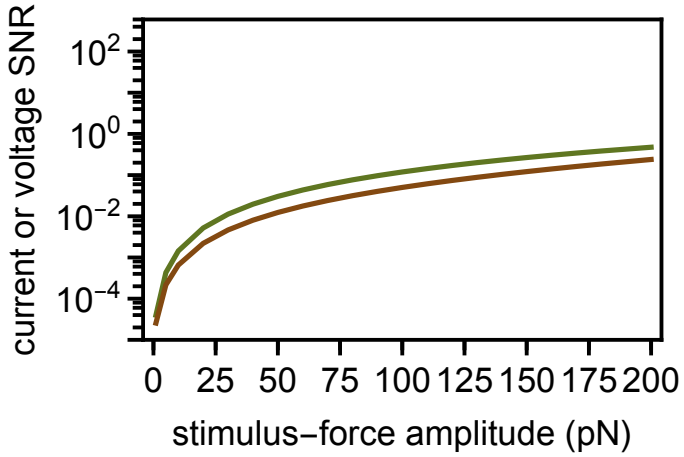

ListLogPlot[{SNRY}, Joined → True, PlotRange → {All, {10^-2, 2 * 10^3}},
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleSNRY, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsSNRY, FrameTicks → {{YTicks, None}, {XTicks, None}}]

ListLogPlot[{SNRP, SNRV}, PlotRange → {All, {10^-5, 6 * 10^2}}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleSNRP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsSNRP,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

Out[ ]=


deflection SNR
10^2
10^0
10^{-2}
0 25 50 75 100 125 150 175 200
stimulus−force amplitude (pN)

Out[ ]=


current or voltage SNR
10^2
10^0
10^{-2}
10^{-4}
0 25 50 75 100 125 150 175 200
stimulus−force amplitude (pN)

In[•]:=
```
(*Fig 8 plots of SNR for a gating swing of 1.75 nm and resting current of 0.5. Set D=
 1.75 and Prest=0.5 in parameters. *)

plotrangeSNRY = {Full, Full};
plotstyleSNRY = {{Thickness → 0.01, Darker[■]}, {Thickness → 0.01, Darker[■]}};
framelabelsSNRY =
  {{"deflection SNR", None}, {"stimulus-force amplitude (pN)", None}};

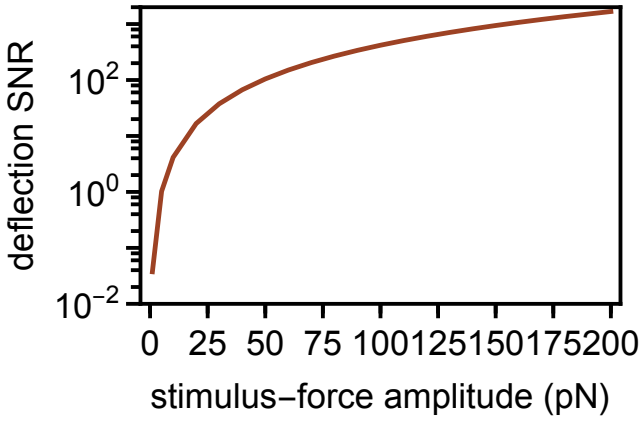
plotrangeSNRP = {{10^-1, 11}, Full};
plotstyleSNRP = {{Thickness → 0.01, Darker[■]}, {Thickness → 0.01, Darker[■]}};
framelabelsSNRP =
  {{"current or voltage SNR", None}, {"stimulus-force amplitude (pN)", None}};


Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1000, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1000, 25}];
XTicks = Union[Xticks1, Xticks3];

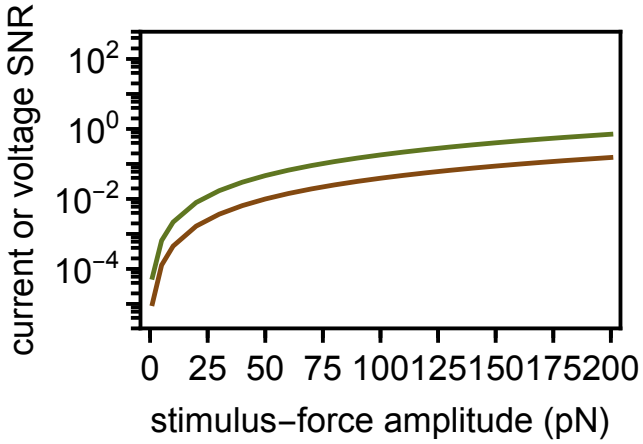

ListLogPlot[{SNRY}, Joined → True, PlotRange → {All, {10^-2, 2 * 10^3}},
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleSNRY, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsSNRY, FrameTicks → {{YTicks, None}, {XTicks, None}}]

ListLogPlot[{SNRP, SNRV}, PlotRange → {All, {2 * 10^-6, 6 * 10^2}}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleSNRP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsSNRP,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

Out[ ]=


deflection SNR
10^2
10^0
10^{-2}
0 25 50 75 100125150175200
stimulus−force amplitude (pN)

Out[ ]=


current or voltage SNR
10^2
10^0
10^{-2}
10^{-4}
0 25 50 75 100125150175200
stimulus−force amplitude (pN)

In[•]:=
```
(*Calculation of linear SNR*)
chi00 = -(kg + I * w * lg) / (kgs^2 - (kg + I * w * lg) * (kgs + kp + I * w * lp));
(* element 0,0 of  susceptibility matrix*)
chi10 = -kgs / (kgs^2 - (kg + I * w * lg) * (kgs + kp + I * w * lp));
(* element 1,0 of  susceptibility matrix*)
pows = FullSimplify[ComplexExpand[Im[chi00]^2]];
(* power in the deflection signal divided by the force*)
powp = FullSimplify[ComplexExpand[Im[chi10]^2]];
(* power in the current signal divided by the force*)
dydy = kbt * kg / (-kgs^2 + kg (kgs + kp)) (* power in the deflection fluctuations*)
dpdp = kbt (kgs + kp) / (-kgs^2 + kg (kgs + kp)) (* power in the current fluctuations*)
snrtemp = FullSimplify[(pows / dydy) /. {kg → Nc * kbt / (Pinf * (1 - Pinf) * D^2),
    lg → Nc * kbt * taoinf / (Pinf * (1 - Pinf) * D^2)}]
snrPtemp = FullSimplify[(powp / dpdp) /. {kg → Nc * kbt / (Pinf * (1 - Pinf) * D^2),
    lg → Nc * kbt * taoinf / (Pinf * (1 - Pinf) * D^2)}]
```

Out[•]=
$$\frac{kbt\ kg}{-kgs^2 + kg\ (kgs + kp)}$$

Out[•]=
$$\frac{kbt\ (kgs + kp)}{-kgs^2 + kg\ (kgs + kp)}$$

Out[•]=
$$\left(Nc\ \left(kbt\ (kgs + kp)\ Nc + D^2\ kgs^2\ (-1 + Pinf)\ Pinf\right)\right.$$
$$\left.w^2\ \left(D^2\ kgs^2\ (-1 + Pinf)\ Pinf\ taoinf - kbt\ lp\ Nc\ \left(1 + taoinf^2\ w^2\right)\right)^2\right) /$$
$$\left(D^4\ kgs^4\ (-1 + Pinf)^2\ Pinf^2 + 2\ D^2\ kbt\ kgs^2\ Nc\ (-1 + Pinf)\ Pinf\ \left(kgs + kp - lp\ taoinf\ w^2\right) + \right.$$
$$\left.kbt^2\ Nc^2\ \left((kgs + kp)^2 + lp^2\ w^2\right)\ \left(1 + taoinf^2\ w^2\right)\right)^2$$

Out[•]=
$$-\left(\left(D^2\ kbt\ kgs^2\ Nc^2\ (-1 + Pinf)\ Pinf\right.\right.$$
$$\left.\left(kbt\ (kgs + kp)\ Nc + D^2\ kgs^2\ (-1 + Pinf)\ Pinf\right)\ \left(lp + (kgs + kp)\ taoinf\right)^2\ w^2\right) /$$
$$\left((kgs + kp)\ \left(D^4\ kgs^4\ (-1 + Pinf)^2\ Pinf^2 + 2\ D^2\ kbt\ kgs^2\ Nc\ (-1 + Pinf)\ Pinf\right.\right.$$
$$\left.\left.\left(kgs + kp - lp\ taoinf\ w^2\right) + kbt^2\ Nc^2\ \left((kgs + kp)^2 + lp^2\ w^2\right)\ \left(1 + taoinf^2\ w^2\right)\right)^2\right)\Bigg)$$

In[ ]:=

```
(*definition of linear snr*)
snr[Nc_, D_, kbt_, kgs_, kp_, Pinf_, taoinf_, lp_, w_] :=
  (Nc (kbt (kgs + kp) Nc + D^2 kgs^2 (-1 + Pinf) Pinf) w^2
     (D^2 kgs^2 (-1 + Pinf) Pinf taoinf - kbt lp Nc (1 + taoinf^2 w^2))^2) /
   (D^4 kgs^4 (-1 + Pinf)^2 Pinf^2 + 2 D^2 kbt kgs^2 Nc (-1 + Pinf) Pinf (kgs + kp - lp taoinf w^2) +
      kbt^2 Nc^2 ((kgs + kp)^2 + lp^2 w^2) (1 + taoinf^2 w^2))^2;
snrp[Nc_, D_, kbt_, kgs_, kp_, Pinf_, taoinf_, lp_, w_] :=
  -((D^2 kbt kgs^2 Nc^2 (-1 + Pinf) Pinf (kbt (kgs + kp) Nc + D^2 kgs^2 (-1 + Pinf) Pinf)
       (lp + (kgs + kp) taoinf)^2 w^2) /
     ((kgs + kp) (D^4 kgs^4 (-1 + Pinf)^2 Pinf^2 + 2 D^2 kbt kgs^2 Nc (-1 + Pinf) Pinf
          (kgs + kp - lp taoinf w^2) + kbt^2 Nc^2 ((kgs + kp)^2 + lp^2 w^2) (1 + taoinf^2 w^2))^2));

(*definition of power in the deflection signal*)
powsf[kgs_, Nc_, kp_, w_, lp_, kbt_, Pinf_, taoinf_, D_] :=
  (kbt^2 Nc^2 w^2 (D^2 kgs^2 (-1 + Pinf) Pinf taoinf - kbt lp Nc (1 + taoinf^2 w^2))^2) /
   (D^4 kgs^4 (-1 + Pinf)^2 Pinf^2 + 2 D^2 kbt kgs^2 Nc (-1 + Pinf) Pinf (kgs + kp - lp taoinf w^2) +
      kbt^2 Nc^2 ((kgs + kp)^2 + lp^2 w^2) (1 + taoinf^2 w^2))^2;

(*definition of power in the current signal*)
powpf[kgs_, Nc_, kp_, w_, lp_, kbt_, Pinf_, taoinf_, D_] :=
  (D^4 kbt^2 kgs^2 Nc^2 (-1 + Pinf)^2 Pinf^2 (lp + (kgs + kp) taoinf)^2 w^2) /
   (D^4 kgs^4 (-1 + Pinf)^2 Pinf^2 + 2 D^2 kbt kgs^2 Nc (-1 + Pinf) Pinf (kgs + kp - lp taoinf w^2) +
      kbt^2 Nc^2 ((kgs + kp)^2 + lp^2 w^2) (1 + taoinf^2 w^2))^2;

(*definition of deflection fluctuations*)
dydyf[kbt_, Nc_, kgs_, kp_, Pinf_, D_] := kbt^2 Nc / (kbt (kgs + kp) Nc + D^2 kgs^2 (-1 + Pinf) Pinf);

(*definition of current fluctuations*)
dpdpf[kbt_, Nc_, kgs_, kp_, Pinf_, D_] := -(D^2 kbt (kgs + kp) (-1 + Pinf) Pinf / (kbt (kgs + kp) Nc + D^2 kgs^2 (-1 + Pinf) Pinf));
```

In[•]:=

```
(*table of values of linear snr,
power in the signal and fluctuation power as a function of D*)

Dvd1 = {1.75, 5, 10, 15, 20, 25, 30, 32, 35, 40, 45, 50, 55, 60, 65,
   70, 73.33323348898698`, 75, 77.56266383324578`, 80, 85, 90, 95, 97};

snrvalsD = Table[{Dvd1[[pp]], (Fzd * 1) ^ 2 * snr[Ncd, Dvd1[[pp]], kbTd, kGSd,
      kPd, Pinfrest, taorest, lPd, 2 * Pi * wstd]}, {pp, 1, Length[Dvd1]}];

snrPvalsD =
  Table[{Dvd1[[pp]], (Fzd * 1) ^ 2 * snrp[Ncd, Dvd1[[pp]], kbTd, kGSd, kPd, Pinfrest,
      taorest, lPd, 2 * Pi * wstd]}, {pp, 1, Length[Dvd1]}];


dydyfD = Table[{Dvd1[[pp]], dydyf[kbTd, Ncd, kGSd, kPd, Pinfrest, Dvd1[[pp]]]},
   {pp, 1, Length[Dvd1]}];

dpdpfD = Table[{Dvd1[[pp]], dpdpf[kbTd, Ncd, kGSd, kPd, Pinfrest, Dvd1[[pp]]]},
   {pp, 1, Length[Dvd1]}];

powsfD = Table[{Dvd1[[pp]], (Fzd * 1) ^ 2 * powsf[kGSd, Ncd, kPd, 2 * Pi * wstd,
      lPd, kbTd, Pinfrest, taorest, Dvd1[[pp]]]}, {pp, 1, Length[Dvd1]}];

powpfD = Table[{Dvd1[[pp]], (Fzd * 1) ^ 2 * powpf[kGSd, Ncd, kPd, 2 * Pi * wstd,
      lPd, kbTd, Pinfrest, taorest, Dvd1[[pp]]]}, {pp, 1, Length[Dvd1]}];
```

In[ ]:=
```
(*plot settings*)

Remove[XTicks]
Remove[YTicks]
TickLength = 0.02;
Xmin = 0;
Xmax = 10 - 1;
Ymin = 0;
Ymax = 20;
deltaY = 2;
plotrangepsd = {{10^(2), 10^6}, {0.5 * 10^(-12), 10^(-2)}};
plotrangecoher = {{10^(2), 10^6}, {-1.05, 1.05}};
framelabels =
  {{"power spectrum (\!\(\*SuperscriptBox[\(nm\), \(2\)]\)/Hz)", None},
   {"noise frequency (Hz)", None}};
TickLength1 = 0.03;

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 0.05, 0.01}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 0.05, 0.01}];
YTicks = Union[Yticks1, Yticks3];

pady = 100;
padx = 70;
pad2 = 30;
fntsize = 18;
```

In[ ]:=
```
(*Fig 3 A-B plots of linear SNR vs gating swing. *)
pady = 100;
padx = 70; plotrangeSNRY = {Full, Full};
plotstyleSNRY = {{Thickness → 0.01, Lighter[■]}, {Thickness → 0.01, Lighter[■]}};
framelabelsSNRY = {{"deflection SNR", None}, {"gating swing (nm)", None}};


plotrangeSNRP = {Full, Full};
plotstyleSNRP = {{Thickness → 0.01, Lighter[■]}, {Thickness → 0.01, Lighter[■]}};
framelabelsSNRP = {{"current SNR", None}, {"gating swing (nm)", None}};

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 100, 20}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 100, 20}];
XTicks = Union[Xticks1, Xticks3];
```

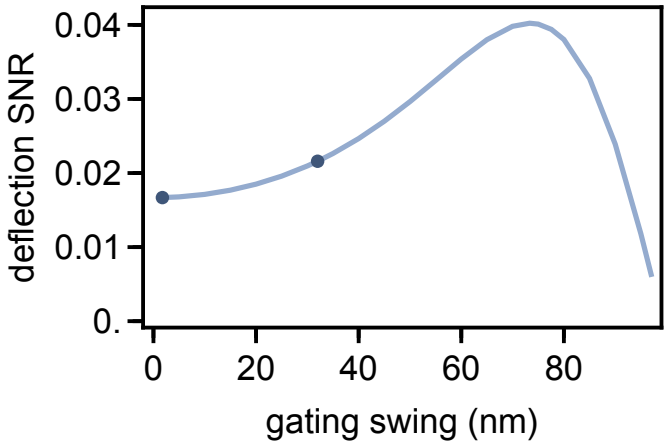
```
ListPlot[{snrvalsD}, Joined → True, PlotRange → {All, All},
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleSNRY, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameLabel → framelabelsSNRY, FrameTicks → {{YTicks, None}, {XTicks, None}}];
Show[%, ListPlot[
  {{1.7462045225885228`, 0.016685804637042244`}, {32, 0.021592631277784675`}},
  PlotMarkers → {Automatic, 10}, PlotStyle → { Darker[■]}]]

ListPlot[{snrPvalsD}, PlotRange → {All, {-0.002, 0.03}}, Joined → True,
  PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
  ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleSNRP,
  Background → White, Frame → {True, True, True, True},
  FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsSNRP,
  FrameTicks → {{YTicks, None}, {XTicks, None}}];
Show[%, ListPlot[
  {{1.7462045225885228`, 0.00001245381035364909 3`}, {32, 0.004716662703128328`}},
  PlotMarkers → {Automatic, 10}, PlotStyle → { Darker[■]}]]
pad2 = 30;

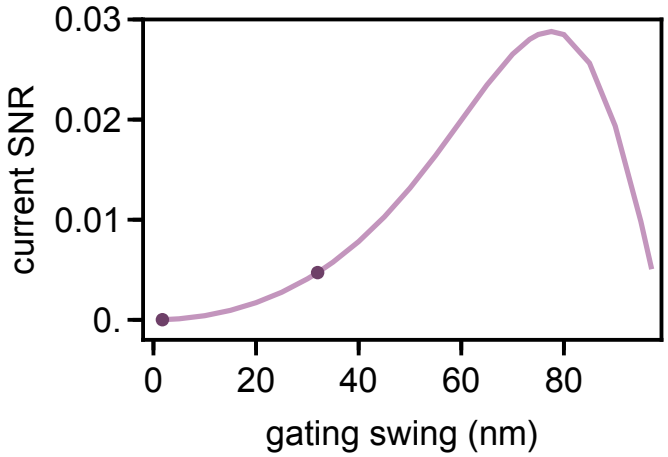
frameticksy = {{{1 / 100 000, Superscript[10, -5], {0, 0.03}},
    {1 / 10 000, Superscript[10, -4], {0, 0.03}}, {1 / 1000, Superscript[10, -3],
     {0, 0.03}}, {1 / 100, Superscript[10, -2], {0, 0.03}}}, None};
frameticksyENLAR =
  {{{1 / 100 000, Superscript[10, -5], {0, 0.03}}, {1 / 10 000, Superscript[10, -4],
     {0, 0.03}}, {1 / 1000, Superscript[10, -3], {0, 0.03}},
    {2 * 10^-5, , {0, 0.02}}, {4 * 10^-5, , {0, 0.02}}, {6 * 10^-5, , {0, 0.02}},
    {8 * 10^-5, , {0, 0.02}}, {2 * 10^-4, , {0, 0.02}}, {4 * 10^-4, , {0, 0.02}},
    {6 * 10^-4, , {0, 0.02}}, {8 * 10^-4, , {0, 0.02}}}, None};
frameticksx =
  {{{1000, Superscript[10, 3], {0, 0.03}}, {10 000, Superscript[10, 4], {0, 0.03}},
    {100 000, Superscript[10, 5], {0, 0.03}}, {2 * 10^3, , {0, 0.02}},
    {4 * 10^3, , {0, 0.02}}, {6 * 10^3, , {0, 0.02}}, {8 * 10^3, , {0, 0.02}},
    {2 * 10^4, , {0, 0.02}}, {4 * 10^4, , {0, 0.02}},
    {6 * 10^4, , {0, 0.02}}, {8 * 10^4, , {0, 0.02}}}, None};
frameticksycoher = {{{0, 0, {0, 0.03}}, {0.2, 0.2, {0, 0.03}}, {0.4, 0.4, {0, 0.03}},
    {0.6, 0.6, {0, 0.03}}, {0.8, 0.8, {0, 0.03}}, {1.0, "1.0", {0, 0.03}}}, None};
frameticksycoherneg =
  {{{0, 0, {0, 0.03}}, {0.5, 0.5, {0, 0.03}}, {1.0, "1.0", {0, 0.03}}, {-0.5, -0.5,
     {0, 0.03}}, {-1.0, "-1.0", {0, 0.03}}, {-1.5, "-1.5", {0, 0.03}}}, None};

fntsize = 18;
```

Out[ ]=



Out[ ]=




```
(*Fig S8 C,D plots of deflection fluctuation power vs gating swing*)

Remove[XTicks]
Remove[YTicks]
TickLength1 = 0.03;

pady = 100;
padx = 70;
pad2 = 30;
fntsize = 18;

fntsize = 18;

plotrangeSNRY = {Full, Full};
```

```
plotstyleSNRY = {{Thickness → 0.01, Lighter[■]}, {Thickness → 0.01, Lighter[■]}};
framelabelsSNRY = {{"deflection SNR", None}, {"quasistatic current", None}};


plotrangeSNRP = {Full, Full};
plotstyleSNRP = {{Thickness → 0.01, Lighter[■]}, {Thickness → 0.01, Lighter[■]}};
framelabelsSNRP = {{"current SNR", None}, {"quasistatic current", None}};

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 100, 20}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 100, 20 / 2}];
XTicks = Union[Xticks1, Xticks3, {{1, 1, {0, 0.03`}}}];

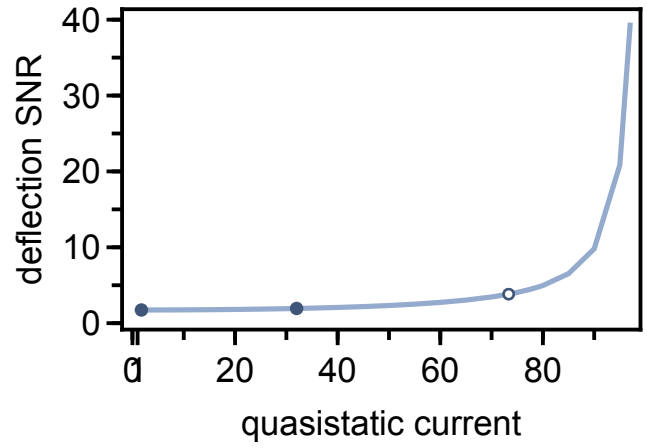
Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 50, 10}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 50, 10 / 2}];
YTicks = Union[Yticks1, Yticks3];


ListPlot[{dydyfD}, Joined → True, PlotRange → {All, All},
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleSNRY, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameLabel → framelabelsSNRY, FrameTicks → {{YTicks, None}, {XTicks, None}}];
Show[%,
  ListPlot[{{1.7462045225885228`, 1.7368208806014414`}, {32, 1.9378588148554803`}},
   PlotMarkers → {Automatic, 10}, PlotStyle → { Darker[■]}]];
Show[%, ListPlot[{{73.33323348898698`, 3.826802641746144`}},
  PlotMarkers → {"OpenMarkers", 5.2}, PlotStyle → { Darker[■]}]]

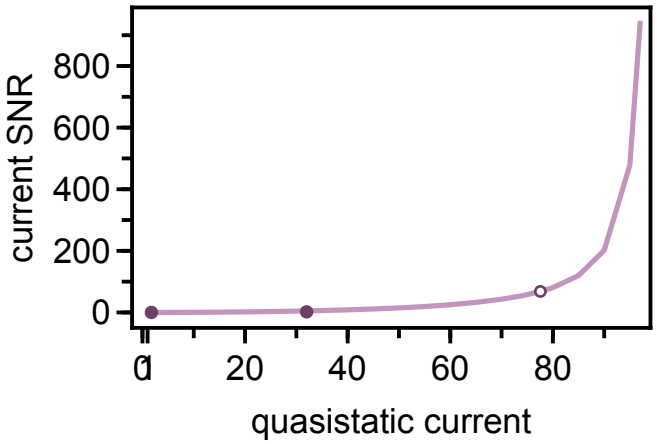

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 900, 200}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 900, 200 / 2}];
YTicks = Union[Yticks1, Yticks3];


ListPlot[{dpdpfD}, PlotRange → {All, {-50, 990}}, Joined → True,
  PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
  ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleSNRP,
  Background → White, Frame → {True, True, True, True},
  FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsSNRP,
  FrameTicks → {{YTicks, None}, {XTicks, None}}];
Show[%, ListPlot[
   {{1.7462045225885228`, 0.013449365463374788`}, {32, 1.9378588148554803`}},
   PlotMarkers → {Automatic, 10}, PlotStyle → { Darker[■]}]];
Show[%, ListPlot[{{77.56266383324578`, 68.211836913011`}},
```

Out[ ]=



Out[ ]=



In[ ]:=
```
(*Fig S8 A,B plots of deflection signal power vs gating swing*)


Remove[XTicks]
Remove[YTicks]
TickLength = 0.02;
TickLength1 = 0.03;



pady = 100;
padx = 70;
pad2 = 30;

fntsize = 18;
```

```
plotrangeSNRY = {Full, Full};
plotstyleSNRY = {{Thickness → 0.01, Lighter[■]}, {Thickness → 0.01, Lighter[■]}};
framelabelsSNRY = {{"deflection SNR", None}, {"quasistatic current", None}};


plotrangeSNRP = {Full, Full};
plotstyleSNRP = {{Thickness → 0.01, Lighter[■]}, {Thickness → 0.01, Lighter[■]}};
framelabelsSNRP = {{"current SNR", None}, {"quasistatic current", None}};

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 100, 20}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 100, 20 / 2}];
XTicks = Union[Xticks1, Xticks3, {{1, 1, {0, 0.03`}}}];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 0.3, 0.05}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 0.3, 0.05}];
YTicks = Union[Yticks1, Yticks3];


ListPlot[{powsfD}, Joined → True, PlotRange → {All, All},
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleSNRY, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameLabel → framelabelsSNRY, FrameTicks → {{YTicks, None}, {XTicks, None}}];
Show[%, ListPlot[
   {{1.7462045225885228`, 0.028980151950550488`}, {32, 0.04184347085757917`}},
   PlotMarkers → {Automatic, 10}, PlotStyle → { Darker[■]}]];

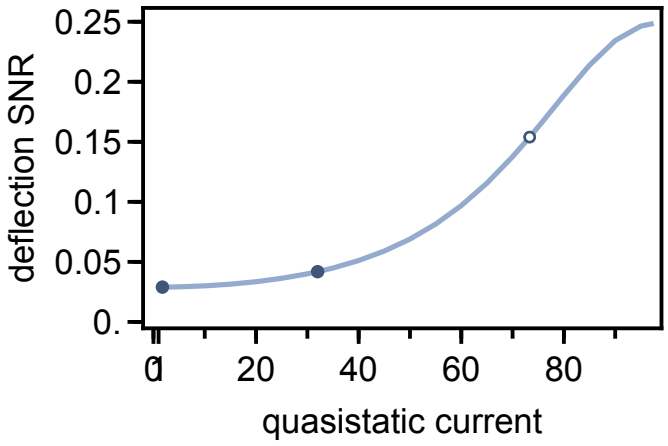
Show[%, ListPlot[{{73.33323348898698`, 0.15394408842649693`}},
  PlotMarkers → {"OpenMarkers", 5.2}, PlotStyle → { Darker[■]}]]


Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 10, 1}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 10, 1}];
YTicks = Union[Yticks1, Yticks3];

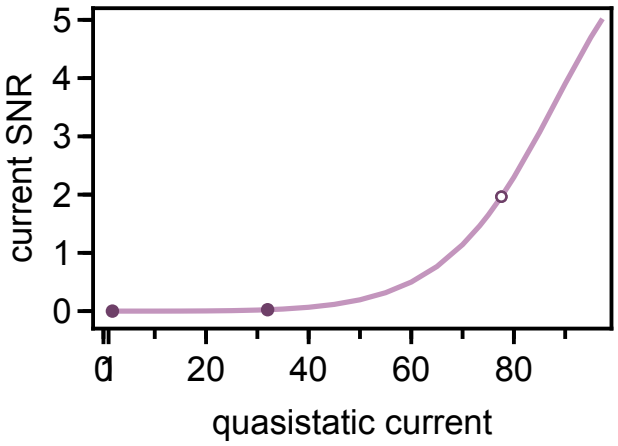

ListPlot[{powpfD}, PlotRange → {All, {-0.3, 5.2}}, Joined → True,
  PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
  ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleSNRP,
  Background → White, Frame → {True, True, True, True},
  FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsSNRP,
  FrameTicks → {{YTicks, None}, {XTicks, None}}];
```

```
Show[%, ListPlot[
   {{1.7462045225885228`, 1.6749584685778744`*^-7}, {32, 0.023769139280711647`}},
   PlotMarkers → {Automatic, 10}, PlotStyle → {Darker[■]}]];
Show[%, ListPlot[{{77.56266383324578`, 1.9643506805456612`}},
  PlotMarkers → {"OpenMarkers", 5.2}, PlotStyle → {Darker[■]}]]
```

Out[ ]=



Out[ ]=



In[ ]:=

In[ ]:=
```
(*table of values of linear snr,
power in the signal and fluctuation power as a function of the resting current*)

Pinfrest1 = {0.1, 0.17, 0.2, 0.25, 0.3, 0.35,
   0.4, 0.45, 0.5, 0.55, 0.6, 0.65, 0.7, 0.75, 0.8, 0.85, 0.9};
ssp1 = Table[Solve[{taorestd1 == taoval1d * (Pinfrest1[[pp]] / Pval1d) ^ (1 - alphad) *
        ((1 - Pinfrest1[[pp]]) / (1 - Pval1d)) ^ alphad},
    taorestd1], {pp, 1, Length[Pinfrest1]}];
taorest1 = taorestd1 /. ssp1;

snrvalsPinf =
  Table[{Pinfrest1[[pp]], (Fzd * 1) ^ 2 * snr[Ncd, Dvd, kbTd, kGSd, kPd, Pinfrest1[[pp]],
       taorest1[[pp]], lPd, 2 * Pi * wstd][[1]]}, {pp, 1, Length[Pinfrest1]}];

snrPvalsPinf =
  Table[{Pinfrest1[[pp]], (Fzd * 1) ^ 2 * snrp[Ncd, Dvd, kbTd, kGSd, kPd, Pinfrest1[[pp]],
       taorest1[[pp]], lPd, 2 * Pi * wstd][[1]]}, {pp, 1, Length[Pinfrest1]}];


dydyfPinf = Table[{Pinfrest1[[pp]], dydyf[kbTd, Ncd, kGSd, kPd, Pinfrest1[[pp]], Dvd]},
   {pp, 1, Length[Pinfrest1]}];

dpdpfPinf = Table[{Pinfrest1[[pp]], dpdpf[kbTd, Ncd, kGSd, kPd, Pinfrest1[[pp]], Dvd]},
   {pp, 1, Length[Pinfrest1]}];

powsfPinf =
  Table[{Pinfrest1[[pp]], (Fzd * 1) ^ 2 * powsf[kGSd, Ncd, kPd, 2 * Pi * wstd, lPd, kbTd,
       Pinfrest1[[pp]], taorest1[[pp]], Dvd][[1]]}, {pp, 1, Length[Pinfrest1]}];

powpfPinf =
  Table[{Pinfrest1[[pp]], (Fzd * 1) ^ 2 * powpf[kGSd, Ncd, kPd, 2 * Pi * wstd, lPd, kbTd,
       Pinfrest1[[pp]], taorest1[[pp]], Dvd][[1]]}, {pp, 1, Length[Pinfrest1]}];
```

In[ ]:=

In[•]:=
```
(*Fig 3 C-D plots of linear SNR vs resting current. *)

plotrangeSNRY = {Full, Full};
plotstyleSNRY = {{Thickness → 0.01, Lighter[■]}, {Thickness → 0.01, Lighter[■]}};
framelabelsSNRY = {{"deflection SNR", None}, {"quasistatic current", None}};


plotrangeSNRP = {{10^-1, 11}, Full};
plotstyleSNRP = {{Thickness → 0.01, Lighter[■]}, {Thickness → 0.01, Lighter[■]}};
framelabelsSNRP = {{"current SNR", None}, {"quasistatic current", None}};

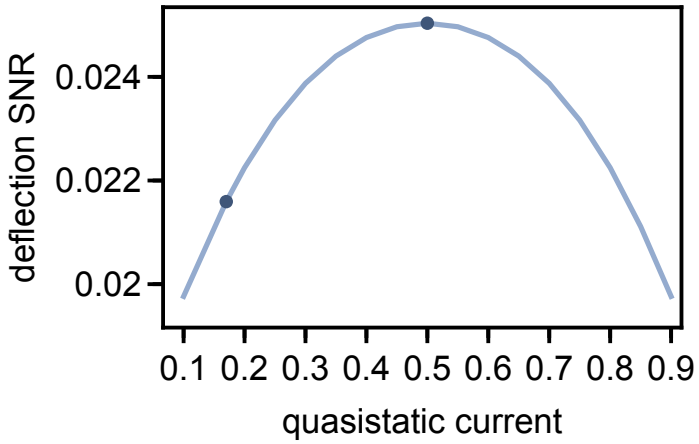
Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 0.025, 0.002}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 0.025, 0.002}];
YTicks = Union[Yticks1, Yticks3];


Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1, 0.1}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1, 0.1}];
XTicks = Union[Xticks1, Xticks3, {{1, 1, {0.1, 0.03`}}, {1, 1, {0.17, 0.03`}}}];

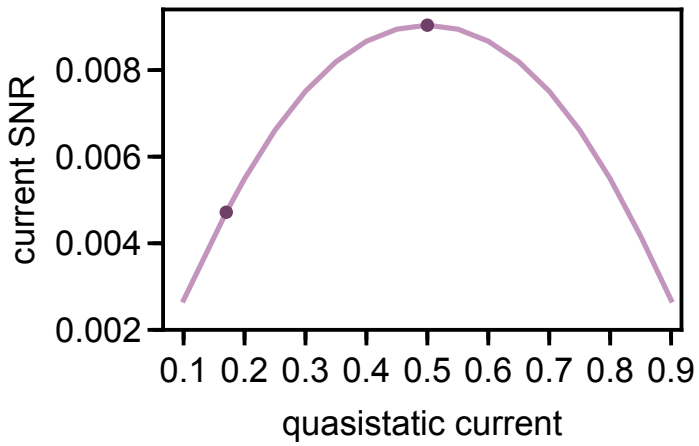

ListPlot[{snrvalsPinf}, Joined → True, PlotRange → {All, All},
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleSNRY, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameLabel → framelabelsSNRY, FrameTicks → {{YTicks, None}, {XTicks, None}}];
Show[%, ListPlot[{{0.17`, 0.021592631277784668`}, {0.5`, 0.025035056449822964`}},
  PlotMarkers → {Automatic, 10}, PlotStyle → { Darker[■]}]]

ListPlot[{snrPvalsPinf}, PlotRange → {All, {0.002, 0.0094}}, Joined → True,
  PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
  ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleSNRP,
  Background → White, Frame → {True, True, True, True},
  FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsSNRP,
  FrameTicks → {{YTicks, None}, {XTicks, None}}];
Show[%, ListPlot[{{0.17`, 0.004716662703128327`}, {0.5`, 0.009036858942153989`}},
  PlotMarkers → {Automatic, 10}, PlotStyle → { Darker[■]}]]
```

Out[ ]=



Out[ ]=




```
(*Calculation of power in signal + power in noise*)

fmult2 = {0, 0.1, 0.2, 0.3, 0.4, 0.5, 1, 2, 3, 4, 5, 10, 20, 30,
    40, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150, 160, 170, 180};
area0 = Table[0, {Length[fmult2]}];
area1 = Table[0, {Length[fmult2]}];
SetSharedVariable[area0]
SetSharedVariable[area1]


(*Calculation of power in the signal by setting noise terms to zero*)
ParallelDo[
SeedRandom[47 741];
  gx =  2 * kbTd * lPd;
  y0in = Yrestd;
```

```
t1 = AbsoluteTime[];
  proc = ItoProcess[
    lPd * ⅆY[t] == (-kPd) * (Y[t] - Yupd) * ⅆt - kGSd * (Y[t] - Yugd - Dvd * Po[t]) * ⅆt + 0 *
        Sqrt[gx] * ⅆn1[t] + Fzd * fmult2[[pp]] * Sin[2 * Pi * wstd * (t - tstartFd)] * ⅆt &&
     taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd] * ⅆPo[t] ==
      (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] - Po[t]) ⅆt +
       0 * Sqrt[taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd]] Sqrt[
         (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] (1 - Clip[Po[t], {0.0, 1.0}]) + (1 - Pinf2[
                Yhd, kGSd, Dvd, kbTd, Ncd]) Clip[Po[t], {0.0, 1.0}]) / Ncd] ⅆn2[t],
    {Y[t], Po[t]}, {{Y, Po}, {y0in, Pinfrest}}, t,
{n1 ≈ WienerProcess[], n2 ≈ WienerProcess[]}];
  totallength = 2 * 0.507;
  factot = 4;
  deltatsima = 0.000001 / factot;
   ap = RandomFunction[proc, {0, totallength, deltatsima}];

  ap0 = ap["PathComponent", 1] - Yrestd;

  ap1 = ap["PathComponent", 2] - Pinfrest;


resampfactor = 4;

  deltatsampl = Round[deltatsima * resampfactor, deltatsima];

  randtime100 = 0.005 * 10;

  stpoint = IntegerPart[randtime100 / deltatsima];

  tsampl = 0.005; (**20**)

  randtimenm = Round[randtime100, deltatsima];
  tsamplnm = Round[tsampl, deltatsima];

  b1 = Table[f[i], {i, randtimenm, tsamplnm + randtimenm, deltatsampl}];
  (*determine length of sampled time series*)
lengthares = Length[b1];

  sizea = lengthares;


  ares0b =
   ap0["Values"][[stpoint ;; Length[ap0["Values"]] - stpoint ;; resampfactor]];
```

```
  ares0bt =
   ap0["Times"][[stpoint ;; Length[ap0["Values"]] - stpoint ;; resampfactor]];
  fousampled0b = ResourceFunction["WelchSpectralEstimate"][ares0b,
    Length[ares0b] / (ares0bt[[Length[ares0b]]] - ares0bt[[1]]), "Window" → None,
    "Detrend" → None, "OverlapOffset" → None, "SegmentSize" → sizea - 1];
  lengthpsd = Length[fousampled0b] - 1;
  meandataplot0b = Table[{0, 0}, {i, 1, lengthpsd}];
  meandataplot0b =
   Table[{fousampled0b[[i]][[1]], fousampled0b[[i]][[2]]}, {i, 1, lengthpsd}];

  ares1b =
   ap1["Values"][[stpoint ;; Length[ap0["Values"]] - stpoint ;; resampfactor]];
  fousampled1b = ResourceFunction["WelchSpectralEstimate"][ares1b,
    Length[ares0b] / (ares0bt[[Length[ares0b]]] - ares0bt[[1]]), "Window" → None,
    "Detrend" → None, "OverlapOffset" → None, "SegmentSize" → sizea - 1];
  meandataplot1b = Table[{0, 0}, {i, 1, lengthpsd}];
  meandataplot1b =
   Table[{fousampled1b[[i]][[1]], fousampled1b[[i]][[2]]}, {i, 1, lengthpsd}];

  (*the power in signal+
   noise is calculated as the variance of the corresponding time series*)
  area0[[pp]] = Variance[ap0];
  area1[[pp]] = Variance[ap1];



  ,
  {pp, 1, Length[fmult2]}];
```

In[•]:=
```
psplf0 =
 Table[{fmult2[[pp]], area0[[pp]] + 1.9630761547464095`}, {pp, 2, Length[fmult2]}]
psplf1 =
 Table[{fmult2[[pp]], area1[[pp]] + 0.004967582098481333`}, {pp, 2, Length[fmult2]}]
```

Out[•]=
```
{{0.1, 1.9639}, {0.2, 1.96638}, {0.3, 1.9705}, {0.4, 1.97628},
 {0.5, 1.9837}, {1, 2.04557}, {2, 2.29307}, {3, 2.70556}, {4, 3.28306},
 {5, 4.02557}, {10, 10.2136}, {20, 34.9722}, {30, 76.246}, {40, 134.007},
 {50, 208.153}, {60, 298.479}, {70, 404.687}, {80, 526.424}, {90, 663.35},
 {100, 815.196}, {110, 981.789}, {120, 1163.04}, {130, 1358.93},
 {140, 1569.45}, {150, 1794.61}, {160, 2034.42}, {170, 2288.9}, {180, 2558.06}}
```

Out[ ]=

```
{{0.1, 0.00496777}, {0.2, 0.00496834}, {0.3, 0.00496929}, {0.4, 0.00497061},
 {0.5, 0.00497231}, {1, 0.00498651}, {2, 0.00504331}, {3, 0.00513806},
 {4, 0.00527089}, {5, 0.00544196}, {10, 0.00688024}, {20, 0.0128333}, {30, 0.0232687},
 {40, 0.0383398}, {50, 0.0573811}, {60, 0.0787673}, {70, 0.100359}, {80, 0.120263},
 {90, 0.137387}, {100, 0.151492}, {110, 0.162896}, {120, 0.172112}, {130, 0.17963},
 {140, 0.185845}, {150, 0.191054}, {160, 0.195478}, {170, 0.199281}, {180, 0.202587}}
```

```
(*Calculation of power in the signal + noise*)
fmult2 = {0, 0.1, 0.2, 0.3, 0.4, 0.5, 1, 2, 3, 4, 5, 10, 20, 30,
   40, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150, 160, 170, 180};
area0 = Table[0, {Length[fmult2]}];
area1 = Table[0, {Length[fmult2]}];
SetSharedVariable[area0]
SetSharedVariable[area1]

ParallelDo[
SeedRandom[47 741];
  gx =  2 * kbTd * lPd;
  y0in = Yrestd;

t1 = AbsoluteTime[];
  proc = ItoProcess[
    lPd * ∂tY[t] == (-kPd) * (Y[t] - Yupd) * ∂t - kGSd * (Y[t] - Yugd - Dvd * Po[t]) * ∂t +
       Sqrt[gx] * ∂tn1[t]  + Fzd * fmult2[[pp]] * Sin[2 * Pi * wstd * (t - tstartFd)] * ∂t &&
     taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd] * ∂tPo[t] ==
      (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] - Po[t]) ∂t +
       Sqrt[taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd]] Sqrt[
         (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] (1 - Clip[Po[t], {0.0, 1.0}]) + (1 - Pinf2[
                 Yhd, kGSd, Dvd, kbTd, Ncd]) Clip[Po[t], {0.0, 1.0}]) / Ncd] ∂tn2[t],
    {Y[t], Po[t]}, {{Y, Po}, {y0in, Pinfrest}}, t,
{n1 ≈ WienerProcess[], n2 ≈ WienerProcess[]}];
  totallength = 2 * 0.507;
  factot = 4;
  deltatsima = 0.000001 / factot;
   ap =  RandomFunction[proc, {0, totallength, deltatsima}];

  ap0 = ap["PathComponent", 1] - Yrestd;

  ap1 = ap["PathComponent", 2] - Pinfrest;


resampfactor = 4;
```

```
  deltatsampl = Round[deltatsima * resampfactor, deltatsima];

  randtime100 = 0.005 * 10;

  stpoint = IntegerPart[randtime100 / deltatsima];

  tsampl = 0.005; (**20**)

  randtimenm = Round[randtime100, deltatsima];
  tsamplnm = Round[tsampl, deltatsima];

  b1 = Table[f[i], {i, randtimenm, tsamplnm + randtimenm, deltatsampl}];
  (*determine length of sampled time series*)
lengthares = Length[b1];

  sizea = lengthares;


  ares0b =
   ap0["Values"][[stpoint ;; Length[ap0["Values"]] - stpoint ;; resampfactor]];
  ares0bt =
   ap0["Times"][[stpoint ;; Length[ap0["Values"]] - stpoint ;; resampfactor]];
  fousampled0b = ResourceFunction["WelchSpectralEstimate"][ares0b,
    Length[ares0b] / (ares0bt[[Length[ares0b]]] - ares0bt[[1]]), "Window" → None,
    "Detrend" → None, "OverlapOffset" → None, "SegmentSize" → sizea - 1];
  lengthpsd = Length[fousampled0b] - 1;
  meandataplot0b = Table[{0, 0}, {i, 1, lengthpsd}];
  meandataplot0b =
   Table[{fousampled0b[[i]][[1]], fousampled0b[[i]][[2]]}, {i, 1, lengthpsd}];

  ares1b =
   ap1["Values"][[stpoint ;; Length[ap0["Values"]] - stpoint ;; resampfactor]];
  fousampled1b = ResourceFunction["WelchSpectralEstimate"][ares1b,
    Length[ares0b] / (ares0bt[[Length[ares0b]]] - ares0bt[[1]]), "Window" → None,
    "Detrend" → None, "OverlapOffset" → None, "SegmentSize" → sizea - 1];
  meandataplot1b = Table[{0, 0}, {i, 1, lengthpsd}];
  meandataplot1b =
   Table[{fousampled1b[[i]][[1]], fousampled1b[[i]][[2]]}, {i, 1, lengthpsd}];

  (*the power in signal+
   noise is calculated as the variance of the corresponding time series*)
  area0[[pp]] = Variance[ap0];
  area1[[pp]] = Variance[ap1];
```

```
  ,
  {pp, 1, Length[fmult2]}];
```

In[ ]:=
```
psf0 = Table[{fmult2[[pp]], area0[[pp]]}, {pp, 2, Length[fmult2]}]
psf1 = Table[{fmult2[[pp]], area1[[pp]]}, {pp, 2, Length[fmult2]}]
```

Out[ ]=
```
{{0.1, 1.96295}, {0.2, 1.96447}, {0.3, 1.96764}, {0.4, 1.97246},
 {0.5, 1.97893}, {1, 2.03605}, {2, 2.27409}, {3, 2.67719}, {4, 3.24536},
 {5, 3.9786}, {10, 10.1212}, {20, 34.7915}, {30, 75.976}, {40, 133.641},
 {50, 207.68}, {60, 297.892}, {70, 403.982}, {80, 525.605}, {90, 662.426},
 {100, 814.175}, {110, 980.681}, {120, 1161.85}, {130, 1357.66},
 {140, 1568.09}, {150, 1793.17}, {160, 2032.9}, {170, 2287.3}, {180, 2556.37}}
```

Out[ ]=
```
{{0.1, 0.00496773}, {0.2, 0.00496826}, {0.3, 0.0049692}, {0.4, 0.00497052},
 {0.5, 0.00497224}, {1, 0.00498677}, {2, 0.00504543}, {3, 0.00514361},
 {4, 0.00528141}, {5, 0.00545898}, {10, 0.00695049}, {20, 0.0130765}, {30, 0.0236628},
 {40, 0.0387221}, {50, 0.0575141}, {60, 0.0784497}, {70, 0.0995044}, {80, 0.1189},
 {90, 0.1356}, {100, 0.14937}, {110, 0.160511}, {120, 0.169513}, {130, 0.176854},
 {140, 0.182919}, {150, 0.187999}, {160, 0.192311}, {170, 0.196018}, {180, 0.199241}}
```

In[ ]:=
```
(*calcualtion of fractional difference between power in signal+
 noise and power in signal + power in noise*)
psdiff0 = Table[{psf0[[pp]][[1]], (psf0[[pp]][[2]] - psplf0[[pp]][[2]]) / psf0[[pp]][[2]]},
   {pp, 1, Length[psf0]}];
psdiff1 = Table[
   {psf1[[pp]][[1]], (psf1[[pp]][[2]] - psplf1[[pp]][[2]]) / psf1[[pp]][[2]]}, {pp, 1, Length[psf1]}];
```

In[ ]:=

```
(*Fig S4 plots of fractional difference between power in signal+
 noise and power in signal + power in noise vs stimulus-force amplitude*)
pady = 120;
padx = 70;
pad2 = 30;
fntsize = 18;
Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 180, 50}];(*/10*)
Xticks2 = Table[{k, Null, {0, 0.02}}, {k, 0, 180, 50}];(*/50*)
XTickshist = Union[Xticks1, Xticks2];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, -0.02, 0.02, 0.005}];(*/10*)
Yticks2 = Table[{k, Null, {0, 0.02}}, {k, -0.02, 0.02, 0.005}];(*/50*)
YTicks = Union[Yticks1, Yticks2];

plotstylepow0 = {{Thickness → 0.01, Lighter[■]}};

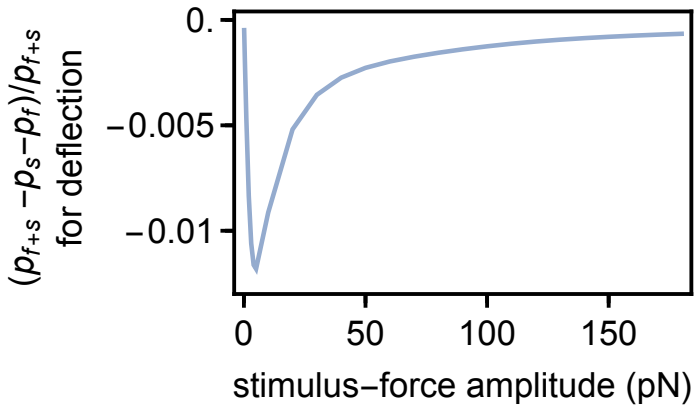
framelabels = {{"(pf+s -ps-pf)/pf+s \n for deflection", None},
   {"stimulus-force amplitude (pN)", None}};
ListPlot[psdiff0, Joined → True, Background → White,
 PlotRange → {{-4, 184}, {-0.013, 0.0004}}, PlotStyle → plotstylepow0,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabels, LabelStyle → {FontSize → 18}, Axes → {False, False},
 PlotStyle → {Black}, FrameTicks → {{YTicks, None}, {XTickshist, None}}]

Yticks1 = Table[{k, k, {0, 0.03}}, {k, -0.02, 0.02, 0.01}];(*/10*)
Yticks2 = Table[{k, Null, {0, 0.02}}, {k, -0.02, 0.02, 0.01}];(*/50*)
YTicks = Union[Yticks1, Yticks2];

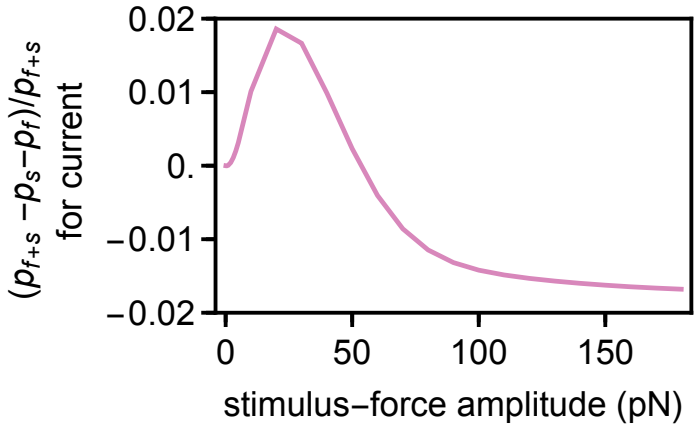
plotstylepow1 = {{Thickness → 0.01, Lighter[■]}};

framelabels = {{"(pf+s -ps-pf)/pf+s \n for current", None},
   {"stimulus-force amplitude (pN)", None}};
ListPlot[psdiff1, Joined → True, Background → White,
 PlotRange → {{-4, 184}, {-0.02, 0.02}}, PlotStyle → plotstylepow1,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabels, LabelStyle → {FontSize → 18}, Axes → {False, False},
 PlotStyle → {Black}, FrameTicks → {{YTicks, None}, {XTickshist, None}}]
```

Out[ ]=


0.
−0.005
−0.01
0
50
100
150
stimulus−force amplitude (pN)
$(p_{f+s} - p_s - p_f)/p_{f+s}$
for deflection

Out[ ]=


0.02
0.01
0.
−0.01
−0.02
0
50
100
150
stimulus−force amplitude (pN)
$(p_{f+s} - p_s - p_f)/p_{f+s}$
for current

In[•]:=
```
(*Calculation of linear SNRs vs sitmulus-force amplitude*)
chi00 = -(kg + I * w * lg) / (kgs^2 - (kg + I * w * lg) * (kgs + kp + I * w * lp));
chi10 = -kgs / (kgs^2 - (kg + I * w * lg) * (kgs + kp + I * w * lp));
pows = FullSimplify[ComplexExpand[Im[chi00]^2]];
powp = FullSimplify[ComplexExpand[Im[chi10]^2]];
dydy = kbt * kg / (-kgs^2 + kg (kgs + kp));
dpdp = kbt (kgs + kp) / (-kgs^2 + kg (kgs + kp));
snrtemp = FullSimplify[(pows / dydy) /. {kg → Nc * kbt / (Pinf * (1 - Pinf) * D^2),
      lg → Nc * kbt * taoinf / (Pinf * (1 - Pinf) * D^2)}];
snrPtemp = FullSimplify[(powp / dpdp) /. {kg → Nc * kbt / (Pinf * (1 - Pinf) * D^2),
      lg → Nc * kbt * taoinf / (Pinf * (1 - Pinf) * D^2)}];

snr[Nc_, D_, kbt_, kgs_, kp_, Pinf_, taoinf_, lp_, w_] :=
  (Nc (kbt (kgs + kp) Nc + D^2 kgs^2 (-1 + Pinf) Pinf) w^2
     (D^2 kgs^2 (-1 + Pinf) Pinf taoinf - kbt lp Nc (1 + taoinf^2 w^2))^2) /
   (D^4 kgs^4 (-1 + Pinf)^2 Pinf^2 + 2 D^2 kbt kgs^2 Nc (-1 + Pinf) Pinf (kgs + kp - lp taoinf w^2) +
      kbt^2 Nc^2 ((kgs + kp)^2 + lp^2 w^2) (1 + taoinf^2 w^2))^2;
snrp[Nc_, D_, kbt_, kgs_, kp_, Pinf_, taoinf_, lp_, w_] :=
  -((D^2 kbt kgs^2 Nc^2 (-1 + Pinf) Pinf (kbt (kgs + kp) Nc + D^2 kgs^2 (-1 + Pinf) Pinf)
       (lp + (kgs + kp) taoinf)^2 w^2) /
     ((kgs + kp) (D^4 kgs^4 (-1 + Pinf)^2 Pinf^2 + 2 D^2 kbt kgs^2 Nc (-1 + Pinf) Pinf
          (kgs + kp - lp taoinf w^2) + kbt^2 Nc^2 ((kgs + kp)^2 + lp^2 w^2) (1 + taoinf^2 w^2))^2));

fmult3 = {0, 1, 5, 10, 20, 30, 40, 50, 60, 70,
   80, 90, 100, 110, 120, 130, 140, 150, 160, 170, 180, 200};
snrvalsF =
  Table[{fmult3[[pp]], (Fzd * fmult3[[pp]])^2 * snr[Ncd, Dvd, kbTd, kGSd, kPd, Pinfrest,
      taorest, lPd, 2 * Pi * wstd]}, {pp, 1, Length[fmult3]}];

snrPvalsF =
  Table[{fmult3[[pp]], (Fzd * fmult3[[pp]])^2 * snrp[Ncd, Dvd, kbTd, kGSd, kPd, Pinfrest,
      taorest, lPd, 2 * Pi * wstd]}, {pp, 1, Length[fmult3]}];
```

In[•]:=
```
(*plot settings*)

Remove[XTicks]
Remove[YTicks]
TickLength = 0.02;
Xmin = 0;
```

```
Xmax = 10 - 1;
Ymin = 0;
Ymax = 20;
deltaY = 2;
plotrangepsd = {{10^(2), 10^6}, {0.5 * 10^(-12), 10^(-2)}};
plotrangecoher = {{10^(2), 10^6}, {-1.05, 1.05}};
framelabels =
  {{"power spectrum (\!\(\*SuperscriptBox[\(nm\), \(2\)]\)/Hz)", None},
   {"noise frequency (Hz)", None}};
TickLength1 = 0.03;
Yticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, -24, 10, 2}];
Yticks2 = Table[{10^k, Null, {0, TickLength1}}, {k, -23, 10, 2}];
TickLength2 = 0.02;
aa = Table[0, {38}, {4}];
For[k = 1, k ≤ 38, k++,
aa[[k]] = Table[{l * 10^(-k + 10), , {0, TickLength2}}, {l, 2, 8, 2}];
]
YTicks = Union[Yticks1, Yticks2, aa[[1]], aa[[2]], aa[[3]],
   aa[[4]], aa[[5]], aa[[6]], aa[[7]], aa[[8]], aa[[9]], aa[[10]], aa[[11]], aa[[12]],
   aa[[13]], aa[[14]], aa[[15]], aa[[16]], aa[[17]], aa[[18]], aa[[19]], aa[[20]],
   aa[[21]], aa[[22]], aa[[23]], aa[[24]], aa[[25]], aa[[26]], aa[[27]], aa[[28]]];

pady = 100;
padx = 70;
pad2 = 30;
fntsize = 18;

frameticksy = {{{1 / 100 000, Superscript[10, -5], {0, 0.03}},
    {1 / 10 000, Superscript[10, -4], {0, 0.03}}, {1 / 1000, Superscript[10, -3],
     {0, 0.03}}, {1 / 100, Superscript[10, -2], {0, 0.03}}}, None};
frameticksyENLAR =
  {{{1 / 100 000, Superscript[10, -5], {0, 0.03}}, {1 / 10 000, Superscript[10, -4],
     {0, 0.03}}, {1 / 1000, Superscript[10, -3], {0, 0.03}},
    {2 * 10^-5, , {0, 0.02}}, {4 * 10^-5, , {0, 0.02}}, {6 * 10^-5, , {0, 0.02}},
    {8 * 10^-5, , {0, 0.02}}, {2 * 10^-4, , {0, 0.02}}, {4 * 10^-4, , {0, 0.02}},
    {6 * 10^-4, , {0, 0.02}}, {8 * 10^-4, , {0, 0.02}}}, None};
frameticksx =
  {{{1000, Superscript[10, 3], {0, 0.03}}, {10 000, Superscript[10, 4], {0, 0.03}},
    {100 000, Superscript[10, 5], {0, 0.03}}, {2 * 10^3, , {0, 0.02}},
    {4 * 10^3, , {0, 0.02}}, {6 * 10^3, , {0, 0.02}}, {8 * 10^3, , {0, 0.02}},
    {2 * 10^4, , {0, 0.02}}, {4 * 10^4, , {0, 0.02}},
    {6 * 10^4, , {0, 0.02}}, {8 * 10^4, , {0, 0.02}}}, None};
frameticksycoher = {{{0, 0, {0, 0.03}}, {0.2, 0.2, {0, 0.03}}, {0.4, 0.4, {0, 0.03}},
    {0.6, 0.6, {0, 0.03}}, {0.8, 0.8, {0, 0.03}}, {1.0, "1.0", {0, 0.03}}}, None};
```

```
frameticksycoherneg =
  {{{0, 0, {0, 0.03}}, {0.5, 0.5, {0, 0.03}}, {1.0, "1.0", {0, 0.03}}, {-0.5, -0.5,
     {0, 0.03}}, {-1.0, "-1.0", {0, 0.03}}, {-1.5, "-1.5", {0, 0.03}}}, None};

fntsize = 18;

(*Fig S11A-B plots of linear SNRs for a gating swing of 32 nm and
   resting current of 0.17. Set D=32 and Prest=0.17 in parameters. *)


plotrangeSNRY = {Full, Full};
plotstyleSNRY = {{Thickness → 0.01, Lighter[■]}, {Thickness → 0.01, Darker[■]}};
framelabelsSNRY =
  {{"deflection SNR", None}, {"stimulus-force amplitude (pN)", None}};

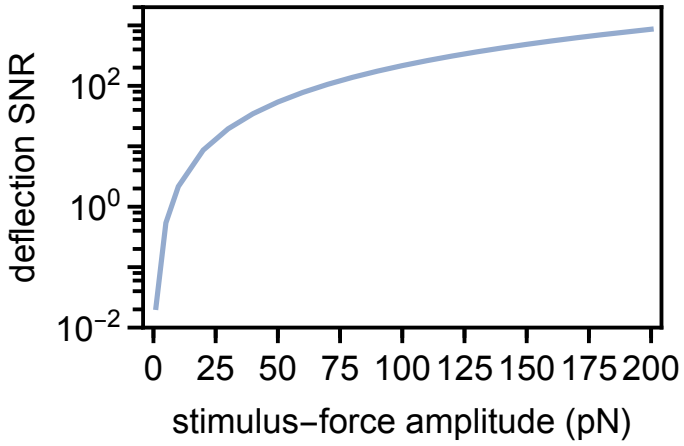

plotrangeSNRP = {{10^-1, 11}, Full};
plotstyleSNRP = {{Thickness → 0.01, Lighter[■]}, {Thickness → 0.01, Lighter[■]}};
framelabelsSNRP =
  {{"current or voltage SNR", None}, {"stimulus-force amplitude (pN)", None}};


Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1000, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1000, 25}];
XTicks = Union[Xticks1, Xticks3];

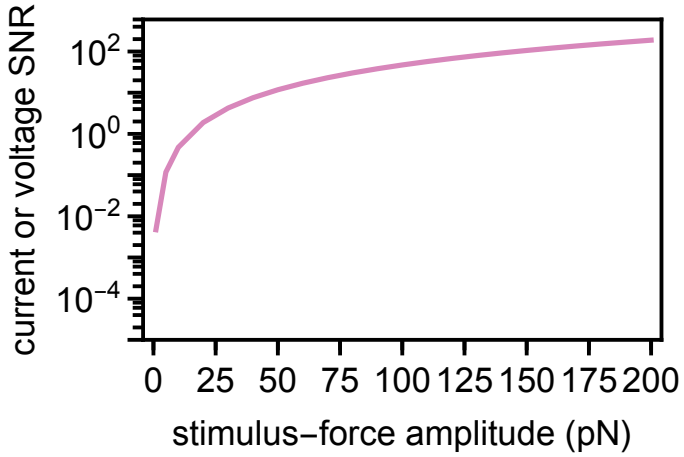

ListLogPlot[{snrvalsF}, Joined → True, PlotRange → {All, {10^-2, 2 * 10^3}},
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleSNRY, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsSNRY, FrameTicks → {{YTicks, None}, {XTicks, None}}]

ListLogPlot[{snrPvalsF}, PlotRange → {All, {10^-5, 6 * 10^2}}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleSNRP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsSNRP,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

Out[ ]=


deflection SNR
10^2
10^0
10^{-2}
0 25 50 75 100 125 150 175 200
stimulus−force amplitude (pN)

Out[ ]=


current or voltage SNR
10^2
10^0
10^{-2}
10^{-4}
0 25 50 75 100 125 150 175 200
stimulus−force amplitude (pN)

```
(*Fig S11C-D plots of linear SNRs for a gating swing of 1.75 nm and
    resting current of 0.17. Set D=1.75 and Prest=0.17 in parameters. *)

plotrangeSNRY = {Full, Full};
plotstyleSNRY = {{Thickness → 0.01, Lighter[■]}, {Thickness → 0.01, Darker[■]}};
framelabelsSNRY =
  {{"deflection SNR", None}, {"stimulus-force amplitude (pN)", None}};

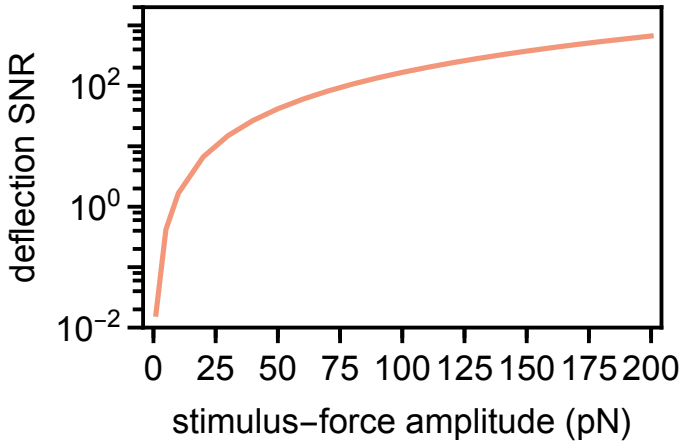
plotrangeSNRP = {{10^-1, 11}, Full};
plotstyleSNRP = {{Thickness → 0.01, Lighter[■]}, {Thickness → 0.01, Darker[■]}};
framelabelsSNRP =
  {{"current or voltage SNR", None}, {"stimulus-force amplitude (pN)", None}};


Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1000, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1000, 25}];
XTicks = Union[Xticks1, Xticks3];

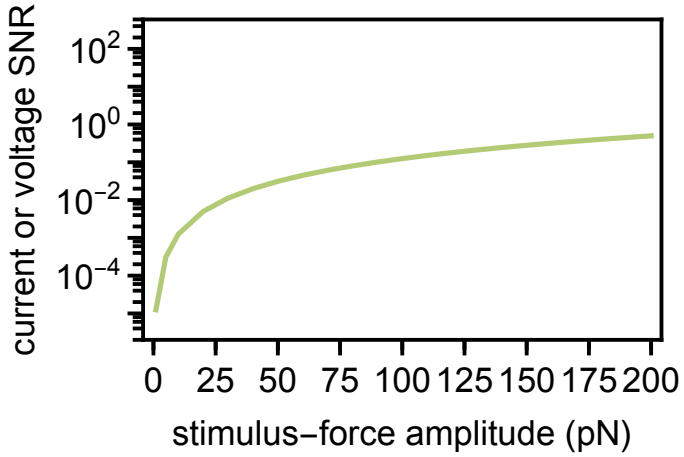

ListLogPlot[{snrvalsF}, Joined → True, PlotRange → {All, {10^-2, 2 * 10^3}},
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleSNRY, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsSNRY, FrameTicks → {{YTicks, None}, {XTicks, None}}]

ListLogPlot[{snrPvalsF}, PlotRange → {All, {2 * 10^-6, 6 * 10^2}}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleSNRP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsSNRP,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

Out[ ]=


deflection SNR
10^2
10^0
10^{-2}
0 25 50 75 100 125 150 175 200
stimulus−force amplitude (pN)

Out[ ]=


current or voltage SNR
10^2
10^0
10^{-2}
10^{-4}
0 25 50 75 100 125 150 175 200
stimulus−force amplitude (pN)

In[•]:=
```
(*Fig S12 plots of linear SNRs for a gating swing of 32 nm
  and resting current of 0.5. Set D=32 and Prest=0.5 in parameters. *)


plotrangeSNRY = {Full, Full};
plotstyleSNRY = {{Thickness → 0.01, Lighter[■]}, {Thickness → 0.01, Darker[■]}};
framelabelsSNRY =
  {{"deflection SNR", None}, {"stimulus-force amplitude (pN)", None}};

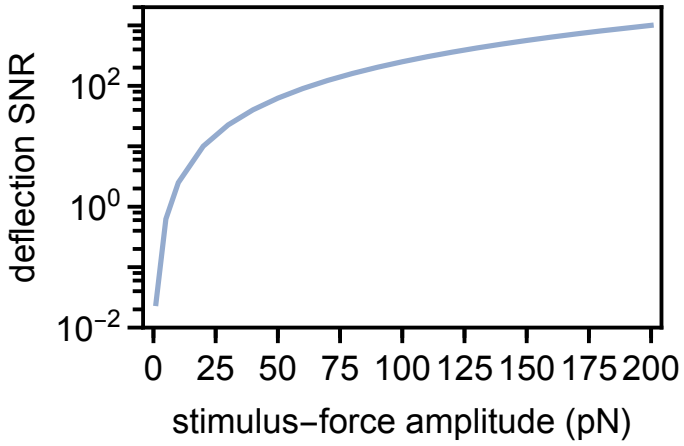

plotrangeSNRP = {{10^-1, 11}, Full};
plotstyleSNRP = {{Thickness → 0.01, Lighter[■]}, {Thickness → 0.01, Lighter[■]}};
framelabelsSNRP =
  {{"current or voltage SNR", None}, {"stimulus-force amplitude (pN)", None}};


Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1000, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1000, 25}];
XTicks = Union[Xticks1, Xticks3];

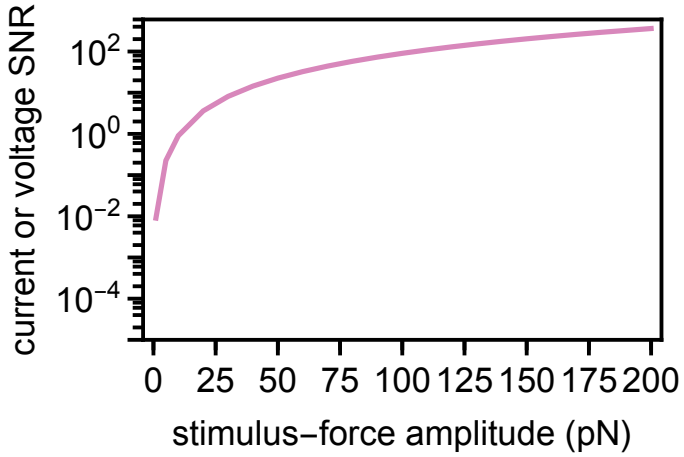

ListLogPlot[{snrvalsF}, Joined → True, PlotRange → {All, {10^-2, 2 * 10^3}},
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleSNRY, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsSNRY, FrameTicks → {{YTicks, None}, {XTicks, None}}]

ListLogPlot[{snrPvalsF}, PlotRange → {All, {10^-5, 6 * 10^2}}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleSNRP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsSNRP,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

Out[ ]=


deflection SNR
10^2
10^0
10^{-2}
0 25 50 75 100 125 150 175 200
stimulus–force amplitude (pN)

Out[ ]=


current or voltage SNR
10^2
10^0
10^{-2}
10^{-4}
0 25 50 75 100 125 150 175 200
stimulus–force amplitude (pN)

In[•]:=
```
(*Fig S13 plots of linear SNRs for a gating swing of 1.75 nm and
  resting current of 0.5. Set D=1.75 and Prest=0.5 in parameters. *)

plotrangeSNRY = {Full, Full};
plotstyleSNRY = {{Thickness → 0.01, Lighter[■]}, {Thickness → 0.01, Darker[■]}};
framelabelsSNRY =
  {{"deflection SNR", None}, {"stimulus-force amplitude (pN)", None}};

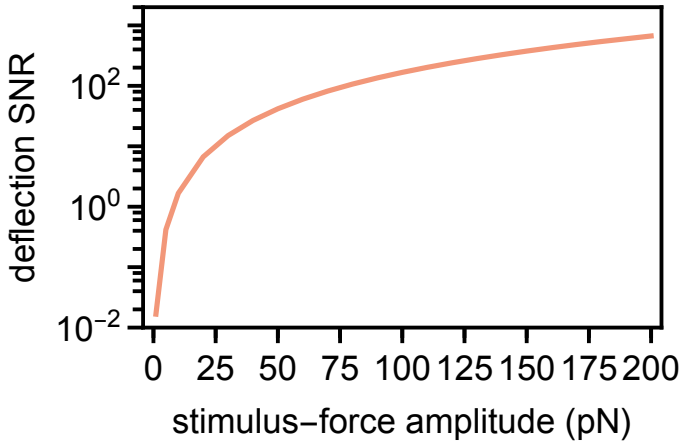
plotrangeSNRP = {{10^-1, 11}, Full};
plotstyleSNRP = {{Thickness → 0.01, Lighter[■]}, {Thickness → 0.01, Darker[■]}};
framelabelsSNRP =
  {{"current or voltage SNR", None}, {"stimulus-force amplitude (pN)", None}};


Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1000, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1000, 25}];
XTicks = Union[Xticks1, Xticks3];

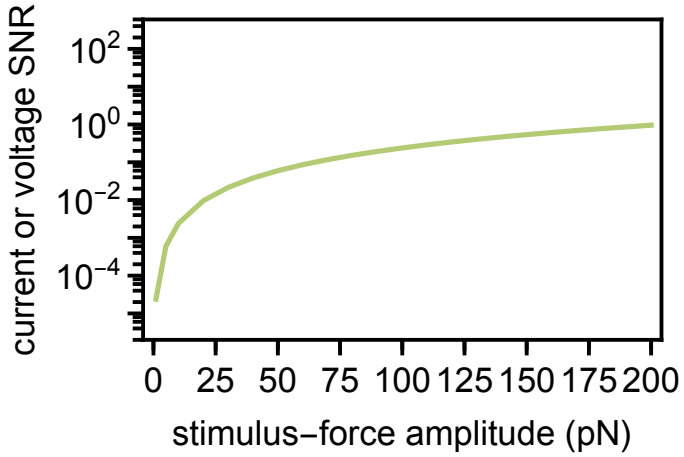

ListLogPlot[{snrvalsF}, Joined → True, PlotRange → {All, {10^-2, 2 * 10^3}},
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleSNRY, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsSNRY, FrameTicks → {{YTicks, None}, {XTicks, None}}]

ListLogPlot[{snrPvalsF}, PlotRange → {All, {2 * 10^-6, 6 * 10^2}}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleSNRP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsSNRP,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

Out[ ]=



Out[ ]=



In[ ]:=

```
(*
Calculation of effect of saturating nonlinearity on white noise signal  +
 sinusoidal signal*)
SeedRandom[47741]
signalampl = 1; (*set to 1 for Fig S16A-B and to 80 for FigS16C-D *)
𝒫 = WhiteNoiseProcess[];
data = 1.39 * RandomFunction[𝒫, {0, 4056002}];
(*white noise signal with same variance and mean as deflection fluctuations*)
Covariance[data, data]
Mean[data]

totallength = 2 * 0.507;
factot = 4;
deltatsima = 0.000001 / factot;
```

```
resampfactor = factot;

(*Calculate power spectrum of white noise signal*)
randtime100 = 0.005 * 10;
randtimenm = Round[randtime100, deltatsima];
stpoint = IntegerPart[randtime100 / deltatsima];
deltatsampl = Round[deltatsima * resampfactor, deltatsima];
tsampl = 0.005;
endpoint = IntegerPart[Length[data["Values"]] - tsampl / deltatsima - stpoint];
tsamplnm = Round[tsampl, deltatsima];
b1 = Table[f[i], {i, randtimenm, tsamplnm + randtimenm, deltatsampl}];
lengthares = Length[b1];
totallength2 = totallength;
ncopiesa = (totallength2 - randtimenm) / (tsamplnm + deltatsampl);
ncopies = IntegerPart[ncopiesa];
ares1 = Table[0, {ncopies}];
sizea = lengthares;
fousampled1 = Table[0, {ncopies}, {lengthares}];
Do[
  ares1[[ll]] = data["Values"][[(ll - 1) * tsampl / deltatsima + stpoint
     ;; IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];

   fousampled1[[ll]] = ResourceFunction["WelchSpectralEstimate"][
   ares1[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
   "OverlapOffset" → None, "SegmentSize" → sizea];
     ,
{ll, ncopies}];
powspmeansampledt1 = Drop[Mean[fousampled1], -1];
lengthpsd = Length[powspmeansampledt1]
meandataplot1 = Table[{0, 0}, {i, 1, lengthpsd}];
meandataplot1 =
  Table[{powspmeansampledt1[[i]][[1]], powspmeansampledt1[[i]][[2]]}, {i, 1, lengthpsd}];

(*plot settings*)
Remove[XTicks]
Remove[YTicks]
TickLength = 0.02;
Xmin = 0;
Xmax = 10 - 1;
Ymin = 0;
Ymax = 20;
deltaY = 2;
plotrangepsd = {{10^(2), 10^6}, {0.5 * 10^(-12), 10^(-2)}};
plotrangecoher = {{10^(2), 10^6}, {-1.05, 1.05}};
```

```
framelabels = {{"input psd (\!\(\*SuperscriptBox[\(nm\), \(2\)]\)/Hz)", None},
   {"noise frequency (Hz)", None}};
TickLength1 = 0.03;
Yticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, -24, 10, 2}];
Yticks2 = Table[{10^k, Null, {0, TickLength1}}, {k, -23, 10, 2}];
TickLength2 = 0.02;
aa = Table[0, {38}, {4}];
For[k = 1, k ≤ 38, k++,
aa[[k]] = Table[{l * 10^(-k + 10), , {0, TickLength2}}, {l, 2, 8, 2}];
]
YTicks = Union[Yticks1, Yticks2, aa[[1]], aa[[2]], aa[[3]],
   aa[[4]], aa[[5]], aa[[6]], aa[[7]], aa[[8]], aa[[9]], aa[[10]], aa[[11]], aa[[12]],
   aa[[13]], aa[[14]], aa[[15]], aa[[16]], aa[[17]], aa[[18]], aa[[19]], aa[[20]],
   aa[[21]], aa[[22]], aa[[23]], aa[[24]], aa[[25]], aa[[26]], aa[[27]], aa[[28]]];

Xticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, 3, 6, 1}];
aa2 = Table[0, {20}, {4}];
For[k = 1, k ≤ 10, k++,
aa2[[k]] = Table[{l * 10^k, , {0, TickLength2}}, {l, 2, 8, 2}];
]

Xticks2 = {{100, 0, {0, 0.03}}};

XTicks = Union[Xticks1, Xticks2, aa2[[2]], aa2[[3]], aa2[[4]], aa2[[5]], aa2[[6]],
   aa2[[7]], aa2[[8]], aa2[[9]], aa2[[10]], aa2[[11]], aa2[[12]], aa2[[13]], aa2[[14]],
   aa2[[15]], aa2[[16]], aa2[[17]], aa2[[18]], aa2[[19]], aa2[[20]], Xticks2];


pady = 100;
padx = 70;
pad2 = 30;
fntsize = 18;


fntsize = 18;


 (*Fig S16A Power of input white noise signal vs frequency*)

plotstyleY = {■};

meandataplot1c = {100, meandataplot1[[1]][[2]]};
plotrange1 = {{80, 6 * 10^5}, {0.2 * 10^(-7), 10^(-3)}};
ListLogLogPlot[{meandataplot1},
```

```
    PlotStyle → plotstyleY, Joined → True, PlotRange → plotrange1,
    ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
    LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
    Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
    FrameTicks → {{YTicks, None}, {XTicks, None}},
    FrameLabel → {"frequency (Hz)",
      "\*SubscriptBox[\(P\), \(Y\)] (\!\(\*SuperscriptBox[\(nm\), \(2\)]\)/Hz)"},
    GridLines → {{5000}, {}}, GridLinesStyle →
     Directive[■, Dashed, Thickness → 0.01 / 2]];
  Show[%, ListLogLogPlot[{meandataplot1c}, PlotStyle → plotstyleY]]
```

Out[ ]=

```
RandomGeneratorState[ Method: ExtendedCA
                      State hash: −8 850 395 944 280 919 728 ]
```

Out[ ]=

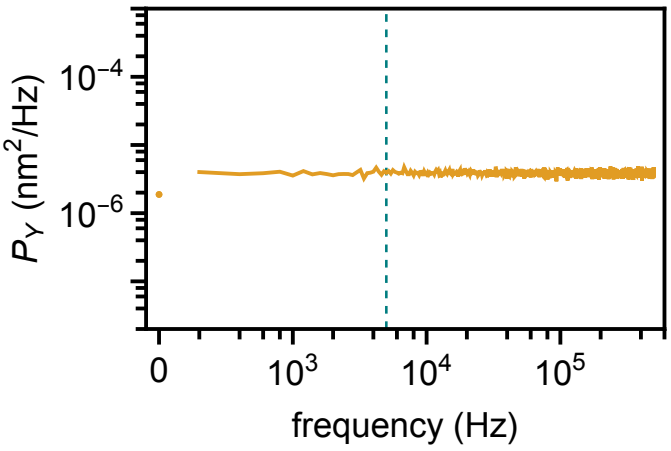
```
1.93224
```

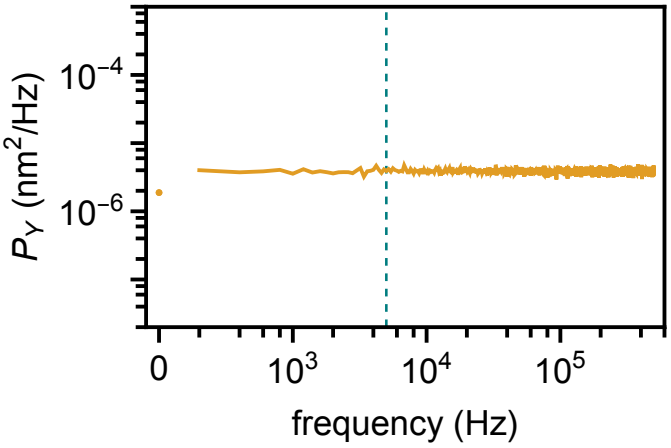
Out[ ]=

```
0.00053573
```

Out[ ]=

```
2501
```





In[ ]:=

```
(*sinusoidal signal with amplitude of 80 nm and frequency wstd =  5 kHz*)
datasin =
  Table[signalampl * (Sin[2 * Pi * wstd * (t - tstartFd)]), {t, 0, 1.014, deltatsima}];

datan = data["Values"];
datan2 = Drop[datan, 2];

(*white noise signal + sinusoidal signal *)
datasinplusnoise = datasin + datan2;



(*calculate power specturm of white noise signal + sinusoidal signal*)
```

```
resampfactor = factot;

randtime100 = 0.005 * 10;
randtimenm = Round[randtime100, deltatsima];
stpoint = IntegerPart[randtime100 / deltatsima];
deltatsampl = Round[deltatsima * resampfactor, deltatsima];
tsampl = 0.005;
endpoint = IntegerPart[Length[datasinplusnoise] - tsampl / deltatsima - stpoint];
tsamplnm = Round[tsampl, deltatsima];
b1 = Table[f[i], {i, randtimenm, tsamplnm + randtimenm, deltatsampl}];
lengthares = Length[b1];

totallength2 = totallength;
ncopiesa = (totallength2 - randtimenm) / (tsamplnm + deltatsampl);
ncopies = IntegerPart[ncopiesa];


ares1 = Table[0, {ncopies}];
sizea = lengthares;


fousampled1 = Table[0, {ncopies}, {lengthares}];

Do[
       ares1[[ll]] = datasinplusnoise[[(ll - 1) * tsampl / deltatsima + stpoint
      ;; IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];

    fousampled1[[ll]] = ResourceFunction["WelchSpectralEstimate"][
    ares1[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
    "OverlapOffset" → None, "SegmentSize" → sizea];


      ,
{ll, ncopies}];

powspmeansampledt1 = Drop[Mean[fousampled1], -1];
meandataplot1 = Table[{0, 0}, {i, 1, lengthpsd}];
meandataplot1 =
  Table[{powspmeansampledt1[[i]][[1]], powspmeansampledt1[[i]][[2]]}, {i, 1, lengthpsd}];


(*Fig S16A Power of input white noise signal + sinusoidal signal vs frequency*)

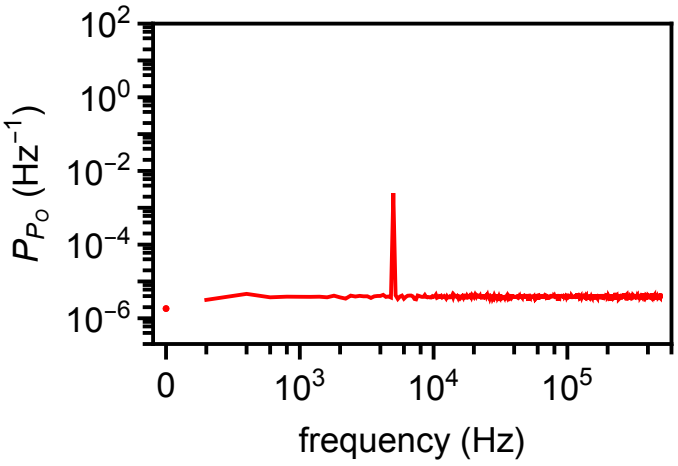
plotstyleY = {Red};
plotstylecps = {■};
```

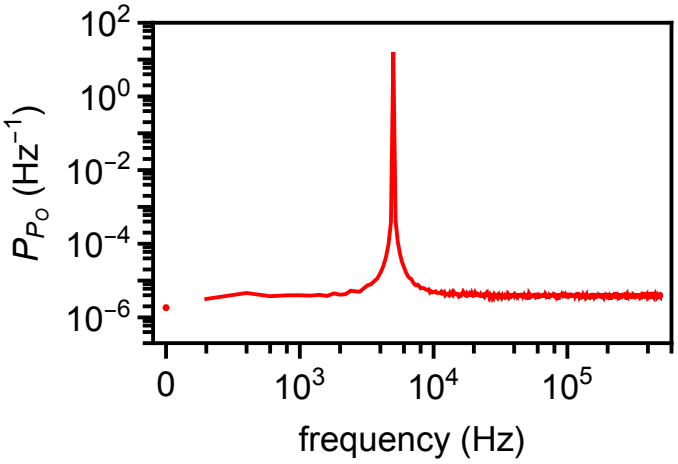
```
plotstyleP = {Red};

meandataplot1c = {100, meandataplot1[[1]][[2]]};

commondirgridlines = {Dashed, Thickness → 0.01 / 2};
yLineStyle = Directive[Darker[■], Dashed, Thickness → 0.01 / 2];

plotrange3 = {{80, 6 * 10^5}, {0.2 * 10^(-6), 10^(2)}};
ListLogLogPlot[{meandataplot1}, PlotStyle → plotstyleP,
  Joined → True, Frame → True, PlotRange → plotrange3,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"frequency (Hz)", "\*SubscriptBox[\(P\), \(P0\)] (Hz^-1)"}];
psd11 = Show[%, ListLogLogPlot[{meandataplot1c}, PlotStyle → plotstyleP]]
```





In[•]:=
```
(*Apply quasistatic current function to white noise signal + sinusoidal signal*)
dataoutsin =
 (1 + Exp[- (datasinplusnoise + Yrestd - Yhd) * kGSd * Dvd / (Ncd * kbTd)]) ^ (-1)
dataoutsin[[1]]

(*Calculation of power of output white noise signal + sinusoidal signal*)
resampfactor = factot;

randtime100 = 0.005 * 10;
randtimenm = Round[randtime100, deltatsima];
stpoint = IntegerPart[randtime100 / deltatsima];
deltatsampl = Round[deltatsima * resampfactor, deltatsima];
tsampl = 0.005;
endpoint = IntegerPart[Length[dataoutsin] - tsampl / deltatsima - stpoint];
tsamplnm = Round[tsampl, deltatsima];
```

```
b1 = Table[f[i], {i, randtimenm, tsamplnm + randtimenm, deltatsampl}];
lengthares = Length[b1];
totallength2 = totallength;
ncopiesa = (totallength2 - randtimenm) / (tsamplnm + deltatsampl);
ncopies = IntegerPart[ncopiesa];

ares1 = Table[0, {ncopies}];
sizea = lengthares;


fousampled1 = Table[0, {ncopies}, {lengthares}];

Do[


    ares1[[ll]] = dataoutsin[[(ll - 1) * tsampl / deltatsima + stpoint ;;
      IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];

    fousampled1[[ll]] = ResourceFunction["WelchSpectralEstimate"][
    ares1[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
    "OverlapOffset" → None, "SegmentSize" → sizea];


      ,
{ll, ncopies}];

powspmeansampledt1 = Drop[Mean[fousampled1], -1];
meandataplot1 = Table[{0, 0}, {i, 1, lengthpsd}];
meandataplot1 =
  Table[{powspmeansampledt1[[i]][[1]], powspmeansampledt1[[i]][[2]]}, {i, 1, lengthpsd}];


 (*Fig S16B Power of white noise signal vs frequency*)
plotstyleY = {Red};
plotstylecps = {■};
plotstyleP = {Red};

meandataplot1c = {100, meandataplot1[[1]][[2]]};

commondirgridlines = {Dashed, Thickness → 0.01 / 2};
yLineStyle = Directive[Darker[■], Dashed, Thickness → 0.01 / 2];

plotrange3 = {{80, 6 * 10^5}, {10^(-10), 10^(-2)}};
ListLogLogPlot[{meandataplot1}, PlotStyle → plotstyleP,
```

```
  Joined → True, Frame → True, PlotRange → plotrange3,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"frequency (Hz)", "\*SubscriptBox[\(P\), \(P0\)] (Hz^-1)"}];
psd11 = Show[%, ListLogLogPlot[{meandataplot1c}, PlotStyle → plotstyleP]]
```

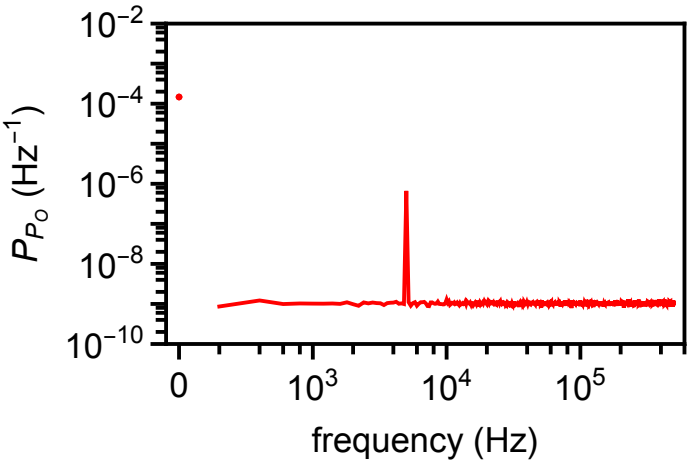
Out[ ]=

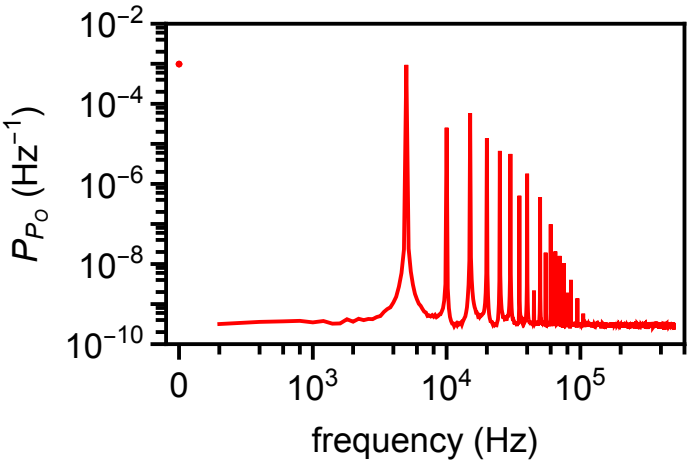
```
{0.175199, 0.134189, 0.175982, 0.213585, 0.183978, 0.187809, 0.15545, 0.183138, 0.198317, 0.217883,
 0.135996, 0.176948, 0.166663, 0.215046, ··· 4055973 ···, 0.183941, 0.117374, 0.203064, 0.204602,
 0.171129, 0.157021, 0.176642, 0.175016, 0.186644, 0.167541, 0.191817, 0.178794, 0.144337, 0.183791}
```

Full expression not available (original memory size: 32.4 MB)

Out[ ]=

```
0.175199
```





In[ ]:=

```
(*Apply quasistatic current function to white noise signal *)
dataoutnoise =
```

```
 (1 + Exp[- (data["Values"] + Yrestd - Yhd) * kGSd * Dvd / (Ncd * kbTd)]) ^ (-1)


(*Calculation of power of output white noise signal *)
resampfactor = factot;

randtime100 = 0.005 * 10;
randtimenm = Round[randtime100, deltatsima];
stpoint = IntegerPart[randtime100 / deltatsima];
deltatsampl = Round[deltatsima * resampfactor, deltatsima];
tsampl = 0.005;
endpoint = IntegerPart[Length[dataoutnoise] - tsampl / deltatsima - stpoint];
tsamplnm = Round[tsampl, deltatsima];
b1 = Table[f[i], {i, randtimenm, tsamplnm + randtimenm, deltatsampl}];
lengthares = Length[b1];
totallength2 = totallength;
ncopiesa = (totallength2 - randtimenm) / (tsamplnm + deltatsampl);
ncopies = IntegerPart[ncopiesa];

ares1 = Table[0, {ncopies}];
sizea = lengthares;


fousampled1 = Table[0, {ncopies}, {lengthares}];

Do[


    ares1[[ll]] = dataoutnoise[[(ll - 1) * tsampl / deltatsima + stpoint
     ;; IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];

    fousampled1[[ll]] = ResourceFunction["WelchSpectralEstimate"][
    ares1[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
    "OverlapOffset" → None, "SegmentSize" → sizea];


      ,
{ll, ncopies}];

powspmeansampledt1 = Drop[Mean[fousampled1], -1];
meandataplot1 = Table[{0, 0}, {i, 1, lengthpsd}];

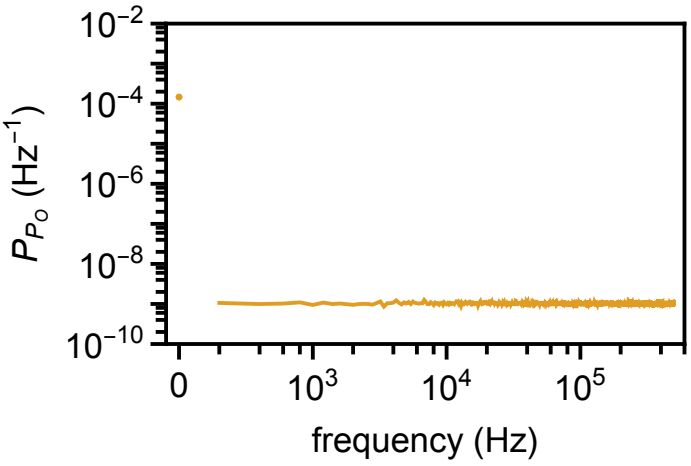
meandataplot1 =
  Table[{powspmeansampledt1[[i]][[1]], powspmeansampledt1[[i]][[2]]}, {i, 1, lengthpsd}];
```

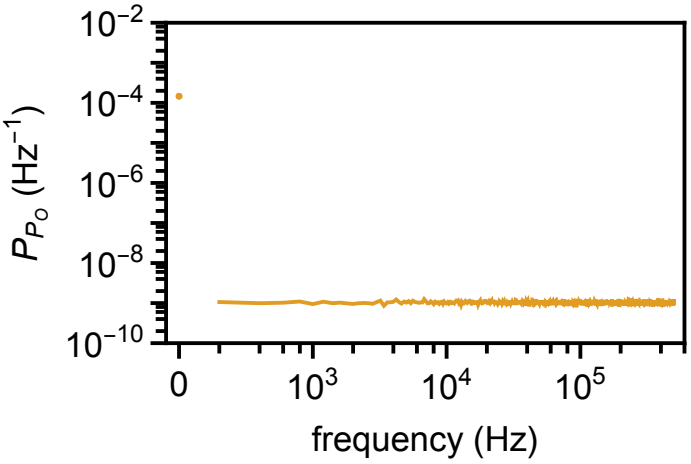
```
(*Fig S16B Power of  outputwhite noise signal vs frequency*)
plotstyleY = {■};
plotstylecps = {■};
plotstyleP = {■};

meandataplot1c = {100, meandataplot1[[1]][[2]]};

commondirgridlines = {Dashed, Thickness → 0.01 / 2};
yLineStyle = Directive[Darker[■], Dashed, Thickness → 0.01 / 2];

plotrange3 = {{80, 6 * 10^5}, {10^(-10), 10^(-2)}};
ListLogLogPlot[{meandataplot1}, PlotStyle → plotstyleP,
  Joined → True, Frame → True, PlotRange → plotrange3,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"frequency (Hz)", "\*SubscriptBox[\(P\), \(P₀\)] (Hz^-1)"}];
psd11 = Show[%, ListLogLogPlot[{meandataplot1c}, PlotStyle → plotstyleP]]
```





In[•]:=
```
(*Calculation of quasistatic current vs stimulus-force and deflection for gating
   swing of 32 nm and resting current of 0.17. Set D=32 and Pinfrest=0.17*)
fval = Table[n, {n, -180, 180, 0.5}];(* table of values of stimulus-force*)
nf = Length[fval];
PYval = Table[0, {nf}];
PFval = Table[0, {nf}];
FYval = Table[0, {nf}];

(*Solve equations of motion numerically to find current,
deflection and stimulus-force as a function of time*)
gx =  2 * kbTd * lPd;
y0in = Yrestd;
Do[sp = NDSolve[{lPd * Y'[t] ==
       (-kPd) * (Y[t] - Yupd) - kGSd * (Y[t] - Yugd - Dvd * Po[t]) + Fzd * fval[[ll]] ,
      taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd] * Po'[t] ==
```

```
      (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] - Po[t]),
     Y[0] == Yrestd, Po[0] == Pinfrest} , {Y, Po}, {t, 10}];
  PYval[[ll]] = {Y[9] /. First[sp], Po[9] /. First[sp]};
  (* deflection as a function of time*)
  PFval[[ll]] = {fval[[ll]], Po[9] /. First[sp]};
  (* current as a function of time*)
  FYval[[ll]] = {Y[9] /. First[sp], fval[[ll]]};(* force as a function of time*)
  ,
{ll, nf}];

(*plot settings*)
framelabels = {{"current", None}, {"stimulus-force amplitude (pN)", None}};
plotrange = {{-500, 500}, {-0.1, 1.1}};

YTicks = {{0.0, "0.0", {0, 0.03`}}, {0.5, 0.5, {0, 0.03`}}, {-0.5, -0.5, {0, 0.03`}},
   {1.0, "1.0", {0, 0.03`}}, {-1.0, "-1.0", {0, 0.03`}}, {0.1, Null, {0, 0.02`}},
   {0.2, Null, {0, 0.02`}}, {0.3, Null, {0, 0.02`}}, {0.4, Null, {0, 0.02`}},
   {0.6, Null, {0, 0.02`}}, {0.7, Null, {0, 0.02`}}, {0.8, Null, {0, 0.02`}},
   {0.9, Null, {0, 0.02`}}, {-0.1, Null, {0, 0.02`}}, {-0.2, Null, {0, 0.02`}},
   {-0.3, Null, {0, 0.02`}}, {-0.4, Null, {0, 0.02`}}, {-0.6, Null, {0, 0.02`}},
   {-0.7, Null, {0, 0.02`}}, {-0.8, Null, {0, 0.02`}}, {-0.9, Null, {0, 0.02`}}};

Xticks1d = Table[{k, k, {0, 0.03}}, {k, -600, 400, 50}];
Xticks2d = Table[{k, Null, {0, 0.02}}, {k, -600, 400, 50 / 2}];
XTickshistd = Union[Xticks1d, Xticks2d];

framelabelsd = {{"current", None}, {"displacement (nm)", None}};
plotrange = {{-510, 340}, {-0.1, 1.1}};

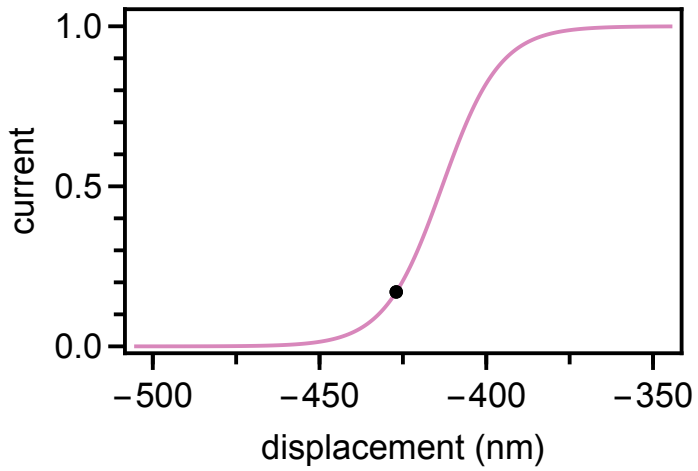
 (*Fig 1D current vs deflection for gating swing of
   32 nm and resting current of 0.17. Set D=32 and Pinfrest=0.17.*)
ListPlot[PYval, PlotRange → {Full, Full},
   Joined → True, Background → White, Frame → {True, True, True, True},
   FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsd,
   PlotRange → plotrange, LabelStyle → {FontSize → 18}, Axes → {False, False},
   PlotStyle → {Lighter[■]}, FrameTicks → {{YTicks, None}, {XTickshistd, None}}] ;
Show[%,
 ListPlot[{{-427, 0.17}}, PlotMarkers → {Automatic, 10}, PlotStyle → { Black}]]

Xticks1 = Table[{k, k, {0, 0.03}}, {k, -3000, 3000, 100}];
Xticks2 = Table[{k, Null, {0, 0.02}}, {k, -3000, 3000, 100 / 2}];
XTickshist = Union[Xticks1, Xticks2];

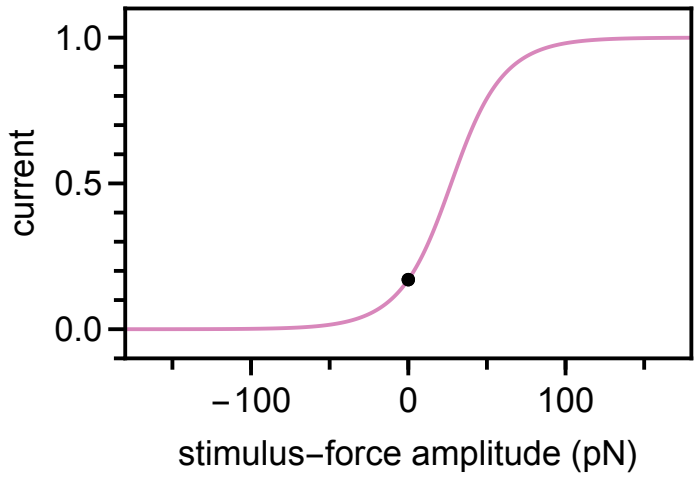
 (*Fig 1C current vs stimulus-force amplitude for gating swing
```

```
   of 32 nm and resting current of 0.17. Set D=32 and Pinfrest=0.17.*)
plotrange = {{-180, 180}, {-0.1, 1.1}};
ListPlot[PFval, Joined → True, Background → White, Frame → {True, True, True, True},
  FrameStyle → Directive[Black, Thick], FrameLabel → framelabels,
  PlotRange → plotrange, LabelStyle → {FontSize → 18}, Axes → {False, False},
  PlotStyle → {Lighter[■]}, FrameTicks → {{YTicks, None}, {XTickshist, None}}] ;
Show[%, ListPlot[{{0, 0.17}}, PlotMarkers → {Automatic, 10}, PlotStyle → { Black}]]
```

Out[ ]=



Out[ ]=




```
(* Calculation of quasistatic current and quasistatic stiffness
  vs IHB deflection for a gating swing of 32 nm. Set D=32*)
fval = Table[n, {n, -700, 700, 0.5}];(* table of values of stimulus-force*)
nf = Length[fval];
PYval = Table[0, {nf}];
PFval = Table[0, {nf}];
FYval = Table[0, {nf}];
```

```
gx = 2 * kbTd * lPd;
y0in = Yrestd;
Do[sp = NDSolve[{lPd * Y'[t] ==
       (-kPd) * (Y[t] - Yupd) - kGSd * (Y[t] - Yugd - Dvd * Po[t]) + Fzd * fval[[ll]] ,
      taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd] * Po'[t] ==
       (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] - Po[t]),
      Y[0] == Yrestd, Po[0] == Pinfrest} , {Y, Po}, {t, 10}];
  PYval[[ll]] = {Y[9] /. First[sp], Po[9] /. First[sp]};
  (* deflection as a function of time*)
  PFval[[ll]] = {fval[[ll]], Po[9] /. First[sp]};
  (* current as a function of time*)
  FYval[[ll]] = {Y[9] /. First[sp], fval[[ll]]};(* force as a function of time*)
  ,
{ll, nf}];

Khbvals = Table[0, Length[FYval] - 5];
Do[

  lm = LinearModelFit[FYval[[ll ;; ll + 5]], x, x];
  Khbvals[[ll]] = {FYval[[ll]][[1]], lm["BestFitParameters"][[2]]};,
  {ll, Length[FYval] - 5}];

framelabels = {{"stiffness (mN/m)", None}, {"deflection (nm)", None}};
plotrange = {{-500, -120}, Full};

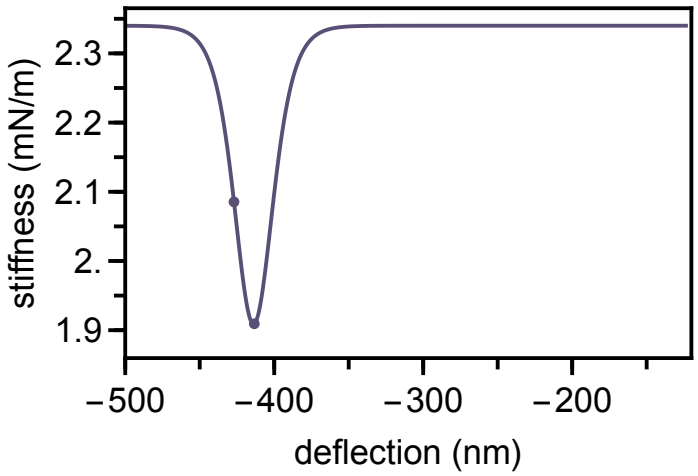

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 1.8, 2.4, 0.1}];
Yticks2 = Table[{k, Null, {0, 0.02}}, {k, 1.8, 2.4, 0.1 / 2}];
Ytickshist = Union[Yticks1, Yticks2];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, -700, 700, 100}];
Xticks2 = Table[{k, Null, {0, 0.02}}, {k, -700, 700, 100 / 2}];
XTickshist = Union[Xticks1, Xticks2];
(* Fig S7B quasisatic stimulus-
  force vs deflection for a gating swing of 32 nm. Set D=32 *)
ListPlot[Khbvals, Joined → True, Background → White, Frame → {True, True, True, True},
  FrameStyle → Directive[Black, Thick], FrameLabel → framelabels,
  PlotRange → plotrange, LabelStyle → {FontSize → 18}, Axes → {False, False},
  PlotStyle → {Darker[■]}, FrameTicks → {{Ytickshist, None}, {XTickshist, None}}] ;
Show[%, ListPlot[{{-427.00000000000006`, 2.0854007254599414`},
   {-413.375527151138`, 1.909275600319795`}},
  PlotMarkers → {Automatic, 8}, PlotStyle → {Darker[■]}]]
```

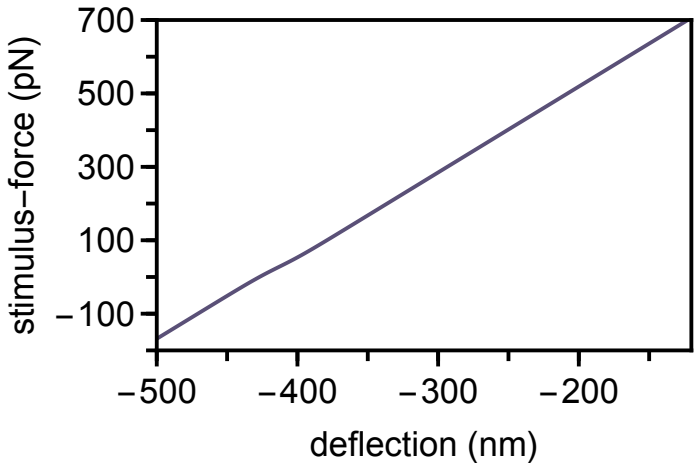
```
Yticks1 = Table[{k, k, {0, 0.03}}, {k, -700, 700, 200}];
Yticks2 = Table[{k, Null, {0, 0.02}}, {k, -700, 700, 200 / 2}];
Yticksc = Union[Yticks1, Yticks2];

framelabels = {{"stimulus-force (pN)", None}, {"deflection (nm)", None}};
(* Fig S7A quasisatic stiffness vs
  deflection for a gating swing of 32 nm. Set D=32 *)
plotrange = {{-500, -120}, {-200, 700}};
ListPlot[FYval, Joined → True, Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabels,
 PlotRange → plotrange, LabelStyle → {FontSize → 18}, Axes → {False, False},
 PlotStyle → {Darker[■]}, FrameTicks → {{Yticksc, None}, {XTickshist, None}}]
```

Out[ ]=



Out[ ]=




```
(*Calculation of quasistatic current vs stimulus-force and deflection for gating
   swing of 32 nm and resting current of 0.5. Set D=32 and Pinfrest=0.5*)
```

```
fval = Table[n, {n, -180, 180, 0.5}];(* table of values of stimulus-force*)
nf = Length[fval];
PYval = Table[0, {nf}];
PFval = Table[0, {nf}];
FYval = Table[0, {nf}];

(*Solve equations of motion numerically to find current,
deflection and stimulus-force as a function of time*)
gx =  2 * kbTd * lPd;
y0in = Yrestd;
Do[sp = NDSolve[{lPd * Y'[t] ==
       (-kPd) * (Y[t] - Yupd) - kGSd * (Y[t] - Yugd - Dvd * Po[t]) + Fzd * fval[[ll]] ,
      taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd] * Po'[t] ==
       (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] - Po[t]),
      Y[0] == Yrestd, Po[0] == Pinfrest} , {Y, Po}, {t, 10}];
  PYval[[ll]] = {Y[9] /. First[sp], Po[9] /. First[sp]};
  (* deflection as a function of time*)
  PFval[[ll]] = {fval[[ll]], Po[9] /. First[sp]};
  (* current as a function of time*)
  FYval[[ll]] = {Y[9] /. First[sp], fval[[ll]]};(* force as a function of time*)
  ,
{ll, nf}];

(*plot settings*)
framelabels = {{"current", None}, {"stimulus-force amplitude (pN)", None}};
plotrange = {{-500, 500}, {-0.1, 1.1}};

YTicks = {{0.0, "0.0", {0, 0.03`}}, {0.5, 0.5, {0, 0.03`}}, {-0.5, -0.5, {0, 0.03`}},
    {1.0, "1.0", {0, 0.03`}}, {-1.0, "-1.0", {0, 0.03`}}, {0.1, Null, {0, 0.02`}},
    {0.2, Null, {0, 0.02`}}, {0.3, Null, {0, 0.02`}}, {0.4, Null, {0, 0.02`}},
    {0.6, Null, {0, 0.02`}}, {0.7, Null, {0, 0.02`}}, {0.8, Null, {0, 0.02`}},
    {0.9, Null, {0, 0.02`}}, {-0.1, Null, {0, 0.02`}}, {-0.2, Null, {0, 0.02`}},
    {-0.3, Null, {0, 0.02`}}, {-0.4, Null, {0, 0.02`}}, {-0.6, Null, {0, 0.02`}},
    {-0.7, Null, {0, 0.02`}}, {-0.8, Null, {0, 0.02`}}, {-0.9, Null, {0, 0.02`}}};

Xticks1d = Table[{k, k, {0, 0.03}}, {k, -600, 400, 50}];
Xticks2d = Table[{k, Null, {0, 0.02}}, {k, -600, 400, 50 / 2}];
XTickshistd = Union[Xticks1d, Xticks2d];

framelabelsd = {{"current", None}, {"displacement (nm)", None}};
plotrange = {{-510, 340}, {-0.1, 1.1}};

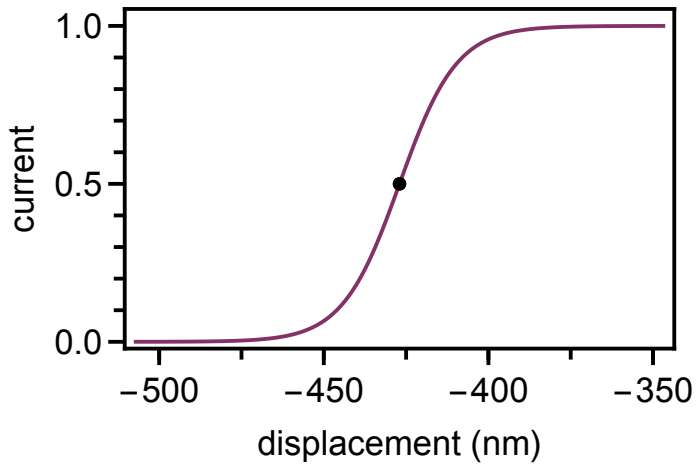
 (*Fig 1D current vs deflection for gating swing of
   32 nm and resting current of 0.5. Set D=32 and Pinfrest=0.5.*)
```

```
ListPlot[PYval, PlotRange → {Full, Full},
  Joined → True, Background → White, Frame → {True, True, True, True},
  FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsd,
  PlotRange → plotrange, LabelStyle → {FontSize → 18}, Axes → {False, False},
  PlotStyle → {Darker[■]}, FrameTicks → {{YTicks, None}, {XTickshistd, None}}] ;
Show[%, ListPlot[{{-427, 0.5}}, PlotMarkers → {Automatic, 10}, PlotStyle → { Black}]]

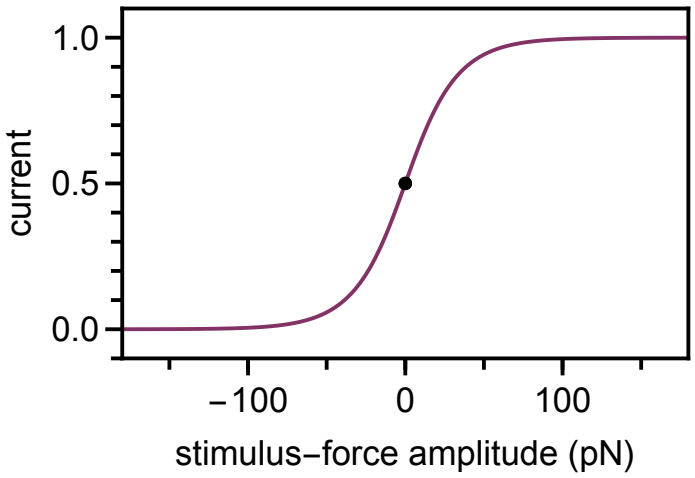
Xticks1 = Table[{k, k, {0, 0.03}}, {k, -3000, 3000, 100}];
Xticks2 = Table[{k, Null, {0, 0.02}}, {k, -3000, 3000, 100 / 2}];
XTickshist = Union[Xticks1, Xticks2];

(*Fig 1C current vs stimulus-force amplitude for gating swing
   of 32 nm and resting current of 0.5. Set D=32 and Pinfrest=0.5.*)
plotrange = {{-180, 180}, {-0.1, 1.1}};
ListPlot[PFval, Joined → True, Background → White, Frame → {True, True, True, True},
  FrameStyle → Directive[Black, Thick], FrameLabel → framelabels,
  PlotRange → plotrange, LabelStyle → {FontSize → 18}, Axes → {False, False},
  PlotStyle → {Darker[■]}, FrameTicks → {{YTicks, None}, {XTickshist, None}}] ;
Show[%, ListPlot[{{0, 0.5}}, PlotMarkers → {Automatic, 10}, PlotStyle → { Black}]]
```

Out[ ]=



Out[ ]=




```
(*Calculation of quasistatic current vs stimulus-force and deflection for gating
   swing of 1.75 nm and resting current of 0.17. Set D=1.75 and Pinfrest=0.17*)
fval = Table[n, {n, -180, 180, 0.5}];(* table of values of stimulus-force*)
nf = Length[fval];
PYval = Table[0, {nf}];
PFval = Table[0, {nf}];
FYval = Table[0, {nf}];

(*Solve equations of motion numerically to find current,
deflection and stimulus-force as a function of time*)
gx =  2 * kbTd * lPd;
y0in = Yrestd;
Do[sp = NDSolve[{lPd * Y'[t] ==
       (-kPd) * (Y[t] - Yupd) - kGSd * (Y[t] - Yugd - Dvd * Po[t]) + Fzd * fval[[ll]] ,
      taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd] * Po'[t] ==
       (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] - Po[t]),
      Y[0] == Yrestd, Po[0] == Pinfrest} , {Y, Po}, {t, 10}];
  PYval[[ll]] = {Y[9] /. First[sp], Po[9] /. First[sp]};
  (* deflection as a function of time*)
  PFval[[ll]] = {fval[[ll]], Po[9] /. First[sp]};
  (* current as a function of time*)
  FYval[[ll]] = {Y[9] /. First[sp], fval[[ll]]};(* force as a function of time*)
  ,
{ll, nf}];

(*plot settings*)
framelabels = {{"current", None}, {"stimulus-force amplitude (pN)", None}};
plotrange = {{-500, 500}, {-0.1, 1.1}};

YTicks = {{0.0, "0.0", {0, 0.03`}}, {0.5, 0.5, {0, 0.03`}}, {-0.5, -0.5, {0, 0.03`}},
```

```
    {1.0, "1.0", {0, 0.03`}}, {-1.0, "-1.0", {0, 0.03`}}, {0.1, Null, {0, 0.02`}},
    {0.2, Null, {0, 0.02`}}, {0.3, Null, {0, 0.02`}}, {0.4, Null, {0, 0.02`}},
    {0.6, Null, {0, 0.02`}}, {0.7, Null, {0, 0.02`}}, {0.8, Null, {0, 0.02`}},
    {0.9, Null, {0, 0.02`}}, {-0.1, Null, {0, 0.02`}}, {-0.2, Null, {0, 0.02`}},
    {-0.3, Null, {0, 0.02`}}, {-0.4, Null, {0, 0.02`}}, {-0.6, Null, {0, 0.02`}},
    {-0.7, Null, {0, 0.02`}}, {-0.8, Null, {0, 0.02`}}, {-0.9, Null, {0, 0.02`}}};

Xticks1d = Table[{k, k, {0, 0.03}}, {k, -600, 400, 50}];
Xticks2d = Table[{k, Null, {0, 0.02}}, {k, -600, 400, 50 / 2}];
XTickshistd = Union[Xticks1d, Xticks2d];

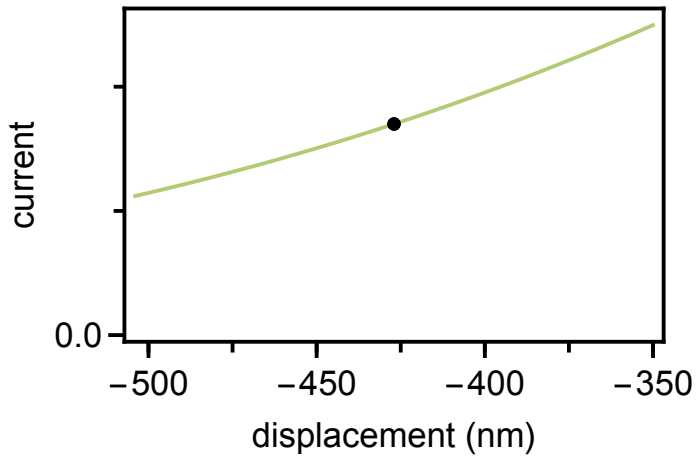
framelabelsd = {{"current", None}, {"displacement (nm)", None}};
plotrange = {{-510, 340}, {-0.1, 1.1}};

(*Fig 1D current vs deflection for gating swing of 1.75
  nm and resting current of 0.17. Set D=1.75 and Pinfrest=0.17.*)
ListPlot[PYval, PlotRange → {Full, Full},
  Joined → True, Background → White, Frame → {True, True, True, True},
  FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsd,
  PlotRange → plotrange, LabelStyle → {FontSize → 18}, Axes → {False, False},
  PlotStyle → {Lighter[■]}, FrameTicks → {{YTicks, None}, {XTickshistd, None}}] ;
Show[%,
 ListPlot[{{-427, 0.17}}, PlotMarkers → {Automatic, 10}, PlotStyle → { Black}]]

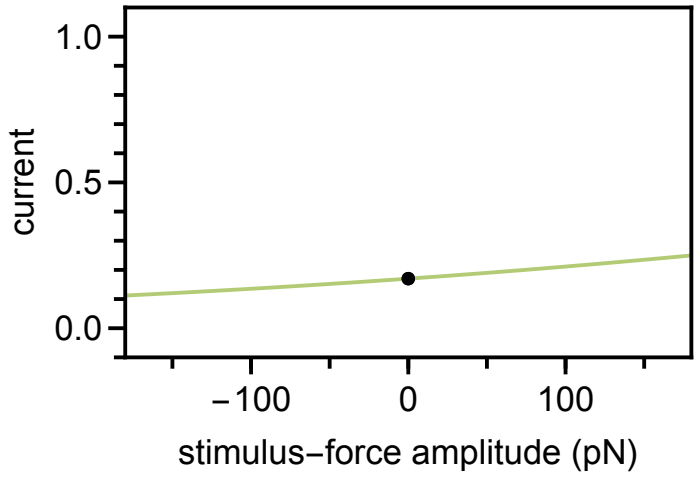
Xticks1 = Table[{k, k, {0, 0.03}}, {k, -3000, 3000, 100}];
Xticks2 = Table[{k, Null, {0, 0.02}}, {k, -3000, 3000, 100 / 2}];
XTickshist = Union[Xticks1, Xticks2];

(*Fig 1C current vs stimulus-force amplitude for gating swing of
   1.75 nm and resting current of 0.17. Set D=1.75 and Pinfrest=0.17.*)
plotrange = {{-180, 180}, {-0.1, 1.1}};
ListPlot[PFval, Joined → True, Background → White, Frame → {True, True, True, True},
  FrameStyle → Directive[Black, Thick], FrameLabel → framelabels,
  PlotRange → plotrange, LabelStyle → {FontSize → 18}, Axes → {False, False},
  PlotStyle → {Lighter[■]}, FrameTicks → {{YTicks, None}, {XTickshist, None}}] ;
Show[%, ListPlot[{{0, 0.17}}, PlotMarkers → {Automatic, 10}, PlotStyle → { Black}]]
```

Out[ ]=



Out[ ]=




```
(* Calculation of quasistatic current and quasistatic stiffness
  vs IHB deflection for a gating swing of 1.75 nm. Set D=1.75*)
fval = Table[n, {n, -700, 700, 0.5}];(* table of values of stimulus-force*)
nf = Length[fval];
PYval = Table[0, {nf}];
PFval = Table[0, {nf}];
FYval = Table[0, {nf}];

gx =  2 * kbTd * lPd;
y0in = Yrestd;
Do[sp = NDSolve[{lPd * Y'[t] ==
       (-kPd) * (Y[t] - Yupd) - kGSd * (Y[t] - Yugd - Dvd * Po[t]) + Fzd * fval[[ll]] ,
      taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd] * Po'[t] ==
       (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] - Po[t]),
      Y[0] == Yrestd, Po[0] == Pinfrest} , {Y, Po}, {t, 10}];
  PYval[[ll]] = {Y[9] /. First[sp], Po[9] /. First[sp]};
```

```
  (* deflection as a function of time*)
  PFval[[ll]] = {fval[[ll]], Po[9] /. First[sp]};
  (* current as a function of time*)
  FYval[[ll]] = {Y[9] /. First[sp], fval[[ll]]};(* force as a function of time*)
  ,
{ll, nf}];

Khbvals = Table[0, Length[FYval] - 5];
Do[

  lm = LinearModelFit[FYval[[ll ;; ll + 5]], x, x];
  Khbvals[[ll]] = {FYval[[ll]][[1]], lm["BestFitParameters"][[2]]};,
  {ll, Length[FYval] - 5}];

framelabels = {{"stiffness (mN/m)", None}, {"deflection (nm)", None}};
plotrange = {{-500, -120}, {1.8, 2.38}};

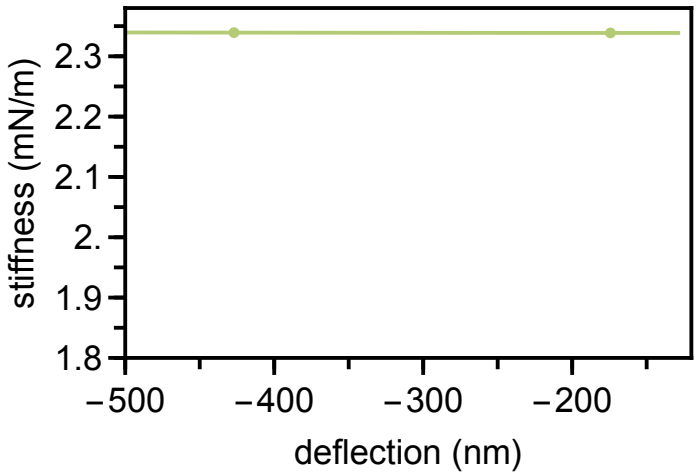

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 1.8, 2.4, 0.1}];
Yticks2 = Table[{k, Null, {0, 0.02}}, {k, 1.8, 2.4, 0.1 / 2}];
Ytickshist = Union[Yticks1, Yticks2];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, -700, 700, 100}];
Xticks2 = Table[{k, Null, {0, 0.02}}, {k, -700, 700, 100 / 2}];
XTickshist = Union[Xticks1, Xticks2];
(* Fig S7B quasisatic stimulus-
  force vs deflection for a gating swing of 1.75 nm. Set D=1.75 *)
ListPlot[Khbvals, Joined → True, Background → White, Frame → {True, True, True, True},
  FrameStyle → Directive[Black, Thick], FrameLabel → framelabels,
  PlotRange → plotrange, LabelStyle → {FontSize → 18}, Axes → {False, False},
  PlotStyle → {Lighter[■]}, FrameTicks → {{Ytickshist, None}, {XTickshist, None}}] ;
Show[%, ListPlot[
  {{-427.`, 2.3392719759408562`}, {-174.32041705433645`, 2.338715823318084`}},
  PlotMarkers → {Automatic, 8}, PlotStyle → {Lighter[■]}]]

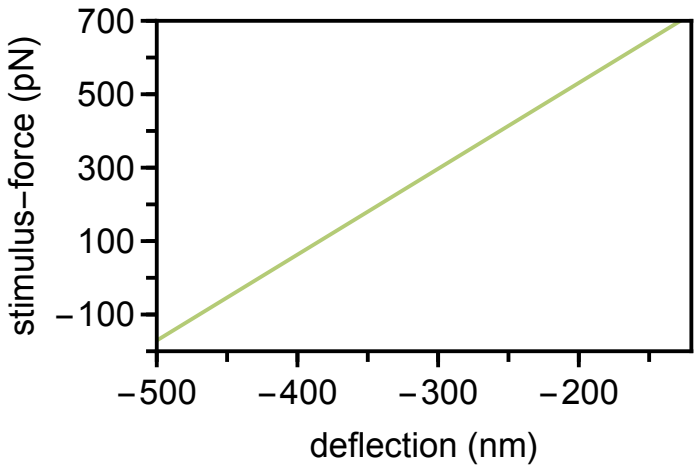
Yticks1 = Table[{k, k, {0, 0.03}}, {k, -700, 700, 200}];
Yticks2 = Table[{k, Null, {0, 0.02}}, {k, -700, 700, 200 / 2}];
Ytickshist = Union[Yticks1, Yticks2];
(* Fig S7B quasisatic stiffness vs
  deflection for a gating swing of 1.75 nm. Set D=1.75 *)
framelabels = {{"stimulus-force (pN)", None}, {"deflection (nm)", None}};
plotrange = {{-500, -120}, {-200, 700}};
ListPlot[FYval, Joined → True, Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabels,
```

```
PlotRange → plotrange, LabelStyle → {FontSize → 18}, Axes → {False, False},
PlotStyle → {Lighter[■]}, FrameTicks → {{Yticksc, None}, {XTickshist, None}}]
```

Out[ ]=



Out[ ]=




```
(* Calculation of quasistatic stiffness vs IHB deflection for a gating
  swing of 1.75 nm for a broader range of IHB deflection. Set D=1.75*)

fval = Table[n, {n, -2500, 2500, 1}];(* table of values of stimulus-force*)
nf = Length[fval];
PYval = Table[0, {nf}];
PFval = Table[0, {nf}];
FYval = Table[0, {nf}];

gx =  2 * kbTd * lPd;
y0in = Yrestd;
Do[sp = NDSolve[{lPd * Y'[t] ==
```

```
        (-kPd) * (Y[t] - Yupd) - kGSd * (Y[t] - Yugd - Dvd * Po[t]) + Fzd * fval[[ll]] ,
       taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd] * Po'[t] ==
        (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] - Po[t]),
       Y[0] == Yrestd, Po[0] == Pinfrest} , {Y, Po}, {t, 10}];
   PYval[[ll]] = {Y[9] /. First[sp], Po[9] /. First[sp]};
   (* deflection as a function of time*)
   PFval[[ll]] = {fval[[ll]], Po[9] /. First[sp]};
   (* current as a function of time*)
   FYval[[ll]] = {Y[9] /. First[sp], fval[[ll]]};(* force as a function of time*)
   ,
{ll, nf}];

Khbvals = Table[0, Length[FYval] - 5];
Do[

   lm = LinearModelFit[FYval[[ll ;; ll + 5]], x, x];
   Khbvals[[ll]] = {FYval[[ll]][[1]], lm["BestFitParameters"][[2]]};,
   {ll, Length[FYval] - 5}];

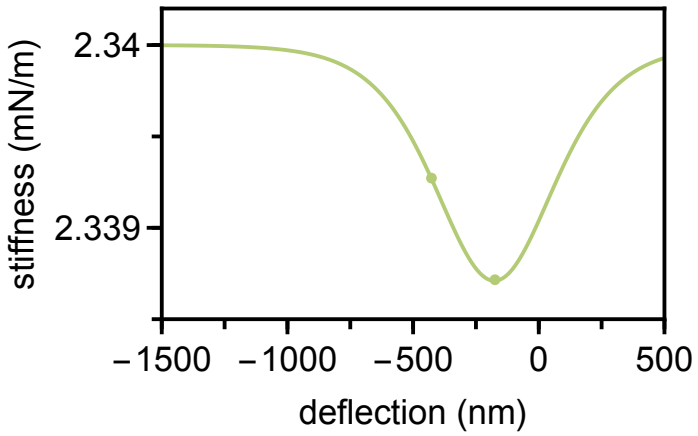
framelabels = {{"stiffness (mN/m)", None}, {"deflection (nm)", None}};
plotrange = {{-1500, 500}, {2.3385, 2.3402}};

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 2.338, 2.340, 0.001}];
Yticks2 = Table[{k, Null, {0, 0.02}}, {k, 2.338, 2.340, 0.001 / 2}];
Ytickshist = Union[Yticks1, Yticks2];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, -1500, 500, 1000 / 2}];
Xticks2 = Table[{k, Null, {0, 0.02}}, {k, -1500, 500, 1000 / 4}];
XTickshist = Union[Xticks1, Xticks2];

ListPlot[Khbvals, Joined → True, Background → White, Frame → {True, True, True, True},
   FrameStyle → Directive[Black, Thick], FrameLabel → framelabels,
   PlotRange → plotrange, LabelStyle → {FontSize → 18}, Axes → {False, False},
   PlotStyle → {Lighter[■]}, FrameTicks → {{Ytickshist, None}, {XTickshist, None}}] ;
Show[%, ListPlot[
   {{-427.`, 2.3392719759408562`}, {-174.32041705433645`, 2.338715823318084`}},
   PlotMarkers → {Automatic, 8}, PlotStyle → {Lighter[■]}]]
```

Out[ ]=




```
(*Calculation of analytical power spectra for default parameters*)
freqs = Table[i, {i, 0, 10^7, 200}];
(*table of frequency values for calculation of power spectra*)
lengthpsd = Length[freqs];
fc = 610; (*corner frequency for cell-
 membrane filter to apply to current psd to derive voltage psd*)
plotstyleY = {Darker[■]};
plotstyleP = {Darker[■]};
plotstyleV = {Darker[■]};

PYY = (4 kbT (kG^2 lP + lG (kGS^2 + lG lP w^2))) / ((kGS^2 - kG (kGS + kP))^2 +
      ((kGS + kP)^2 lG^2 + 2 kGS^2 lG lP + kG^2 lP^2) w^2 + lG^2 lP^2 w^4);
(*analytical expression of deflection psd*)
PYG = (4 kbT kGS ((kGS + kP) lG + kG lP)) /
    ((kGS^2 - kG (kGS + kP))^2 + ((kGS + kP)^2 lG^2 + 2 kGS^2 lG lP + kG^2 lP^2) w^2 +
      lG^2 lP^2 w^4); (*analytical expression of current psd*)
PGG = (4 kbT (2 kGS kP lG + kGS^2 (lG + lP) + lG (kP^2 + lP^2 w^2))) /
    ((kGS^2 - kG (kGS + kP))^2 + ((kGS + kP)^2 lG^2 + 2 kGS^2 lG lP + kG^2 lP^2) w^2 +
      lG^2 lP^2 w^4); (*analytical expression of cross-psd*)


theorPSDtaplota0 = Table[{freqs[[i]], PYY /.
      {w → (2 * Pi * freqs[[i]]), lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
       kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2), kbT → kbTd,
       lP → lPd, kGS → kGSd, kP → kPd}}, {i, 1, lengthpsd}];
(*table of values of deflection psd*)
theorPSDtaplota1 = Table[{freqs[[i]], (PGG / (Dvd^2)) /.
      {w → (2 * Pi * freqs[[i]]), lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
```

```
        kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
        kbT → kbTd, lP → lPd, kGS → kGSd, kP → kPd}},
    {i, 1, lengthpsd}]; (*table of values of current psd*)
theorCPSDtaplota = Table[{freqs[[i]], (PYG / (Dvd)) /.
      {w → (2 * Pi * freqs[[i]]), lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
        kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
        kbT → kbTd, lP → lPd, kGS → kGSd, kP → kPd}},
    {i, 1, lengthpsd}]; (*table of values of cross-psd*)
filtersq1v = Table[
    {freqs[[i]], Abs[1 / (1 + I * theorPSDtaplota1[[i]][[1]] / fc)]^2}, {i, 1, lengthpsd}];
theorPSDvolt = Table[{freqs[[i]], (filtersq1v[[i]][[2]] * PGG / (Dvd^2)) /.
      {w → (2 * Pi * freqs[[i]]), lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
        kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
        kbT → kbTd, lP → lPd, kGS → kGSd, kP → kPd}},
    {i, 1, lengthpsd}]; (*table of values of voltage psd*)
```

In[•]:=

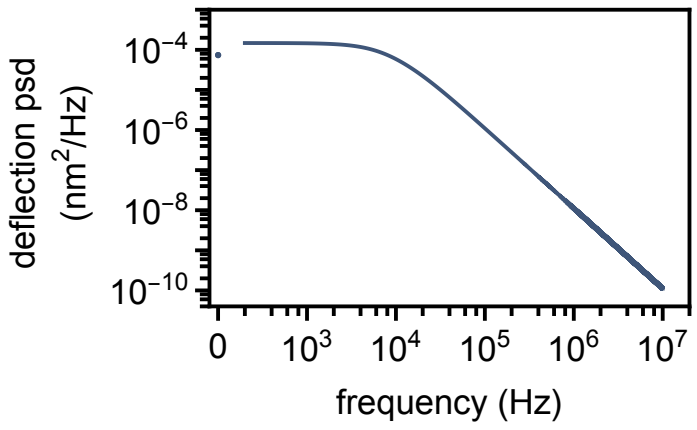
```
(*Fig S18 Deflection,
current and voltage power spectra and coherency vs frequency for default parameters*)
plotrange = {{-180, 180}, {-0.1, 1.1}};
meandataplot0c = {100, theorPSDtaplota0[[1]][[2]] / 2};
plotrange1 = {{80, 2 * 10^7}, {0.4 * 10^(-10), 10^(-3)}};
ListLogLogPlot[{theorPSDtaplota0},
  PlotStyle → plotstyleY, Joined → True, PlotRange → plotrange1,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"frequency (Hz)",
    "deflection psd \n (\!\(\*SuperscriptBox[\(nm\), \(2\)]\)/Hz)"}];
Show[%, ListLogLogPlot[{meandataplot0c}, PlotStyle → plotstyleY]]

meandataplot1c = {100, theorPSDtaplota1[[1]][[2]] / 2};
plotrange3 = {{80, 2 * 10^7}, {10^(-13), 10^(-5)}};
ListLogLogPlot[{theorPSDtaplota1}, PlotStyle → plotstyleP,
  Joined → True, Frame → True, PlotRange → plotrange3,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"frequency (Hz)", "current psd \!\(\*TemplateBox[{RowBox[{\"(\",
      \"Hz\"}], RowBox[{\"-\", \"1\"}]},\"Superscript\"]\))"}];
Show[%, ListLogLogPlot[{meandataplot1c}, PlotStyle → plotstyleP]]
```

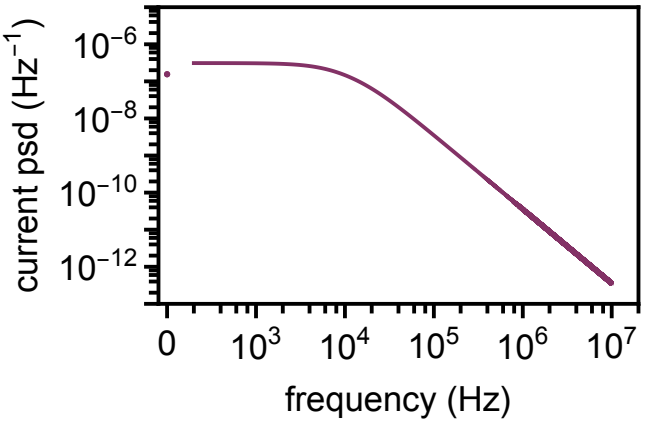
```
theorcoher01 = Table[{freqs[[i]], ((PYG / (Dvd) / Sqrt[PYY * PGG / (Dvd^2)])) /.
      {w → (2 * Pi * freqs[[i]]), lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
       kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2), kbT → kbTd,
       lP → lPd, kGS → kGSd, kP → kPd}}, {i, 1, lengthpsd}];
coher01c = {100, theorcoher01[[1]][[2]]};
plotrange4 = {{80, 2 * 10^7}, {-0.55, 1.05}};
ListLogLinearPlot[{theorcoher01}, PlotStyle → plotstyleY,
  Joined → True, Frame → True, PlotRange → plotrange4,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {frameticksycoherneg, {XTicks, None}},
  FrameLabel → {"frequency (Hz)", "coherency"}];
Show[%, ListLogLinearPlot[{coher01c}, PlotStyle → plotstyleY]]

meandataplot1vc = {100, theorPSDvolt[[1]][[2]] / 2};
plotrange4 = {{80, 2 * 10^7}, {10^(-22), 10^(-6)}};
ListLogLogPlot[{theorPSDvolt}, PlotStyle → plotstyleV, Joined → True, Frame → True,
  PlotRange → plotrange4, ImagePadding → {{pady, pad2}, {padx, pad2}},
  ImageSize → 400, LabelStyle → {FontSize → fntsize}, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"frequency (Hz)", "voltage psd \!\(\*TemplateBox[{RowBox[{\"(\",
      \"Hz\"}], RowBox[{\"-\", \"1\"}]},\"Superscript\"]\))"}];
Show[%, ListLogLogPlot[{meandataplot1vc}, PlotStyle → plotstyleV]]
```

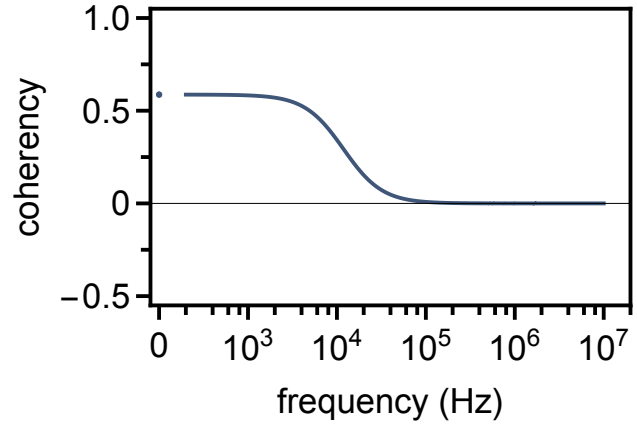
Out[ ]=



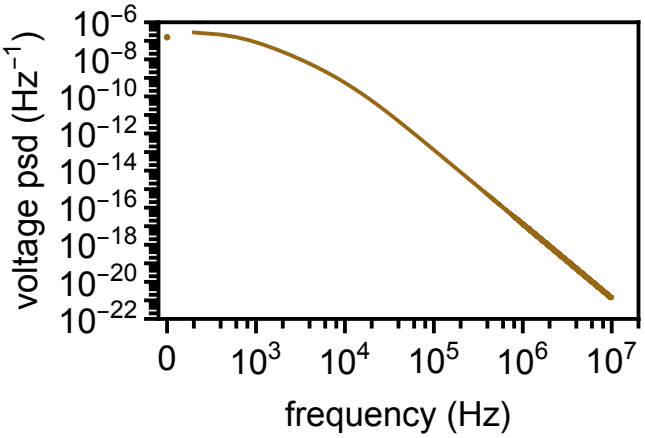
Out[ ]=



Out[ ]=



Out[ ]=




```
(*Calculation of analytical power spectra for pivot damping coefficient =
 default value /100. Set lPd=0.0369535/100*)
```

```
freqs = Table[i, {i, 0, 10^7, 200}];
(*table of frequency values for calculation of power spectra*)
lengthpsd = Length[freqs];
fc = 610; (*corner frequency for cell-
 membrane filter to apply to current psd to derive voltage psd*)
plotstyleY = {Lighter[■]};
plotstyleP = {Lighter[■]};
plotstyleV = {Lighter[■]};

PYY = (4 kbT (kG^2 lP + lG (kGS^2 + lG lP w^2))) / ((kGS^2 - kG (kGS + kP))^2 +
      ((kGS + kP)^2 lG^2 + 2 kGS^2 lG lP + kG^2 lP^2) w^2 + lG^2 lP^2 w^4);
(*analytical expression of deflection psd*)
PYG = (4 kbT kGS ((kGS + kP) lG + kG lP)) /
    ((kGS^2 - kG (kGS + kP))^2 + ((kGS + kP)^2 lG^2 + 2 kGS^2 lG lP + kG^2 lP^2) w^2 +
      lG^2 lP^2 w^4); (*analytical expression of current psd*)
PGG = (4 kbT (2 kGS kP lG + kGS^2 (lG + lP) + lG (kP^2 + lP^2 w^2))) /
    ((kGS^2 - kG (kGS + kP))^2 + ((kGS + kP)^2 lG^2 + 2 kGS^2 lG lP + kG^2 lP^2) w^2 +
      lG^2 lP^2 w^4); (*analytical expression of voltage psd*)

theorPSDtaplota0 = Table[{freqs[[i]], PYY /.
      {w → (2 * Pi * freqs[[i]]), lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
       kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2), kbT → kbTd,
       lP → lPd, kGS → kGSd, kP → kPd}}, {i, 1, lengthpsd}];
(*table of values of deflection psd*)
theorPSDtaplota1 = Table[{freqs[[i]], (PGG / (Dvd^2)) /.
      {w → (2 * Pi * freqs[[i]]), lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
       kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
       kbT → kbTd, lP → lPd, kGS → kGSd, kP → kPd}},
   {i, 1, lengthpsd}]; (*table of values of current psd*)
theorCPSDtaplota = Table[{freqs[[i]], (PYG / (Dvd)) /.
      {w → (2 * Pi * freqs[[i]]), lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
       kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
       kbT → kbTd, lP → lPd, kGS → kGSd, kP → kPd}},
   {i, 1, lengthpsd}]; (*table of values of cross-psd*)
filtersq1v = Table[
   {freqs[[i]], Abs[1 / (1 + I * theorPSDtaplota1[[i]][[1]] / fc)]^2}, {i, 1, lengthpsd}];
theorPSDvolt = Table[{freqs[[i]], (filtersq1v[[i]][[2]] * PGG / (Dvd^2)) /.
      {w → (2 * Pi * freqs[[i]]), lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
       kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
       kbT → kbTd, lP → lPd, kGS → kGSd, kP → kPd}},
   {i, 1, lengthpsd}]; (*table of values of voltage psd*)

(*Fig S18 Deflection, current and voltage power spectra and coherency
  vs frequency for pivot damping coefficient = default value /100.*)
```

```
plotrange = {{-180, 180}, {-0.1, 1.1}};
meandataplot0c = {100, theorPSDtaplota0[[1]][[2]] / 2};
plotrange1 = {{80, 2 * 10^7}, {0.4 * 10^(-10), 10^(-3)}};
ListLogLogPlot[{theorPSDtaplota0},
  PlotStyle → plotstyleY, Joined → True, PlotRange → plotrange1,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"frequency (Hz)",
    "deflection psd \n (\!\(\*SuperscriptBox[\(nm\), \(2\)]\)/Hz)"}];
Show[%, ListLogLogPlot[{meandataplot0c}, PlotStyle → plotstyleY]]

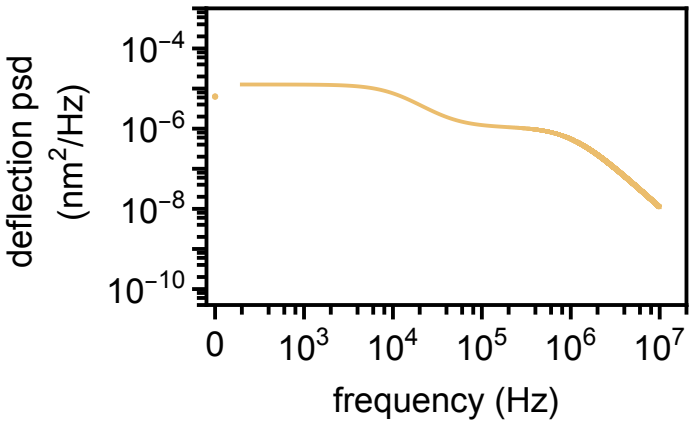
meandataplot1c = {100, theorPSDtaplota1[[1]][[2]] / 2};
plotrange3 = {{80, 2 * 10^7}, {10^(-13), 10^(-5)}};
ListLogLogPlot[{theorPSDtaplota1}, PlotStyle → plotstyleP,
  Joined → True, Frame → True, PlotRange → plotrange3,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"frequency (Hz)", "current psd \!\(\*TemplateBox[{RowBox[{\"(\",
      \"Hz\"}], RowBox[{\"-\", \"1\"}]},\"Superscript\"]\))"}];
Show[%, ListLogLogPlot[{meandataplot1c}, PlotStyle → plotstyleP]]

theorcoher01 = Table[{freqs[[i]], ((PYG / (Dvd) / Sqrt[PYY * PGG / (Dvd^2)])) /.
      {w → (2 * Pi * freqs[[i]]), lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
       kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2), kbT → kbTd,
       lP → lPd, kGS → kGSd, kP → kPd}}, {i, 1, lengthpsd}];
coher01c = {100, theorcoher01[[1]][[2]]};
plotrange4 = {{80, 2 * 10^7}, {-0.55, 1.05}};
ListLogLinearPlot[{theorcoher01}, PlotStyle → plotstyleY,
  Joined → True, Frame → True, PlotRange → plotrange4,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {frameticksycoherneg, {XTicks, None}},
  FrameLabel → {"frequency (Hz)", "coherency"}];
Show[%, ListLogLinearPlot[{coher01c}, PlotStyle → plotstyleY]]

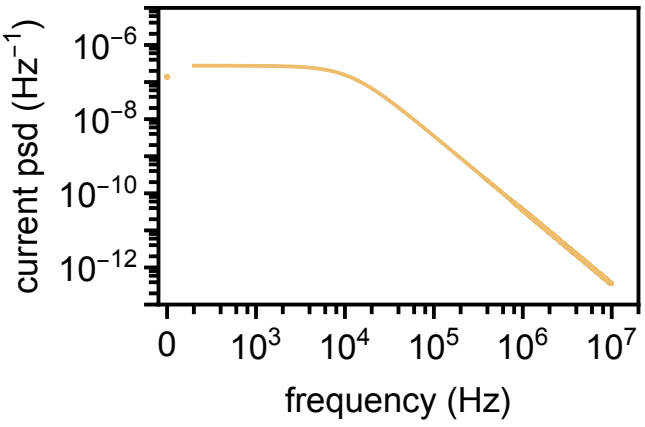
meandataplot1vc = {100, theorPSDvolt[[1]][[2]] / 2};
plotrange4 = {{80, 2 * 10^7}, {10^(-22), 10^(-6)}};
ListLogLogPlot[{theorPSDvolt}, PlotStyle → plotstyleV, Joined → True, Frame → True,
  PlotRange → plotrange4, ImagePadding → {{pady, pad2}, {padx, pad2}},
```

```
    ImageSize → 400, LabelStyle → {FontSize → fntsize}, Background → White,
    Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
    FrameTicks → {{YTicks, None}, {XTicks, None}},
    FrameLabel → {"frequency (Hz)", "voltage psd \!\(\*TemplateBox[{RowBox[{\"(\",
        \"Hz\"}], RowBox[{\"-\", \"1\"}]},\"Superscript\"]\))"}];
Show[%, ListLogLogPlot[{meandataplot1vc}, PlotStyle → plotstyleV]]
```

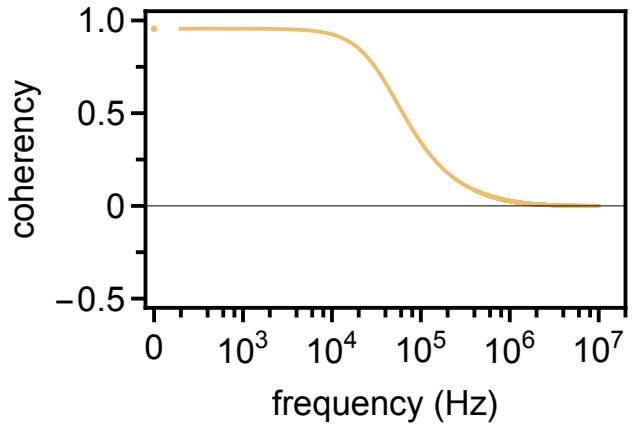
Out[ ]=



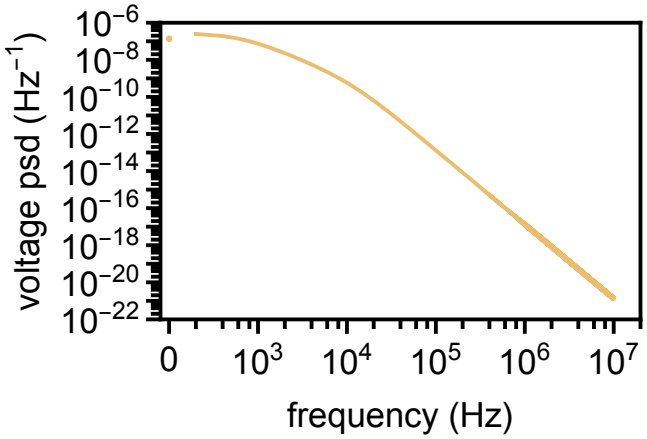
Out[ ]=



Out[ ]=



Out[ ]=



In[ ]:=

```
(*Calculation of analytical power spectra for channel gating timescale =
 default value /100. Set taoval1d=10^(-5)/100*)

freqs = Table[i, {i, 0, 10^7, 200}];
(*table of frequency values for calculation of power spectra*)
lengthpsd = Length[freqs];
fc = 610; (*corner frequency for cell-
 membrane filter to apply to current psd to derive voltage psd*)
plotstyleY = {Lighter[■]};
plotstyleP = {Lighter[■]};
plotstyleV = {Lighter[■]};

PYY = (4 kbT (kG^2 lP + lG (kGS^2 + lG lP w^2))) / ((kGS^2 - kG (kGS + kP))^2 +
      ((kGS + kP)^2 lG^2 + 2 kGS^2 lG lP + kG^2 lP^2) w^2 + lG^2 lP^2 w^4);
(*analytical expression of deflection psd*)
PYG = (4 kbT kGS ((kGS + kP) lG + kG lP)) /
```

```
    ((kGS^2 - kG (kGS + kP))^2 + ((kGS + kP)^2 lG^2 + 2 kGS^2 lG lP + kG^2 lP^2) w^2 +
      lG^2 lP^2 w^4); (*analytical expression of current psd*)
PGG = (4 kbT (2 kGS kP lG + kGS^2 (lG + lP) + lG (kP^2 + lP^2 w^2))) /
    ((kGS^2 - kG (kGS + kP))^2 + ((kGS + kP)^2 lG^2 + 2 kGS^2 lG lP + kG^2 lP^2) w^2 +
      lG^2 lP^2 w^4); (*analytical expression of voltage psd*)

theorPSDtaplota0 = Table[{freqs[[i]], PYY /.
      {w → (2 * Pi * freqs[[i]]), lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
       kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2), kbT → kbTd,
       lP → lPd, kGS → kGSd, kP → kPd}}, {i, 1, lengthpsd}];
(*table of values of deflection psd*)
theorPSDtaplota1 = Table[{freqs[[i]], (PGG / (Dvd^2)) /.
      {w → (2 * Pi * freqs[[i]]), lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
       kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
       kbT → kbTd, lP → lPd, kGS → kGSd, kP → kPd}},
   {i, 1, lengthpsd}]; (*table of values of current psd*)
theorCPSDtaplota = Table[{freqs[[i]], (PYG / (Dvd)) /.
      {w → (2 * Pi * freqs[[i]]), lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
       kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
       kbT → kbTd, lP → lPd, kGS → kGSd, kP → kPd}},
   {i, 1, lengthpsd}]; (*table of values of cross-psd*)
filtersq1v = Table[
   {freqs[[i]], Abs[1 / (1 + I * theorPSDtaplota1[[i]][[1]] / fc)]^2}, {i, 1, lengthpsd}];
theorPSDvolt = Table[{freqs[[i]], (filtersq1v[[i]][[2]] * PGG / (Dvd^2)) /.
      {w → (2 * Pi * freqs[[i]]), lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
       kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
       kbT → kbTd, lP → lPd, kGS → kGSd, kP → kPd}},
   {i, 1, lengthpsd}]; (*table of values of voltage psd*)
```

In[•]:=

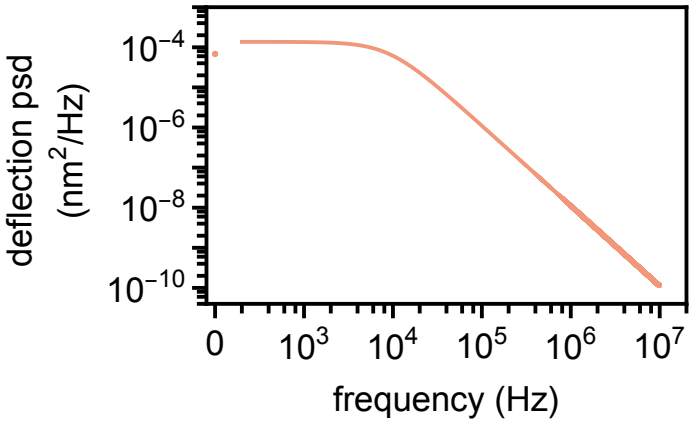
```
(*Fig S18 Deflection,
current and voltage power spectra and coherency vs frequency for channel
  gating timescale = default value /100. Set taoval1d=10^(-5)/100.*)
plotrange = {{-180, 180}, {-0.1, 1.1}};
meandataplot0c = {100, theorPSDtaplota0[[1]][[2]] / 2};
plotrange1 = {{80, 2 * 10^7}, {0.4 * 10^(-10), 10^(-3)}};
ListLogLogPlot[{theorPSDtaplota0},
  PlotStyle → plotstyleY, Joined → True, PlotRange → plotrange1,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"frequency (Hz)",
    "deflection psd \n (\!\(\*SuperscriptBox[\(nm\), \(2\)]\)/Hz)"}];
Show[%, ListLogLogPlot[{meandataplot0c}, PlotStyle → plotstyleY]]
```

```
meandataplot1c = {100, theorPSDtaplota1[[1]][[2]] / 2};
plotrange3 = {{80, 2 * 10^7}, {10^(-13), 10^(-5)}};
ListLogLogPlot[{theorPSDtaplota1}, PlotStyle → plotstyleP,
  Joined → True, Frame → True, PlotRange → plotrange3,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"frequency (Hz)", "current psd \!\(\*TemplateBox[{RowBox[{\"(\",
      \"Hz\"}], RowBox[{\"-\", \"1\"}]},\"Superscript\"]\))"}];
Show[%, ListLogLogPlot[{meandataplot1c}, PlotStyle → plotstyleP]]

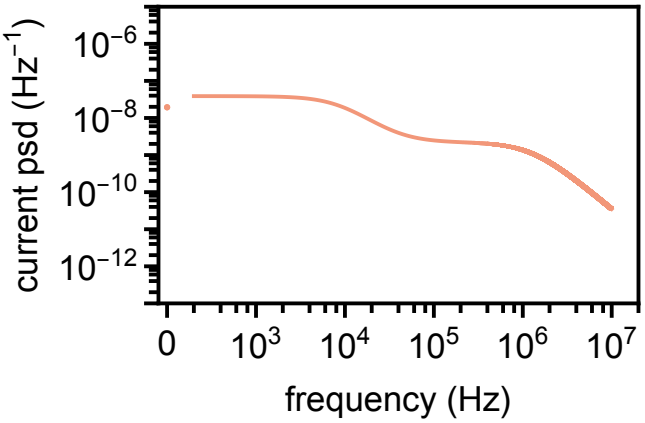
theorcoher01 = Table[{freqs[[i]], ((PYG / (Dvd) / Sqrt[PYY * PGG / (Dvd^2)])) /.
     {w → (2 * Pi * freqs[[i]]), lG → Ncd * taorest * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2),
      kG → Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2), kbT → kbTd,
      lP → lPd, kGS → kGSd, kP → kPd}}, {i, 1, lengthpsd}];
coher01c = {100, theorcoher01[[1]][[2]]};
plotrange4 = {{80, 2 * 10^7}, {-0.55, 1.05}};
ListLogLinearPlot[{theorcoher01}, PlotStyle → plotstyleY,
  Joined → True, Frame → True, PlotRange → plotrange4,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {frameticksycoherneg, {XTicks, None}},
  FrameLabel → {"frequency (Hz)", "coherency"}];
Show[%, ListLogLinearPlot[{coher01c}, PlotStyle → plotstyleY]]

meandataplot1vc = {100, theorPSDvolt[[1]][[2]] / 2};
plotrange4 = {{80, 2 * 10^7}, {10^(-22), 10^(-6)}};
ListLogLogPlot[{theorPSDvolt}, PlotStyle → plotstyleV, Joined → True, Frame → True,
  PlotRange → plotrange4, ImagePadding → {{pady, pad2}, {padx, pad2}},
  ImageSize → 400, LabelStyle → {FontSize → fntsize}, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"frequency (Hz)", "voltage psd \!\(\*TemplateBox[{RowBox[{\"(\",
      \"Hz\"}], RowBox[{\"-\", \"1\"}]},\"Superscript\"]\))"}];
Show[%, ListLogLogPlot[{meandataplot1vc}, PlotStyle → plotstyleV]]
```

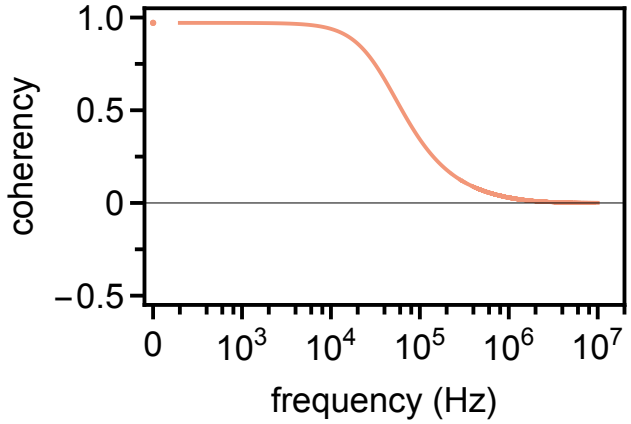
Out[ ]=


deflection psd
(nm²/Hz)
10⁻⁴
10⁻⁶
10⁻⁸
10⁻¹⁰
0
10³
10⁴
10⁵
10⁶
10⁷
frequency (Hz)

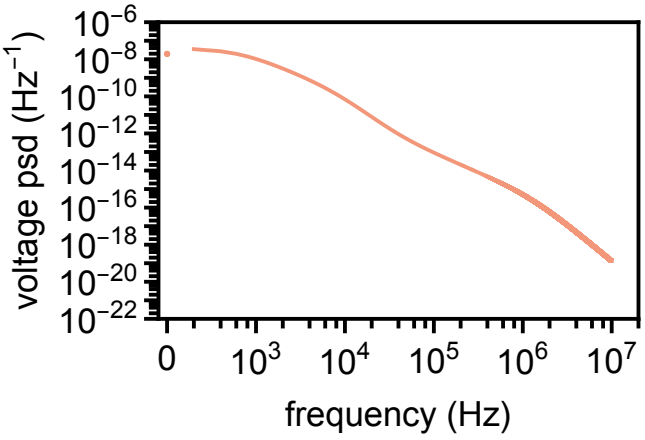
Out[ ]=


current psd (Hz⁻¹)
10⁻⁶
10⁻⁸
10⁻¹⁰
10⁻¹²
0
10³
10⁴
10⁵
10⁶
10⁷
frequency (Hz)

Out[ ]=


coherency
1.0
0.5
0
−0.5
0
10³
10⁴
10⁵
10⁶
10⁷
frequency (Hz)

Out[ ]=



In[ ]:=

```
(*Calculation of fluctuation power with and without fluctuation
 suppression. Gating swing can be set to 1.75 nm or 32 nm. *)
t1 = AbsoluteTime[];
fmult = {0, 1, 5, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150,
   160, 170, 172, 173, 175, 177, 178, 179, 180}; (*table of values of force-
 stimulus amplitude for calculation of fluctuation power*)
areaback0 = Table[0, {Length[fmult]}];
areatot0 = Table[0, {Length[fmult]}];
areaback1 = Table[0, {Length[fmult]}];
areatot1 = Table[0, {Length[fmult]}];
areaback1filt = Table[0, {Length[fmult]}];
areatot1filt = Table[0, {Length[fmult]}];
SetSharedVariable[areaback0];
SetSharedVariable[areatot0];
SetSharedVariable[areaback1];
SetSharedVariable[areatot1];
SetSharedVariable[areaback1filt];
SetSharedVariable[areatot1filt];
SetSharedVariable[fmult];



ParallelDo[
  SeedRandom[47 741];

gx =  2 * kbTd * lPd;
  y0in = Yrestd;

  proc = ItoProcess[
    lPd * dY[t] == (-kPd) * (Y[t] - Yupd) * dt - kGSd * (Y[t] - Yugd - Dvd * Po[t]) * dt +
```

```
        Sqrt[gx] * ⅆn1[t] + fmult[[pp]] * Fzd * Sin[2 * Pi * wstd * (t - tstartFd)] * ⅆt &&
      taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd] * ⅆPo[t] ==
       (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] - Po[t]) ⅆt +
        Sqrt[taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd]] Sqrt[
          (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] (1 - Clip[Po[t], {0.0, 1.0}]) + (1 - Pinf2[
                  Yhd, kGSd, Dvd, kbTd, Ncd]) Clip[Po[t], {0.0, 1.0}]) / Ncd] ⅆn2[t],
    {Y[t], Po[t]}, {{Y, Po}, {y0in, Pinfrest}}, t,
{n1 ≈ WienerProcess[], n2 ≈ WienerProcess[]}];

  totallength = 2 * 0.507;
  factot = 4;
  deltatsima = 0.000001 / factot;
   ap = RandomFunction[proc, {0, totallength, deltatsima}];

  ap0 = ap["PathComponent", 1] - Yrestd;

  ap1 = ap["PathComponent", 2] - Pinfrest;
  resampfactor = factot;

randtime100 = 0.005 * 10;
randtimenm = Round[randtime100, deltatsima];
stpoint = IntegerPart[randtime100 / deltatsima];

deltatsampl = Round[deltatsima * resampfactor, deltatsima];
tsampl = 0.005;
endpoint = IntegerPart[
    Length[ap["PathComponent", 1]["Values"]] - tsampl / deltatsima - stpoint];
tsamplnm = Round[tsampl, deltatsima];
b1 = Table[f[i], {i, randtimenm, tsamplnm + randtimenm, deltatsampl}];
lengthares = Length[b1];

Clear[ares];
totallength2 = totallength;

ncopiesa = (totallength2 - randtimenm) / (tsamplnm + deltatsampl);
ncopies = IntegerPart[ncopiesa];
ares0 = Table[0, {ncopies}]; (*a resampled*)
ares1 = Table[0, {ncopies}];
sizea = lengthares;

fousampled0 = Table[0, {ncopies}, {lengthares}];
fousampled1 = Table[0, {ncopies}, {lengthares}];
cfousampled = Table[0, {ncopies}, {lengthares}];
Do[
```

```
    ares0[[ll]] = ap0["Values"][[(ll - 1) * tsampl / deltatsima + stpoint ;;
       IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];
     ares1[[ll]] = ap1["Values"][[(ll - 1) * tsampl / deltatsima + stpoint ;;
       IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];

        fousampled0[[ll]] = ResourceFunction["WelchSpectralEstimate"][
      ares0[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
      "OverlapOffset" → None, "SegmentSize" → sizea];
     cfousampled[[ll]] = ResourceFunction["WelchSpectralEstimate"][
      ares1[[ll]], ares0[[ll]], sizea / tsampl, "Window" → None,
      "Detrend" → None, "OverlapOffset" → None, "SegmentSize" → sizea];
     fousampled1[[ll]] = ResourceFunction["WelchSpectralEstimate"][
      ares1[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
      "OverlapOffset" → None, "SegmentSize" → sizea];
    ,

{ll, ncopies}];
  powspmeansampledt0 = Drop[Mean[fousampled0], -1];

  powspmeansampledt1 = Drop[Mean[fousampled1], -1];

cpowspmeansampledt = Drop[Re[Mean[cfousampled]], -1];
  lengthpsd = Length[powspmeansampledt0];


  meandataplot0 = Table[{0, 0}, {i, 1, lengthpsd}];
  meandataplot1 = Table[{0, 0}, {i, 1, lengthpsd}];
meandataplot1cpsd = Table[{0, 0}, {i, 1, lengthpsd}];


  meandataplot0 =
   Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt0[[i]][[2]]}, {i, 1, lengthpsd}];
  meandataplot1 =
   Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt1[[i]][[2]]}, {i, 1, lengthpsd}];
meandataplot1cpsd =
   Table[{cpowspmeansampledt[[i]][[1]], cpowspmeansampledt[[i]][[2]]}, {i, 1, lengthpsd}];
  fc = 610;
  filtersq1 = Table[{meandataplot0[[i]][[1]],
     Abs[1 / (1 + I * meandataplot1[[i]][[1]] / fc)]^2}, {i, 1, Length[meandataplot0]}];
  meandataplot1filt = Table[{meandataplot0[[i]][[1]],
     filtersq1[[i]][[2]] * meandataplot1[[i]][[2]]}, {i, 1, Length[filtersq1]}];

  (*calculation of background area for deflection psd*)
```

```
data0 = Table[meandataplot0[[nn]][[2]], {nn, 2, Length[meandataplot0]}];
dataest0 = MedianFilter[data0, 15]; (*use 15 point median filter
 for estimate of background data from freq 2.8 kHz upward (point 15) *)
dataestwfreq0 = Table[{meandataplot0[[nn + 1]][[1]], dataest0[[nn]]},
   {nn, 8 + 7, Length[meandataplot0] - 8 + 7}]; (*for frequencies lower than 2.8 kHz,
the background psd is set equal to the actual psd*)
data0part = Table[meandataplot0[[nn]][[2]], {nn, 2, 8 + 7}];
dataest0part = data0part;
dataestwfreq0part =
 Table[{meandataplot0[[nn + 1]][[1]], dataest0part[[nn]]}, {nn, 1, 8 + 7 - 1}];
dataest0tot = Union[
   {{meandataplot0[[1]][[1]], meandataplot0[[2]][[2]]}}, dataestwfreq0part, dataestwfreq0];
dataest0tot[[1]][[2]] = 0.00007091605149763008`;
(*DC value is set equal to the corresponding value for the unstimulated psd*)
areaback0[[pp]] = {fmult[[pp]],
   Integrate[Interpolation[dataest0tot, InterpolationOrder → 1][x], {x, 0, 5 * 10^5}]};
(*area of the background deflection psd is obtained using
 a linear interpolation between points*)
areatot0[[pp]] = {fmult[[pp]],
   Integrate[Interpolation[meandataplot0, InterpolationOrder → 1][x], {x, 0, 5 * 10^5}]};
(*area of stimulated psd is obtained using a linear interpolation between points*)

(*calculation of background area for current psd*)
data1 = Table[meandataplot1[[nn]][[2]], {nn, 2, Length[meandataplot1]}];
dataest1 = MedianFilter[data1, 15];
dataestwfreq1 = Table[{meandataplot1[[nn + 1]][[1]], dataest1[[nn]]},
   {nn, 8 + 7, Length[meandataplot1] - 8 + 7}];
data1part = Table[meandataplot1[[nn]][[2]], {nn, 2, 8 + 7}];
dataest1part = data1part;
dataestwfreq1part =
 Table[{meandataplot1[[nn + 1]][[1]], dataest1part[[nn]]}, {nn, 1, 8 + 7 - 1}];
dataest1tot1 = Union[
   {{meandataplot1[[1]][[1]], meandataplot1[[2]][[2]]}}, dataestwfreq1part, dataestwfreq1];
areaback1[[pp]] =
 {fmult[[pp]], Integrate[Interpolation[dataest1tot1, InterpolationOrder → 1][x],
    {x, 200, 5 * 10^5}] + meandataplot1[[2]][[2]] * meandataplot1[[2]][[1]] / 2};
areatot1[[pp]] = {fmult[[pp]],
   Integrate[Interpolation[meandataplot1, InterpolationOrder → 1][x], {x, 0, 5 * 10^5}]};

(*calculation of background area of voltage psd*)
data1filt = Table[meandataplot1filt[[nn]][[2]], {nn, 2, Length[meandataplot1filt]}];
dataest1filt = MedianFilter[data1filt, 15];
dataestwfreq1filt = Table[{meandataplot1filt[[nn + 1]][[1]], dataest1filt[[nn]]},
   {nn, 8 + 7, Length[meandataplot1filt] - 8 + 7}];
```

```
  data1filtpart = Table[meandataplot1filt[[nn]][[2]], {nn, 2, 8 + 7}];
  dataest1filtpart = data1filtpart;
  dataestwfreq1filtpart =
   Table[{meandataplot1filt[[nn + 1]][[1]], dataest1filtpart[[nn]]}, {nn, 1, 8 + 7 - 1}];
  dataest1totfilt = Union[{{meandataplot1filt[[1]][[1]], meandataplot1filt[[2]][[2]]}},
    dataestwfreq1filtpart, dataestwfreq1filt];
  areaback1filt[[pp]] =
   {fmult[[pp]], Integrate[Interpolation[dataest1totfilt, InterpolationOrder → 1][x],
      {x, 200, 5 * 10^5}]} + meandataplot1filt[[2]][[2]] * meandataplot1filt[[2]][[1]] / 2;
  areatot1filt[[pp]] = {fmult[[pp]], Integrate[
     Interpolation[meandataplot1filt, InterpolationOrder → 1][x], {x, 0, 5 * 10^5}]};

,
  {pp, 1, Length[fmult]}];

t2 = AbsoluteTime[];
t2 - t1
```

Out[ ]=

```
108.530217
```

In[•]:=
```
areaback0const = Table[{fmult[[i]], areatot0[[1]][[2]]}, {i, 1, Length[fmult]}];
(*deflection fluctuation power without fluctuation suppression*)
areaback0norm =
  Table[{fmult[[i]], areaback0[[i]][[2]] / areatot0[[i]][[2]]}, {i, 1, Length[fmult]}];
(*deflection fluctuation power fraction*)
areaback0constnorm =
  Table[{fmult[[i]], areatot0[[1]][[2]] / areatot0[[i]][[2]]}, {i, 1, Length[fmult]}];
(*deflection fluctuation power fraction without fluctuation suppression*)


areaback1const = Table[{fmult[[i]], areatot1[[1]][[2]]}, {i, 1, Length[fmult]}];
(*current fluctuation power without fluctuation suppression*)
areaback1norm =
  Table[{fmult[[i]], areaback1[[i]][[2]] / areatot1[[i]][[2]]}, {i, 1, Length[fmult]}];
(*current fluctuation power fraction*)
areaback1constnorm =
  Table[{fmult[[i]], areatot1[[1]][[2]] / areatot1[[i]][[2]]}, {i, 1, Length[fmult]}];
(*current fluctuation power fraction without fluctuation suppression*)


areaback1filtconst = Table[{fmult[[i]], areaback1filt[[1]][[2]]}, {i, 1, Length[fmult]}];
(*voltage fluctuation power without fluctuation suppression*)
areaback1filtnorm = Table[
   {fmult[[i]], areaback1filt[[i]][[2]] / areatot1filt[[i]][[2]]}, {i, 1, Length[fmult]}];
(*voltage fluctuation power fraction*)
areaback1filtnormconst =
  Table[{fmult[[i]], areatot1filt[[1]][[2]] / areatot1filt[[i]][[2]]}, {i, 1, Length[fmult]}];
(*voltage fluctuation power fraction without fluctuation suppression*)
```

In[ ]:=
```
(*plot settings*)
TickLength1 = 0.03;
Yticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, -24, 10, 2}];
Yticks2 = Table[{10^k, Null, {0, TickLength1}}, {k, -23, 10, 2}];
TickLength2 = 0.02;
aa = Table[0, {38}, {4}];
For[k = 1, k ≤ 38, k++,
aa[[k]] = Table[{l * 10^(-k + 10), , {0, TickLength2}}, {l, 2, 8, 2}];
]
YTickslog = Union[Yticks1, Yticks2, aa[[1]], aa[[2]], aa[[3]],
    aa[[4]], aa[[5]], aa[[6]], aa[[7]], aa[[8]], aa[[9]], aa[[10]], aa[[11]], aa[[12]],
    aa[[13]], aa[[14]], aa[[15]], aa[[16]], aa[[17]], aa[[18]], aa[[19]], aa[[20]],
    aa[[21]], aa[[22]], aa[[23]], aa[[24]], aa[[25]], aa[[26]], aa[[27]], aa[[28]]];

pady = 120;
padx = 70;
pad2 = 30;
fntsize = 18;
```

In[ ]:=
```
(*Fig S15 for gating swing of 32 nm. Set D=32 nm.*)
(*Fig S15A-B Deflection fluctuation power and deflection fluctuation power
  fraction vs stimulus-force amplitude with and without noise reduction*)
plotrangeYdef = {{0, kGSd * 52 / 100}, {0, 5}};
framelabelsYdef = {{"deflection \n fluctuation power (nm^2)", None},
    {"stimulus-force amplitude (pN)", None}};
Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 180, 50}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 180, 50 / 2}];
XTicks = Union[Xticks1, Xticks3];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 2, 0.25}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 2, 0.25 / 2}];
YTicks = Union[Yticks1, Yticks3];

plotstyleYdef = {{Thickness → 0.01, Lighter[■]},
    {Thickness → 0.01, Lighter[■], Dashed}};(* Lighter[■]■*)

ListPlot[{areaback0, areaback0const}, PlotRange → {All, {1.5, 2.02}},
 Joined → True, ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleYdef, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
```

```
 FrameLabel → framelabelsYdef, FrameTicks → {{YTicks, None}, {XTicks, None}}]

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1, 0.2}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1, 0.2 / 2}];
YTicks = Union[Yticks1, Yticks3];

framelabelsYdef = {{"deflection fluctuation \n power fraction ", None},
    {"stimulus-force amplitude (pN)", None}};

ListLogPlot[{areaback0norm, areaback0constnorm}, PlotRange → {All, All},
 Joined → True, ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleYdef, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsYdef, FrameTicks → {{YTickslog, None}, {XTicks, None}}]

(*Fig S15C-D Current fluctuation power and current fluctuation power
  fraction vs stimulus-force amplitude with and without noise reduction*)
Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 0.005, 0.001}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 0.005, 0.001}];
YTicks = Union[Yticks1, Yticks3];

plotstyleYdef = {{Thickness → 0.01, Lighter[■]},
    {Thickness → 0.01, Lighter[■], Dashed}}; (*Lighter[■]■*)
framelabelsYdef = {{"current \n fluctuation power", None},
    {"stimulus-force amplitude (pN)", None}};

ListPlot[{areaback1, areaback1const}, PlotRange → {All, {-0.0005, 0.006}},
 Joined → True, ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleYdef, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsYdef, FrameTicks → {{YTicks, None}, {XTicks, None}}]

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1, 0.2}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1, 0.2 / 2}];
YTicks = Union[Yticks1, Yticks3];


framelabelsYdef = {{"current fluctuation \n power fraction ", None},
    {"stimulus-force amplitude (pN)", None}};

ListLogPlot[{areaback1norm, areaback1constnorm}, PlotRange → {All, All},
 Joined → True, ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleYdef, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
```

```
 FrameLabel → framelabelsYdef, FrameTicks → {{YTickslog, None}, {XTicks, None}}]

(*Fig S15C-D Voltage fluctuation power and voltage fluctuation power
  fraction vs stimulus-force amplitude with and without noise reduction*)
Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 0.00035, 0.00010}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 0.00035, 0.00010}];
YTicks = Union[Yticks1, Yticks3];

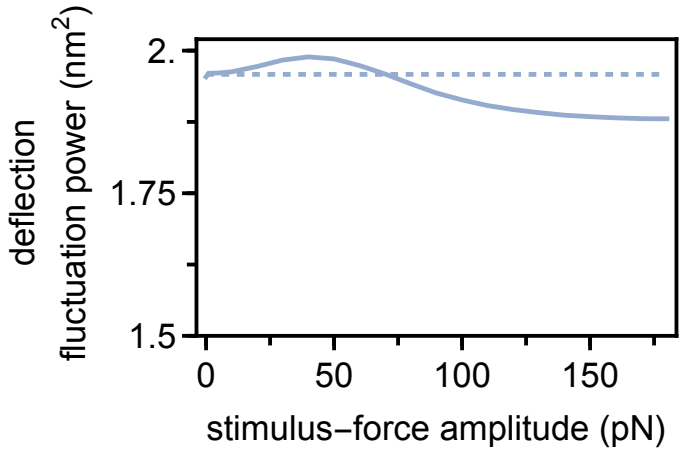
plotstyleYdef = {{Thickness → 0.01, ■}, {Thickness → 0.01, ■, Dashed}};(*■ ■*)
framelabelsYdef = {{"voltage \n fluctuation power", None},
   {"stimulus-force amplitude (pN)", None}};
ListPlot[{areaback1filt, areaback1filtconst},
 PlotRange → {All, {-0.00003, 0.00037}}, Joined → True,
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleYdef, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsYdef, FrameTicks → {{YTicks, None}, {XTicks, None}}]

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1, 0.2}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1, 0.2 / 2}];
YTicks = Union[Yticks1, Yticks3];

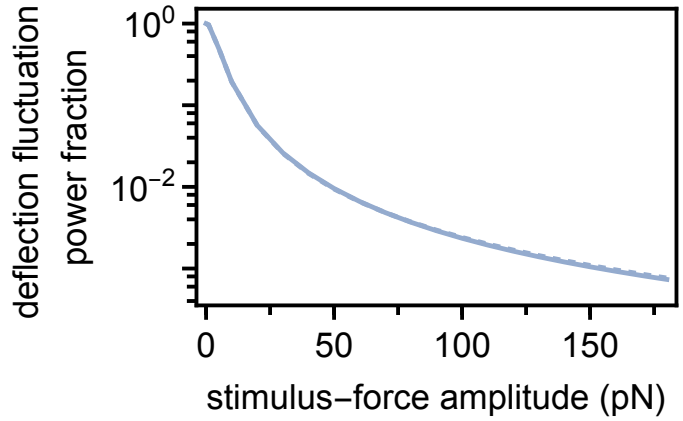

framelabelsYdef = {{"voltage fluctuation \n power fraction ", None},
   {"stimulus-force amplitude (pN)", None}};

ListLogPlot[{areaback1filtnorm, areaback1filtnormconst}, PlotRange → {All, All},
 Joined → True, ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleYdef, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsYdef, FrameTicks → {{YTickslog, None}, {XTicks, None}}]
```

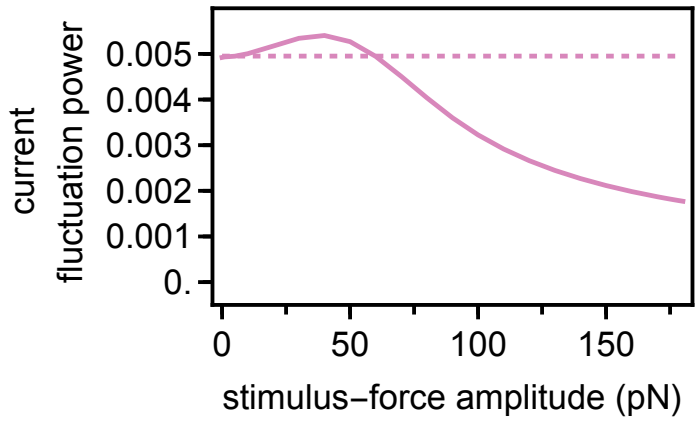
Out[ ]=


deflection
fluctuation power (nm^2)
2.
1.75
1.5
0
50
100
150
stimulus–force amplitude (pN)

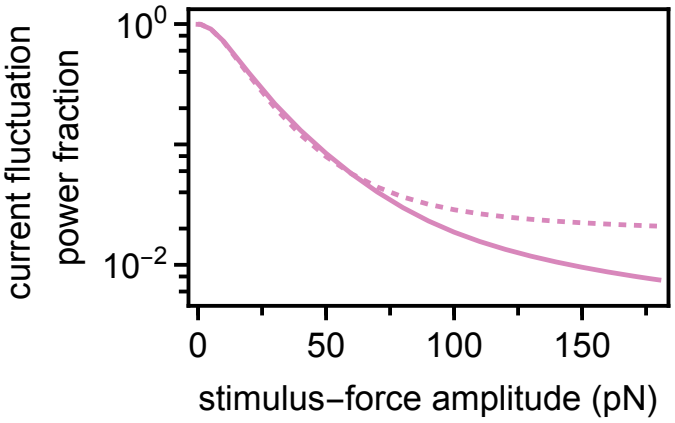
Out[ ]=


deflection fluctuation
power fraction
10^0
10^{-2}
0
50
100
150
stimulus–force amplitude (pN)

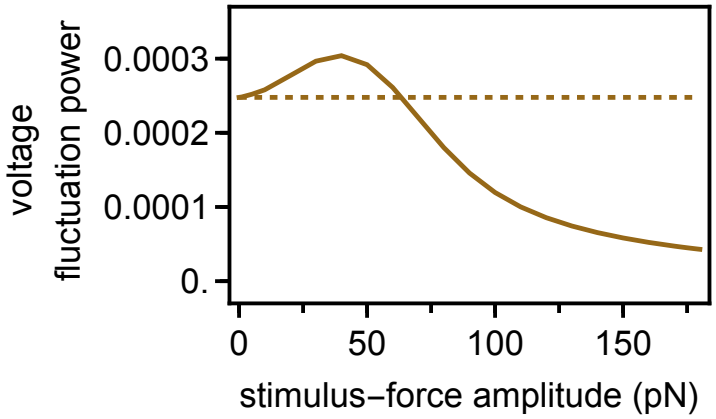
Out[ ]=


current
fluctuation power
0.005
0.004
0.003
0.002
0.001
0.
0
50
100
150
stimulus–force amplitude (pN)

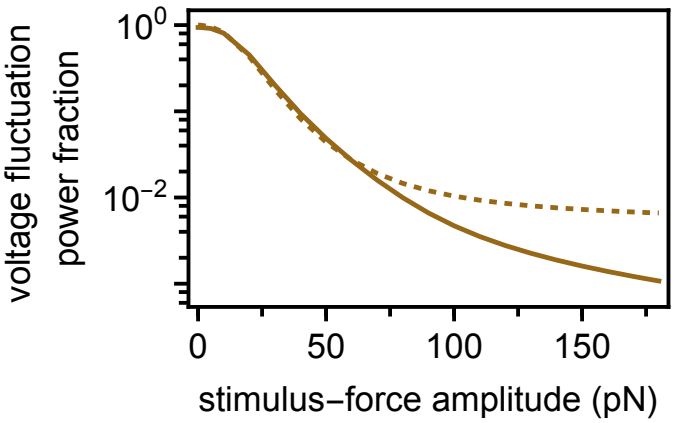
Out[ ]=



Out[ ]=



Out[ ]=



In[ ]:=
```
(*Fig S15 for gating swing of 1.75 nm. Set D=1.75 nm.*)
(*Fig S15A-B Deflection fluctuation power and deflection fluctuation power
```

```
  fraction vs stimulus-force amplitude with and without noise reduction*)
plotrangeYdef = {{0, kGSd * 52 / 100}, {0, 5}};
framelabelsYdef = {{"deflection \n fluctuation power (nm^2)", None},
    {"stimulus-force amplitude (pN)", None}};
Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 180, 50}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 180, 50 / 2}];
XTicks = Union[Xticks1, Xticks3];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 2, 0.25}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 2, 0.25 / 2}];
YTicks = Union[Yticks1, Yticks3];

plotstyleYdef = {{Thickness → 0.01, Lighter[■]},
    {Thickness → 0.01, Lighter[■], Dashed}};(* Lighter[■]■*)

ListPlot[{areaback0, areaback0const}, PlotRange → {All, {1.5, 2.02}},
 Joined → True, ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleYdef, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsYdef, FrameTicks → {{YTicks, None}, {XTicks, None}}]

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1, 0.2}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1, 0.2 / 2}];
YTicks = Union[Yticks1, Yticks3];

framelabelsYdef = {{"deflection fluctuation \n power fraction ", None},
    {"stimulus-force amplitude (pN)", None}};

ListLogPlot[{areaback0norm, areaback0constnorm}, PlotRange → {All, All},
 Joined → True, ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleYdef, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsYdef, FrameTicks → {{YTickslog, None}, {XTicks, None}}]

(*Fig S15C-D Current fluctuation power and current fluctuation power
  fraction vs stimulus-force amplitude with and without noise reduction*)
Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 0.005, 0.001}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 0.005, 0.001}];
YTicks = Union[Yticks1, Yticks3];

plotstyleYdef = {{Thickness → 0.01, Lighter[■]},
    {Thickness → 0.01, Lighter[■], Dashed}}; (*Lighter[■]■*)
framelabelsYdef = {{"current \n fluctuation power", None},
```

```
   {"stimulus-force amplitude (pN)", None}};

ListPlot[{areaback1, areaback1const}, PlotRange → {All, {-0.0005, 0.006}},
 Joined → True, ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleYdef, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsYdef, FrameTicks → {{YTicks, None}, {XTicks, None}}]

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1, 0.2}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1, 0.2 / 2}];
YTicks = Union[Yticks1, Yticks3];


framelabelsYdef = {{"current fluctuation \n power fraction ", None},
   {"stimulus-force amplitude (pN)", None}};

ListLogPlot[{areaback1norm, areaback1constnorm}, PlotRange → {All, All},
 Joined → True, ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleYdef, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsYdef, FrameTicks → {{YTickslog, None}, {XTicks, None}}]

(*Fig S15C-D Voltage fluctuation power and voltage fluctuation power
  fraction vs stimulus-force amplitude with and without noise reduction*)
Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 0.00035, 0.00010}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 0.00035, 0.00010}];
YTicks = Union[Yticks1, Yticks3];

plotstyleYdef = {{Thickness → 0.01, ■}, {Thickness → 0.01, ■, Dashed}};(*■ ■*)
framelabelsYdef = {{"voltage \n fluctuation power", None},
   {"stimulus-force amplitude (pN)", None}};
ListPlot[{areaback1filt, areaback1filtconst},
 PlotRange → {All, {-0.00003, 0.00037}}, Joined → True,
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleYdef, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsYdef, FrameTicks → {{YTicks, None}, {XTicks, None}}]

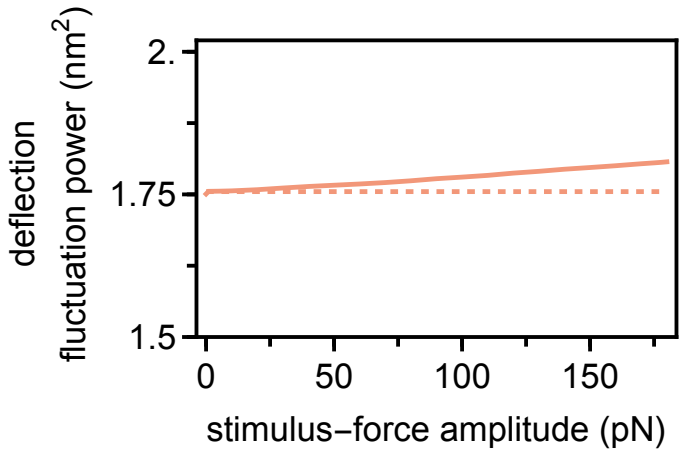
Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1, 0.2}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1, 0.2 / 2}];
YTicks = Union[Yticks1, Yticks3];


framelabelsYdef = {{"voltage fluctuation \n power fraction ", None},
```

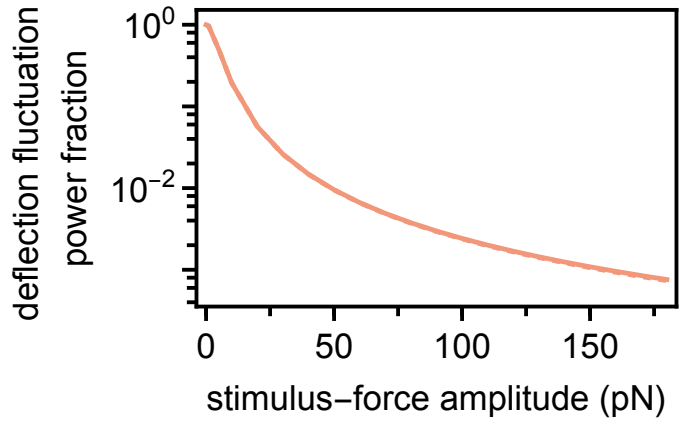
```
    {"stimulus-force amplitude (pN)", None}};

ListLogPlot[{areaback1filtnorm, areaback1filtnormconst}, PlotRange → {All, All},
 Joined → True, ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleYdef, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsYdef, FrameTicks → {{YTickslog, None}, {XTicks, None}}]
```

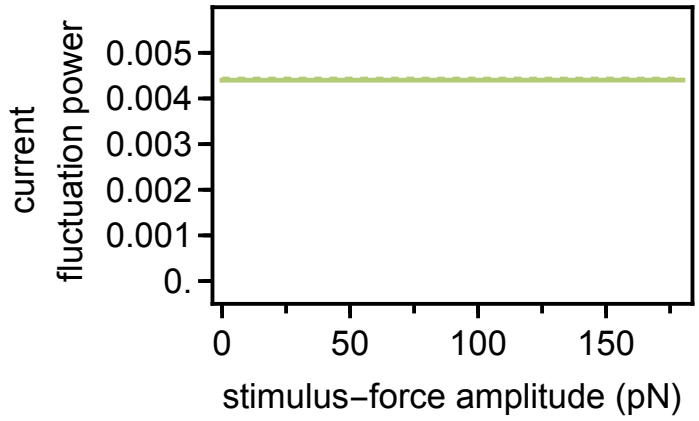
Out[ ]=



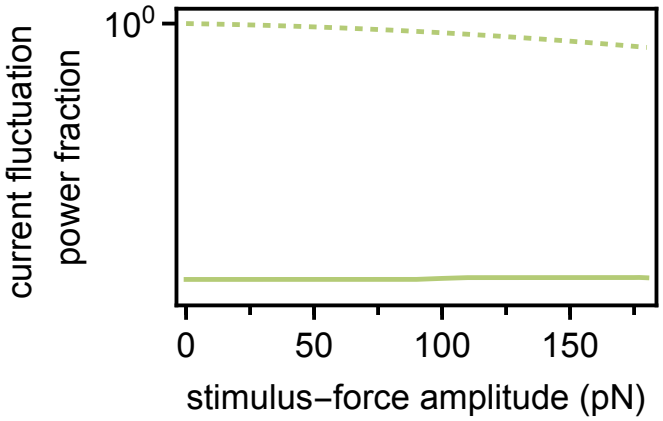
Out[ ]=



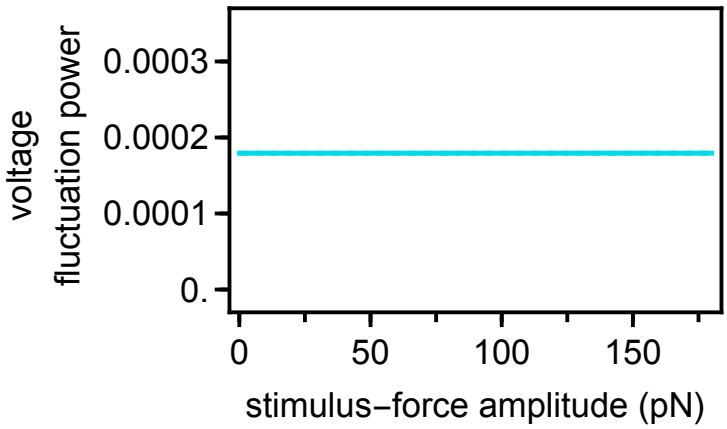
Out[ ]=


current
fluctuation power
0.005
0.004
0.003
0.002
0.001
0.
0
50
100
150
stimulus−force amplitude (pN)

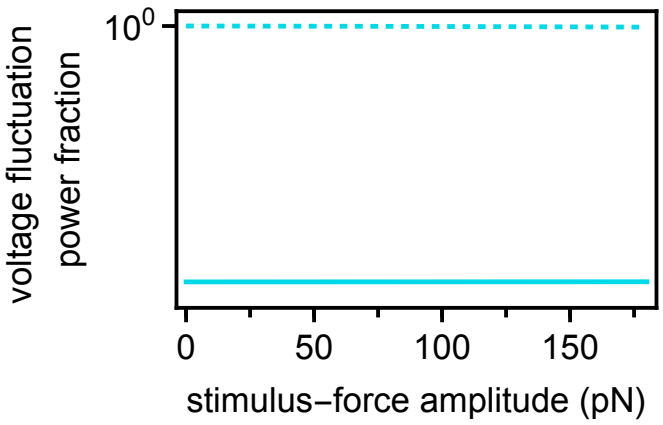
Out[ ]=


current fluctuation
power fraction
10^0
0
50
100
150
stimulus−force amplitude (pN)

Out[ ]=


voltage
fluctuation power
0.0003
0.0002
0.0001
0.
0
50
100
150
stimulus−force amplitude (pN)

Out[ ]=



In[ ]:=
```
(*Calculation of background psd from stimulated ps for the case of a stimulus-
 force amplitude of 180 pN*)
t1 = AbsoluteTime[];
fmult = 180; (*table of values of force-
 stimulus amplitude for calculation of fluctuation power*)



SeedRandom[47 741];

gx =  2 * kbTd * lPd;
y0in = Yrestd;

proc = ItoProcess[
    lPd * ⅆY[t] == (-kPd) * (Y[t] - Yupd) * ⅆt - kGSd * (Y[t] - Yugd - Dvd * Po[t]) * ⅆt +
       Sqrt[gx] * ⅆn1[t] + fmult * Fzd * Sin[2 * Pi * wstd * (t - tstartFd)] * ⅆt &&
     taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd] * ⅆPo[t] ==
      (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] - Po[t]) ⅆt +
       Sqrt[taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd]] Sqrt[
         (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] (1 - Clip[Po[t], {0.0, 1.0}]) + (1 - Pinf2[
                 Yhd, kGSd, Dvd, kbTd, Ncd]) Clip[Po[t], {0.0, 1.0}]) / Ncd] ⅆn2[t],
   {Y[t], Po[t]}, {{Y, Po}, {y0in, Pinfrest}}, t,
{n1 ≈ WienerProcess[], n2 ≈ WienerProcess[]}];

totallength = 2 * 0.507;
```

```
factot = 4;
deltatsima = 0.000001 / factot;
 ap =  RandomFunction[proc, {0, totallength, deltatsima}];

ap0 = ap["PathComponent", 1] - Yrestd;

ap1 = ap["PathComponent", 2] - Pinfrest;
resampfactor = factot;

randtime100 = 0.005 * 10;
randtimenm = Round[randtime100, deltatsima];
stpoint = IntegerPart[randtime100 / deltatsima];

deltatsampl = Round[deltatsima * resampfactor, deltatsima];
tsampl = 0.005;
endpoint = IntegerPart[
    Length[ap["PathComponent", 1]["Values"]] - tsampl / deltatsima - stpoint];
tsamplnm = Round[tsampl, deltatsima];
b1 = Table[f[i], {i, randtimenm, tsamplnm + randtimenm, deltatsampl}];
lengthares = Length[b1];

Clear[ares];
totallength2 = totallength;

ncopiesa = (totallength2 - randtimenm) / (tsamplnm + deltatsampl);
ncopies = IntegerPart[ncopiesa];
ares0 = Table[0, {ncopies}]; (*a resampled*)
ares1 = Table[0, {ncopies}];
sizea = lengthares;

fousampled0 = Table[0, {ncopies}, {lengthares}];
fousampled1 = Table[0, {ncopies}, {lengthares}];
cfousampled = Table[0, {ncopies}, {lengthares}];
Do[

  ares0[[ll]] = ap0["Values"][[(ll - 1) * tsampl / deltatsima + stpoint
     ;; IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];
    ares1[[ll]] = ap1["Values"][[(ll - 1) * tsampl / deltatsima + stpoint
     ;; IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];

      fousampled0[[ll]] = ResourceFunction["WelchSpectralEstimate"][
    ares0[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
    "OverlapOffset" → None, "SegmentSize" → sizea];
    cfousampled[[ll]] = ResourceFunction["WelchSpectralEstimate"][
```

```
    ares1[[ll]], ares0[[ll]], sizea / tsampl, "Window" → None,
    "Detrend" → None, "OverlapOffset" → None, "SegmentSize" → sizea];
    fousampled1[[ll]] = ResourceFunction["WelchSpectralEstimate"][
    ares1[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
    "OverlapOffset" → None, "SegmentSize" → sizea];
  ,
{ll, ncopies}];
powspmeansampledt0 = Drop[Mean[fousampled0], -1];

powspmeansampledt1 = Drop[Mean[fousampled1], -1];

cpowspmeansampledt = Drop[Re[Mean[cfousampled]], -1];
lengthpsd = Length[powspmeansampledt0];


meandataplot0 = Table[{0, 0}, {i, 1, lengthpsd}];
meandataplot1 = Table[{0, 0}, {i, 1, lengthpsd}];
meandataplot1cpsd = Table[{0, 0}, {i, 1, lengthpsd}];


meandataplot0 =
  Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt0[[i]][[2]]}, {i, 1, lengthpsd}];
meandataplot1 =
  Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt1[[i]][[2]]}, {i, 1, lengthpsd}];
meandataplot1cpsd =
  Table[{cpowspmeansampledt[[i]][[1]], cpowspmeansampledt[[i]][[2]]}, {i, 1, lengthpsd}];
fc = 610;
filtersq1 =
  Table[{meandataplot0[[i]][[1]], Abs[1 / (1 + I * meandataplot1[[i]][[1]] / fc)]^2},
   {i, 1, Length[meandataplot0]}];
meandataplot1filt = Table[{meandataplot0[[i]][[1]],
    filtersq1[[i]][[2]] * meandataplot1[[i]][[2]]}, {i, 1, Length[filtersq1]}];

(*Calculation of fluctuation power with and without fluctuation
 suppression. Gating swing can be set to 1.75 nm or 32 nm. *)
t1 = AbsoluteTime[];
fmult = 180; (*table of values of force-
 stimulus amplitude for calculation of fluctuation power*)



SeedRandom[47 741];
```

```
gx = 2 * kbTd * lPd;
y0in = Yrestd;

proc = ItoProcess[
   lPd * dY[t] == (-kPd) * (Y[t] - Yupd) * dt - kGSd * (Y[t] - Yugd - Dvd * Po[t]) * dt +
      Sqrt[gx] * dn1[t] + fmult * Fzd * Sin[2 * Pi * wstd * (t - tstartFd)] * dt &&
    taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd] * dPo[t] ==
     (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] - Po[t]) dt +
      Sqrt[taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd]] Sqrt[
        (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] (1 - Clip[Po[t], {0.0, 1.0}]) + (1 - Pinf2[
               Yhd, kGSd, Dvd, kbTd, Ncd]) Clip[Po[t], {0.0, 1.0}]) / Ncd] dn2[t],
   {Y[t], Po[t]}, {{Y, Po}, {y0in, Pinfrest}}, t,
{n1 ≈ WienerProcess[], n2 ≈ WienerProcess[]}];

totallength = 2 * 0.507;
factot = 4;
deltatsima = 0.000001 / factot;
ap = RandomFunction[proc, {0, totallength, deltatsima}];

ap0 = ap["PathComponent", 1] - Yrestd;

ap1 = ap["PathComponent", 2] - Pinfrest;
resampfactor = factot;

randtime100 = 0.005 * 10;
randtimenm = Round[randtime100, deltatsima];
stpoint = IntegerPart[randtime100 / deltatsima];

deltatsampl = Round[deltatsima * resampfactor, deltatsima];
tsampl = 0.005;
endpoint = IntegerPart[
   Length[ap["PathComponent", 1]["Values"]] - tsampl / deltatsima - stpoint];
tsamplnm = Round[tsampl, deltatsima];
b1 = Table[f[i], {i, randtimenm, tsamplnm + randtimenm, deltatsampl}];
lengthares = Length[b1];

Clear[ares];
totallength2 = totallength;

ncopiesa = (totallength2 - randtimenm) / (tsamplnm + deltatsampl);
ncopies = IntegerPart[ncopiesa];
ares0 = Table[0, {ncopies}]; (*a resampled*)
ares1 = Table[0, {ncopies}];
sizea = lengthares;
```

```
fousampled0 = Table[0, {ncopies}, {lengthares}];
fousampled1 = Table[0, {ncopies}, {lengthares}];
cfousampled = Table[0, {ncopies}, {lengthares}];
Do[

  ares0[[ll]] = ap0["Values"][[(ll - 1) * tsampl / deltatsima + stpoint
     ;; IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];
    ares1[[ll]] = ap1["Values"][[(ll - 1) * tsampl / deltatsima + stpoint
     ;; IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];

      fousampled0[[ll]] = ResourceFunction["WelchSpectralEstimate"][
    ares0[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
    "OverlapOffset" → None, "SegmentSize" → sizea];
    cfousampled[[ll]] = ResourceFunction["WelchSpectralEstimate"][
    ares1[[ll]], ares0[[ll]], sizea / tsampl, "Window" → None,
    "Detrend" → None, "OverlapOffset" → None, "SegmentSize" → sizea];
    fousampled1[[ll]] = ResourceFunction["WelchSpectralEstimate"][
    ares1[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
    "OverlapOffset" → None, "SegmentSize" → sizea];
  ,

{ll, ncopies}];
powspmeansampledt0 = Drop[Mean[fousampled0], -1];

powspmeansampledt1 = Drop[Mean[fousampled1], -1];

cpowspmeansampledt = Drop[Re[Mean[cfousampled]], -1];
lengthpsd = Length[powspmeansampledt0];


meandataplot0 = Table[{0, 0}, {i, 1, lengthpsd}];
meandataplot1 = Table[{0, 0}, {i, 1, lengthpsd}];
meandataplot1cpsd = Table[{0, 0}, {i, 1, lengthpsd}];


meandataplot0 =
  Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt0[[i]][[2]]}, {i, 1, lengthpsd}];
meandataplot1 =
  Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt1[[i]][[2]]}, {i, 1, lengthpsd}];
meandataplot1cpsd =
  Table[{cpowspmeansampledt[[i]][[1]], cpowspmeansampledt[[i]][[2]]}, {i, 1, lengthpsd}];
fc = 610;
filtersq1 =
```

```
  Table[{meandataplot0[[i]][[1]], Abs[1 / (1 + I * meandataplot1[[i]][[1]] / fc)]^2},
   {i, 1, Length[meandataplot0]}];
meandataplot1filt = Table[{meandataplot0[[i]][[1]],
    filtersq1[[i]][[2]] * meandataplot1[[i]][[2]]}, {i, 1, Length[filtersq1]}];

(*calculation of background area for deflection psd*)
data0 = Table[meandataplot0[[nn]][[2]], {nn, 2, Length[meandataplot0]}];
dataest0 = MedianFilter[data0, 15]; (*use 15 point median filter
 for estimate of background data from freq 2.8 kHz upward (point 15) *)
dataestwfreq0 =
  Table[{meandataplot0[[nn + 1]][[1]], dataest0[[nn]]}, {nn, 8 + 7, Length[meandataplot0] - 8 + 7}];
data0part = Table[meandataplot0[[nn]][[2]], {nn, 2, 8 + 7}];
dataest0part = data0part;
dataestwfreq0part = Table[{meandataplot0[[nn + 1]][[1]], dataest0part[[nn]]}, {nn, 1, 8 + 7 - 1}];
dataest0tot =
  Union[{{meandataplot0[[1]][[1]], meandataplot0[[2]][[2]]}}, dataestwfreq0part, dataestwfreq0];
dataest0tot[[1]][[2]] = 0.00007091605149763008`;
(*DC value is set equal to the corresponding value for the unstimulated psd*)
areaback0 = {fmult,
   Integrate[Interpolation[dataest0tot, InterpolationOrder → 1][x], {x, 0, 5 * 10^5}]};
(*area of the background deflection psd is obtained using a
 linear interpolation between points*)
areatot0 = {fmult,
   Integrate[Interpolation[meandataplot0, InterpolationOrder → 1][x], {x, 0, 5 * 10^5}]};
(*area of stimulated psd is obtained using a linear interpolation between points*)

(*calculation of background area for current psd*)
data1 = Table[meandataplot1[[nn]][[2]], {nn, 2, Length[meandataplot1]}];
dataest1 = MedianFilter[data1, 15];
dataestwfreq1 =
  Table[{meandataplot1[[nn + 1]][[1]], dataest1[[nn]]}, {nn, 8 + 7, Length[meandataplot1] - 8 + 7}];
data1part = Table[meandataplot1[[nn]][[2]], {nn, 2, 8 + 7}];
dataest1part = data1part;
dataestwfreq1part = Table[{meandataplot1[[nn + 1]][[1]], dataest1part[[nn]]}, {nn, 1, 8 + 7 - 1}];
dataest1tot1 =
  Union[{{meandataplot1[[1]][[1]], meandataplot1[[2]][[2]]}}, dataestwfreq1part, dataestwfreq1];
areaback1 = {fmult, Integrate[Interpolation[dataest1tot1, InterpolationOrder → 1][x],
     {x, 200, 5 * 10^5}] + meandataplot1[[2]][[2]] * meandataplot1[[2]][[1]] / 2};
areatot1 = {fmult,
   Integrate[Interpolation[meandataplot1, InterpolationOrder → 1][x], {x, 0, 5 * 10^5}]};

(*calculation of background area of voltage psd*)
data1filt = Table[meandataplot1filt[[nn]][[2]], {nn, 2, Length[meandataplot1filt]}];
dataest1filt = MedianFilter[data1filt, 15];
```

```
dataestwfreq1filt = Table[{meandataplot1filt[[nn + 1]][[1]], dataest1filt[[nn]]},
    {nn, 8 + 7, Length[meandataplot1filt] - 8 + 7}];
data1filtpart = Table[meandataplot1filt[[nn]][[2]], {nn, 2, 8 + 7}];
dataest1filtpart = data1filtpart;
dataestwfreq1filtpart =
  Table[{meandataplot1filt[[nn + 1]][[1]], dataest1filtpart[[nn]]}, {nn, 1, 8 + 7 - 1}];
dataest1totfilt = Union[{{meandataplot1filt[[1]][[1]], meandataplot1filt[[2]][[2]]}},
    dataestwfreq1filtpart, dataestwfreq1filt];
areaback1filt =
  {fmult, Integrate[Interpolation[dataest1totfilt, InterpolationOrder → 1][x],
      {x, 200, 5 * 10^5}]} + meandataplot1filt[[2]][[2]] * meandataplot1filt[[2]][[1]] / 2;
areatot1filt = {fmult, Integrate[
    Interpolation[meandataplot1filt, InterpolationOrder → 1][x], {x, 0, 5 * 10^5}]};



t2 = AbsoluteTime[];
t2 - t1
```

Out[ ]=

```
14.541474
```

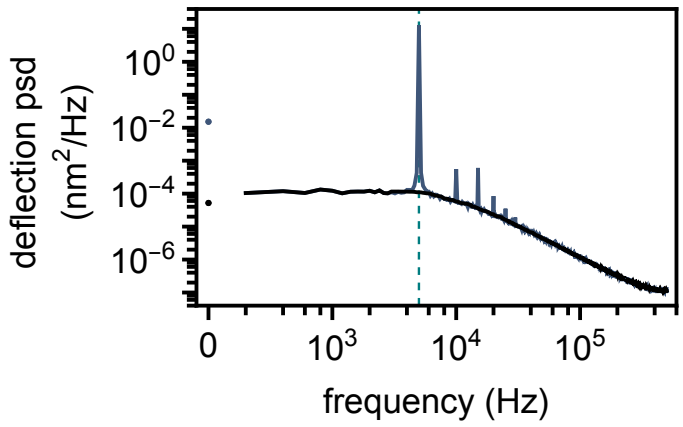
In[ ]:=

```
(*plot settings*)
plotstyleY = {Darker[■]};
plotstyleP = {Darker[■]};
plotstyleV = {Darker[■]};
plotrangeP = {{80, 6 * 10^5}, {10^(-10), 2 * 10^(-3)}};
```

In[ ]:=

```
(*Fig S14A Deflection psd and background psd vs frequency*)
meandataplot0c = {100, meandataplot0[[1]][[2]]};
meandataplot10back = {100, dataest0tot[[2]][[2]] / 2};
plotrangeY = {{80, 6 * 10^5}, {0.4 * 10^(-7), 4 * 10^(1)}};
ListLogLogPlot[{meandataplot0},
  PlotStyle → plotstyleY, Joined → True, PlotRange → plotrangeY,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"frequency (Hz)",
    "deflection psd \n (\!\(\*SuperscriptBox[\(nm\), \(2\)]\)/Hz)"}, GridLines →
   {{5000}, {}}, GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLogPlot[{meandataplot0c}, PlotStyle → plotstyleY]];
Show[%,
  ListLogLogPlot[dataest0tot, PlotStyle → Black, PlotRange → All, Joined → True]];
Show[%, ListLogLogPlot[{meandataplot10back}, PlotStyle → Black]]
```

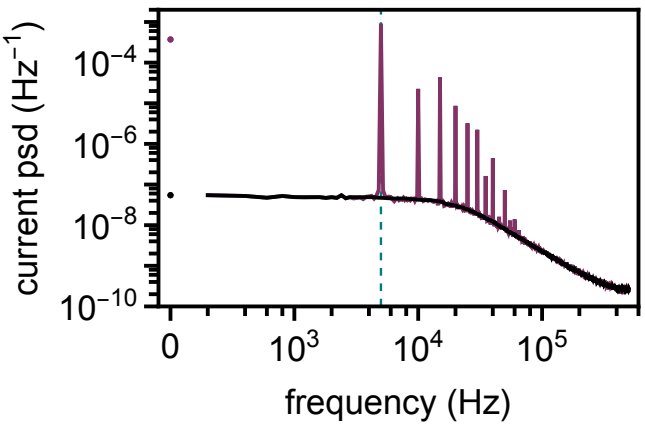
Out[ ]=



In[ ]:=
```
(*Fig S14B Current psd and background psd vs frequency*)
meandataplot1c = {100, meandataplot1[[1]][[2]]};
meandataplot1backc = {100, meandataplot1[[2]][[2]]};
ListLogLogPlot[{meandataplot1}, PlotStyle → plotstyleP,
  Joined → True, Frame → True, PlotRange → plotrangeP,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"frequency (Hz)", "current psd \!\(\*TemplateBox[{RowBox[{\"(\",
      \"Hz\"}], RowBox[{\"-\", \"1\"}]},\"Superscript\"]\))"}, GridLines →
   {{5000}, {}}, GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
psd11 = Show[%, ListLogLogPlot[{meandataplot1c}, PlotStyle → plotstyleP]];

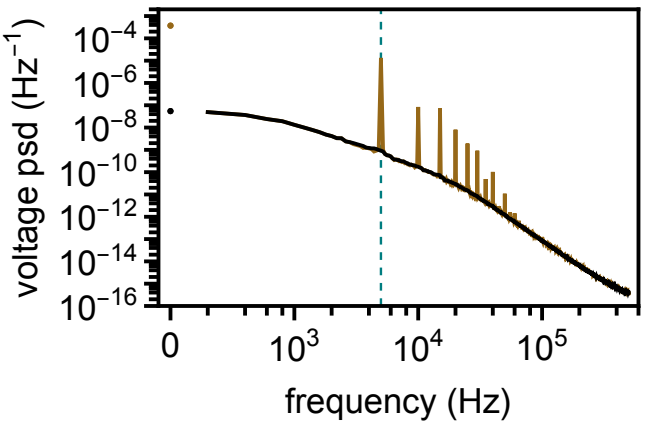
Show[%,
  ListLogLogPlot[dataest1tot1, PlotStyle → Black, PlotRange → All, Joined → True]];
Show[%, ListLogLogPlot[{meandataplot1backc}, PlotStyle → Black]]
```

Out[ ]=



In[ ]:=
```
(*Fig S14C Voltage psd and background psd vs frequency*)
meandataplot1v = {100, meandataplot1filt[[1]][[2]]};
plotrangeV = {{80, 6 * 10^5}, {10^(-16), 2 * 10^(-3)}};
ListLogLogPlot[{meandataplot1filt}, PlotStyle → plotstyleV, Joined → True,
  Frame → True, PlotRange → plotrangeV, ImagePadding → {{pady, pad2}, {padx, pad2}},
  ImageSize → 400, LabelStyle → {FontSize → fntsize}, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"frequency (Hz)", "voltage psd \!\(\*TemplateBox[{RowBox[{\"(\",
      \"Hz\"}], RowBox[{\"-\", \"1\"}]},\"Superscript\"]\))"}, GridLines →
   {{5000}, {}}, GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLogPlot[{meandataplot1v}, PlotStyle → plotstyleV]];
Show[%,
  ListLogLogPlot[dataest1totfilt, PlotStyle → Black, PlotRange → All, Joined → True]];
Show[%, ListLogLogPlot[{meandataplot1backc}, PlotStyle → Black]]
```

Out[ ]=




```
(*Calculation of CSNR*)
(*Calculation of PSD without noise reduction by setting stimulus-
 force amplitude to 0*)
fmult = {0};

CNRYDC = Table[0, {Length[fmult]}];
CNRYst = Table[0, {Length[fmult]}];
CNRPDC = Table[0, {Length[fmult]}];
CNRPst = Table[0, {Length[fmult]}];
deltafval = Table[0, {Length[fmult]}];
deltaf1val = Table[0, {Length[fmult]}];
meandataplot0base = Table[0, 1];
meandataplot1base  = Table[0, 1];
meandataplot1filtbase  = Table[0, 1];
```

```
SetSharedVariable[CNRYDC];
SetSharedVariable[CNRYst];
SetSharedVariable[CNRPDC];
SetSharedVariable[CNRPst];
SetSharedVariable[fmult];
SetSharedVariable[deltafval];
SetSharedVariable[deltaf1val];
SetSharedVariable[Ncd];
SetSharedVariable[meandataplot0base];
SetSharedVariable[meandataplot1base];
SetSharedVariable[meandataplot1filtbase];
fun = ResourceFunction["WelchSpectralEstimate"];
SetSharedFunction[fun];


t1 = AbsoluteTime[];
pp = 1;

SeedRandom[47 741];

gx =  2 * kbTd * lPd;
y0in = Yrestd;

proc = ItoProcess[
    lPd * ⅆY[t] == (-kPd) * (Y[t] - Yupd) * ⅆt - kGSd * (Y[t] - Yugd - Dvd * Po[t]) * ⅆt +
       Sqrt[gx] * ⅆn1[t] + Fzd * fmult[[pp]] * Sin[2 * Pi * wstd1 * (t - tstartFd)] * ⅆt &&
     taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd] * ⅆPo[t] ==
      (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] - Po[t]) ⅆt +
       Sqrt[taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd]] Sqrt[
         (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] (1 - Clip[Po[t], {0.0, 1.0}]) + (1 - Pinf2[
                 Yhd, kGSd, Dvd, kbTd, Ncd]) Clip[Po[t], {0.0, 1.0}]) / Ncd] ⅆn2[t],
   {Y[t], Po[t]}, {{Y, Po}, {y0in, Pinfrest}}, t,
 {n1 ≈ WienerProcess[], n2 ≈ WienerProcess[]}];

totallength = 2 * 0.507;
fact = 4;
deltatsima = 0.000001 / fact;
ap =  RandomFunction[proc, {0, totallength, deltatsima}];

ap0 = ap["PathComponent", 1] - Yrestd;

ap1 = ap["PathComponent", 2] - Pinfrest;
```

```
resampfactor = fact;
randtime100 = 0.005 * 10;
randtimenm = Round[randtime100, deltatsima];
stpoint = IntegerPart[randtime100 / deltatsima];

deltatsampl = Round[deltatsima * resampfactor, deltatsima];
tsampl = 0.005;
endpoint = IntegerPart[
    Length[ap["PathComponent", 1]["Values"]] - tsampl / deltatsima - stpoint];
tsamplnm = Round[tsampl, deltatsima];
b1 = Table[f[i], {i, randtimenm, tsamplnm + randtimenm, deltatsampl}];
lengthares = Length[b1];

Clear[ares];
totallength2 = totallength;

ncopiesa = (totallength2 - randtimenm) / (tsamplnm + deltatsampl);
ncopies = IntegerPart[ncopiesa];

ares0 = Table[0, {ncopies}];
ares1 = Table[0, {ncopies}];
sizea = lengthares;

fousampled0 = Table[0, {ncopies}, {lengthares}];
fousampled1 = Table[0, {ncopies}, {lengthares}];
cfousampled = Table[0, {ncopies}, {lengthares}];

Do[

  ares0[[ll]] = ap0["Values"][[(ll - 1) * tsampl / deltatsima + stpoint
      ;; IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];
    ares1[[ll]] = ap1["Values"][[(ll - 1) * tsampl / deltatsima + stpoint
     ;; IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];

    fousampled0[[ll]] = fun[ares0[[ll]], sizea / tsampl, "Window" → None,
    "Detrend" → None, "OverlapOffset" → None, "SegmentSize" → sizea];
    fousampled1[[ll]] = fun[ares1[[ll]], sizea / tsampl, "Window" → None,
    "Detrend" → None, "OverlapOffset" → None, "SegmentSize" → sizea];,
{ll, ncopies}];

powspmeansampledt0 = Mean[fousampled0];
powspmeansampledt1 = Mean[fousampled1];
lengthpsd = Length[powspmeansampledt0];
```

```
meandataplot0base =
  Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt0[[i]][[2]]}, {i, 1, lengthpsd}];
meandataplot1base =
  Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt1[[i]][[2]]}, {i, 1, lengthpsd}];

fc = 610;
filtersq1 =
  Table[{meandataplot0base[[i]][[1]], Abs[1 / (1 + I * meandataplot1base[[i]][[1]] / fc)]^2},
   {i, 1, Length[meandataplot0base]}];
meandataplot1filtbase = Table[{meandataplot0base[[i]][[1]],
    filtersq1[[i]][[2]] * meandataplot1base[[i]][[2]]}, {i, 1, Length[filtersq1]}];


t2 = AbsoluteTime[];
t2 - t1


(*Setting of parameters for CSNRs calcualtion
  for a resting current of 0.17. Set Pinfrest=0.17*)
fmult = {0, 0.1, 0.5, 1, 2, 3, 5, 10, 15, 20, 30, 40,
   50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150, 160, 170, 180};

CNRYDC = Table[0, {Length[fmult]}];
CNRYst = Table[0, {Length[fmult]}];
CNRPDC = Table[0, {Length[fmult]}];
CNRPst = Table[0, {Length[fmult]}];
CNRVDC = Table[0, {Length[fmult]}];
CNRVst = Table[0, {Length[fmult]}];
deltafval = Table[0, {Length[fmult]}];
deltaf1val = Table[0, {Length[fmult]}];

SetSharedVariable[CNRYDC];
SetSharedVariable[CNRYst];
SetSharedVariable[CNRPDC];
SetSharedVariable[CNRPst];
SetSharedVariable[CNRVDC];
SetSharedVariable[CNRVst];
SetSharedVariable[fmult];
SetSharedVariable[deltafval];
SetSharedVariable[deltaf1val];
SetSharedVariable[Ncd];
SetSharedVariable[meandataplot0base];
```

```
SetSharedVariable[meandataplot1base];
fun = ResourceFunction["WelchSpectralEstimate"];
SetSharedFunction[fun];

(* banwidths for a gating swing of 32 nm and a resting current of 0.17 (Eq.47 )*)
(*freqpeaklc=Table[4819.985420669219`,{Length[fmult]}];
freqpeakrc=Table[5180.014579330781`,{Length[fmult]}];
freqnoisel1c=Table[1819.985420669219`,{Length[fmult]}];
freqnoiser1c=Table[2000.`,{Length[fmult]}];
freqnoisel2c=Table[8000.`,{Length[fmult]}];
freqnoiser2c=Table[8180.014579330781`,{Length[fmult]}];

freqpeakl=Table[4819.993969867439`,{Length[fmult]}];
freqpeakr=Table[5180.006030132561`,{Length[fmult]}];
freqnoisel1=Table[1819.9939698674389`,{Length[fmult]}];
freqnoiser1=Table[2000.`,{Length[fmult]}];
freqnoisel2=Table[8000.`,{Length[fmult]}];
freqnoiser2=Table[8180.006030132561`,{Length[fmult]}];

freqpeaklv=Table[4819.984200018502`,{Length[fmult]}];
freqpeakrv=Table[5180.015799981498`,{Length[fmult]}];
freqnoisel1v=Table[1819.985420669219`,{Length[fmult]}];
freqnoiser1v=Table[2000.`,{Length[fmult]}];
freqnoisel2v=Table[8000.`,{Length[fmult]}];
freqnoiser2v=Table[8180.015799981498`,{Length[fmult]}];*)

(* banwidths for a gating swing of 1.75 nm and a resting current of 0.17 (Eq.47 )*)

(*freqpeaklc=Table[4815.479245694055`,{Length[fmult]}];
freqpeakrc=Table[5184.520754305945`,{Length[fmult]}];
freqnoisel1c=Table[1815.479245694055`,{Length[fmult]}];
freqnoiser1c=Table[2000.`,{Length[fmult]}];
freqnoisel2c=Table[8000.`,{Length[fmult]}];
freqnoiser2c=Table[8184.520754305945`,{Length[fmult]}];

freqpeakl=Table[4819.993981889589`,{Length[fmult]}];
freqpeakr=Table[5180.006018110411`,{Length[fmult]}];
freqnoisel1=Table[1819.9939818895891`,{Length[fmult]}];
freqnoiser1=Table[2000.`,{Length[fmult]}];
freqnoisel2=Table[8000.`,{Length[fmult]}];
freqnoiser2=Table[8180.006018110411`,{Length[fmult]}];

freqpeaklv=Table[4819.984200018502`,{Length[fmult]}];
freqpeakrv=Table[5184.909542819152`,{Length[fmult]}];
```

```
freqnoisel1v=Table[1815.0904571808478`,{Length[fmult]}];
freqnoiser1v=Table[2000.`,{Length[fmult]}];
freqnoisel2v=Table[8000.`,{Length[fmult]}];
freqnoiser2v=Table[8184.909542819152`,{Length[fmult]}];*)
```

```
t1 = AbsoluteTime[];

ParallelDo[
SeedRandom[47 741];

  gx =  2 * kbTd * lPd;
  y0in = Yrestd;


  proc = ItoProcess[
    lPd * ⅆY[t] == (-kPd) * (Y[t] - Yupd) * ⅆt - kGSd * (Y[t] - Yugd - Dvd * Po[t]) * ⅆt +
       Sqrt[gx] * ⅆn1[t] + Fzd * fmult[[pp]] * Sin[2 * Pi * wstd * (t - tstartFd)] * ⅆt &&
     taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd] * ⅆPo[t] ==
      (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] - Po[t]) ⅆt +
       Sqrt[taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd]] Sqrt[
         (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] (1 - Clip[Po[t], {0.0, 1.0}]) + (1 - Pinf2[
                 Yhd, kGSd, Dvd, kbTd, Ncd]) Clip[Po[t], {0.0, 1.0}]) / Ncd] ⅆn2[t],
    {Y[t], Po[t]}, {{Y, Po}, {y0in, Pinfrest}}, t,
{n1 ≈ WienerProcess[], n2 ≈ WienerProcess[]}];


totallength = 2 * 0.507;
  factot = 4;
  deltatsima = 0.000001 / factot;
   ap = RandomFunction[proc, {0, totallength, deltatsima}];


  ap0 = ap["PathComponent", 1] - Yrestd;

ap1 = ap["PathComponent", 2] - Pinfrest;

resampfactor = factot;
```

```
randtime100 = 0.005 * 10;
randtimenm = Round[randtime100, deltatsima];
stpoint = IntegerPart[randtime100 / deltatsima];

deltatsampl = Round[deltatsima * resampfactor, deltatsima];
tsampl = 0.005;
endpoint = IntegerPart[
    Length[ap["PathComponent", 1]["Values"]] - tsampl / deltatsima - stpoint];
tsamplnm = Round[tsampl, deltatsima];
b1 = Table[f[i], {i, randtimenm, tsamplnm + randtimenm, deltatsampl}];
lengthares = Length[b1];

Clear[ares];
totallength2 = totallength;

ncopiesa = (totallength2 - randtimenm) / (tsamplnm + deltatsampl);
ncopies = IntegerPart[ncopiesa];

  ares0 = Table[0, {ncopies}];
ares1 = Table[0, {ncopies}];
sizea = lengthares;

fousampled0 = Table[0, {ncopies}, {lengthares}];
fousampled1 = Table[0, {ncopies}, {lengthares}];
cfousampled = Table[0, {ncopies}, {lengthares}];
  Do[

   ares0[[ll]] = ap0["Values"][[(ll - 1) * tsampl / deltatsima + stpoint ;;
      IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];
    ares1[[ll]] = ap1["Values"][[(ll - 1) * tsampl / deltatsima + stpoint ;;
      IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];

       fousampled0[[ll]] = ResourceFunction["WelchSpectralEstimate"][
     ares0[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
     "OverlapOffset" → None, "SegmentSize" → sizea];
    cfousampled[[ll]] = ResourceFunction["WelchSpectralEstimate"][
     ares1[[ll]], ares0[[ll]], sizea / tsampl, "Window" → None,
     "Detrend" → None, "OverlapOffset" → None, "SegmentSize" → sizea];
    fousampled1[[ll]] = ResourceFunction["WelchSpectralEstimate"][
     ares1[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
     "OverlapOffset" → None, "SegmentSize" → sizea];
    ,
```

```
{ll, ncopies}];
  powspmeansampledt0 = Drop[Mean[fousampled0], -1];

  powspmeansampledt1 = Drop[Mean[fousampled1], -1];


lengthpsd = Length[powspmeansampledt0];

meandataplot0 = Table[{0, 0}, {i, 1, lengthpsd}];
  meandataplot1 = Table[{0, 0}, {i, 1, lengthpsd}];

  meandataplot0 =
   Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt0[[i]][[2]]}, {i, 1, lengthpsd}];
  meandataplot1 =
   Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt1[[i]][[2]]}, {i, 1, lengthpsd}];

  fc = 610;
  filtersq1 = Table[{meandataplot0[[i]][[1]],
     Abs[1 / (1 + I * meandataplot1[[i]][[1]] / fc)]^2}, {i, 1, Length[meandataplot0]}];
  meandataplot1filt = Table[{meandataplot0[[i]][[1]],
     filtersq1[[i]][[2]] * meandataplot1[[i]][[2]]}, {i, 1, Length[filtersq1]}];

  (*Interpolation of 1 order of defleciton, current and voltage
   psd for calculation of area under corresponding bandwidths *)
  ff0 = Interpolation[meandataplot0, InterpolationOrder → 1];
  ff1 = Interpolation[meandataplot1, InterpolationOrder → 1];
  ff1filt = Interpolation[meandataplot1filt, InterpolationOrder → 1];

  (*Interpolation of 1 order of defleciton,
  current and voltage unstimulated psd *)
  ff0base = Interpolation[meandataplot0base, InterpolationOrder → 1];
  ff1base = Interpolation[meandataplot1base, InterpolationOrder → 1];
  ff1filtbase = Interpolation[meandataplot1filtbase, InterpolationOrder → 1];

  n1 = 26; (*vector position for psd corresponding to a frequency of 5 kHz*)

  (*stimulus-
   frequency deflection CSNR and DC deflection CSNR caclualted as ratio of
    areas of corresponding bandwidths for the case with noise reduction*)
  CNRYst[[pp]] = Integrate[ff0[x], {x, freqpeakl[[pp]], freqpeakr[[pp]]}] /
     (Integrate[ff0[x], {x, freqnoisel1[[pp]], freqnoiser1[[pp]]}] +
       Integrate[ff0[x], {x, freqnoisel2[[pp]], freqnoiser2[[pp]]}]) - 1;
  CNRYDC[[pp]] = 2 * (meandataplot0[[1]][[2]]) / meandataplot0[[2]][[2]] - 1;
```

```
  (*(*stimulus-
   frequency deflection CSNR and DC deflection CSNR caclualted as ratio of
    areas of corresponding bandwidths for the case without noise reduction*)
  CNRYst[[pp]]=Integrate[ff0[x],{x,freqpeakl[[pp]],freqpeakr[[pp]]}]/
     (Integrate[ff0base[x],{x,freqnoisel1[[pp]],freqnoiser1[[pp]]}]+
       Integrate[ff0base[x],{x,freqnoisel2[[pp]],freqnoiser2[[pp]]}])-1;
  CNRYDC[[pp]]=2*(meandataplot0[[1]][[2]])/meandataplot0base[[2]][[2]]-1;*)

  (*stimulus-frequency current CSNR and DC current CSNR caclualted as ratio of
    areas of corresponding bandwidths for the case with noise reduction*)
  CNRPst[[pp]] = Integrate[ff1[x], {x, freqpeaklc[[pp]], freqpeakrc[[pp]]}] /
     (Integrate[ff1[x], {x, freqnoisel1c[[pp]], freqnoiser1c[[pp]]}] +
       Integrate[ff1[x], {x, freqnoisel2c[[pp]], freqnoiser2c[[pp]]}]) - 1;
  CNRPDC[[pp]] = 2 * (meandataplot1[[1]][[2]]) / meandataplot1[[2]][[2]] - 1;

  (*(*stimulus-frequency current CSNR and DC current CSNR caclualted as ratio of
    areas of corresponding bandwidths for the case without noise reduction*)
  CNRPst[[pp]]=Integrate[ff1[x],{x,freqpeaklc[[pp]],freqpeakrc[[pp]]}]/
     (Integrate[ff1base[x],{x,freqnoisel1c[[pp]],freqnoiser1c[[pp]]}]+
       Integrate[ff1base[x],{x,freqnoisel2c[[pp]],freqnoiser2c[[pp]]}])-1;
  CNRPDC[[pp]]=2*(meandataplot1[[1]][[2]])/meandataplot1base[[2]][[2]]-1;*)


  (*stimulus-frequency voltage CSNR and DC voltage CSNR caclualted as ratio of
    areas of corresponding bandwidths for the case with noise reduction *)
  CNRVst[[pp]] = Integrate[ff1filt[x], {x, freqpeaklv[[pp]], freqpeakrv[[pp]]}] /
     (Integrate[ff1filt[x], {x, freqnoisel1v[[pp]], freqnoiser1v[[pp]]}] +
       Integrate[ff1filt[x], {x, freqnoisel2v[[pp]], freqnoiser2v[[pp]]}]) - 1;
  CNRVDC[[pp]] = 2 * (meandataplot1filt[[1]][[2]]) / meandataplot1filt[[2]][[2]] - 1;

  (*(*stimulus-frequency voltage CSNR and DC voltage CSNR caclualted as ratio of
    areas of corresponding bandwidths for the case without noise reduction *)
  CNRVst[[pp]]=Integrate[ff1filt[x],{x,freqpeaklv[[pp]],freqpeakrv[[pp]]}]/
     (Integrate[ff1filtbase[x],{x,freqnoisel1v[[pp]],freqnoiser1v[[pp]]}]+
       Integrate[ff1filtbase[x],{x,freqnoisel2v[[pp]],freqnoiser2v[[pp]]}])-1;
  CNRVDC[[pp]]=2*(meandataplot1filt[[1]][[2]])/meandataplot1filtbase[[2]][[2]]-1;*)

  ,
  {pp, 1, Length[fmult]}];

 t2 = AbsoluteTime[];
t2 - t1
```

```
(*Fig 9 CSNR vs stimulus-
  force amplitude for a gating swing of 32 nm and a resting current of 0.17. Set D=
 32 and Pinfrest=0.17. Case with noise reduction*)

(*table of values of CSNRs*)
CNRYstval = Table[{fmult[[i]], CNRYst[[i]]}, {i, 1, Length[fmult]}];
CNRYDCval = Table[{fmult[[i]], CNRYDC[[i]]}, {i, 1, Length[fmult]}];
CNRPstval = Table[{fmult[[i]], CNRPst[[i]]}, {i, 1, Length[fmult]}];
CNRPDCval = Table[{fmult[[i]], CNRPDC[[i]]}, {i, 1, Length[fmult]}];
CNRVstval = Table[{fmult[[i]], CNRVst[[i]]}, {i, 1, Length[fmult]}];
CNRVDCval = Table[{fmult[[i]], CNRVDC[[i]]}, {i, 1, Length[fmult]}];

(*plot settings*)
plotstyleY = {Lighter[■]};
plotstyleP = {Lighter[■]};
plotstyleV = {Lighter[■]};

Remove[YTicksb]
Remove[XTicks]
Remove[YTicks]

TickLength = 0.02;
TickLength1 = 0.03;
Yticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, -24, 1, 2}];
Yticks2 = Table[{10^k, Null, {0, TickLength1}}, {k, -23, 1, 2}];
TickLength2 = 0.02;
aa = Table[0, {22}, {4}];
For[k = 1, k ≤ 22, k++,
aa[[k]] = Table[{l * 10^-k, , {0, TickLength2}}, {l, 2, 8, 2}];
]
YTicks = Union[Yticks1, Yticks2, aa[[1]], aa[[2]], aa[[3]], aa[[4]], aa[[5]],
    aa[[6]], aa[[7]], aa[[8]], aa[[9]], aa[[10]], aa[[11]], aa[[12]], aa[[13]], aa[[14]],
    aa[[15]], aa[[16]], aa[[17]], aa[[18]], aa[[19]], aa[[20]], aa[[21]], aa[[22]]];

Xticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, 0, 6, 1}];
aa2 = Table[0, {20}, {4}];
For[k = 1, k ≤ 10, k++,
```

```
aa2[[k]] = Table[{l * 10^k, , {0, TickLength2}}, {l, 2, 8, 2}];
]
XTicks = Union[Xticks1, aa2[[1]], aa2[[2]], aa2[[3]], aa2[[4]], aa2[[5]],
    aa2[[6]], aa2[[7]], aa2[[8]], aa2[[9]], aa2[[10]], aa2[[11]], aa2[[12]], aa2[[13]],
    aa2[[14]], aa2[[15]], aa2[[16]], aa2[[17]], aa2[[18]], aa2[[19]], aa2[[20]]];

pady = 120;
padx = 70;
pad2 = 30;
fntsize = 18;

(*FigA insert CSNR deflection stimulus*)

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1, 0.5}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1, 0.5}];
Xticks4 = Table[{k, k, {0, 0.03}}, {k, 0.1, 0.1, 0.1}];
XTicks = Union[Xticks1, Xticks3, Xticks4];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 2, 2}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 2, 1}];
YTicksb = Union[Yticks1, Yticks3];


framelabelsCSNRY = {{"stimulus-frequency \n deflection CSNR", None},
   {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRYstval}, PlotRange → {{0.1, 1}, {-0.25, 2.1}}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRY,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigA CSNR deflection stimulus*)

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 400 000, 20 000}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 400 000, 20 000 / 2}];
YTicksb = Union[Yticks1, Yticks3];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 180, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 180, 25}];
XTicks = Union[Xticks1, Xticks3];
```

```
framelabelsCSNRY = {{"stimulus-frequency \n deflection CSNR", None},
   {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRYstval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRY,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]



(*FigB insert CSNR deflection DC*)

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 3, 1}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 3, 1}];
YTicksb = Union[Yticks1, Yticks3];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 30, 5}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 30, 5}];
XTicks = Union[Xticks1, Xticks3];


framelabelsCSNRY =
  {{"DC deflection CSNR", None}, {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRYDCval}, PlotRange → {{0, 25}, {0, 3}}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRY,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigB CSNR deflection DC*)


Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 180, 25}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 180, 25}];
YTicksb = Union[Yticks1, Yticks3];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 400, 100}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 400, 100 / 2}];
YTicksb = Union[Yticks1, Yticks3];

framelabelsCSNRY =
```

```
  {{"DC deflection CSNR", None}, {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRYDC}, PlotRange → {Full, Full}, Joined → True, PlotRange → plotrangeSNRP,
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsCSNRY, FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigC insert CSNR current ST*)

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 3, 1}];(*50000 200*)
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 3, 1 / 2}];(*25000 100*)
YTicksb = Union[Yticks1, Yticks3];


Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 3, 1}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 3, 0.5}];
Xticks4 = Table[{k, k, {0, 0.03}}, {k, 0.1, 0.1, 0.1}];
XTicks = Union[Xticks1, Xticks3, Xticks4];


framelabelsCSNRP = {{"stimulus-frequency \n current CSNR", None},
   {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRPstval}, PlotRange → {{0.1, 3}, {-0.25, 2}}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRP,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]

(*FigC CSNR current ST*)


Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 50 000, 4000}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 50 000, 4000 / 2}];
YTicks = Union[Yticks1, Yticks3];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 200, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 200, 25}];
XTicks = Union[Xticks1, Xticks3];


framelabelsCSNRP = {{"stimulus-frequency \n current CSNR", None},
   {"stimlusu-force amplitude (pN)", None}};
```

```
ListPlot[{CNRPstval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRP,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]


(*FigD insert CSNR current DC*)

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 30, 5}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 30, 5}];
Xticks4 = Table[{k, k, {0, 0.03}}, {k, 1, 1, 1}];
XTicks = Union[Xticks1, Xticks3, Xticks4];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 5, 1}];(*50000 200*)
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 5, 1}];(*25000 100*)
YTicksb = Union[Yticks1, Yticks3];

framelabelsCSNRP =
  {{"DC current CSNR", None}, {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRPDCval}, PlotRange → {{1, 15}, {-0.2, 3}}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRP,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigD CSNR current DC*)


Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 16000, 4000}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 16000, 2000}];
YTicks = Union[Yticks1, Yticks3];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 180, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 180, 25}];
XTicks = Union[Xticks1, Xticks3];



framelabelsCSNRP =
  {{"DC current CSNR", None}, {"stimulus-force amplitude (pN)", None}};
```

```
ListPlot[{CNRPDCval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRP,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]


(*FigE insert CSNR voltage stimulus*)

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 10, 5}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 10, 5 / 2}];
Xticks4 = Table[{k, k, {0, 0.03}}, {k, 1, 1, 1}];
XTicks = Union[Xticks1, Xticks3, Xticks4];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, -3, 5, 1}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, -3, 5, 1}];
YTicksb = Union[Yticks1, Yticks3];

framelabelsSNRP = {{"stimulus-frequency \n voltage CSNR", None},
   {"stimulus-force amplitude (pN)", None}};

ListPlot[{CNRVstval}, PlotRange → {{0.2, 8}, {-1, 3}}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleV,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsSNRP,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigE CSNR voltage stimulus*)


Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 5000, 2 * 500}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 5000, 2 * 500 / 2}];
YTicks = Union[Yticks1, Yticks3];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 200, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 200, 25}];
XTicks = Union[Xticks1, Xticks3];


framelabelsCSNRV = {{"stimulus-frequency \n voltage CSNR", None},
   {"stimlusu-force amplitude (pN)", None}};
```

```
ListPlot[{CNRVstval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleV,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRV,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]

(*FigF insert CSNR voltage DC*)

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 10, 5}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 10, 5}];
Xticks4 = Table[{k, k, {0, 0.03}}, {k, 1, 1, 1}];
XTicks = Union[Xticks1, Xticks3, Xticks4];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 5, 1}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 5, 1}];
YTicksb = Union[Yticks1, Yticks3];

framelabelsCSNRV =
  {{"DC voltage CSNR", None}, {"stimulus-force amplitude (pN)", None}};

ListPlot[{CNRVDCval}, PlotRange → {{1, 12}, {-0.2, 3}}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleV,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRV,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigF CSNR voltage DC*)

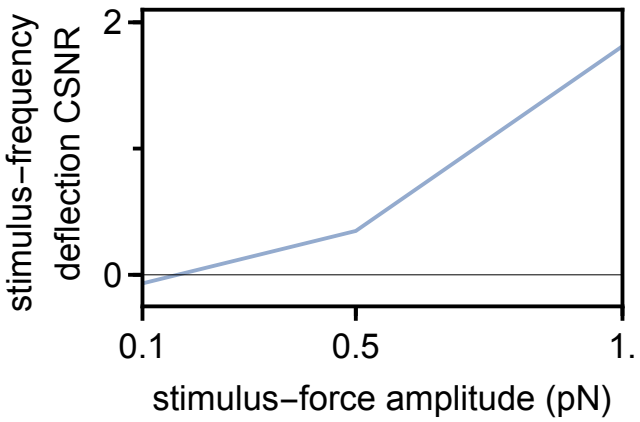
Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 20 000, 5000}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 20 000, 5000 / 2}];
YTicks = Union[Yticks1, Yticks3];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 180, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 180, 25}];
XTicks = Union[Xticks1, Xticks3];

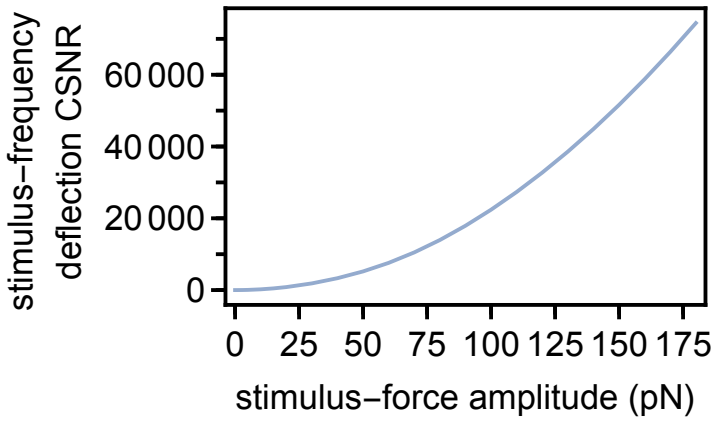
framelabelsCSNRV =
  {{"DC voltage CSNR", None}, {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRVDCval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleV,
```

```
Background → White, Frame → {True, True, True, True},
FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRV,
FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

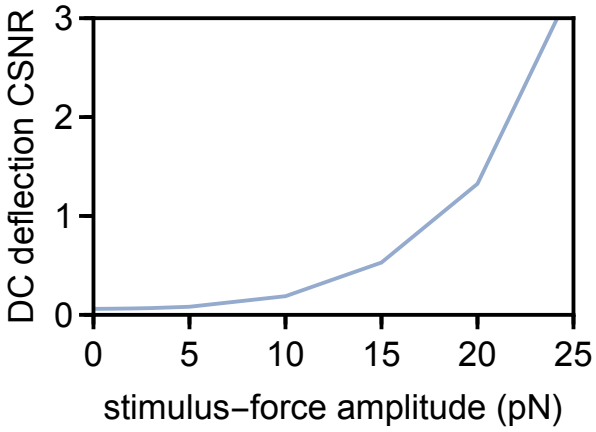
Out[ ]=



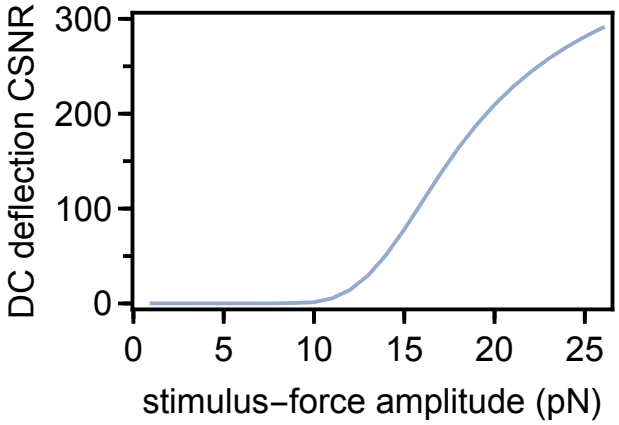
Out[ ]=



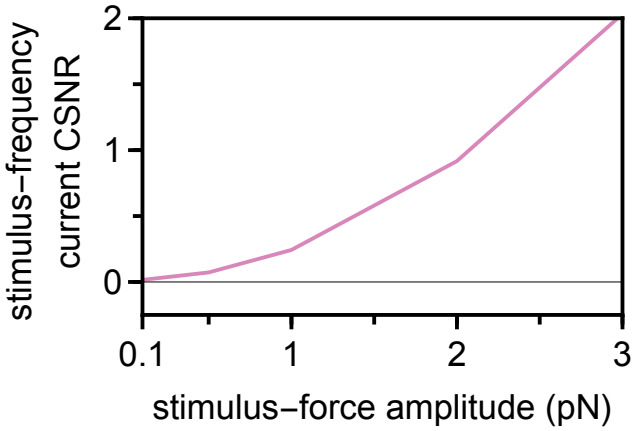
Out[ ]=


DC deflection CSNR
3
2
1
0
0
5
10
15
20
25
stimulus–force amplitude (pN)

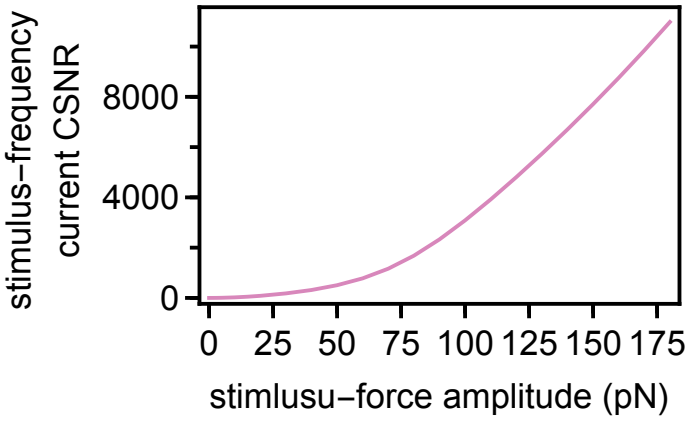
Out[ ]=


DC deflection CSNR
300
200
100
0
0
5
10
15
20
25
stimulus–force amplitude (pN)

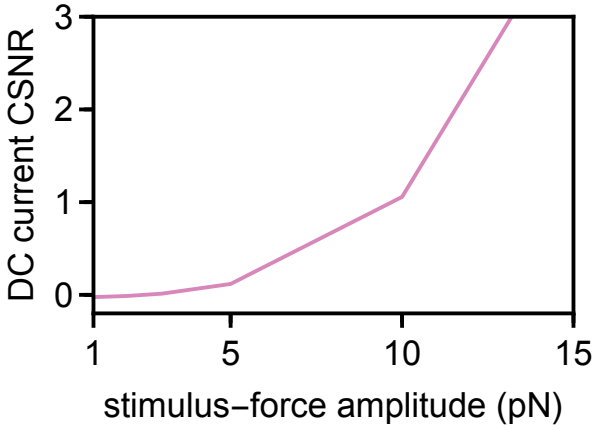
Out[ ]=


stimulus–frequency
current CSNR
2
1
0
0.1
1
2
3
stimulus–force amplitude (pN)

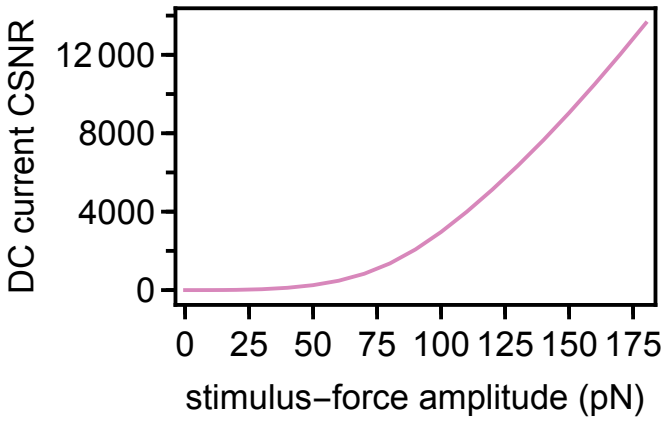
Out[ ]=


stimulus–frequency current CSNR
8000
4000
0
0 25 50 75 100 125 150 175
stimlusu–force amplitude (pN)

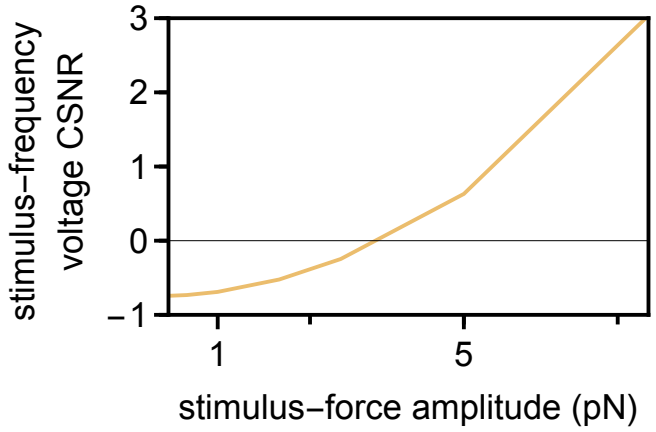
Out[ ]=


DC current CSNR
3
2
1
0
1 5 10 15
stimulus–force amplitude (pN)

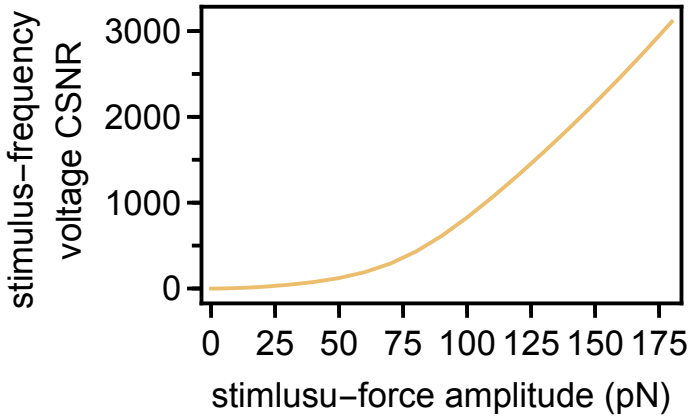
Out[ ]=


DC current CSNR
12 000
8000
4000
0
0 25 50 75 100 125 150 175
stimulus–force amplitude (pN)

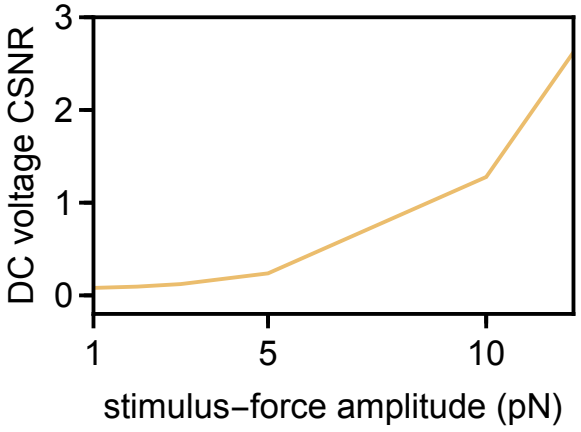
Out[ ]=


stimulus−frequency
voltage CSNR
3
2
1
0
−1
1
5
stimulus−force amplitude (pN)

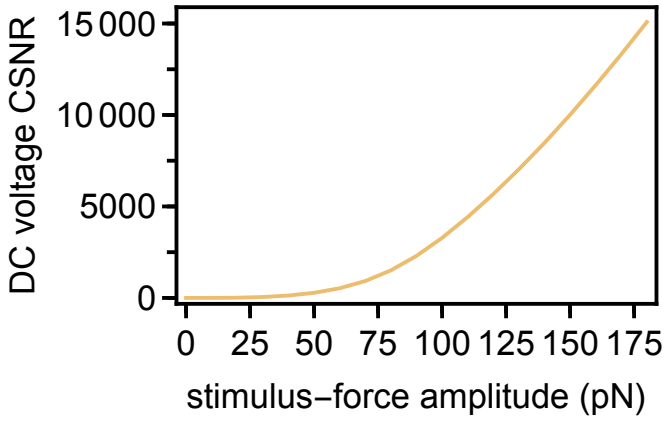
Out[ ]=


stimulus−frequency
voltage CSNR
3000
2000
1000
0
0 25 50 75 100 125 150 175
stimlusu−force amplitude (pN)

Out[ ]=


DC voltage CSNR
3
2
1
0
1
5
10
stimulus−force amplitude (pN)

Out[ ]=



In[ ]:=
```
(*Fig 9 CSNR vs stimulus-
  force amplitude for a gating swing of 32 nm and a resting current of 0.17. Set D=
 32 and Pinfrest=0.17. Case without noise reduction*)

(*table of values of CSNRs*)
CNRYstval = Table[{fmult[[i]], CNRYst[[i]]}, {i, 1, Length[fmult]}];
CNRYDCval = Table[{fmult[[i]], CNRYDC[[i]]}, {i, 1, Length[fmult]}];
CNRPstval = Table[{fmult[[i]], CNRPst[[i]]}, {i, 1, Length[fmult]}];
CNRPDCval = Table[{fmult[[i]], CNRPDC[[i]]}, {i, 1, Length[fmult]}];
CNRVstval = Table[{fmult[[i]], CNRVst[[i]]}, {i, 1, Length[fmult]}];
CNRVDCval = Table[{fmult[[i]], CNRVDC[[i]]}, {i, 1, Length[fmult]}];

(*plot settings*)
plotstyleY = {Darker[■]};
plotstyleP = {Darker[■]};
plotstyleV = {Darker[■]};

Remove[YTicksb]
Remove[XTicks]
Remove[YTicks]

TickLength = 0.02;
TickLength1 = 0.03;
Yticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, -24, 1, 2}];
Yticks2 = Table[{10^k, Null, {0, TickLength1}}, {k, -23, 1, 2}];
TickLength2 = 0.02;
aa = Table[0, {22}, {4}];
For[k = 1, k ≤ 22, k++,
aa[[k]] = Table[{l * 10^-k, , {0, TickLength2}}, {l, 2, 8, 2}];
]
```

```
YTicks = Union[Yticks1, Yticks2, aa[[1]], aa[[2]], aa[[3]], aa[[4]], aa[[5]],
   aa[[6]], aa[[7]], aa[[8]], aa[[9]], aa[[10]], aa[[11]], aa[[12]], aa[[13]], aa[[14]],
   aa[[15]], aa[[16]], aa[[17]], aa[[18]], aa[[19]], aa[[20]], aa[[21]], aa[[22]]];

Xticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, 0, 6, 1}];
aa2 = Table[0, {20}, {4}];
For[k = 1, k ≤ 10, k++,
aa2[[k]] = Table[{l * 10^k, , {0, TickLength2}}, {l, 2, 8, 2}];
]
XTicks = Union[Xticks1, aa2[[1]], aa2[[2]], aa2[[3]], aa2[[4]], aa2[[5]],
   aa2[[6]], aa2[[7]], aa2[[8]], aa2[[9]], aa2[[10]], aa2[[11]], aa2[[12]], aa2[[13]],
   aa2[[14]], aa2[[15]], aa2[[16]], aa2[[17]], aa2[[18]], aa2[[19]], aa2[[20]]];

pady = 120;
padx = 70;
pad2 = 30;
fntsize = 18;

(*FigA insert CSNR deflection stimulus*)

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1, 0.5}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1, 0.5}];
Xticks4 = Table[{k, k, {0, 0.03}}, {k, 0.1, 0.1, 0.1}];
XTicks = Union[Xticks1, Xticks3, Xticks4];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 2, 2}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 2, 1}];
YTicksb = Union[Yticks1, Yticks3];


framelabelsCSNRY = {{"stimulus-frequency \n deflection CSNR", None},
   {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRYstval}, PlotRange → {{0.1, 1}, {-0.25, 2.1}}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRY,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigA CSNR deflection stimulus*)

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 400 000, 20 000}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 400 000, 20 000 / 2}];
```

```
YTicksb = Union[Yticks1, Yticks3];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 180, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 180, 25}];
XTicks = Union[Xticks1, Xticks3];


framelabelsCSNRY = {{"stimulus-frequency \n deflection CSNR", None},
   {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRYstval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRY,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]



(*FigB insert CSNR deflection DC*)

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 3, 1}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 3, 1}];
YTicksb = Union[Yticks1, Yticks3];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 30, 5}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 30, 5}];
XTicks = Union[Xticks1, Xticks3];


framelabelsCSNRY =
  {{"DC deflection CSNR", None}, {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRYDCval}, PlotRange → {{0, 25}, {0, 3}}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRY,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigB CSNR deflection DC*)


Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 180, 25}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 180, 25}];
```

```
YTicksb = Union[Yticks1, Yticks3];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 400, 100}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 400, 100 / 2}];
YTicksb = Union[Yticks1, Yticks3];

framelabelsCSNRY =
  {{"DC deflection CSNR", None}, {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRYDC}, PlotRange → {Full, Full}, Joined → True, PlotRange → plotrangeSNRP,
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsCSNRY, FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigC insert CSNR current ST*)

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 3, 1}];(*50000 200*)
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 3, 1 / 2}];(*25000 100*)
YTicksb = Union[Yticks1, Yticks3];


Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 3, 1}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 3, 0.5}];
Xticks4 = Table[{k, k, {0, 0.03}}, {k, 0.1, 0.1, 0.1}];
XTicks = Union[Xticks1, Xticks3, Xticks4];


framelabelsCSNRP = {{"stimulus-frequency \n current CSNR", None},
   {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRPstval}, PlotRange → {{0.1, 3}, {-0.25, 2}}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRP,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]

(*FigC CSNR current ST*)


Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 50 000, 2000}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 50 000, 2000 / 2}];
YTicks = Union[Yticks1, Yticks3];
```

```
Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 200, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 200, 25}];
XTicks = Union[Xticks1, Xticks3];


framelabelsCSNRP = {{"stimulus-frequency \n current CSNR", None},
   {"stimlusu-force amplitude (pN)", None}};
ListPlot[{CNRPstval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRP,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]


(*FigD insert CSNR current DC*)

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 30, 5}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 30, 5}];
Xticks4 = Table[{k, k, {0, 0.03}}, {k, 1, 1, 1}];
XTicks = Union[Xticks1, Xticks3, Xticks4];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 5, 1}];(*50000 200*)
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 5, 1}];(*25000 100*)
YTicksb = Union[Yticks1, Yticks3];

framelabelsCSNRP =
  {{"DC current CSNR", None}, {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRPDCval}, PlotRange → {{1, 15}, {-0.2, 3}}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRP,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigD CSNR current DC*)


Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 16 000, 1000}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 16 000, 1000 / 2}];
YTicks = Union[Yticks1, Yticks3];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 180, 25}];
```

```
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 180, 25}];
XTicks = Union[Xticks1, Xticks3];



framelabelsCSNRP =
  {{"DC current CSNR", None}, {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRPDCval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRP,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]


(*FigE insert CSNR voltage stimulus*)

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 10, 5}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 10, 5 / 2}];
Xticks4 = Table[{k, k, {0, 0.03}}, {k, 1, 1, 1}];
XTicks = Union[Xticks1, Xticks3, Xticks4];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, -3, 5, 1}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, -3, 5, 1}];
YTicksb = Union[Yticks1, Yticks3];

framelabelsSNRP = {{"stimulus-frequency \n voltage CSNR", None},
   {"stimulus-force amplitude (pN)", None}};

ListPlot[{CNRVstval}, PlotRange → {{0.2, 8}, {-1, 3}}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleV,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsSNRP,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigE CSNR voltage stimulus*)


Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 5000, 500}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 5000, 500 / 2}];
YTicks = Union[Yticks1, Yticks3];
```

```
Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 200, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 200, 25}];
XTicks = Union[Xticks1, Xticks3];


framelabelsCSNRV = {{"stimulus-frequency \n voltage CSNR", None},
   {"stimlusu-force amplitude (pN)", None}};
ListPlot[{CNRVstval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleV,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRV,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]

(*FigF insert CSNR voltage DC*)

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 10, 5}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 10, 5}];
Xticks4 = Table[{k, k, {0, 0.03}}, {k, 1, 1, 1}];
XTicks = Union[Xticks1, Xticks3, Xticks4];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 5, 1}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 5, 1}];
YTicksb = Union[Yticks1, Yticks3];

framelabelsCSNRV =
  {{"DC voltage CSNR", None}, {"stimulus-force amplitude (pN)", None}};

ListPlot[{CNRVDCval}, PlotRange → {{1, 12}, {-0.2, 3}}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleV,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRV,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigF CSNR voltage DC*)

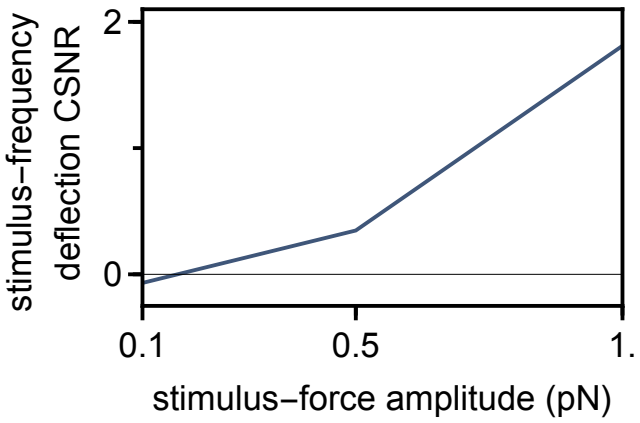
Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 20 000, 1000}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 20 000, 1000 / 2}];
YTicks = Union[Yticks1, Yticks3];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 180, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 180, 25}];
```

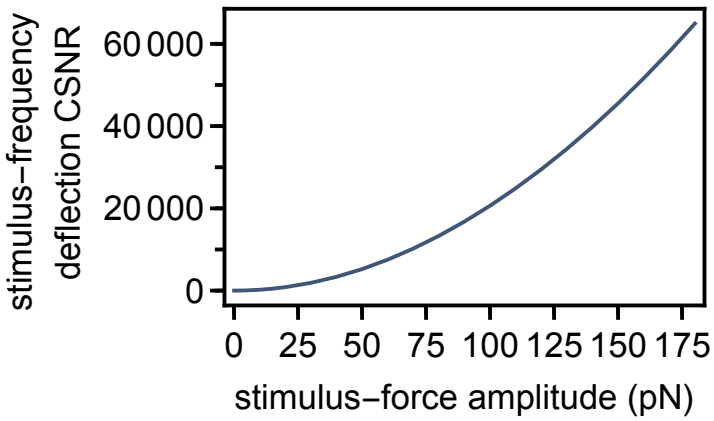
```
XTicks = Union[Xticks1, Xticks3];

framelabelsCSNRV =
  {{"DC voltage CSNR", None}, {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRVDCval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleV,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRV,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

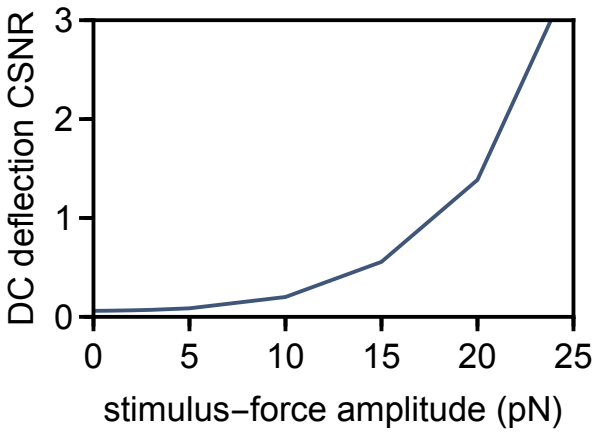
Out[ ]=



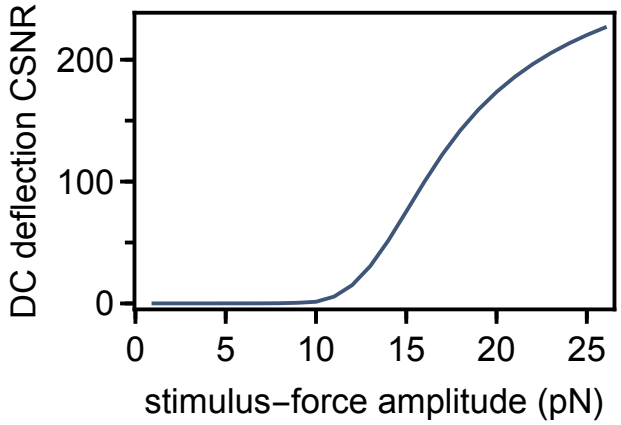
Out[ ]=



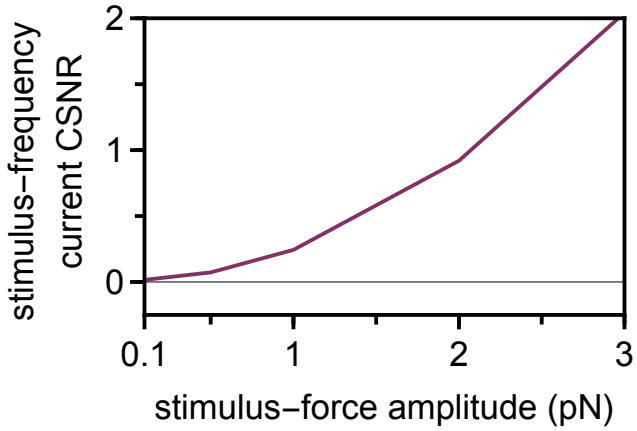
Out[ ]=


DC deflection CSNR
3
2
1
0
0
5
10
15
20
25
stimulus–force amplitude (pN)

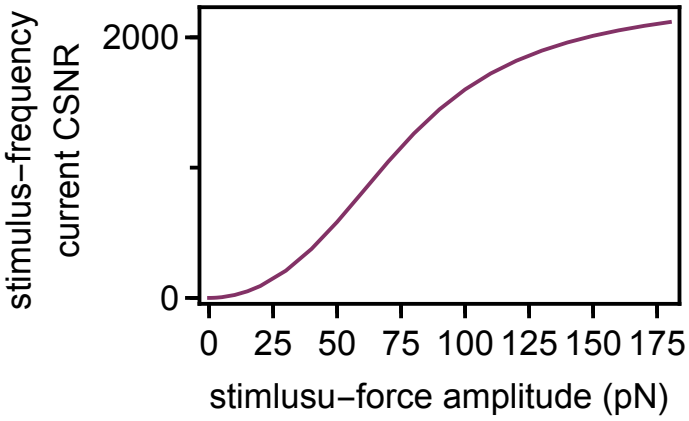
Out[ ]=


DC deflection CSNR
200
100
0
0
5
10
15
20
25
stimulus–force amplitude (pN)

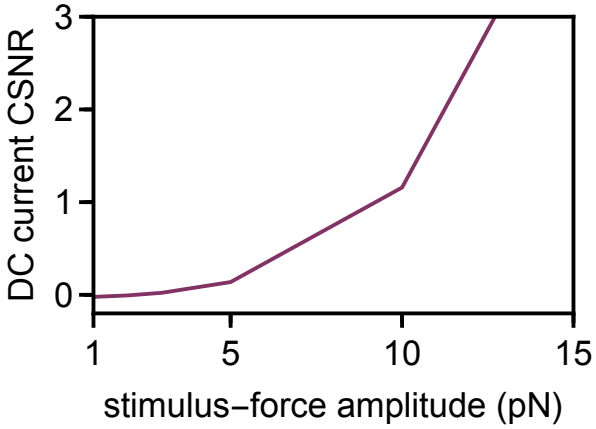
Out[ ]=


stimulus–frequency
current CSNR
2
1
0
0.1
1
2
3
stimulus–force amplitude (pN)

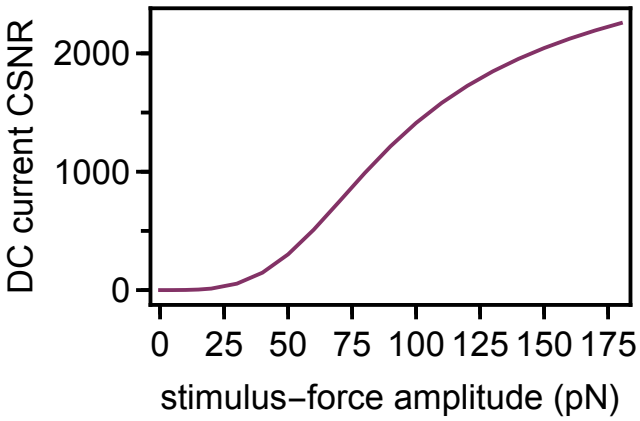
Out[ ]=


stimulus−frequency
current CSNR
2000
0
0 25 50 75 100 125 150 175
stimlusu−force amplitude (pN)

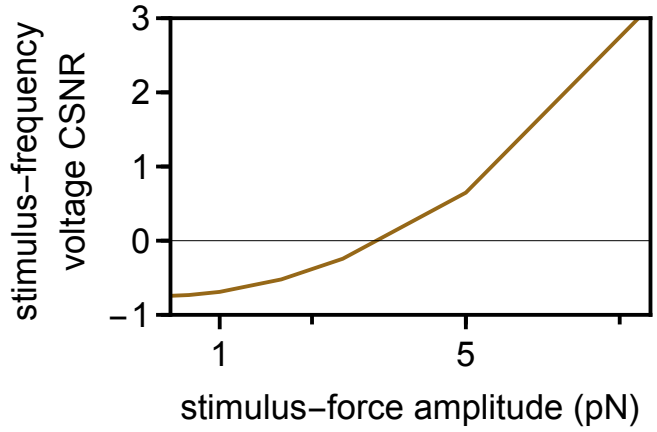
Out[ ]=


DC current CSNR
3
2
1
0
1 5 10 15
stimulus−force amplitude (pN)

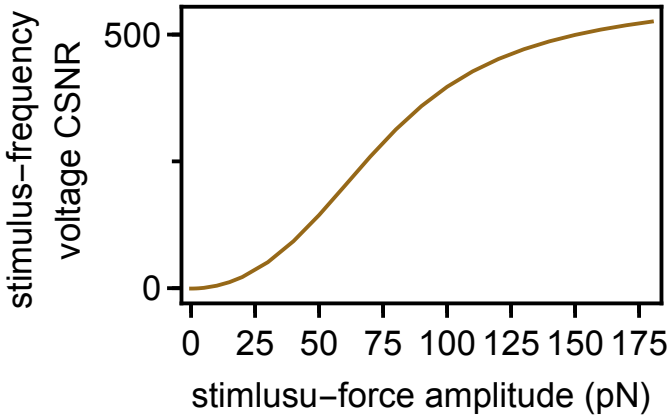
Out[ ]=


DC current CSNR
2000
1000
0
0 25 50 75 100 125 150 175
stimulus−force amplitude (pN)

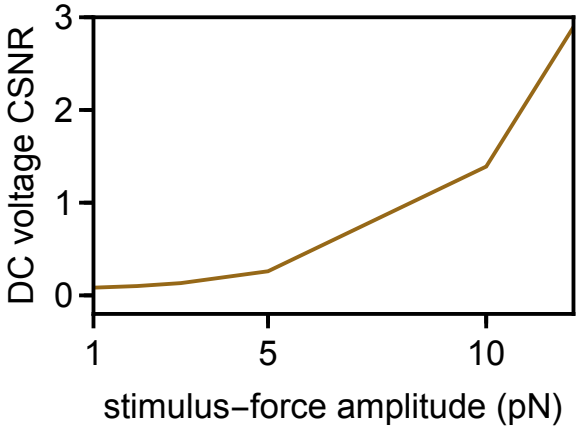
Out[ ]=


stimulus−frequency
voltage CSNR
3
2
1
0
−1
1
5
stimulus−force amplitude (pN)

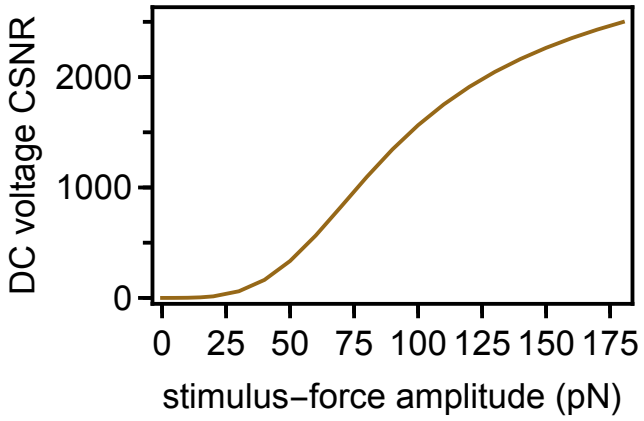
Out[ ]=


stimulus−frequency
voltage CSNR
500
0
0 25 50 75 100 125 150 175
stimlusu−force amplitude (pN)

Out[ ]=


DC voltage CSNR
3
2
1
0
1
5
10
stimulus−force amplitude (pN)

Out[ ]=



In[ ]:=
```
(*Fig 9 CSNR vs stimulus-
  force amplitude for a gating swing of 1.75 nm and a resting current of 0.17. Set D=
 1.75 and Pinfrest=0.17. Case with noise reduction*)

(*table of values of CSNRs*)
CNRYstval = Table[{fmult[[i]], CNRYst[[i]]}, {i, 1, Length[fmult]}];
CNRYDCval = Table[{fmult[[i]], CNRYDC[[i]]}, {i, 1, Length[fmult]}];
CNRPstval = Table[{fmult[[i]], CNRPst[[i]]}, {i, 1, Length[fmult]}];
CNRPDCval = Table[{fmult[[i]], CNRPDC[[i]]}, {i, 1, Length[fmult]}];
CNRVstval = Table[{fmult[[i]], CNRVst[[i]]}, {i, 1, Length[fmult]}];
CNRVDCval = Table[{fmult[[i]], CNRVDC[[i]]}, {i, 1, Length[fmult]}];

(*plot settings*)
plotstyleY = {Lighter[■]};
plotstyleP = {Lighter[■]};
plotstyleV = {Lighter[■]};

Remove[YTicksb]
Remove[XTicks]
Remove[YTicks]

TickLength = 0.02;
TickLength1 = 0.03;
Yticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, -24, 1, 2}];
Yticks2 = Table[{10^k, Null, {0, TickLength1}}, {k, -23, 1, 2}];
TickLength2 = 0.02;
aa = Table[0, {22}, {4}];
For[k = 1, k ≤ 22, k++,
aa[[k]] = Table[{l * 10^-k, , {0, TickLength2}}, {l, 2, 8, 2}];
]
```

```
YTicks = Union[Yticks1, Yticks2, aa[[1]], aa[[2]], aa[[3]], aa[[4]], aa[[5]],
   aa[[6]], aa[[7]], aa[[8]], aa[[9]], aa[[10]], aa[[11]], aa[[12]], aa[[13]], aa[[14]],
   aa[[15]], aa[[16]], aa[[17]], aa[[18]], aa[[19]], aa[[20]], aa[[21]], aa[[22]]];

Xticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, 0, 6, 1}];
aa2 = Table[0, {20}, {4}];
For[k = 1, k ≤ 10, k++,
aa2[[k]] = Table[{l * 10^k, , {0, TickLength2}}, {l, 2, 8, 2}];
]
XTicks = Union[Xticks1, aa2[[1]], aa2[[2]], aa2[[3]], aa2[[4]], aa2[[5]],
   aa2[[6]], aa2[[7]], aa2[[8]], aa2[[9]], aa2[[10]], aa2[[11]], aa2[[12]], aa2[[13]],
   aa2[[14]], aa2[[15]], aa2[[16]], aa2[[17]], aa2[[18]], aa2[[19]], aa2[[20]]];

pady = 120;
padx = 70;
pad2 = 30;
fntsize = 18;

(*FigA insert CSNR deflection stimulus*)

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1, 0.5}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1, 0.5}];
Xticks4 = Table[{k, k, {0, 0.03}}, {k, 0.1, 0.1, 0.1}];
XTicks = Union[Xticks1, Xticks3, Xticks4];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 2, 2}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 2, 1}];
YTicksb = Union[Yticks1, Yticks3];


framelabelsCSNRY = {{"stimulus-frequency \n deflection CSNR", None},
   {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRYstval}, PlotRange → {{0.1, 1}, {-0.25, 2.1}}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRY,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigA CSNR deflection stimulus*)

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 400 000, 20 000}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 400 000, 20 000 / 2}];
```

```
YTicksb = Union[Yticks1, Yticks3];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 180, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 180, 25}];
XTicks = Union[Xticks1, Xticks3];


framelabelsCSNRY = {{"stimulus-frequency \n deflection CSNR", None},
   {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRYstval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRY,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]



(*FigB insert CSNR deflection DC*)

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 3, 1}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 3, 1}];
YTicksb = Union[Yticks1, Yticks3];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 30, 5}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 30, 5}];
XTicks = Union[Xticks1, Xticks3];


framelabelsCSNRY =
  {{"DC deflection CSNR", None}, {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRYDCval}, PlotRange → {{0, 25}, {0, 3}}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRY,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigB CSNR deflection DC*)


Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 180, 25}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 180, 25}];
```

```
YTicksb = Union[Yticks1, Yticks3];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 400, 100}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 400, 100 / 2}];
YTicksb = Union[Yticks1, Yticks3];

framelabelsCSNRY =
  {{"DC deflection CSNR", None}, {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRYDC}, PlotRange → {Full, Full}, Joined → True, PlotRange → plotrangeSNRP,
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsCSNRY, FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigC insert CSNR current ST*)

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 3, 1}];(*50000 200*)
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 3, 1 / 2}];(*25000 100*)
YTicksb = Union[Yticks1, Yticks3];


Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 3, 1}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 3, 0.5}];
Xticks4 = Table[{k, k, {0, 0.03}}, {k, 0.1, 0.1, 0.1}];
XTicks = Union[Xticks1, Xticks3, Xticks4];


framelabelsCSNRP = {{"stimulus-frequency \n current CSNR", None},
   {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRPstval}, PlotRange → {{0.1, 3}, {-0.25, 2}}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRP,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]

(*FigC CSNR current ST*)


Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 50 000, 4000}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 50 000, 4000 / 2}];
YTicks = Union[Yticks1, Yticks3];
```

```
Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 200, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 200, 25}];
XTicks = Union[Xticks1, Xticks3];


framelabelsCSNRP = {{"stimulus-frequency \n current CSNR", None},
   {"stimlusu-force amplitude (pN)", None}};
ListPlot[{CNRPstval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRP,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]


(*FigD insert CSNR current DC*)

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 30, 5}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 30, 5}];
Xticks4 = Table[{k, k, {0, 0.03}}, {k, 1, 1, 1}];
XTicks = Union[Xticks1, Xticks3, Xticks4];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 5, 1}];(*50000 200*)
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 5, 1}];(*25000 100*)
YTicksb = Union[Yticks1, Yticks3];

framelabelsCSNRP =
  {{"DC current CSNR", None}, {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRPDCval}, PlotRange → {{1, 15}, {-0.2, 3}}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRP,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigD CSNR current DC*)


Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 16000, 4000}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 16000, 2000}];
YTicks = Union[Yticks1, Yticks3];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 180, 25}];
```

```
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 180, 25}];
XTicks = Union[Xticks1, Xticks3];



framelabelsCSNRP =
  {{"DC current CSNR", None}, {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRPDCval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRP,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]


(*FigE insert CSNR voltage stimulus*)

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 10, 5}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 10, 5 / 2}];
Xticks4 = Table[{k, k, {0, 0.03}}, {k, 1, 1, 1}];
XTicks = Union[Xticks1, Xticks3, Xticks4];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, -3, 5, 1}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, -3, 5, 1}];
YTicksb = Union[Yticks1, Yticks3];

framelabelsSNRP = {{"stimulus-frequency \n voltage CSNR", None},
   {"stimulus-force amplitude (pN)", None}};

ListPlot[{CNRVstval}, PlotRange → {{0.2, 8}, {-1, 3}}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleV,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsSNRP,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigE CSNR voltage stimulus*)


Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 5000, 2 * 500}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 5000, 2 * 500 / 2}];
YTicks = Union[Yticks1, Yticks3];
```

```
Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 200, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 200, 25}];
XTicks = Union[Xticks1, Xticks3];


framelabelsCSNRV = {{"stimulus-frequency \n voltage CSNR", None},
   {"stimlusu-force amplitude (pN)", None}};
ListPlot[{CNRVstval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleV,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRV,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]

(*FigF insert CSNR voltage DC*)

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 10, 5}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 10, 5}];
Xticks4 = Table[{k, k, {0, 0.03}}, {k, 1, 1, 1}];
XTicks = Union[Xticks1, Xticks3, Xticks4];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 5, 1}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 5, 1}];
YTicksb = Union[Yticks1, Yticks3];

framelabelsCSNRV =
  {{"DC voltage CSNR", None}, {"stimulus-force amplitude (pN)", None}};

ListPlot[{CNRVDCval}, PlotRange → {{1, 12}, {-0.2, 3}}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleV,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRV,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigF CSNR voltage DC*)

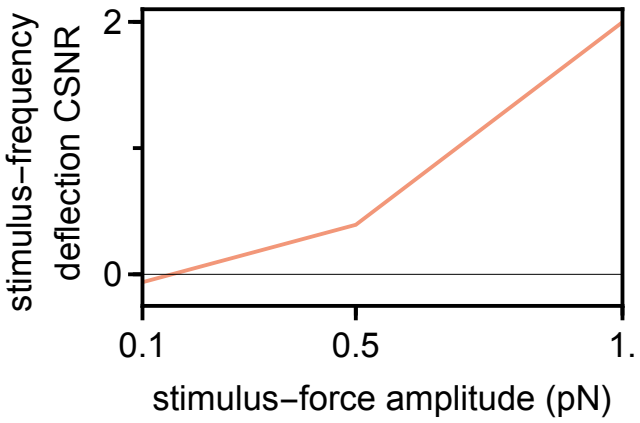
Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 20 000, 5000}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 20 000, 5000 / 2}];
YTicks = Union[Yticks1, Yticks3];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 180, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 180, 25}];
```

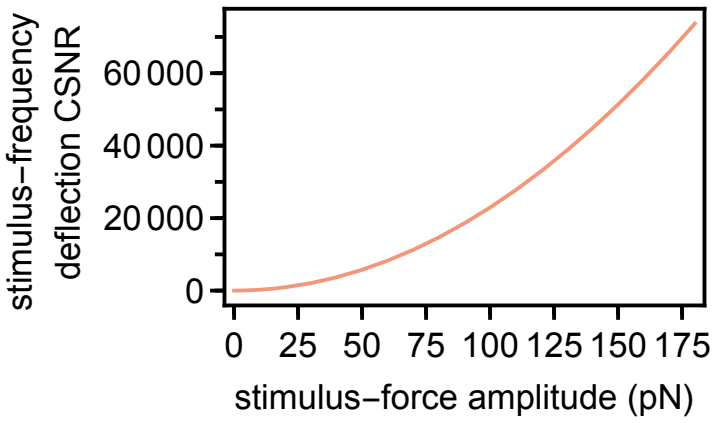
```
XTicks = Union[Xticks1, Xticks3];

framelabelsCSNRV =
  {{"DC voltage CSNR", None}, {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRVDCval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleV,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRV,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

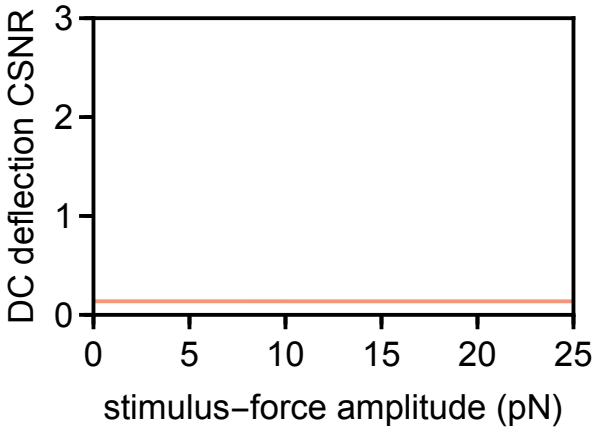
Out[ ]=



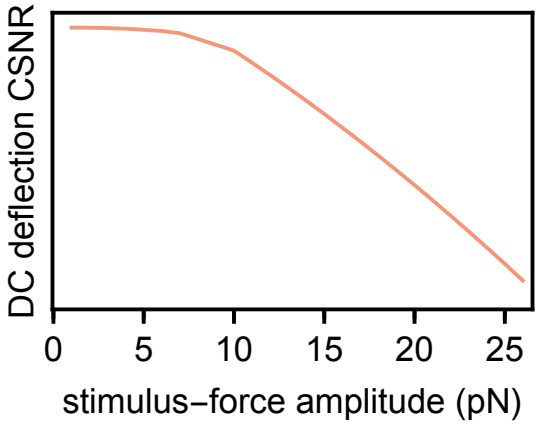
Out[ ]=



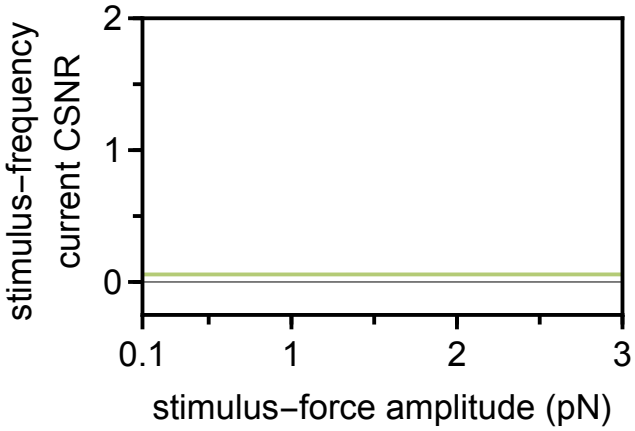
Out[ ]=


DC deflection CSNR
3
2
1
0
0
5
10
15
20
25
stimulus−force amplitude (pN)

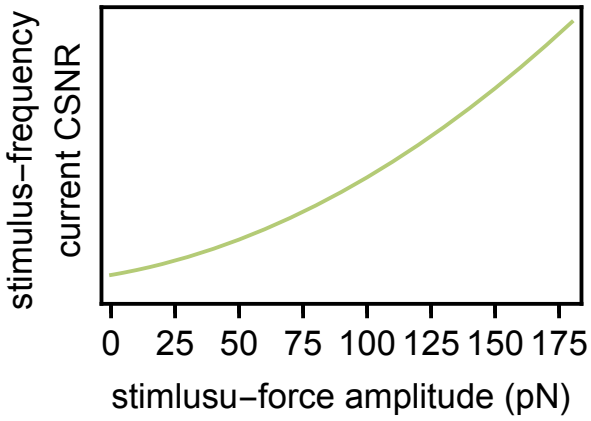
Out[ ]=


DC deflection CSNR
0
5
10
15
20
25
stimulus−force amplitude (pN)

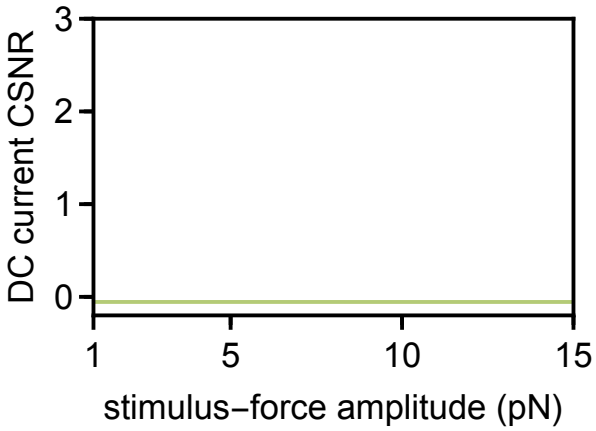
Out[ ]=


stimulus−frequency
current CSNR
2
1
0
0.1
1
2
3
stimulus−force amplitude (pN)

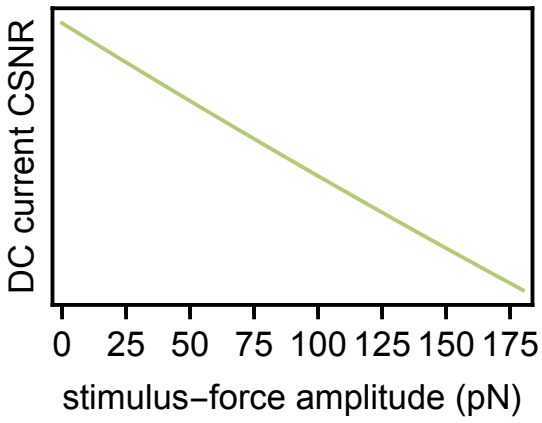
Out[ ]=


stimulus−frequency
current CSNR
0
25
50
75
100
125
150
175
stimlusu−force amplitude (pN)

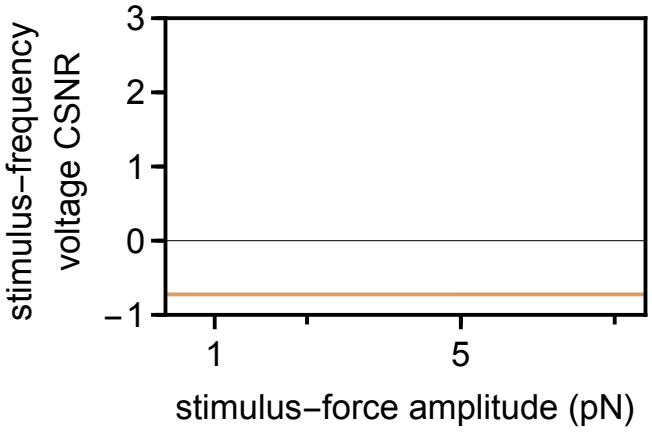
Out[ ]=


DC current CSNR
3
2
1
0
1
5
10
15
stimulus−force amplitude (pN)

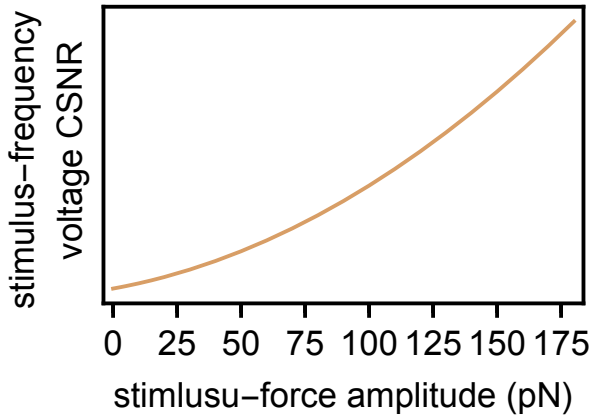
Out[ ]=


DC current CSNR
0
25
50
75
100
125
150
175
stimulus−force amplitude (pN)

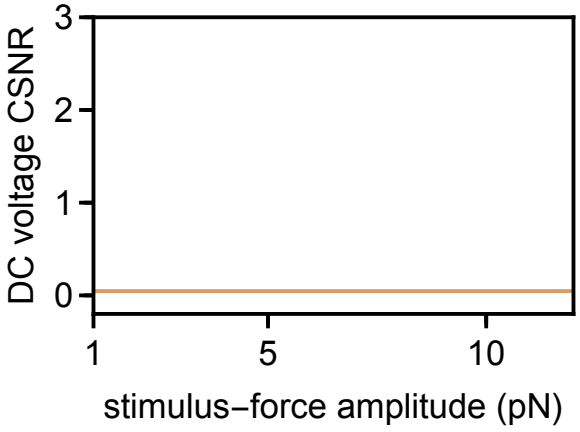
Out[ ]=


stimulus−frequency
voltage CSNR
3
2
1
0
−1
1
5
stimulus−force amplitude (pN)

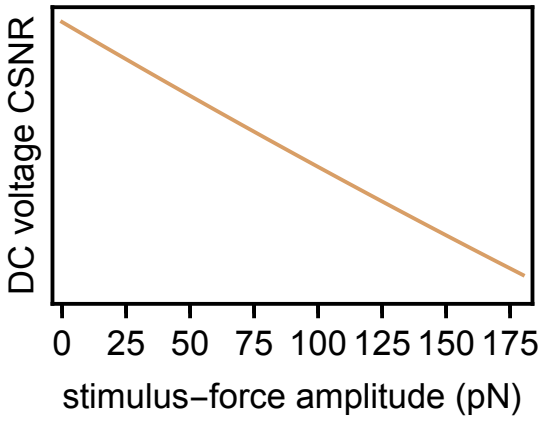
Out[ ]=


stimulus−frequency
voltage CSNR
0
25
50
75
100
125
150
175
stimlusu−force amplitude (pN)

Out[ ]=


DC voltage CSNR
3
2
1
0
1
5
10
stimulus−force amplitude (pN)

Out[ ]=



In[ ]:=
```
(*Fig 9 CSNR vs stimulus-
  force amplitude for a gating swing of 1.75 nm and a resting current of 0.17. Set D=
 1.75 and Pinfrest=0.17. Case without noise reduction*)

(*table of values of CSNRs*)
CNRYstval = Table[{fmult[[i]], CNRYst[[i]]}, {i, 1, Length[fmult]}];
CNRYDCval = Table[{fmult[[i]], CNRYDC[[i]]}, {i, 1, Length[fmult]}];
CNRPstval = Table[{fmult[[i]], CNRPst[[i]]}, {i, 1, Length[fmult]}];
CNRPDCval = Table[{fmult[[i]], CNRPDC[[i]]}, {i, 1, Length[fmult]}];
CNRVstval = Table[{fmult[[i]], CNRVst[[i]]}, {i, 1, Length[fmult]}];
CNRVDCval = Table[{fmult[[i]], CNRVDC[[i]]}, {i, 1, Length[fmult]}];

(*plot settings*)
plotstyleY = {Darker[■]};
plotstyleP = {Darker[■]};
plotstyleV = {Darker[■]};

Remove[YTicksb]
Remove[XTicks]
Remove[YTicks]

TickLength = 0.02;
TickLength1 = 0.03;
Yticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, -24, 1, 2}];
Yticks2 = Table[{10^k, Null, {0, TickLength1}}, {k, -23, 1, 2}];
TickLength2 = 0.02;
aa = Table[0, {22}, {4}];
For[k = 1, k ≤ 22, k++,
aa[[k]] = Table[{l * 10^-k, , {0, TickLength2}}, {l, 2, 8, 2}];
]
```

```
YTicks = Union[Yticks1, Yticks2, aa[[1]], aa[[2]], aa[[3]], aa[[4]], aa[[5]],
   aa[[6]], aa[[7]], aa[[8]], aa[[9]], aa[[10]], aa[[11]], aa[[12]], aa[[13]], aa[[14]],
   aa[[15]], aa[[16]], aa[[17]], aa[[18]], aa[[19]], aa[[20]], aa[[21]], aa[[22]]];

Xticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, 0, 6, 1}];
aa2 = Table[0, {20}, {4}];
For[k = 1, k ≤ 10, k++,
aa2[[k]] = Table[{l * 10^k, , {0, TickLength2}}, {l, 2, 8, 2}];
]
XTicks = Union[Xticks1, aa2[[1]], aa2[[2]], aa2[[3]], aa2[[4]], aa2[[5]],
   aa2[[6]], aa2[[7]], aa2[[8]], aa2[[9]], aa2[[10]], aa2[[11]], aa2[[12]], aa2[[13]],
   aa2[[14]], aa2[[15]], aa2[[16]], aa2[[17]], aa2[[18]], aa2[[19]], aa2[[20]]];

pady = 120;
padx = 70;
pad2 = 30;
fntsize = 18;

(*FigA insert CSNR deflection stimulus*)

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1, 0.5}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1, 0.5}];
Xticks4 = Table[{k, k, {0, 0.03}}, {k, 0.1, 0.1, 0.1}];
XTicks = Union[Xticks1, Xticks3, Xticks4];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 2, 2}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 2, 1}];
YTicksb = Union[Yticks1, Yticks3];


framelabelsCSNRY = {{"stimulus-frequency \n deflection CSNR", None},
   {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRYstval}, PlotRange → {{0.1, 1}, {-0.25, 2.1}}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRY,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigA CSNR deflection stimulus*)

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 400 000, 20 000}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 400 000, 20 000 / 2}];
```

```
YTicksb = Union[Yticks1, Yticks3];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 180, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 180, 25}];
XTicks = Union[Xticks1, Xticks3];


framelabelsCSNRY = {{"stimulus-frequency \n deflection CSNR", None},
   {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRYstval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRY,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]



(*FigB insert CSNR deflection DC*)

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 3, 1}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 3, 1}];
YTicksb = Union[Yticks1, Yticks3];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 30, 5}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 30, 5}];
XTicks = Union[Xticks1, Xticks3];


framelabelsCSNRY =
  {{"DC deflection CSNR", None}, {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRYDCval}, PlotRange → {{0, 25}, {0, 3}}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRY,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigB CSNR deflection DC*)


Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 180, 25}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 180, 25}];
```

```
YTicksb = Union[Yticks1, Yticks3];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 400, 100}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 400, 100 / 2}];
YTicksb = Union[Yticks1, Yticks3];

framelabelsCSNRY =
  {{"DC deflection CSNR", None}, {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRYDC}, PlotRange → {Full, Full}, Joined → True, PlotRange → plotrangeSNRP,
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsCSNRY, FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigC insert CSNR current ST*)

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 3, 1}];(*50000 200*)
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 3, 1 / 2}];(*25000 100*)
YTicksb = Union[Yticks1, Yticks3];


Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 3, 1}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 3, 0.5}];
Xticks4 = Table[{k, k, {0, 0.03}}, {k, 0.1, 0.1, 0.1}];
XTicks = Union[Xticks1, Xticks3, Xticks4];


framelabelsCSNRP = {{"stimulus-frequency \n current CSNR", None},
   {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRPstval}, PlotRange → {{0.1, 3}, {-0.25, 2}}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRP,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]

(*FigC CSNR current ST*)


Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 50 000, 2000}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 50 000, 2000 / 2}];
YTicks = Union[Yticks1, Yticks3];
```

```
Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 200, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 200, 25}];
XTicks = Union[Xticks1, Xticks3];


framelabelsCSNRP = {{"stimulus-frequency \n current CSNR", None},
   {"stimlusu-force amplitude (pN)", None}};
ListPlot[{CNRPstval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRP,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]


(*FigD insert CSNR current DC*)

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 30, 5}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 30, 5}];
Xticks4 = Table[{k, k, {0, 0.03}}, {k, 1, 1, 1}];
XTicks = Union[Xticks1, Xticks3, Xticks4];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 5, 1}];(*50000 200*)
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 5, 1}];(*25000 100*)
YTicksb = Union[Yticks1, Yticks3];

framelabelsCSNRP =
  {{"DC current CSNR", None}, {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRPDCval}, PlotRange → {{1, 15}, {-0.2, 3}}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRP,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigD CSNR current DC*)


Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 16 000, 1000}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 16 000, 1000 / 2}];
YTicks = Union[Yticks1, Yticks3];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 180, 25}];
```

```
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 180, 25}];
XTicks = Union[Xticks1, Xticks3];



framelabelsCSNRP =
  {{"DC current CSNR", None}, {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRPDCval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRP,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]


(*FigE insert CSNR voltage stimulus*)

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 10, 5}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 10, 5 / 2}];
Xticks4 = Table[{k, k, {0, 0.03}}, {k, 1, 1, 1}];
XTicks = Union[Xticks1, Xticks3, Xticks4];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, -3, 5, 1}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, -3, 5, 1}];
YTicksb = Union[Yticks1, Yticks3];

framelabelsSNRP = {{"stimulus-frequency \n voltage CSNR", None},
   {"stimulus-force amplitude (pN)", None}};

ListPlot[{CNRVstval}, PlotRange → {{0.2, 8}, {-1, 3}}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleV,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsSNRP,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigE CSNR voltage stimulus*)


Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 5000, 500}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 5000, 500 / 2}];
YTicks = Union[Yticks1, Yticks3];
```

```
Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 200, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 200, 25}];
XTicks = Union[Xticks1, Xticks3];


framelabelsCSNRV = {{"stimulus-frequency \n voltage CSNR", None},
   {"stimlusu-force amplitude (pN)", None}};
ListPlot[{CNRVstval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleV,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRV,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]

(*FigF insert CSNR voltage DC*)

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 10, 5}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 10, 5}];
Xticks4 = Table[{k, k, {0, 0.03}}, {k, 1, 1, 1}];
XTicks = Union[Xticks1, Xticks3, Xticks4];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 5, 1}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 5, 1}];
YTicksb = Union[Yticks1, Yticks3];

framelabelsCSNRV =
  {{"DC voltage CSNR", None}, {"stimulus-force amplitude (pN)", None}};

ListPlot[{CNRVDCval}, PlotRange → {{1, 12}, {-0.2, 3}}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleV,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRV,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigF CSNR voltage DC*)

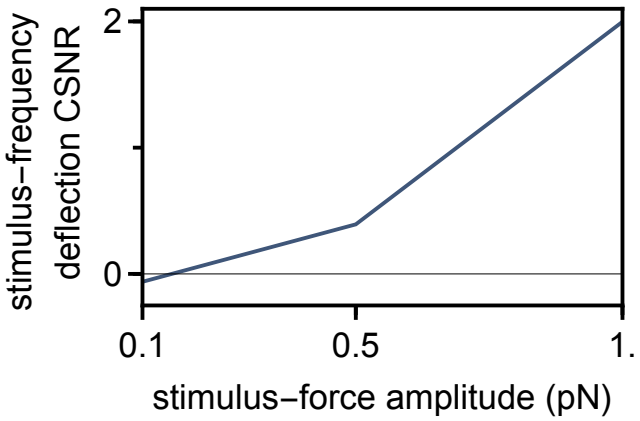
Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 20 000, 1000}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 20 000, 1000 / 2}];
YTicks = Union[Yticks1, Yticks3];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 180, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 180, 25}];
```

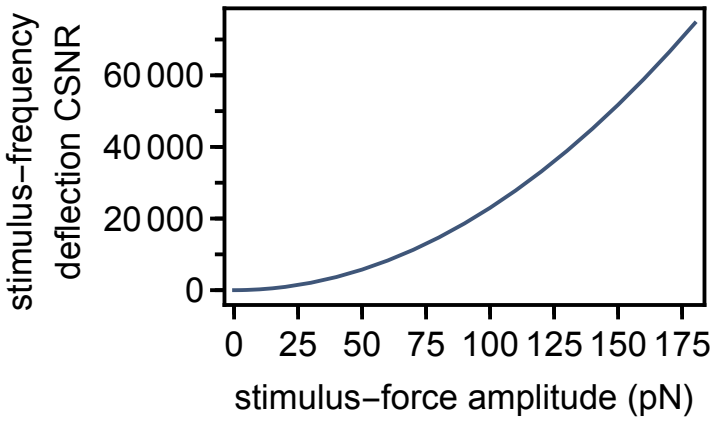
```
XTicks = Union[Xticks1, Xticks3];

framelabelsCSNRV =
  {{"DC voltage CSNR", None}, {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRVDCval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleV,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRV,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

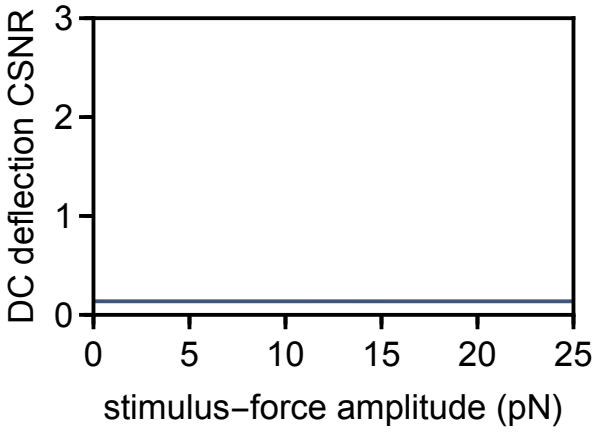
Out[ ]=



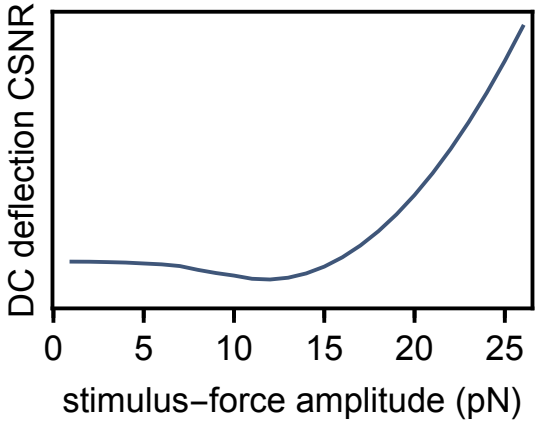
Out[ ]=



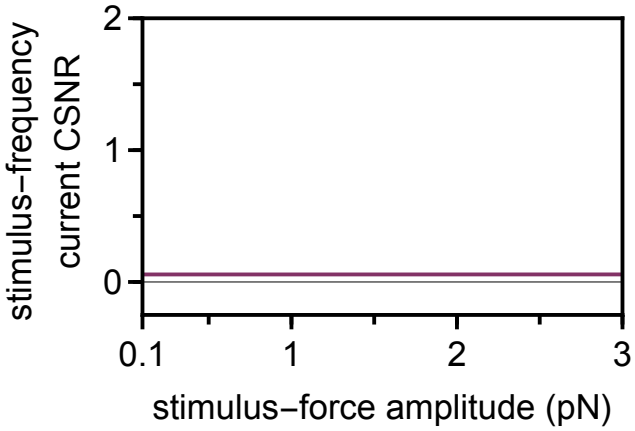
Out[ ]=


DC deflection CSNR
3
2
1
0
0
5
10
15
20
25
stimulus–force amplitude (pN)

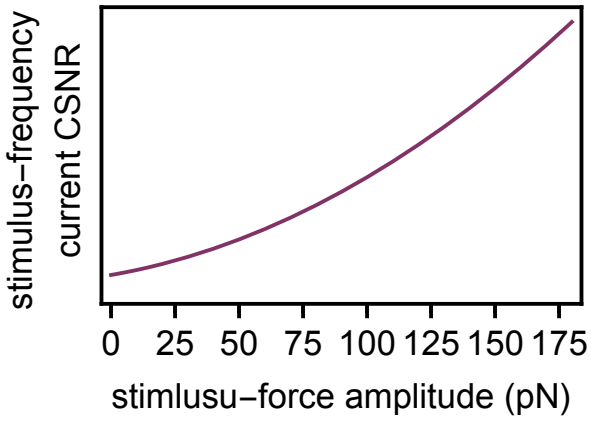
Out[ ]=


DC deflection CSNR
0
5
10
15
20
25
stimulus–force amplitude (pN)

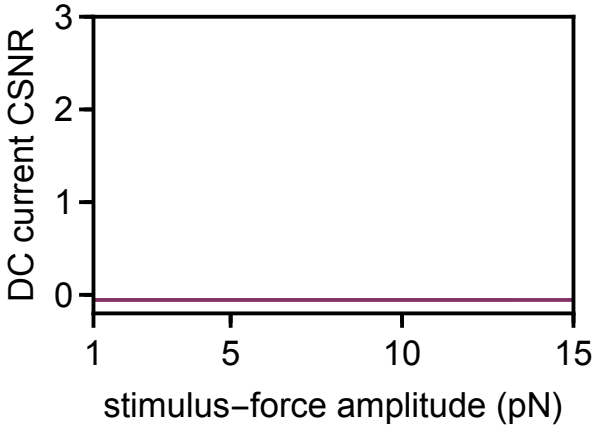
Out[ ]=


stimulus–frequency
current CSNR
2
1
0
0.1
1
2
3
stimulus–force amplitude (pN)

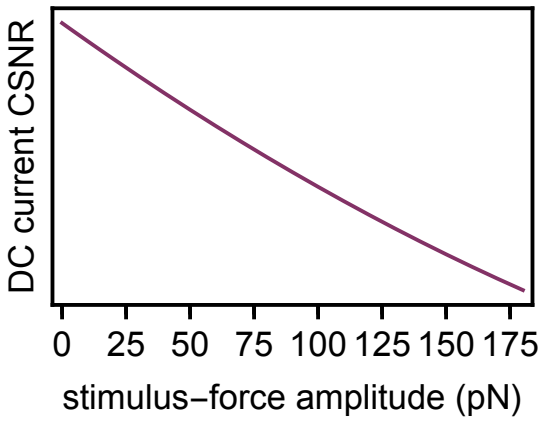
Out[ ]=


stimulus−frequency
current CSNR
0
25
50
75
100
125
150
175
stimlusu−force amplitude (pN)

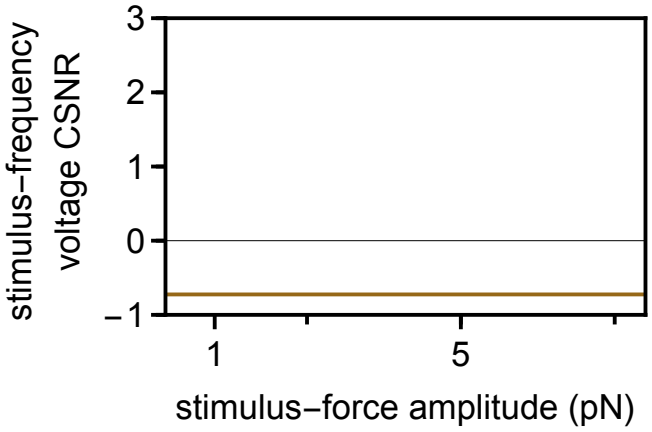
Out[ ]=


DC current CSNR
3
2
1
0
1
5
10
15
stimulus−force amplitude (pN)

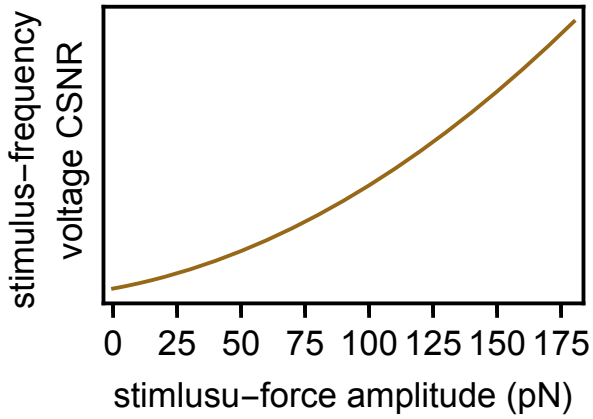
Out[ ]=


DC current CSNR
0
25
50
75
100
125
150
175
stimulus−force amplitude (pN)

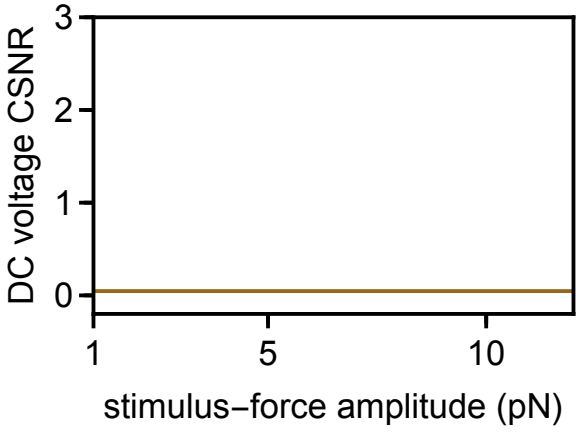
Out[ ]=


stimulus−frequency
voltage CSNR
3
2
1
0
−1
1
5
stimulus−force amplitude (pN)

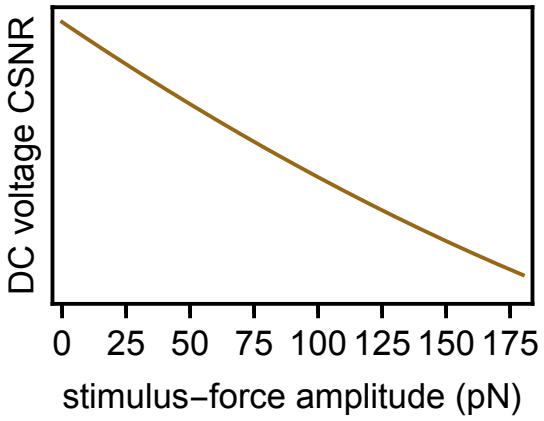
Out[ ]=


stimulus−frequency
voltage CSNR
0
25
50
75
100
125
150
175
stimlusu−force amplitude (pN)

Out[ ]=


DC voltage CSNR
3
2
1
0
1
5
10
stimulus−force amplitude (pN)

Out[ ]=




```
(*Setting of parameters for CSNRs calcualtion
  for a resting current of 0.5. Set Pinfrest=0.5*)
(*32 nm 0.5*)

(* banwidths for a gating swing of 32 nm and a resting current of 0.5. (Eq.47 )*)
(*freqpeaklc=Table[4816.58373426373`,{Length[fmult]}];
freqpeakrc=Table[5183.41626573627`,{Length[fmult]}];
freqnoisel1c=Table[1816.5837342637296`,{Length[fmult]}];
freqnoiser1c=Table[2000.`,{Length[fmult]}];
freqnoisel2c=Table[8000.`,{Length[fmult]}];
freqnoiser2c=Table[8183.41626573627`,{Length[fmult]}];

freqpeakl=Table[4819.993979013689`,{Length[fmult]}];
freqpeakr=Table[5180.006020986311`,{Length[fmult]}];
freqnoisel1=Table[1819.9939790136887`,{Length[fmult]}];
freqnoiser1=Table[2000.`,{Length[fmult]}];
freqnoisel2=Table[8000.`,{Length[fmult]}];
freqnoiser2=Table[8180.006020986311`,{Length[fmult]}];

freqpeaklv=Table[4819.984817581482`,{Length[fmult]}];
freqpeakrv=Table[5180.015182418518`,{Length[fmult]}];
freqnoisel1v=Table[1819.9848175814823`,{Length[fmult]}];
freqnoiser1v=Table[2000.`,{Length[fmult]}];
freqnoisel2v=Table[8000.`,{Length[fmult]}];
freqnoiser2v=Table[8180.015182418518`,{Length[fmult]}];*)

(* banwidths for a gating swing of 1.75 nm and a resting current of 0.5. (Eq.47 )*)
(*freqpeaklc=Table[4816.58373426373`,{Length[fmult]}];
freqpeakrc=Table[5183.41626573627`,{Length[fmult]}];
freqnoisel1c=Table[1816.5837342637296`,{Length[fmult]}];
```

```
freqnoiser1c=Table[2000.`,{Length[fmult]}];
freqnoisel2c=Table[8000.`,{Length[fmult]}];
freqnoiser2c=Table[8183.41626573627`,{Length[fmult]}];

freqpeakl=Table[4819.993979013689`,{Length[fmult]}];
freqpeakr=Table[5180.006020986311`,{Length[fmult]}];
freqnoisel1=Table[1819.9939790136887`,{Length[fmult]}];
freqnoiser1=Table[2000.`,{Length[fmult]}];
freqnoisel2=Table[8000.`,{Length[fmult]}];
freqnoiser2=Table[8180.006020986311`,{Length[fmult]}];

freqpeaklv=Table[4816.291841529541`,{Length[fmult]}];
freqpeakrv=Table[5183.708158470459`,{Length[fmult]}];
freqnoisel1v=Table[1816.2918415295408`,{Length[fmult]}];
freqnoiser1v=Table[2000.`,{Length[fmult]}];
freqnoisel2v=Table[8000.`,{Length[fmult]}];
freqnoiser2v=Table[8183.708158470459`,{Length[fmult]}];*)



t1 = AbsoluteTime[];

ParallelDo[
SeedRandom[47 741];

  gx =  2 * kbTd * lPd;
  y0in = Yrestd;



  proc = ItoProcess[
    lPd * ⅆY[t] == (-kPd) * (Y[t] - Yupd) * ⅆt - kGSd * (Y[t] - Yugd - Dvd * Po[t]) * ⅆt +
       Sqrt[gx] * ⅆn1[t] + Fzd * fmult⟦pp⟧ * Sin[2 * Pi * wstd * (t - tstartFd)] * ⅆt &&
     taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd] * ⅆPo[t] ==
      (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] - Po[t]) ⅆt +
       Sqrt[taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd]] Sqrt[
         (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] (1 - Clip[Po[t], {0.0, 1.0}]) + (1 - Pinf2[
                 Yhd, kGSd, Dvd, kbTd, Ncd]) Clip[Po[t], {0.0, 1.0}]) / Ncd] ⅆn2[t],
    {Y[t], Po[t]}, {{Y, Po}, {y0in, Pinfrest}}, t,
{n1 ≈ WienerProcess[], n2 ≈ WienerProcess[]}];


totallength = 2 * 0.507;
```

```
  factot = 4;
  deltatsima = 0.000001 / factot;
   ap = RandomFunction[proc, {0, totallength, deltatsima}];



  ap0 = ap["PathComponent", 1] - Yrestd;

ap1 = ap["PathComponent", 2] - Pinfrest;

resampfactor = factot;


randtime100 = 0.005 * 10;
randtimenm = Round[randtime100, deltatsima];
stpoint = IntegerPart[randtime100 / deltatsima];

deltatsampl = Round[deltatsima * resampfactor, deltatsima];
tsampl = 0.005;
endpoint = IntegerPart[
     Length[ap["PathComponent", 1]["Values"]] - tsampl / deltatsima - stpoint];
tsamplnm = Round[tsampl, deltatsima];
b1 = Table[f[i], {i, randtimenm, tsamplnm + randtimenm, deltatsampl}];
lengthares = Length[b1];

Clear[ares];
totallength2 = totallength;

ncopiesa = (totallength2 - randtimenm) / (tsamplnm + deltatsampl);
ncopies = IntegerPart[ncopiesa];

  ares0 = Table[0, {ncopies}];
ares1 = Table[0, {ncopies}];
sizea = lengthares;

fousampled0 = Table[0, {ncopies}, {lengthares}];
fousampled1 = Table[0, {ncopies}, {lengthares}];
cfousampled = Table[0, {ncopies}, {lengthares}];
  Do[

   ares0[[ll]] = ap0["Values"][[(ll - 1) * tsampl / deltatsima + stpoint ;;
      IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];
    ares1[[ll]] = ap1["Values"][[(ll - 1) * tsampl / deltatsima + stpoint ;;
```

```
      IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor];

       fousampled0[[ll]] = ResourceFunction["WelchSpectralEstimate"][
     ares0[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
     "OverlapOffset" → None, "SegmentSize" → sizea];
    cfousampled[[ll]] = ResourceFunction["WelchSpectralEstimate"][
     ares1[[ll]], ares0[[ll]], sizea / tsampl, "Window" → None,
     "Detrend" → None, "OverlapOffset" → None, "SegmentSize" → sizea];
    fousampled1[[ll]] = ResourceFunction["WelchSpectralEstimate"][
     ares1[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
     "OverlapOffset" → None, "SegmentSize" → sizea];
    ,
{ll, ncopies}];
  powspmeansampledt0 = Drop[Mean[fousampled0], -1];

  powspmeansampledt1 = Drop[Mean[fousampled1], -1];


lengthpsd = Length[powspmeansampledt0];

meandataplot0 = Table[{0, 0}, {i, 1, lengthpsd}];
  meandataplot1 = Table[{0, 0}, {i, 1, lengthpsd}];

  meandataplot0 =
   Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt0[[i]][[2]]}, {i, 1, lengthpsd}];
  meandataplot1 =
   Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt1[[i]][[2]]}, {i, 1, lengthpsd}];

  fc = 610;
  filtersq1 = Table[{meandataplot0[[i]][[1]],
     Abs[1 / (1 + I * meandataplot1[[i]][[1]] / fc)]^2}, {i, 1, Length[meandataplot0]}];
  meandataplot1filt = Table[{meandataplot0[[i]][[1]],
     filtersq1[[i]][[2]] * meandataplot1[[i]][[2]]}, {i, 1, Length[filtersq1]}];

  (*Interpolation of 1 order of defleciton, current and voltage
   psd for calculation of area under corresponding bandwidths *)
  ff0 = Interpolation[meandataplot0, InterpolationOrder → 1];
  ff1 = Interpolation[meandataplot1, InterpolationOrder → 1];
  ff1filt = Interpolation[meandataplot1filt, InterpolationOrder → 1];

  (*Interpolation of 1 order of defleciton,
  current and voltage unstimulated psd *)
  ff0base = Interpolation[meandataplot0base, InterpolationOrder → 1];
  ff1base = Interpolation[meandataplot1base, InterpolationOrder → 1];
```

```
ff1filtbase = Interpolation[meandataplot1filtbase, InterpolationOrder → 1];

n1 = 26; (*vector position for psd corresponding to a frequency of 5 kHz*)

(*stimulus-
 frequency deflection CSNR and DC deflection CSNR caclualted as ratio of
  areas of corresponding bandwidths for the case with noise reduction*)
CNRYst[[pp]] = Integrate[ff0[x], {x, freqpeakl[[pp]], freqpeakr[[pp]]}] /
   (Integrate[ff0[x], {x, freqnoisel1[[pp]], freqnoiser1[[pp]]}] +
     Integrate[ff0[x], {x, freqnoisel2[[pp]], freqnoiser2[[pp]]}]) - 1;
CNRYDC[[pp]] = 2 * (meandataplot0[[1]][[2]]) / meandataplot0[[2]][[2]] - 1;

(*(*stimulus-
 frequency deflection CSNR and DC deflection CSNR caclualted as ratio of
  areas of corresponding bandwidths for the case without noise reduction*)
CNRYst[[pp]]=Integrate[ff0[x],{x,freqpeakl[[pp]],freqpeakr[[pp]]}]/
   (Integrate[ff0base[x],{x,freqnoisel1[[pp]],freqnoiser1[[pp]]}]+
     Integrate[ff0base[x],{x,freqnoisel2[[pp]],freqnoiser2[[pp]]}])-1;
CNRYDC[[pp]]=2*(meandataplot0[[1]][[2]])/meandataplot0base[[2]][[2]]-1;*)

(*stimulus-frequency current CSNR and DC current CSNR caclualted as ratio of
  areas of corresponding bandwidths for the case with noise reduction*)
CNRPst[[pp]] = Integrate[ff1[x], {x, freqpeaklc[[pp]], freqpeakrc[[pp]]}] /
   (Integrate[ff1[x], {x, freqnoisel1c[[pp]], freqnoiser1c[[pp]]}] +
     Integrate[ff1[x], {x, freqnoisel2c[[pp]], freqnoiser2c[[pp]]}]) - 1;
CNRPDC[[pp]] = 2 * (meandataplot1[[1]][[2]]) / meandataplot1[[2]][[2]] - 1;

(*(*stimulus-frequency current CSNR and DC current CSNR caclualted as ratio of
  areas of corresponding bandwidths for the case without noise reduction*)
CNRPst[[pp]]=Integrate[ff1[x],{x,freqpeaklc[[pp]],freqpeakrc[[pp]]}]/
   (Integrate[ff1base[x],{x,freqnoisel1c[[pp]],freqnoiser1c[[pp]]}]+
     Integrate[ff1base[x],{x,freqnoisel2c[[pp]],freqnoiser2c[[pp]]}])-1;
CNRPDC[[pp]]=2*(meandataplot1[[1]][[2]])/meandataplot1base[[2]][[2]]-1;*)

(*stimulus-frequency voltage CSNR and DC voltage CSNR caclualted as ratio of
  areas of corresponding bandwidths for the case with noise reduction*)
CNRVst[[pp]] = Integrate[ff1filt[x], {x, freqpeaklv[[pp]], freqpeakrv[[pp]]}] /
   (Integrate[ff1filt[x], {x, freqnoisel1v[[pp]], freqnoiser1v[[pp]]}] +
     Integrate[ff1filt[x], {x, freqnoisel2v[[pp]], freqnoiser2v[[pp]]}]) - 1;
CNRVDC[[pp]] = 2 * (meandataplot1filt[[1]][[2]]) / meandataplot1filt[[2]][[2]] - 1;

(*(*stimulus-frequency voltage CSNR and DC voltage CSNR caclualted as ratio of
  areas of corresponding bandwidths for the case without noise reduction*)
CNRVst[[pp]]=Integrate[ff1filt[x],{x,freqpeaklv[[pp]],freqpeakrv[[pp]]}]/
```

```
      (Integrate[ff1filtbase[x],{x,freqnoisel1v[[pp]],freqnoiser1v[[pp]]}]+
        Integrate[ff1filtbase[x],{x,freqnoisel2v[[pp]],freqnoiser2v[[pp]]}])-1;
  CNRVDC[[pp]]=2*(meandataplot1filtbase[[1]][[2]])/meandataplot1filtbase[[2]][[2]]-1;*)

  ,
  {pp, 1, Length[fmult]}];

t2 = AbsoluteTime[];
t2 - t1


(*Fig S17 CSNR vs stimulus-
  force amplitude for a gating swing of 32 nm and a resting current of 0.5. Set D=
 32 and Pinfrest=0.5. Case with noise reduction*)

(*table of values of CSNRs*)
CNRYstval = Table[{fmult[[i]], CNRYst[[i]]}, {i, 1, Length[fmult]}];
CNRYDCval = Table[{fmult[[i]], CNRYDC[[i]]}, {i, 1, Length[fmult]}];
CNRPstval = Table[{fmult[[i]], CNRPst[[i]]}, {i, 1, Length[fmult]}];
CNRPDCval = Table[{fmult[[i]], CNRPDC[[i]]}, {i, 1, Length[fmult]}];
CNRVstval = Table[{fmult[[i]], CNRVst[[i]]}, {i, 1, Length[fmult]}];
CNRVDCval = Table[{fmult[[i]], CNRVDC[[i]]}, {i, 1, Length[fmult]}];

(*plot settings*)
plotstyleY = {Lighter[■]};
plotstyleP = {Lighter[■]};
plotstyleV = {Lighter[■]};

Remove[YTicksb]
Remove[XTicks]
Remove[YTicks]

TickLength = 0.02;
TickLength1 = 0.03;
Yticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, -24, 1, 2}];
Yticks2 = Table[{10^k, Null, {0, TickLength1}}, {k, -23, 1, 2}];
TickLength2 = 0.02;
aa = Table[0, {22}, {4}];
For[k = 1, k ≤ 22, k++,
aa[[k]] = Table[{l * 10^-k, , {0, TickLength2}}, {l, 2, 8, 2}];
]
YTicks = Union[Yticks1, Yticks2, aa[[1]], aa[[2]], aa[[3]], aa[[4]], aa[[5]],
    aa[[6]], aa[[7]], aa[[8]], aa[[9]], aa[[10]], aa[[11]], aa[[12]], aa[[13]], aa[[14]],
    aa[[15]], aa[[16]], aa[[17]], aa[[18]], aa[[19]], aa[[20]], aa[[21]], aa[[22]]];
```

```
Xticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, 0, 6, 1}];
aa2 = Table[0, {20}, {4}];
For[k = 1, k ≤ 10, k++,
aa2[[k]] = Table[{l * 10^k, , {0, TickLength2}}, {l, 2, 8, 2}];
]
XTicks = Union[Xticks1, aa2[[1]], aa2[[2]], aa2[[3]], aa2[[4]], aa2[[5]],
   aa2[[6]], aa2[[7]], aa2[[8]], aa2[[9]], aa2[[10]], aa2[[11]], aa2[[12]], aa2[[13]],
   aa2[[14]], aa2[[15]], aa2[[16]], aa2[[17]], aa2[[18]], aa2[[19]], aa2[[20]]];

pady = 120;
padx = 70;
pad2 = 30;
fntsize = 18;

(*FigA insert CSNR deflection stimulus*)

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0.2, 1, 0.4}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0.2, 1, 0.4}];
XTicks = Union[Xticks1, Xticks3];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 2, 2}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 2, 1}];
YTicksb = Union[Yticks1, Yticks3];


framelabelsCSNRY = {{"stimulus-frequency \n deflection CSNR", None},
   {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRYstval}, PlotRange → {{0.2, 1}, {-0.25, 2.1}}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRY,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigA CSNR deflection stimulus*)

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 400 000, 20 000}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 400 000, 20 000 / 2}];
YTicksb = Union[Yticks1, Yticks3];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 180, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 180, 25}];
```

```
XTicks = Union[Xticks1, Xticks3];


framelabelsCSNRY = {{"stimulus-frequency \n deflection CSNR", None},
   {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRYstval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRY,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]



(*FigB insert CSNR current ST*)

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 3, 1}];(*50000 200*)
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 3, 1 / 2}];(*25000 100*)
YTicksb = Union[Yticks1, Yticks3];


Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 3, 1}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 3, 0.5}];
Xticks4 = Table[{k, k, {0, 0.03}}, {k, 0.2, 0.2, 0.2}];
XTicks = Union[Xticks1, Xticks3, Xticks4];


framelabelsCSNRP = {{"stimulus-frequency \n current CSNR", None},
   {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRPstval}, PlotRange → {{0.2, 3}, {-0.25, 2}}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRP,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]

(*FigB CSNR current ST*)


Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 50 000, 5000}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 50 000, 5000 / 2}];
YTicks = Union[Yticks1, Yticks3];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 200, 25}];
```

```
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 200, 25}];
XTicks = Union[Xticks1, Xticks3];


framelabelsCSNRP = {{"stimulus-frequency \n current CSNR", None},
   {"stimlusu-force amplitude (pN)", None}};
ListPlot[{CNRPstval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRP,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]


(*FigC insert CSNR voltage stimulus*)

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 8, 2}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 8, 2 / 2}];
Xticks4 = Table[{k, k, {0, 0.03}}, {k, 0.2, 0.2, 0.2}];
XTicks = Union[Xticks1, Xticks3, Xticks4];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, -3, 5, 1}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, -3, 5, 1}];
YTicksb = Union[Yticks1, Yticks3];

framelabelsSNRP = {{"stimulus-frequency \n voltage CSNR", None},
   {"stimulus-force amplitude (pN)", None}};

ListPlot[{CNRVstval}, PlotRange → {{0.2, 8}, {-1, 3}}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleV,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsSNRP,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigC CSNR voltage stimulus*)

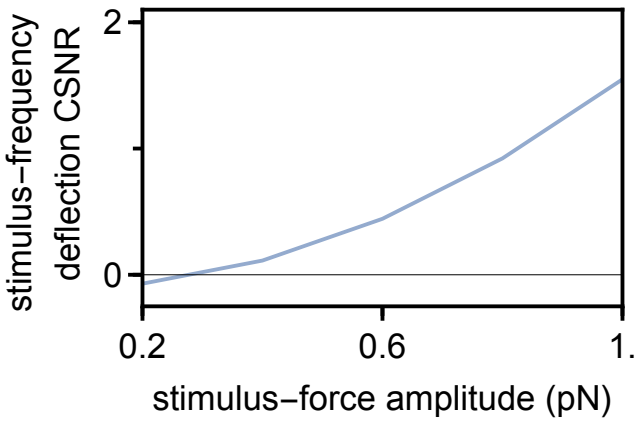

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 5000, 2 * 500}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 5000, 2 * 500 / 2}];
YTicks = Union[Yticks1, Yticks3];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 200, 25}];
```

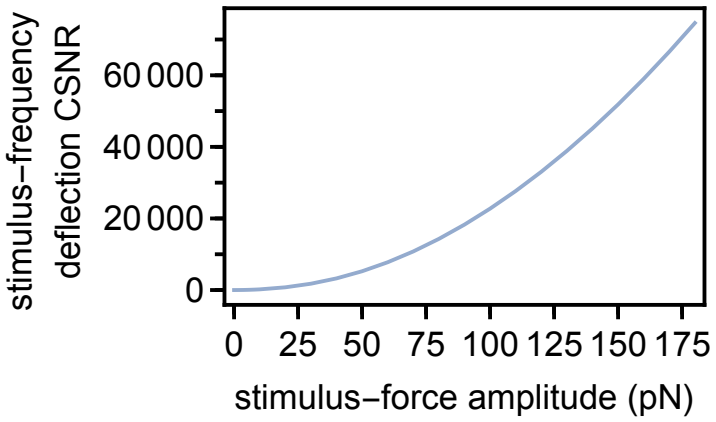
```
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 200, 25}];
XTicks = Union[Xticks1, Xticks3];


framelabelsCSNRV = {{"stimulus-frequency \n voltage CSNR", None},
   {"stimlusu-force amplitude (pN)", None}};
ListPlot[{CNRVstval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleV,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRV,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

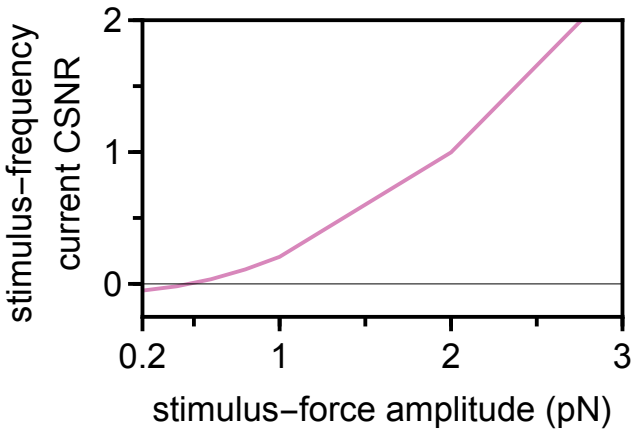
Out[ ]=



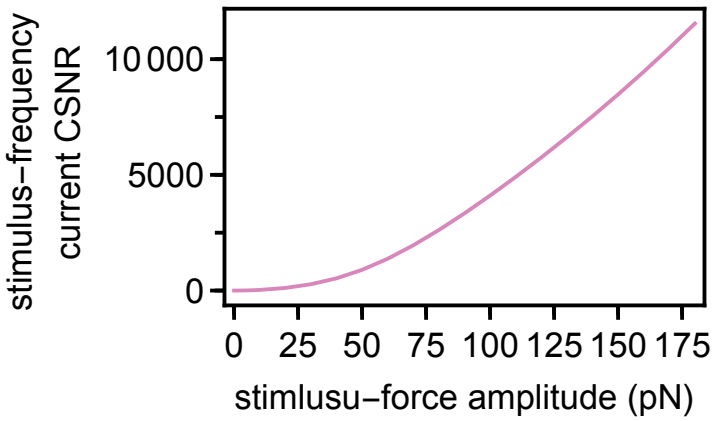
Out[ ]=



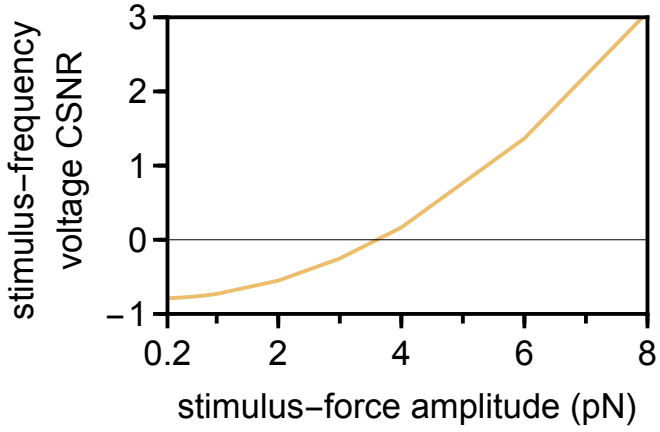
Out[ ]=


stimulus–frequency
current CSNR
2
1
0
0.2
1
2
3
stimulus–force amplitude (pN)

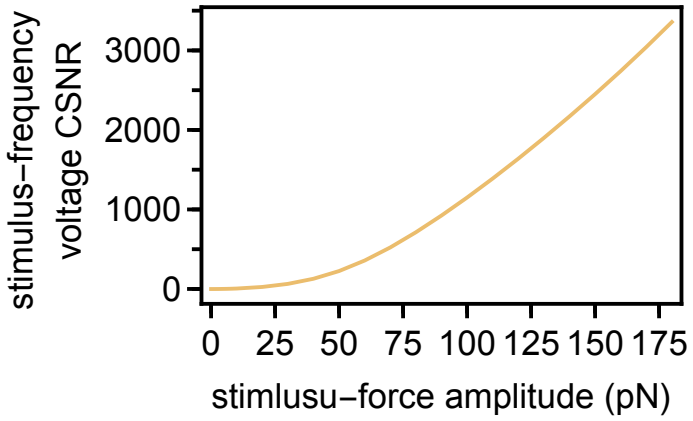
Out[ ]=


stimulus–frequency
current CSNR
10 000
5000
0
0
25
50
75
100
125
150
175
stimlusu–force amplitude (pN)

Out[ ]=


stimulus–frequency
voltage CSNR
3
2
1
0
–1
0.2
2
4
6
8
stimulus–force amplitude (pN)

Out[ ]=



In[ ]:=
```
(*Fig S17 CSNR vs stimulus-
  force amplitude for a gating swing of 32 nm and a resting current of 0.5. Set D=
 32 and Pinfrest=0.5. Case without noise reduction*)

(*table of values of CSNRs*)
CNRYstval = Table[{fmult[[i]], CNRYst[[i]]}, {i, 1, Length[fmult]}];
CNRYDCval = Table[{fmult[[i]], CNRYDC[[i]]}, {i, 1, Length[fmult]}];
CNRPstval = Table[{fmult[[i]], CNRPst[[i]]}, {i, 1, Length[fmult]}];
CNRPDCval = Table[{fmult[[i]], CNRPDC[[i]]}, {i, 1, Length[fmult]}];
CNRVstval = Table[{fmult[[i]], CNRVst[[i]]}, {i, 1, Length[fmult]}];
CNRVDCval = Table[{fmult[[i]], CNRVDC[[i]]}, {i, 1, Length[fmult]}];

(*plot settings*)
plotstyleY = {Darker[■]};
plotstyleP = {Darker[■]};
plotstyleV = {Darker[■]};

Remove[YTicksb]
Remove[XTicks]
Remove[YTicks]

TickLength = 0.02;
TickLength1 = 0.03;
Yticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, -24, 1, 2}];
Yticks2 = Table[{10^k, Null, {0, TickLength1}}, {k, -23, 1, 2}];
TickLength2 = 0.02;
aa = Table[0, {22}, {4}];
For[k = 1, k ≤ 22, k++,
aa[[k]] = Table[{l * 10^-k, , {0, TickLength2}}, {l, 2, 8, 2}];
]
```

```
YTicks = Union[Yticks1, Yticks2, aa[[1]], aa[[2]], aa[[3]], aa[[4]], aa[[5]],
   aa[[6]], aa[[7]], aa[[8]], aa[[9]], aa[[10]], aa[[11]], aa[[12]], aa[[13]], aa[[14]],
   aa[[15]], aa[[16]], aa[[17]], aa[[18]], aa[[19]], aa[[20]], aa[[21]], aa[[22]]];

Xticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, 0, 6, 1}];
aa2 = Table[0, {20}, {4}];
For[k = 1, k ≤ 10, k++,
aa2[[k]] = Table[{l * 10^k, , {0, TickLength2}}, {l, 2, 8, 2}];
]
XTicks = Union[Xticks1, aa2[[1]], aa2[[2]], aa2[[3]], aa2[[4]], aa2[[5]],
   aa2[[6]], aa2[[7]], aa2[[8]], aa2[[9]], aa2[[10]], aa2[[11]], aa2[[12]], aa2[[13]],
   aa2[[14]], aa2[[15]], aa2[[16]], aa2[[17]], aa2[[18]], aa2[[19]], aa2[[20]]];

pady = 120;
padx = 70;
pad2 = 30;
fntsize = 18;

(*FigA insert CSNR deflection stimulus*)

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0.2, 1, 0.4}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0.2, 1, 0.4}];
XTicks = Union[Xticks1, Xticks3];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 2, 2}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 2, 1}];
YTicksb = Union[Yticks1, Yticks3];


framelabelsCSNRY = {{"stimulus-frequency \n deflection CSNR", None},
   {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRYstval}, PlotRange → {{0.2, 1}, {-0.25, 2.1}}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRY,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigA CSNR deflection stimulus*)

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 400000, 20000}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 400000, 20000 / 2}];
YTicksb = Union[Yticks1, Yticks3];
```

```
Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 180, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 180, 25}];
XTicks = Union[Xticks1, Xticks3];


framelabelsCSNRY = {{"stimulus-frequency \n deflection CSNR", None},
   {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRYstval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRY,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]



(*FigC insert CSNR current ST*)

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 3, 1}];(*50000 200*)
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 3, 1 / 2}];(*25000 100*)
YTicksb = Union[Yticks1, Yticks3];


Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 3, 1}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 3, 0.5}];
Xticks4 = Table[{k, k, {0, 0.03}}, {k, 0.2, 0.2, 0.2}];
XTicks = Union[Xticks1, Xticks3, Xticks4];


framelabelsCSNRP = {{"stimulus-frequency \n current CSNR", None},
   {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRPstval}, PlotRange → {{0.2, 3}, {-0.25, 2}}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRP,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]

(*FigC CSNR current ST*)


Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 50 000, 500}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 50 000, 500 / 2}];
```

```
YTicks = Union[Yticks1, Yticks3];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 200, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 200, 25}];
XTicks = Union[Xticks1, Xticks3];


framelabelsCSNRP = {{"stimulus-frequency \n current CSNR", None},
   {"stimlusu-force amplitude (pN)", None}};
ListPlot[{CNRPstval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRP,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]


(*FigE insert CSNR voltage stimulus*)

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 8, 2}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 8, 2 / 2}];
Xticks4 = Table[{k, k, {0, 0.03}}, {k, 0.2, 0.2, 0.2}];
XTicks = Union[Xticks1, Xticks3, Xticks4];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, -3, 5, 1}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, -3, 5, 1}];
YTicksb = Union[Yticks1, Yticks3];

framelabelsSNRP = {{"stimulus-frequency \n voltage CSNR", None},
   {"stimulus-force amplitude (pN)", None}};

ListPlot[{CNRVstval}, PlotRange → {{0.2, 8}, {-1, 3}}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleV,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsSNRP,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigE CSNR voltage stimulus*)

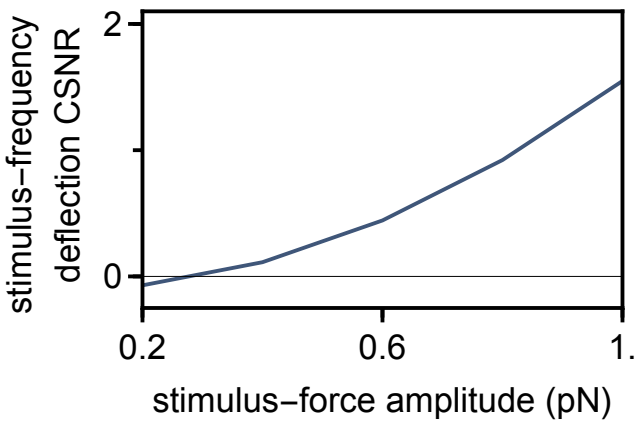

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 5000, 100}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 5000, 100 / 2}];
```

```
YTicks = Union[Yticks1, Yticks3];

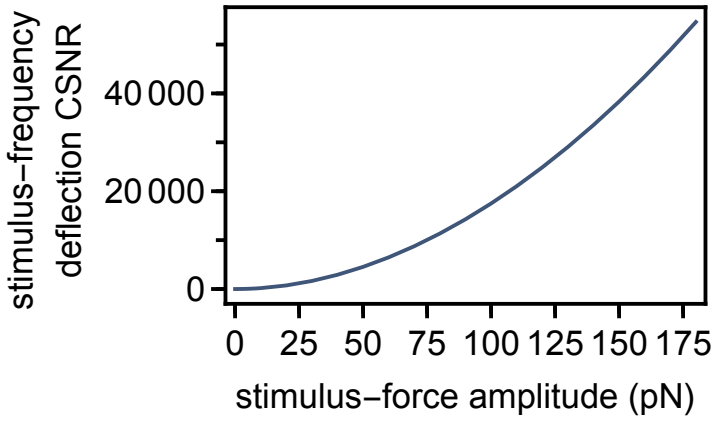
Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 200, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 200, 25}];
XTicks = Union[Xticks1, Xticks3];


framelabelsCSNRV = {{"stimulus-frequency \n voltage CSNR", None},
   {"stimlusu-force amplitude (pN)", None}};
ListPlot[{CNRVstval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleV,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRV,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

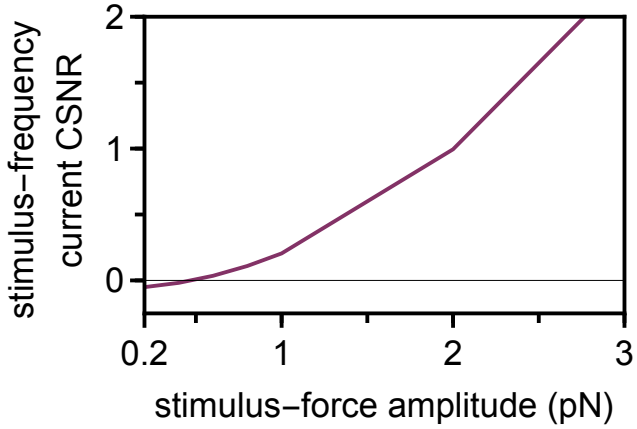
Out[ ]=



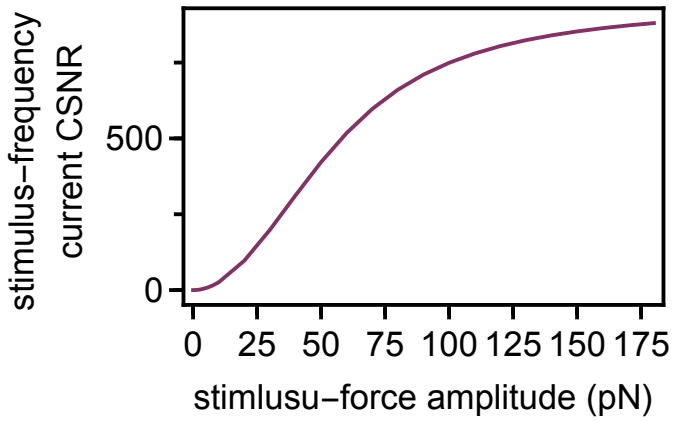
Out[ ]=


stimulus−frequency
deflection CSNR
40 000
20 000
0
0 25 50 75 100 125 150 175
stimulus−force amplitude (pN)

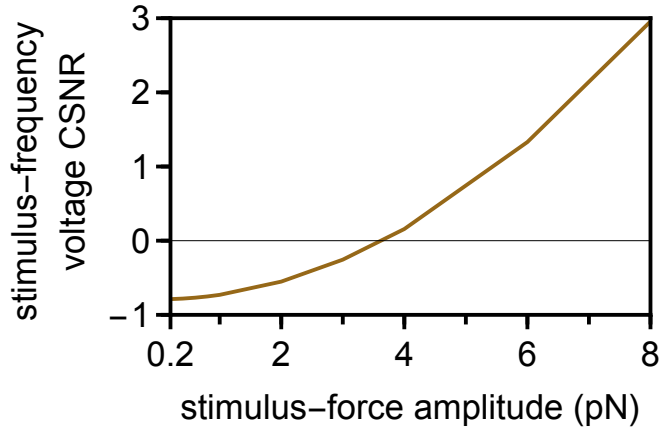
Out[ ]=


stimulus−frequency
current CSNR
2
1
0
0.2 1 2 3
stimulus−force amplitude (pN)

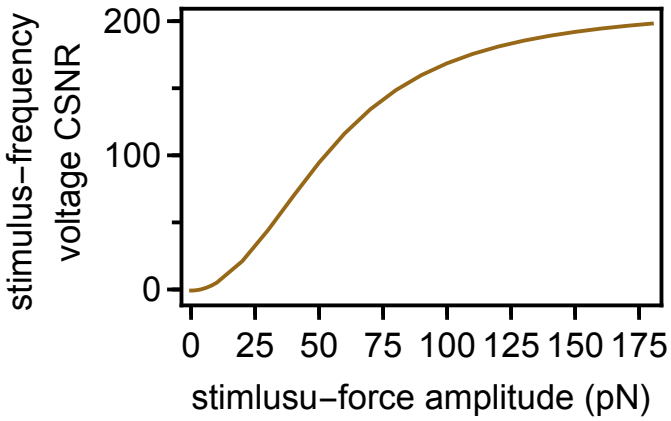
Out[ ]=


stimulus−frequency
current CSNR
500
0
0 25 50 75 100 125 150 175
stimlusu−force amplitude (pN)

Out[ ]=



Out[ ]=



In[ ]:=
```
(*Fig S17 CSNR vs stimulus-
  force amplitude for a gating swing of 1.75 nm and a resting current of 0.5. Set D=
 1.75 and Pinfrest=0.5. Case with noise reduction*)

(*table of values of CSNRs*)
CNRYstval = Table[{fmult[[i]], CNRYst[[i]]}, {i, 1, Length[fmult]}];
CNRYDCval = Table[{fmult[[i]], CNRYDC[[i]]}, {i, 1, Length[fmult]}];
CNRPstval = Table[{fmult[[i]], CNRPst[[i]]}, {i, 1, Length[fmult]}];
CNRPDCval = Table[{fmult[[i]], CNRPDC[[i]]}, {i, 1, Length[fmult]}];
CNRVstval = Table[{fmult[[i]], CNRVst[[i]]}, {i, 1, Length[fmult]}];
CNRVDCval = Table[{fmult[[i]], CNRVDC[[i]]}, {i, 1, Length[fmult]}];

(*plot settings*)
plotstyleY = {Lighter[■]};
plotstyleP = {Lighter[■]};
plotstyleV = {Lighter[■]};
```

```
Remove[YTicksb]
Remove[XTicks]
Remove[YTicks]

TickLength = 0.02;
TickLength1 = 0.03;
Yticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, -24, 1, 2}];
Yticks2 = Table[{10^k, Null, {0, TickLength1}}, {k, -23, 1, 2}];
TickLength2 = 0.02;
aa = Table[0, {22}, {4}];
For[k = 1, k ≤ 22, k++,
aa[[k]] = Table[{l * 10^-k, , {0, TickLength2}}, {l, 2, 8, 2}];
]
YTicks = Union[Yticks1, Yticks2, aa[[1]], aa[[2]], aa[[3]], aa[[4]], aa[[5]],
   aa[[6]], aa[[7]], aa[[8]], aa[[9]], aa[[10]], aa[[11]], aa[[12]], aa[[13]], aa[[14]],
   aa[[15]], aa[[16]], aa[[17]], aa[[18]], aa[[19]], aa[[20]], aa[[21]], aa[[22]]];

Xticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, 0, 6, 1}];
aa2 = Table[0, {20}, {4}];
For[k = 1, k ≤ 10, k++,
aa2[[k]] = Table[{l * 10^k, , {0, TickLength2}}, {l, 2, 8, 2}];
]
XTicks = Union[Xticks1, aa2[[1]], aa2[[2]], aa2[[3]], aa2[[4]], aa2[[5]],
   aa2[[6]], aa2[[7]], aa2[[8]], aa2[[9]], aa2[[10]], aa2[[11]], aa2[[12]], aa2[[13]],
   aa2[[14]], aa2[[15]], aa2[[16]], aa2[[17]], aa2[[18]], aa2[[19]], aa2[[20]]];

pady = 120;
padx = 70;
pad2 = 30;
fntsize = 18;

(*FigA insert CSNR deflection stimulus*)

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0.2, 1, 0.4}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0.2, 1, 0.4}];
XTicks = Union[Xticks1, Xticks3];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 2, 2}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 2, 1}];
YTicksb = Union[Yticks1, Yticks3];


framelabelsCSNRY = {{"stimulus-frequency \n deflection CSNR", None},
```

```
   {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRYstval}, PlotRange → {{0.2, 1}, {-0.25, 2.1}}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRY,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigA CSNR deflection stimulus*)

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 400 000, 20 000}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 400 000, 20 000 / 2}];
YTicksb = Union[Yticks1, Yticks3];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 180, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 180, 25}];
XTicks = Union[Xticks1, Xticks3];


framelabelsCSNRY = {{"stimulus-frequency \n deflection CSNR", None},
   {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRYstval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRY,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]



(*FigB insert CSNR current ST*)

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 3, 1}];(*50000 200*)
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 3, 1 / 2}];(*25000 100*)
YTicksb = Union[Yticks1, Yticks3];


Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 3, 1}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 3, 0.5}];
Xticks4 = Table[{k, k, {0, 0.03}}, {k, 0.2, 0.2, 0.2}];
XTicks = Union[Xticks1, Xticks3, Xticks4];
```

```
framelabelsCSNRP = {{"stimulus-frequency \n current CSNR", None},
   {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRPstval}, PlotRange → {{0.2, 3}, {-0.25, 2}}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRP,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]

(*FigB CSNR current ST*)


Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 50, 20}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 50, 20 / 2}];
YTicks = Union[Yticks1, Yticks3];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 200, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 200, 25}];
XTicks = Union[Xticks1, Xticks3];


framelabelsCSNRP = {{"stimulus-frequency \n current CSNR", None},
   {"stimlusu-force amplitude (pN)", None}};
ListPlot[{CNRPstval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRP,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]


(*FigC insert CSNR voltage stimulus*)

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 8, 2}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 8, 2 / 2}];
Xticks4 = Table[{k, k, {0, 0.03}}, {k, 0.2, 0.2, 0.2}];
XTicks = Union[Xticks1, Xticks3, Xticks4];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, -3, 5, 1}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, -3, 5, 1}];
YTicksb = Union[Yticks1, Yticks3];

framelabelsSNRP = {{"stimulus-frequency \n voltage CSNR", None},
   {"stimulus-force amplitude (pN)", None}};
```

```
ListPlot[{CNRVstval}, PlotRange → {{0.2, 8}, {-1, 3}}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleV,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsSNRP,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]

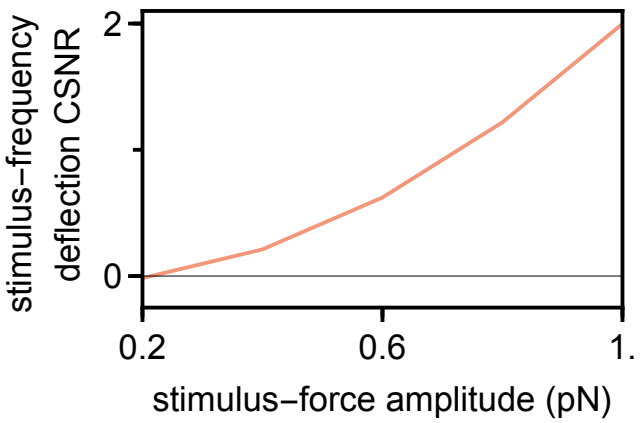

(*FigC CSNR voltage stimulus*)


Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 50, 5}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 50, 5 / 2}];
YTicks = Union[Yticks1, Yticks3];

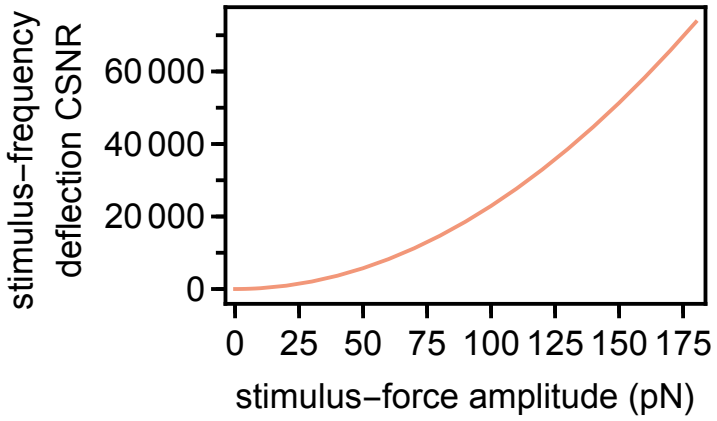
Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 200, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 200, 25}];
XTicks = Union[Xticks1, Xticks3];


framelabelsCSNRV = {{"stimulus-frequency \n voltage CSNR", None},
   {"stimlusu-force amplitude (pN)", None}};
ListPlot[{CNRVstval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleV,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRV,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

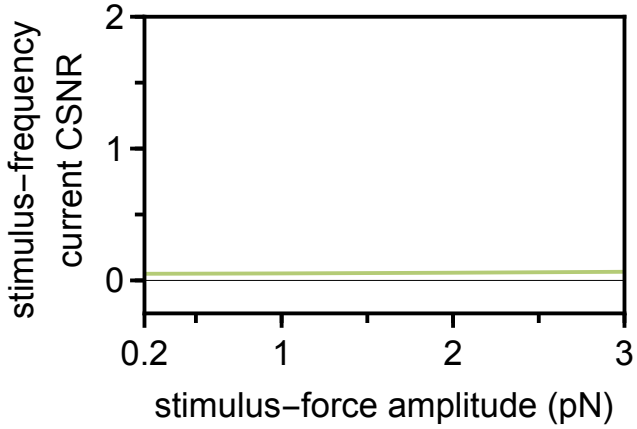
Out[ ]=



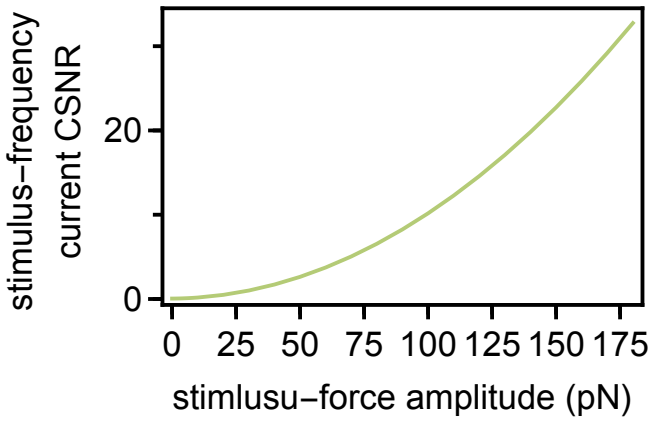
Out[ ]=


stimulus–frequency
deflection CSNR
60 000
40 000
20 000
0
0 25 50 75 100 125 150 175
stimulus–force amplitude (pN)

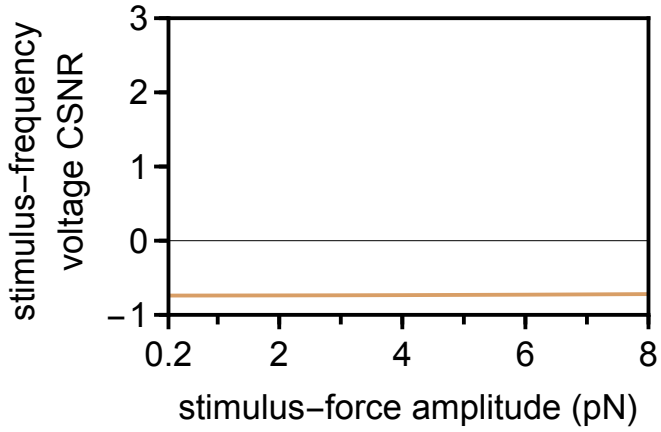
Out[ ]=


stimulus–frequency
current CSNR
2
1
0
0.2 1 2 3
stimulus–force amplitude (pN)

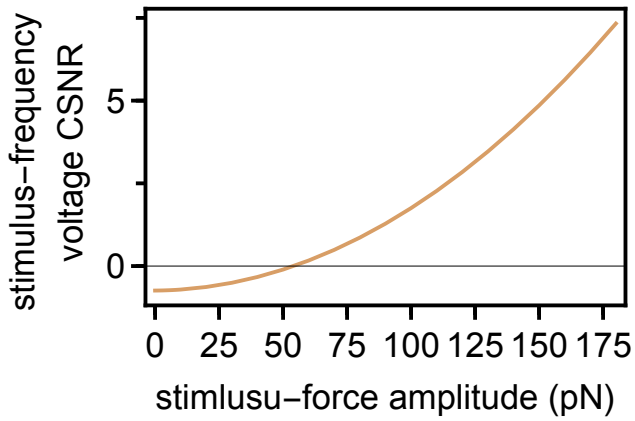
Out[ ]=


stimulus–frequency
current CSNR
20
0
0 25 50 75 100 125 150 175
stimlusu–force amplitude (pN)

Out[ ]=



Out[ ]=



In[ ]:=
```
(*Fig S17 CSNR vs stimulus-
  force amplitude for a gating swing of 1.75 nm and a resting current of 0.5. Set D=
 1.75 and Pinfrest=0.5. Case without noise reduction*)

(*table of values of CSNRs*)
CNRYstval = Table[{fmult[[i]], CNRYst[[i]]}, {i, 1, Length[fmult]}];
CNRYDCval = Table[{fmult[[i]], CNRYDC[[i]]}, {i, 1, Length[fmult]}];
CNRPstval = Table[{fmult[[i]], CNRPst[[i]]}, {i, 1, Length[fmult]}];
CNRPDCval = Table[{fmult[[i]], CNRPDC[[i]]}, {i, 1, Length[fmult]}];
CNRVstval = Table[{fmult[[i]], CNRVst[[i]]}, {i, 1, Length[fmult]}];
CNRVDCval = Table[{fmult[[i]], CNRVDC[[i]]}, {i, 1, Length[fmult]}];

(*plot settings*)
plotstyleY = {Darker[■]};
plotstyleP = {Darker[■]};
plotstyleV = {Darker[■]};
```

```
Remove[YTicksb]
Remove[XTicks]
Remove[YTicks]

TickLength = 0.02;
TickLength1 = 0.03;
Yticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, -24, 1, 2}];
Yticks2 = Table[{10^k, Null, {0, TickLength1}}, {k, -23, 1, 2}];
TickLength2 = 0.02;
aa = Table[0, {22}, {4}];
For[k = 1, k ≤ 22, k++,
aa[[k]] = Table[{l * 10^-k, , {0, TickLength2}}, {l, 2, 8, 2}];
]
YTicks = Union[Yticks1, Yticks2, aa[[1]], aa[[2]], aa[[3]], aa[[4]], aa[[5]],
   aa[[6]], aa[[7]], aa[[8]], aa[[9]], aa[[10]], aa[[11]], aa[[12]], aa[[13]], aa[[14]],
   aa[[15]], aa[[16]], aa[[17]], aa[[18]], aa[[19]], aa[[20]], aa[[21]], aa[[22]]];

Xticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, 0, 6, 1}];
aa2 = Table[0, {20}, {4}];
For[k = 1, k ≤ 10, k++,
aa2[[k]] = Table[{l * 10^k, , {0, TickLength2}}, {l, 2, 8, 2}];
]
XTicks = Union[Xticks1, aa2[[1]], aa2[[2]], aa2[[3]], aa2[[4]], aa2[[5]],
   aa2[[6]], aa2[[7]], aa2[[8]], aa2[[9]], aa2[[10]], aa2[[11]], aa2[[12]], aa2[[13]],
   aa2[[14]], aa2[[15]], aa2[[16]], aa2[[17]], aa2[[18]], aa2[[19]], aa2[[20]]];

pady = 120;
padx = 70;
pad2 = 30;
fntsize = 18;

(*FigA insert CSNR deflection stimulus*)

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0.2, 1, 0.4}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0.2, 1, 0.4}];
XTicks = Union[Xticks1, Xticks3];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 2, 2}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 2, 1}];
YTicksb = Union[Yticks1, Yticks3];


framelabelsCSNRY = {{"stimulus-frequency \n deflection CSNR", None},
```

```
   {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRYstval}, PlotRange → {{0.2, 1}, {-0.25, 2.1}}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRY,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]


(*FigA CSNR deflection stimulus*)

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 400 000, 20 000}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 400 000, 20 000 / 2}];
YTicksb = Union[Yticks1, Yticks3];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 180, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 180, 25}];
XTicks = Union[Xticks1, Xticks3];


framelabelsCSNRY = {{"stimulus-frequency \n deflection CSNR", None},
   {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRYstval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleY,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRY,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]



(*FigB insert CSNR current ST*)

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 3, 1}];(*50000 200*)
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 3, 1 / 2}];(*25000 100*)
YTicksb = Union[Yticks1, Yticks3];


Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 3, 1}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 3, 0.5}];
Xticks4 = Table[{k, k, {0, 0.03}}, {k, 0.2, 0.2, 0.2}];
XTicks = Union[Xticks1, Xticks3, Xticks4];
```

```
framelabelsCSNRP = {{"stimulus-frequency \n current CSNR", None},
   {"stimulus-force amplitude (pN)", None}};
ListPlot[{CNRPstval}, PlotRange → {{0.2, 3}, {-0.25, 2}}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRP,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]

(*FigB CSNR current ST*)


Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 50, 20}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 50, 20 / 2}];
YTicks = Union[Yticks1, Yticks3];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 200, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 200, 25}];
XTicks = Union[Xticks1, Xticks3];


framelabelsCSNRP = {{"stimulus-frequency \n current CSNR", None},
   {"stimlusu-force amplitude (pN)", None}};
ListPlot[{CNRPstval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRP,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]


(*FigC insert CSNR voltage stimulus*)

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 8, 2}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 8, 2 / 2}];
Xticks4 = Table[{k, k, {0, 0.03}}, {k, 0.2, 0.2, 0.2}];
XTicks = Union[Xticks1, Xticks3, Xticks4];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, -3, 5, 1}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, -3, 5, 1}];
YTicksb = Union[Yticks1, Yticks3];

framelabelsSNRP = {{"stimulus-frequency \n voltage CSNR", None},
   {"stimulus-force amplitude (pN)", None}};
```

```
ListPlot[{CNRVstval}, PlotRange → {{0.2, 8}, {-1, 3}}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleV,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsSNRP,
 FrameTicks → {{YTicksb, None}, {XTicks, None}}]

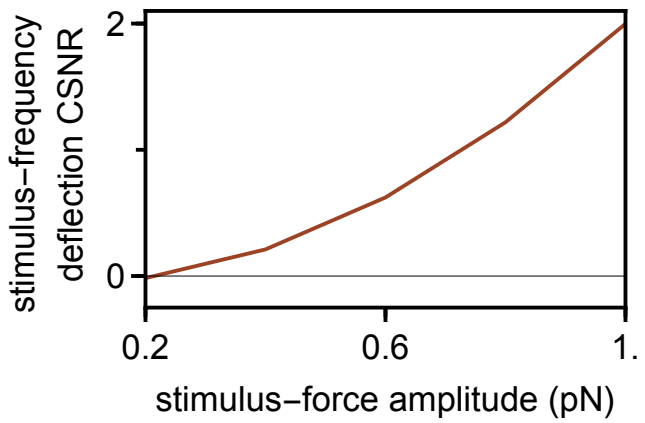

(*FigC CSNR voltage stimulus*)


Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 50, 5}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 50, 5 / 2}];
YTicks = Union[Yticks1, Yticks3];

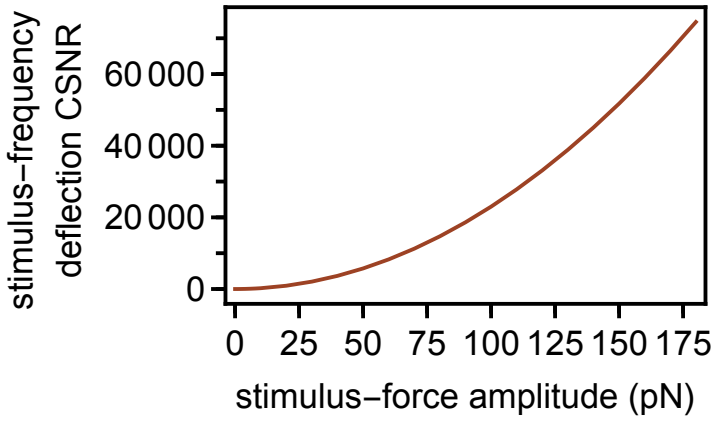
Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 200, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 200, 25}];
XTicks = Union[Xticks1, Xticks3];


framelabelsCSNRV = {{"stimulus-frequency \n voltage CSNR", None},
   {"stimlusu-force amplitude (pN)", None}};
ListPlot[{CNRVstval}, PlotRange → {Full, Full}, Joined → True,
 PlotRange → plotrangeSNRY, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleV,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsCSNRV,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

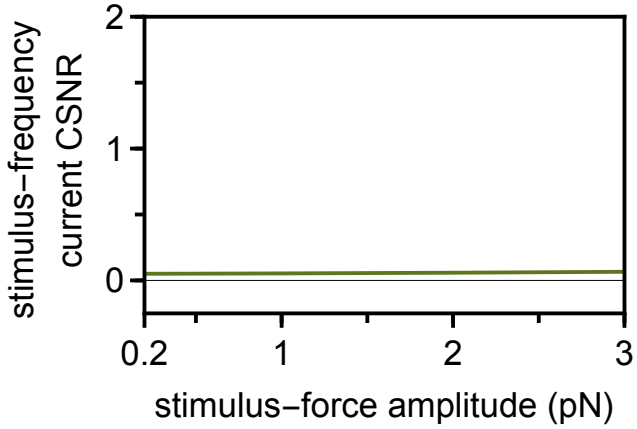
Out[ ]=



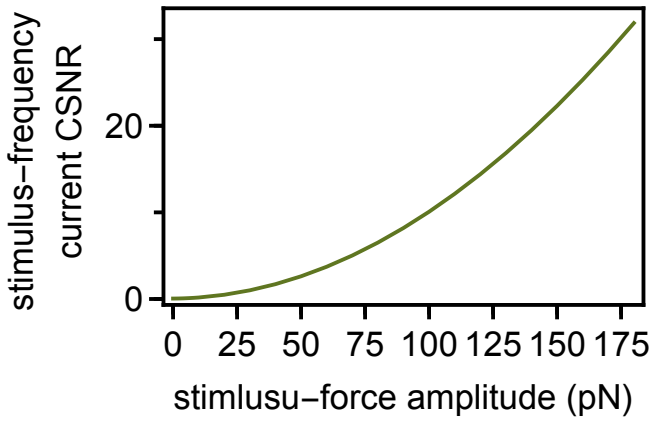
Out[ ]=


stimulus–frequency
deflection CSNR
60 000
40 000
20 000
0
0 25 50 75 100 125 150 175
stimulus–force amplitude (pN)

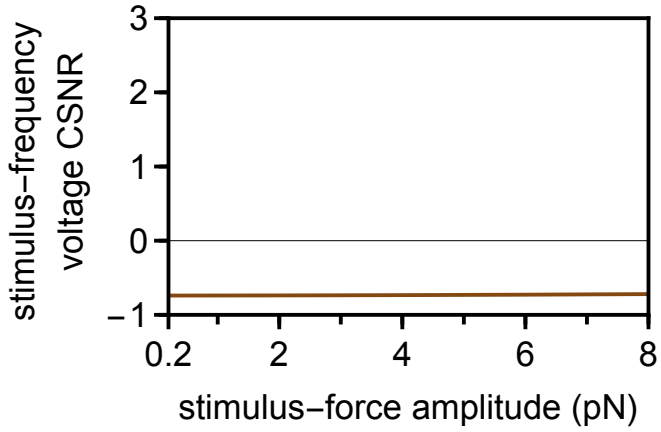
Out[ ]=


stimulus–frequency
current CSNR
2
1
0
0.2 1 2 3
stimulus–force amplitude (pN)

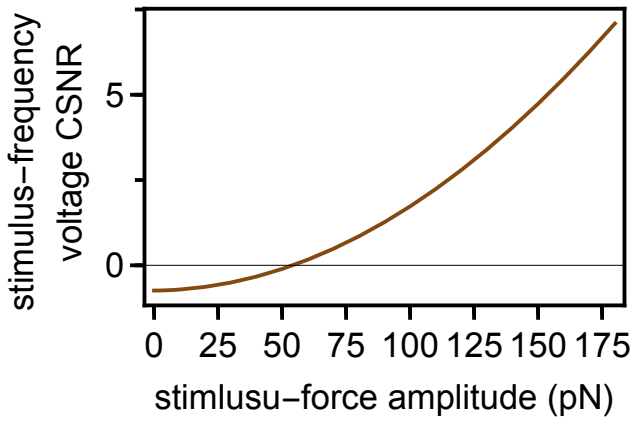
Out[ ]=


stimulus–frequency
current CSNR
20
0
0 25 50 75 100 125 150 175
stimlusu–force amplitude (pN)

Out[ ]=



Out[ ]=




```
(*Calculation of peak JNDs and fractional JNDs*)
t1 = AbsoluteTime[];

fmult = {0, 0.1, 0.2, 0.3, 0.4, 0.5, 1, 5, 10, 20, 30, 40, 50, 60, 70, 80, 90,
    100, 110, 120, 130, 140, 150, 160, 170, 172, 173, 175, 177, 178, 179, 180};

areaback0 = Table[0, {Length[fmult]}];
areatot0 = Table[0, {Length[fmult]}];
areaback1 = Table[0, {Length[fmult]}];
areatot1 = Table[0, {Length[fmult]}];
areaback1filt = Table[0, {Length[fmult]}];
areatot1filt = Table[0, {Length[fmult]}];
SetSharedVariable[areaback0];
SetSharedVariable[areatot0];
```

```
SetSharedVariable[areaback1];
SetSharedVariable[areatot1];
SetSharedVariable[areaback1filt];
SetSharedVariable[areatot1filt];
SetSharedVariable[fmult];



ParallelDo[

  SeedRandom[47 741];

gx =  2 * kbTd * lPd;
  y0in = Yrestd;



  proc = ItoProcess[
    lPd * ⅆY[t] == (-kPd) * (Y[t] - Yupd) * ⅆt - kGSd * (Y[t] - Yugd - Dvd * Po[t]) * ⅆt +
       Sqrt[gx] * ⅆn1[t] + fmult[[pp]] * Fzd * Sin[2 * Pi * wstd * (t - tstartFd)] * ⅆt &&
     taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd] * ⅆPo[t] ==
      (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] - Po[t]) ⅆt +
       Sqrt[taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd]] Sqrt[
         (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] (1 - Clip[Po[t], {0.0, 1.0}]) + (1 - Pinf2[
                 Yhd, kGSd, Dvd, kbTd, Ncd]) Clip[Po[t], {0.0, 1.0}]) / Ncd] ⅆn2[t],
    {Y[t], Po[t]}, {{Y, Po}, {y0in, Pinfrest}}, t,
{n1 ≈ WienerProcess[], n2 ≈ WienerProcess[]}];

  totallength = 2 * 0.507;
  factot = 4;
  deltatsima = 0.000001 / factot; (* /100 /4;*)
   ap =  RandomFunction[proc, {0, totallength, deltatsima}];

  ap0 = ap["PathComponent", 1] - Yrestd;

  ap1 = ap["PathComponent", 2] - Pinfrest;
  resampfactor = factot;


randtime100 = 0.005 * 10;
randtimenm = Round[randtime100, deltatsima];
stpoint = IntegerPart[randtime100 / deltatsima];

deltatsampl = Round[deltatsima * resampfactor, deltatsima];
```

```
tsampl = 0.005;
endpoint = IntegerPart[
    Length[ap["PathComponent", 1]["Values"]] - tsampl / deltatsima - stpoint];
tsamplnm = Round[tsampl, deltatsima];
b1 = Table[f[i], {i, randtimenm, tsamplnm + randtimenm, deltatsampl}];
lengthares = Length[b1];

Clear[ares];
totallength2 = totallength;
ncopiesa = (totallength2 - randtimenm) / (tsamplnm + deltatsampl);
ncopies = IntegerPart[ncopiesa];

ares0 = Table[0, {ncopies}];
ares1 = Table[0, {ncopies}];
sizea = lengthares;

fousampled0 = Table[0, {ncopies}, {lengthares}];
fousampled1 = Table[0, {ncopies}, {lengthares}];
cfousampled = Table[0, {ncopies}, {lengthares}];

Do[

   ares0[[ll]] = ap0["Values"][[(ll - 1) * tsampl / deltatsima + stpoint ;;
      IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];
    ares1[[ll]] = ap1["Values"][[(ll - 1) * tsampl / deltatsima + stpoint ;;
      IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];




       fousampled0[[ll]] = ResourceFunction["WelchSpectralEstimate"][
     ares0[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
     "OverlapOffset" → None, "SegmentSize" → sizea];
    cfousampled[[ll]] = ResourceFunction["WelchSpectralEstimate"][
     ares1[[ll]], ares0[[ll]], sizea / tsampl, "Window" → None,
     "Detrend" → None, "OverlapOffset" → None, "SegmentSize" → sizea];
    fousampled1[[ll]] = ResourceFunction["WelchSpectralEstimate"][
     ares1[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
     "OverlapOffset" → None, "SegmentSize" → sizea];

      ,
{ll, ncopies}];
  powspmeansampledt0 = Drop[Mean[fousampled0], -1];
```

```
  powspmeansampledt1 = Drop[Mean[fousampled1], -1];

cpowspmeansampledt = Drop[Re[Mean[cfousampled]], -1];
  lengthpsd = Length[powspmeansampledt0];


  meandataplot0 = Table[{0, 0}, {i, 1, lengthpsd}];
  meandataplot1 = Table[{0, 0}, {i, 1, lengthpsd}];
meandataplot1cpsd = Table[{0, 0}, {i, 1, lengthpsd}];


  meandataplot0 =
   Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt0[[i]][[2]]}, {i, 1, lengthpsd}];
  meandataplot1 =
   Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt1[[i]][[2]]}, {i, 1, lengthpsd}];
meandataplot1cpsd =
   Table[{cpowspmeansampledt[[i]][[1]], cpowspmeansampledt[[i]][[2]]}, {i, 1, lengthpsd}];
  fc = 610;
  filtersq1 = Table[{meandataplot0[[i]][[1]],
     Abs[1 / (1 + I * meandataplot1[[i]][[1]] / fc)]^2}, {i, 1, Length[meandataplot0]}];
  meandataplot1filt = Table[{meandataplot0[[i]][[1]],
     filtersq1[[i]][[2]] * meandataplot1[[i]][[2]]}, {i, 1, Length[filtersq1]}];

  data0 = Table[meandataplot0[[nn]][[2]], {nn, 2, Length[meandataplot0]}];
  dataest0 = MedianFilter[data0, 15];
  dataestwfreq0 = Table[{meandataplot0[[nn + 1]][[1]], dataest0[[nn]]},
    {nn, 8 + 7, Length[meandataplot0] - 8 + 7}];
  data0part = Table[meandataplot0[[nn]][[2]], {nn, 2, 8 + 7}];
  dataest0part = data0part;
  dataestwfreq0part =
   Table[{meandataplot0[[nn + 1]][[1]], dataest0part[[nn]]}, {nn, 1, 8 + 7 - 1}];
  dataest0tot = Union[
    {{meandataplot0[[1]][[1]], meandataplot0[[2]][[2]]}}, dataestwfreq0part, dataestwfreq0];
  dataest0tot[[1]][[2]] = 0.00007091605149763008`;
  areaback0[[pp]] = {fmult[[pp]],
    Integrate[Interpolation[dataest0tot, InterpolationOrder → 1][x], {x, 0, 5 * 10^5}]};
  areatot0[[pp]] = {fmult[[pp]],
    Integrate[Interpolation[meandataplot0, InterpolationOrder → 1][x], {x, 0, 5 * 10^5}]};


  data1 = Table[meandataplot1[[nn]][[2]], {nn, 2, Length[meandataplot1]}];
  dataest1 = MedianFilter[data1, 15];
  dataestwfreq1 = Table[{meandataplot1[[nn + 1]][[1]], dataest1[[nn]]},
    {nn, 8 + 7, Length[meandataplot1] - 8 + 7}];
```

```
  data1part = Table[meandataplot1[[nn]][[2]], {nn, 2, 8 + 7}];
  dataest1part = data1part;
  dataestwfreq1part =
   Table[{meandataplot1[[nn + 1]][[1]], dataest1part[[nn]]}, {nn, 1, 8 + 7 - 1}];
  dataest1tot1 = Union[
    {{meandataplot1[[1]][[1]], meandataplot1[[2]][[2]]}}, dataestwfreq1part, dataestwfreq1];
  areaback1[[pp]] =
   {fmult[[pp]], Integrate[Interpolation[dataest1tot1, InterpolationOrder → 1][x],
      {x, 200, 5 * 10^5}] + meandataplot1[[2]][[2]] * meandataplot1[[2]][[1]] / 2};
  areatot1[[pp]] = {fmult[[pp]],
    Integrate[Interpolation[meandataplot1, InterpolationOrder → 1][x], {x, 0, 5 * 10^5}]};

  data1filt = Table[meandataplot1filt[[nn]][[2]], {nn, 2, Length[meandataplot1filt]}];
  dataest1filt = MedianFilter[data1filt, 15];
  dataestwfreq1filt = Table[{meandataplot1filt[[nn + 1]][[1]], dataest1filt[[nn]]},
    {nn, 8 + 7, Length[meandataplot1filt] - 8 + 7}];
  data1filtpart = Table[meandataplot1filt[[nn]][[2]], {nn, 2, 8 + 7}];
  dataest1filtpart = data1filtpart;
  dataestwfreq1filtpart =
   Table[{meandataplot1filt[[nn + 1]][[1]], dataest1filtpart[[nn]]}, {nn, 1, 8 + 7 - 1}];
  dataest1totfilt = Union[{{meandataplot1filt[[1]][[1]], meandataplot1filt[[2]][[2]]}},
    dataestwfreq1filtpart, dataestwfreq1filt];
  areaback1filt[[pp]] =
   {fmult[[pp]], Integrate[Interpolation[dataest1totfilt, InterpolationOrder → 1][x],
      {x, 200, 5 * 10^5}]} + meandataplot1filt[[2]][[2]] * meandataplot1filt[[2]][[1]] / 2;
  areatot1filt[[pp]] = {fmult[[pp]], Integrate[
     Interpolation[meandataplot1filt, InterpolationOrder → 1][x], {x, 0, 5 * 10^5}]};
,
  {pp, 1, Length[fmult]}];

t2 = AbsoluteTime[];
t2 - t1



areaback0fit = Interpolation[areaback0, InterpolationOrder → 1];
areatot0fit = Interpolation[areatot0, InterpolationOrder → 1];
areapeacks0 = Table[0, Length[areatot0]];
fracjndpeaks0 = Table[0, Length[areatot0]];

(*deflection power of the peaks is derived respect to the background psd*)
Do[areapeacks0[[pp]] = {areatot0[[pp]][[1]], areatot0[[pp]][[2]] - areaback0[[pp]][[2]]},
 {pp, 1, Length[areatot0]}]
```

```
areapeacks0fit = Interpolation[areapeacks0, InterpolationOrder → 1];

solpeaks0 = Table[0, Length[areatot0]];
jndpeaks0 = Table[0, Length[areatot0]];
(*derivation of the value of the force that is
 required to increase the deflection power of the peaks by
 a factor equal to the fluctuation power at that level *)
Do[
 solpeaks0[[pp]] =
   FindRoot[areapeacks0fit[x] == areatot0[[pp]][[2]], {x, 1, Max[fmult]}][[1]][[2]];
 , {pp, 1, Length[areatot0]}]
(*jnd is defined as the difference of the
 force required to increase the deflection power*)
Do[
 jndpeaks0[[pp]] = {areatot0[[pp]][[1]], solpeaks0[[pp]] - areatot0[[pp]][[1]]},
 {pp, 1, Length[areatot0]}]

(*fractional deflection peak jnd*)
Do[
 fracjndpeaks0[[pp]] = {areatot0[[pp]][[1]],
   (solpeaks0[[pp]] - areatot0[[pp]][[1]]) / (areatot0[[pp]][[1]])}, {pp, 2, Length[areatot1]}]
fracjndpeaks0 = Drop[fracjndpeaks0, 1]


(*derivation of current power of the peaks*)
areaback1fit = Interpolation[areaback1, InterpolationOrder → 1];
areatot1fit = Interpolation[areatot1, InterpolationOrder → 1];
areapeacks1 = Table[0, Length[areatot1]];
fracjndpeaks1 = Table[0, Length[areatot1]];

(*current power of the peaks is derived respect to the background psd*)
Do[areapeacks1[[pp]] = {areatot1[[pp]][[1]], areatot1[[pp]][[2]] - areaback1[[pp]][[2]]},
 {pp, 1, Length[areatot1]}]
areapeacks1fit = Interpolation[areapeacks1, InterpolationOrder → 1];

solpeaks1 = Table[0, Length[areatot1]];
jndpeaks1 = Table[0, Length[areatot1]];
(*derivation of the value of the force that is required to increase the current
 power of the peaks by a factor equal to the fluctuation power at that level *)

Do[
 solpeaks1[[pp]] =
   FindRoot[areapeacks1fit[x] == areatot1[[pp]][[2]], {x, 1, Max[fmult]}][[1]][[2]];
 , {pp, 1, Length[areatot1]}]
```

```
(*jnd is defined as the difference of the
 force required to increase the current power*)

Do[
 jndpeaks1[[pp]] = {areatot1[[pp]][[1]], solpeaks1[[pp]] - areatot1[[pp]][[1]]},
 {pp, 1, Length[areatot1]}]
jndpeaks1 = Drop[jndpeaks1, 1]

(*fractional current peak jnd*)
Do[
 fracjndpeaks1[[pp]] = {areatot1[[pp]][[1]],
   (solpeaks1[[pp]] - areatot1[[pp]][[1]]) / (areatot1[[pp]][[1]])}, {pp, 2, Length[areatot1]}]
fracjndpeaks1 = Drop[fracjndpeaks1, 1]




(*derivation of voltage power of the peaks*)
areaback1filtfit = Interpolation[areaback1filt, InterpolationOrder → 1];
areatot1filtfit = Interpolation[areatot1filt, InterpolationOrder → 1];
areapeacks1filt = Table[0, Length[areatot1filt]];

(*voltage power of the peaks is derived respect to the background psd*)
Do[areapeacks1filt[[pp]] =
  {areatot1filt[[pp]][[1]], areatot1filt[[pp]][[2]] - areaback1filt[[pp]][[2]]},
 {pp, 1, Length[areatot1filt]}]
ListPlot[{areatot1filt, areapeacks1filt}]
areapeacks1filtfit = Interpolation[areapeacks1filt, InterpolationOrder → 1];

solpeaks1filt = Table[0, Length[areatot1filt]];
jndpeaks1filt = Table[0, Length[areatot1filt]];
fracjndpeaks1filt = Table[0, Length[areatot1filt]];

(*derivation of the value of the force that is required to increase the voltage
 power of the peaks by a factor equal to the fluctuation power at that level *)

Do[
 solpeaks1filt[[pp]] =
   FindRoot[areapeacks1filtfit[x] == areatot1filt[[pp]][[2]], {x, 1, Max[fmult]}][[1]][[2]];
 , {pp, 1, Length[areatot1filt]}]

(*jnd is defined as the difference of the
 force required to increase the voltage power*)
```

```
Do[
 jndpeaks1filt[[pp]] = {areatot1filt[[pp]][[1]], solpeaks1filt[[pp]] - areatot1filt[[pp]][[1]]},
 {pp, 1, Length[areatot1filt]}]
ListPlot[jndpeaks1filt, Joined → True]

(*fractional voltage peak jnd*)
Do[
 fracjndpeaks1filt[[pp]] =
  {areatot1filt[[pp]][[1]], (solpeaks1filt[[pp]] - areatot1filt[[pp]][[1]]) /
    (areatot1filt[[pp]][[1]])}, {pp, 2, Length[areatot1]}]
ListPlot[fracjndpeaks1filt, Joined → True]
fracjndpeaks1filt = Drop[fracjndpeaks1filt, 1]



(*definition of deflection peak jnd and
 fractional deflection peak jnd without noise reduction*)
areatotwithnored0 = Table[0, Length[areatot0]];
fracjndpeaks0nored = Table[0, Length[areatot0]];

(*deflection power of the peaks is derived respect to the unstimulated psd*)
Do[
 areatotwithnored0[[pp]] = {areatot0[[pp]][[1]], areapeacks0[[pp]][[2]] + areatot0[[1]][[2]]};
 , {pp, 1, Length[areatot0]}]

ListPlot[{areatot0, areatotwithnored0}]
ListLogPlot[{areatotwithnored0, areapeacks0}]

solpeaks0nored = Table[0, Length[areatot0]];
jndpeaks0nored = Table[0, Length[areatot0]];
Do[
 solpeaks0nored[[pp]] =
   Solve[areapeacks0fit[x] == areatotwithnored0[[pp]][[2]], x][[1]][[1]][[2]];
 , {pp, 1, Length[areatot0]}]

Do[
 jndpeaks0nored[[pp]] = {areatot0[[pp]][[1]],
   solpeaks0nored[[pp]] - areatotwithnored0[[pp]][[1]]}, {pp, 1, Length[areatot0]}]
ListPlot[jndpeaks0nored, Joined → True]


Do[
 fracjndpeaks0nored[[pp]] =
```

```
  {areatotwithnored0[[pp]][[1]], (solpeaks0nored[[pp]] - areatotwithnored0[[pp]][[1]]) /
    (areatotwithnored0[[pp]][[1]])}, {pp, 2, Length[areatot1]}]
ListPlot[fracjndpeaks0nored, Joined → True]
fracjndpeaks0nored = Drop[fracjndpeaks0nored, 1]



(*definition of current peak jnd and
 fractional current peak jnd without noise reduction*)

areatotwithnored1 = Table[0, Length[areatot1]];
fracjndpeaks1nored = Table[0, Length[areatot1]];

(*current power of the peaks is derived respect to the unstimulated psd*)
Do[
 areatotwithnored1[[pp]] = {areatot1[[pp]][[1]], areapeacks1[[pp]][[2]] + areatot1[[1]][[2]]};
 , {pp, 1, Length[areatot1]}]

ListPlot[{areatot1, areatotwithnored1}]
ListLogPlot[{areatotwithnored1, areapeacks1}]

solpeaks1nored = Table[0, Length[areatot1]];
jndpeaks1nored = Table[0, Length[areatot1]];
Do[
 solpeaks1nored[[pp]] =
   Solve[areapeacks1fit[x] == areatotwithnored1[[pp]][[2]], x][[1]][[1]][[2]];
 , {pp, 1, Length[areatot1]}]

Do[
 jndpeaks1nored[[pp]] = {areatot1[[pp]][[1]],
   solpeaks1nored[[pp]] - areatotwithnored1[[pp]][[1]]}, {pp, 1, Length[areatot1]}]
jndpeaks1nored = Drop[jndpeaks1nored, 1]
ListPlot[jndpeaks1nored, Joined → True]

Do[
 fracjndpeaks1nored[[pp]] =
  {areatotwithnored1[[pp]][[1]], (solpeaks1nored[[pp]] - areatotwithnored1[[pp]][[1]]) /
    (areatotwithnored1[[pp]][[1]])}, {pp, 2, Length[areatot1]}]
ListPlot[fracjndpeaks1nored, Joined → True]
fracjndpeaks1nored = Drop[fracjndpeaks1nored, 1]


(*definition of voltage peak jnd and
```

```
 fractional current peak jnd without noise reduction*)

areatotwithnored1filt = Table[0, Length[areatot1filt]];
fracjndpeaks1noredfilt = Table[0, Length[areatot1filt]];

(*voltage power of the peaks is derived respect to the unstimulated psd*)

Do[
 areatotwithnored1filt[[pp]] =
   {areatot1filt[[pp]][[1]], areapeacks1filt[[pp]][[2]] + areatot1filt[[1]][[2]]};
 , {pp, 1, Length[areatot1]}]

ListPlot[{areatot1filt, areatotwithnored1filt}]
ListLogPlot[{areatotwithnored1filt, areapeacks1filt}]

solpeaks1noredfilt = Table[0, Length[areatot1]];
jndpeaks1noredfilt = Table[0, Length[areatot1]];
Do[
 solpeaks1noredfilt[[pp]] =
   Solve[areapeacks1filtfit[x] == areatotwithnored1filt[[pp]][[2]], x][[1]][[1]][[2]];
 , {pp, 1, Length[areatot1]}]

Do[
 jndpeaks1noredfilt[[pp]] =
  {areatot1[[pp]][[1]], solpeaks1noredfilt[[pp]] - areatotwithnored1filt[[pp]][[1]]},
 {pp, 1, Length[areatot1]}]
ListPlot[jndpeaks1noredfilt, Joined → True]

Do[
 fracjndpeaks1noredfilt[[pp]] = {areatotwithnored1filt[[pp]][[1]],
   (solpeaks1noredfilt[[pp]] - areatotwithnored1filt[[pp]][[1]]) /
    (areatotwithnored1filt[[pp]][[1]])}, {pp, 2, Length[areatot1]}]
ListPlot[fracjndpeaks1noredfilt, Joined → True]
fracjndpeaks1noredfilt = Drop[fracjndpeaks1noredfilt, 1]
```

In[ ]:=

```
(*Fig10 peak JND and fractional peak JND vs stimulus-force amplitude*)

(*plot settings*)
TickLength1 = 0.03;
Yticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, -24, 10, 1}];
Yticks2 = Table[{10^k, Null, {0, TickLength1}}, {k, -23, 10, 2}];
```

```
TickLength2 = 0.02;

aa = Table[0, {38}, {4}];
For[k = 1, k ≤ 38, k++,
aa[[k]] = Table[{l * 10^(-k + 10), , {0, TickLength2}}, {l, 2, 8, 2}];
]
YTickslog = Union[Yticks1, Yticks2, aa[[1]], aa[[2]], aa[[3]],
   aa[[4]], aa[[5]], aa[[6]], aa[[7]], aa[[8]], aa[[9]], aa[[10]], aa[[11]], aa[[12]],
   aa[[13]], aa[[14]], aa[[15]], aa[[16]], aa[[17]], aa[[18]], aa[[19]], aa[[20]],
   aa[[21]], aa[[22]], aa[[23]], aa[[24]], aa[[25]], aa[[26]], aa[[27]], aa[[28]]];

pady = 120;
padx = 70;
pad2 = 30;
fntsize = 18;
plotrangeYdef = {{0, kGSd * 52 / 100}, {0, 5}};
framelabelsYdef = {{"deflection \n peak JND (pN)", None},
   {"stimulus-force amplitude (pN)", None}};
Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 180, 50}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 180, 50 / 2}];
XTicks = Union[Xticks1, Xticks3];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 20, 2}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 20, 2 / 2}];
YTicks = Union[Yticks1, Yticks3];

plotstyleYdef =
  {{Thickness → 0.01, Lighter[■]}, {Thickness → 0.01, Lighter[■], Dashed}};

(*Fig10C peak deflection JND*)
ListLogPlot[{jndpeaks0, jndpeaks0nored},
 PlotRange → {{-5, 185}, {4 * 10^-2, 4 * 10^1}}, Joined → True,
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleYdef, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsYdef, FrameTicks → {{YTickslog, None}, {XTicks, None}}]

(*Fig10D fractional peak deflection JND*)
Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 100, 20}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 100, 20 / 2}];
YTicks = Union[Yticks1, Yticks3];

framelabelsYdef = {{"fractional defleciton \n peak JND ", None},
   {"stimulus-force amplitude (pN)", None}};
```

```
ListLogPlot[{fracjndpeaks0, fracjndpeaks0nored},
 PlotRange → {{-5, 185}, {2 * 10^-4, 10^3}}, Joined → True,
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleYdef, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsYdef, FrameTicks → {{YTickslog, None}, {XTicks, None}}]


(*Fig10C peak current JND*)
Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 500, 50}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 500, 50 / 2}];
YTicks = Union[Yticks1, Yticks3];

plotstyleYdef = {{Thickness → 0.01, Lighter[■]},
    {Thickness → 0.01, Lighter[■], Dashed}}; (*Lighter[■]■*)
framelabelsYdef =
  {{"current peak \n JND (pN)", None}, {"stimulus-force amplitude (pN)", None}};
ListLogPlot[{jndpeaks1, jndpeaks1nored}, PlotRange → {{-5, 180}, {10^0, 4 * 10^2}},
 Joined → True, ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleYdef, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsYdef, FrameTicks → {{YTickslog, None}, {XTicks, None}}]

(*Fig10D fractional peak current JND*)
framelabelsYdef =
  {{"fractional peak \n JND ", None}, {"stimulus-force amplitude (pN)", None}};

ListLogPlot[{fracjndpeaks1, fracjndpeaks1nored},
 PlotRange → {{-5, 185}, {10^-4, 3 * 10^2}}, Joined → True,
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleYdef, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsYdef, FrameTicks → {{YTickslog, None}, {XTicks, None}}]

(*Fig10C peak voltage JND*)
Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 500, 100}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 500, 50 / 2}];
YTicks = Union[Yticks1, Yticks3];


plotstyleYdef = {{Thickness → 0.01, ■}, {Thickness → 0.01, ■, Dashed}};
framelabelsYdef =
  {{"peak JND (pN)", None}, {"stimulus-force amplitude (pN)", None}};
```

```
ListLogPlot[{jndpeaks1filt, jndpeaks1noredfilt},
 PlotRange → {{-5, 185}, {2 * 10^-1, 6 * 10^2}}, Joined → True,
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleYdef, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsYdef, FrameTicks → {{YTickslog, None}, {XTicks, None}}]

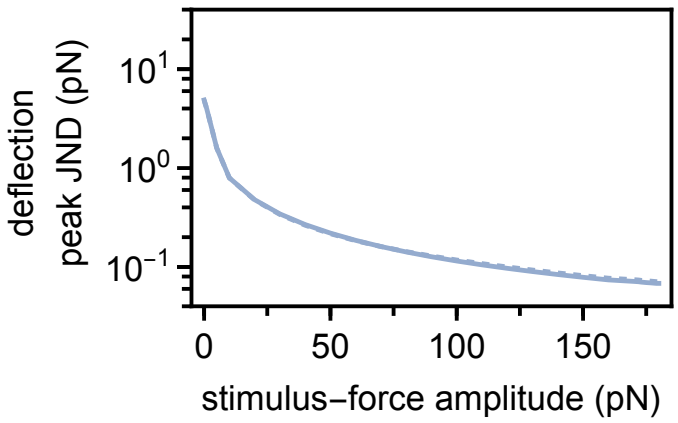
(*Fig10D fractional peak voltage JND*)

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1, 0.2}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1, 0.2 / 2}];
YTicks = Union[Yticks1, Yticks3];

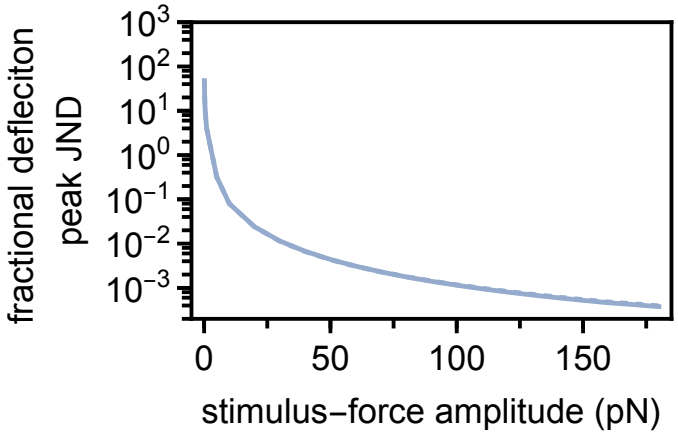

framelabelsYdef =
  {{"fractional voltage \n JND ", None}, {"stimulus-force amplitude (pN)", None}};

ListLogPlot[{fracjndpeaks1filt, fracjndpeaks1noredfilt},
 PlotRange → {{-5, 185}, {10^-3, 0.8 * 10^3}}, Joined → True,
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleYdef, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsYdef, FrameTicks → {{YTickslog, None}, {XTicks, None}}]
```

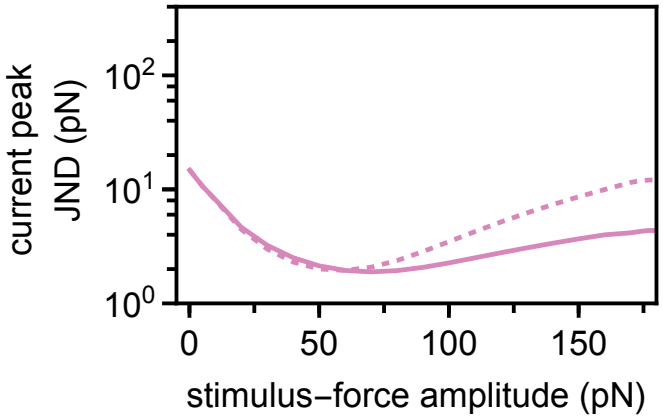
Out[ ]=



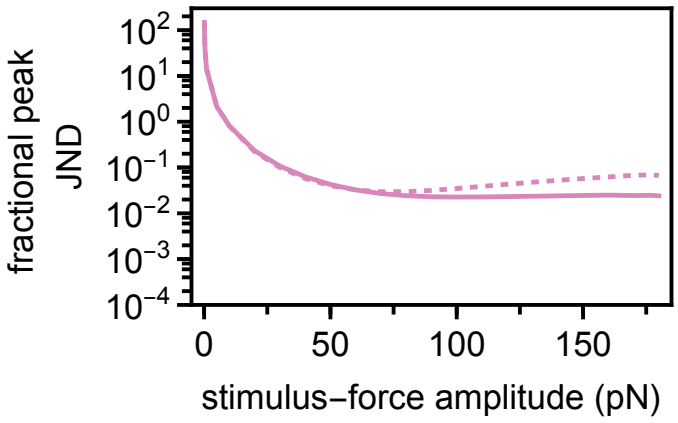
Out[ ]=


fractional defleciton
peak JND
10^3
10^2
10^1
10^0
10^{-1}
10^{-2}
10^{-3}
0
50
100
150
stimulus–force amplitude (pN)

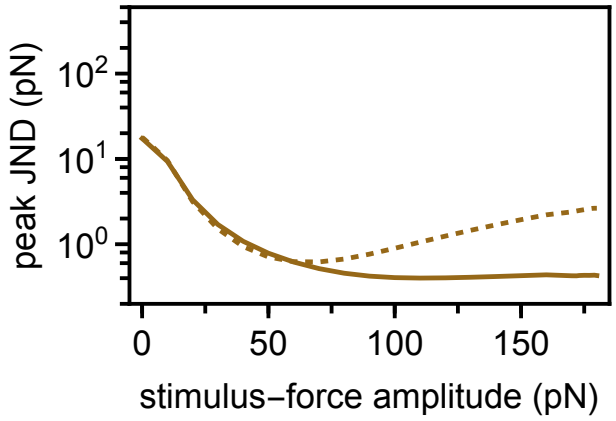
Out[ ]=


current peak
JND (pN)
10^2
10^1
10^0
0
50
100
150
stimulus–force amplitude (pN)

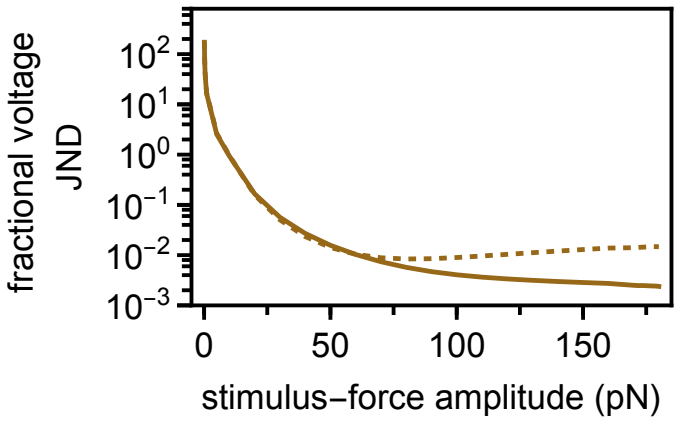
Out[ ]=


fractional peak
JND
10^2
10^1
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
0
50
100
150
stimulus–force amplitude (pN)

Out[ ]=



Out[ ]=



In[ ]:=

```
(*Fig10 peak JND and fractional peak JND vs stimulus-
  force amplitude for a gating swing of 1.75 nm. Set D=1.75 nm.*)

(*plot settings*)
TickLength1 = 0.03;
Yticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, -24, 10, 1}];
Yticks2 = Table[{10^k, Null, {0, TickLength1}}, {k, -23, 10, 2}];
TickLength2 = 0.02;

aa = Table[0, {38}, {4}];
For[k = 1, k ≤ 38, k++,
aa[[k]] = Table[{l * 10^(-k + 10), , {0, TickLength2}}, {l, 2, 8, 2}];
]
```

```
YTickslog = Union[Yticks1, Yticks2, aa[[1]], aa[[2]], aa[[3]],
   aa[[4]], aa[[5]], aa[[6]], aa[[7]], aa[[8]], aa[[9]], aa[[10]], aa[[11]], aa[[12]],
   aa[[13]], aa[[14]], aa[[15]], aa[[16]], aa[[17]], aa[[18]], aa[[19]], aa[[20]],
   aa[[21]], aa[[22]], aa[[23]], aa[[24]], aa[[25]], aa[[26]], aa[[27]], aa[[28]]];

pady = 120;
padx = 70;
pad2 = 30;
fntsize = 18;
plotrangeYdef = {{0, kGSd * 52 / 100}, {0, 5}};
framelabelsYdef = {{"deflection \n peak JND (pN)", None},
   {"stimulus-force amplitude (pN)", None}};
Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 180, 50}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 180, 50 / 2}];
XTicks = Union[Xticks1, Xticks3];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 20, 2}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 20, 2 / 2}];
YTicks = Union[Yticks1, Yticks3];

plotstyleYdef =
  {{Thickness → 0.01, Lighter[■]}, {Thickness → 0.01, Lighter[■], Dashed}};

(*Fig10C peak deflection JND*)
ListLogPlot[{jndpeaks0}, PlotRange → {{-5, 185}, {4 * 10^-2, 4 * 10^1}},
 Joined → True, ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleYdef, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsYdef, FrameTicks → {{YTickslog, None}, {XTicks, None}}]

(*Fig10D fractional peak deflection JND*)
Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 100, 20}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 100, 20 / 2}];
YTicks = Union[Yticks1, Yticks3];

framelabelsYdef = {{"fractional defleciton \n peak JND ", None},
   {"stimulus-force amplitude (pN)", None}};

ListLogPlot[{fracjndpeaks0}, PlotRange → {{-5, 185}, {2 * 10^-4, 10^3}},
 Joined → True, ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleYdef, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsYdef, FrameTicks → {{YTickslog, None}, {XTicks, None}}]
```

```
(*Fig10C peak current JND*)
Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 500, 50}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 500, 50 / 2}];
YTicks = Union[Yticks1, Yticks3];

plotstyleYdef = {{Thickness → 0.01, Lighter[■]},
    {Thickness → 0.01, Lighter[■], Dashed}}; (*Lighter[■]■*)
framelabelsYdef =
  {{"current peak \n JND (pN)", None}, {"stimulus-force amplitude (pN)", None}};
ListLogPlot[{jndpeaks1}, PlotRange → {{-5, 180}, {10^0, 4 * 10^2}},
 Joined → True, ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleYdef, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsYdef, FrameTicks → {{YTickslog, None}, {XTicks, None}}]

(*Fig10D fractional peak current JND*)
framelabelsYdef =
  {{"fractional peak \n JND ", None}, {"stimulus-force amplitude (pN)", None}};

ListLogPlot[{fracjndpeaks1}, PlotRange → {{-5, 185}, {10^-4, 3 * 10^2}},
 Joined → True, ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleYdef, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsYdef, FrameTicks → {{YTickslog, None}, {XTicks, None}}]

(*Fig10C peak voltage JND*)
Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 500, 100}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 500, 50 / 2}];
YTicks = Union[Yticks1, Yticks3];


plotstyleYdef = {{Thickness → 0.01, ■}, {Thickness → 0.01, ■, Dashed}};
framelabelsYdef =
  {{"peak JND (pN)", None}, {"stimulus-force amplitude (pN)", None}};
ListLogPlot[{jndpeaks1filt}, PlotRange → {{-5, 185}, {2 * 10^-1, 6 * 10^2}},
 Joined → True, ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleYdef, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsYdef, FrameTicks → {{YTickslog, None}, {XTicks, None}}]

(*Fig10D fractional peak voltage JND*)

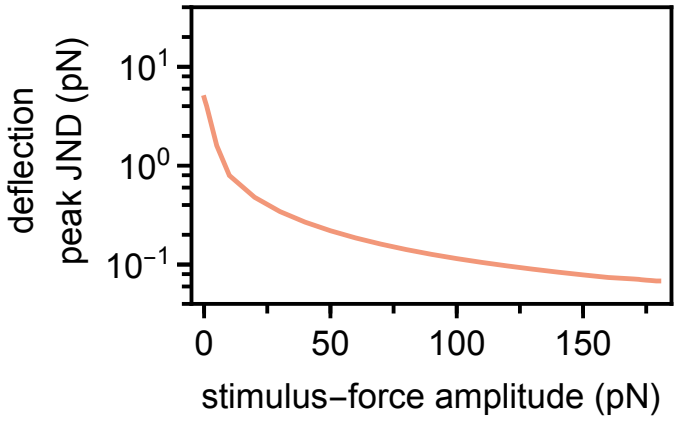
Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1, 0.2}];
```

```
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1, 0.2 / 2}];
YTicks = Union[Yticks1, Yticks3];

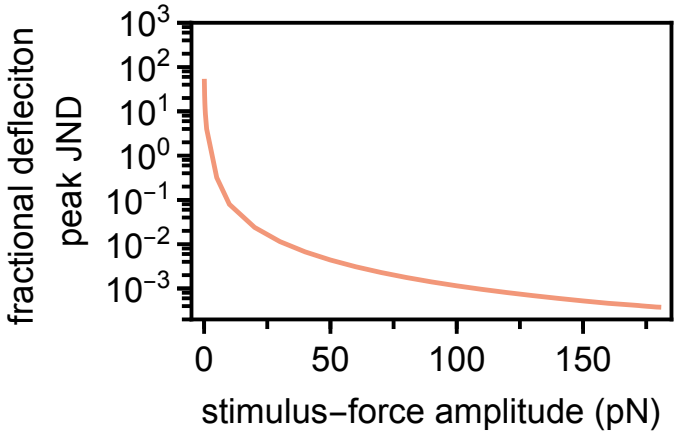

framelabelsYdef =
  {{"fractional voltage \n JND ", None}, {"stimulus-force amplitude (pN)", None}};

ListLogPlot[{fracjndpeaks1filt}, PlotRange → {{-5, 185}, {10^-3, 0.8 * 10^3}},
 Joined → True, ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleYdef, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsYdef, FrameTicks → {{YTickslog, None}, {XTicks, None}}]
```

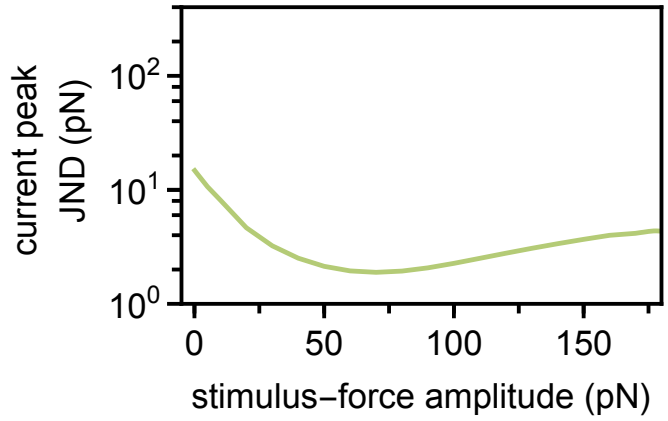
Out[ ]=



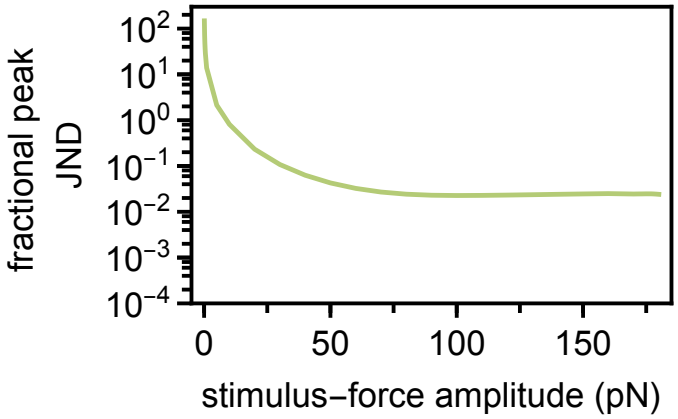
Out[ ]=



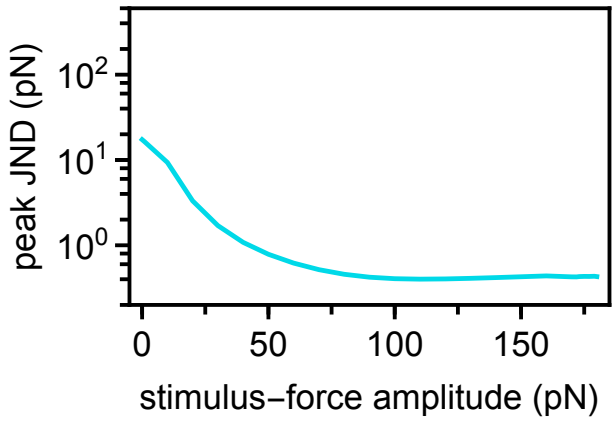
Out[ ]=


current peak
JND (pN)
10^2
10^1
10^0
0
50
100
150
stimulus–force amplitude (pN)

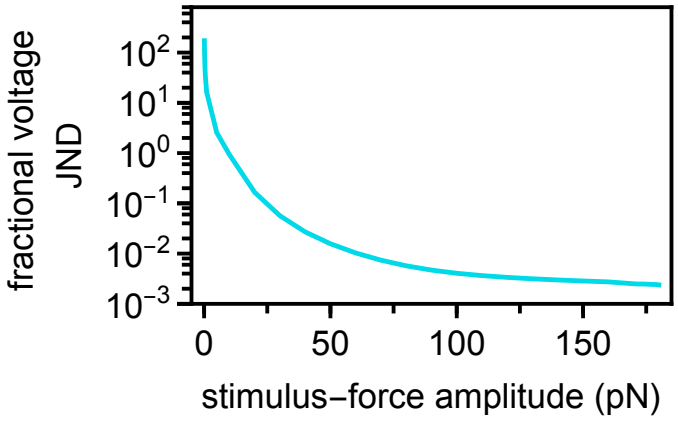
Out[ ]=


fractional peak
JND
10^2
10^1
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
0
50
100
150
stimulus–force amplitude (pN)

Out[ ]=


peak JND (pN)
10^2
10^1
10^0
0
50
100
150
stimulus–force amplitude (pN)

Out[ ]=


fractional voltage JND
10^2
10^1
10^0
10^{-1}
10^{-2}
10^{-3}
0
50
100
150
stimulus−force amplitude (pN)

In[ ]:=

```
(*Calculation of deflection psd with channels blocked*)
SeedRandom[47741] (* set seed for reproducibility of results *)

gx = 2 * kbTd * lPd;
y0in = Yrestd;

t1 = AbsoluteTime[];

(*define Ito process for simulations of deflection.
  Fzd multiplied by 180 for stimulated psd,
0 for unstimulated psd. For channels blocked set Po[t]=1.*)
proc =
 ItoProcess[lPd * ⅆY[t] == (-kPd) * (Y[t] - Yupd) * ⅆt - kGSd * (Y[t] - Yugd - Dvd * 1) * ⅆt +
     Sqrt[gx] * ⅆn1[t] + Fzd * fmult * Sin[2 * Pi * wstd * (t - tstartFd)] * ⅆt &&
   taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd] * ⅆPo[t] ==
    (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] - Po[t]) ⅆt +
     Sqrt[taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd]]
      Sqrt[(Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] (1 - Clip[Po[t], {0.0, 1.0}]) + (1 - Pinf2[
               Yhd, kGSd, Dvd, kbTd, Ncd]) Clip[Po[t], {0.0, 1.0}]) / Ncd] ⅆn2[t],
  {Y[t], Po[t]}, {{Y, Po}, {y0in, Pinfrest}}, t,
 {n1 ≈ WienerProcess[], n2 ≈ WienerProcess[]}]

totallength = 1.014;(*length of simulation*)
factot = 4;
deltatsima = 0.000001 / factot;(*time-step of simulation*)
 ap = RandomFunction[proc, {0, totallength, deltatsima}];
(*simulation of deflection and curent*)

 t2 = AbsoluteTime[];
t2 - t1
```

Out[ ]=

```
RandomGeneratorState[ Method: ExtendedCA
                      State hash: -8833868631830390932 ]
```

Out[ ]=

$$\text{ItoProcess}\Bigg[\Big\{\Big\{0. + 27.061\,(180\,000.\,\text{Sin}[31\,415.9\,t] - 1872.\,(407. + Y[t]) - 468.\,(480.44 + Y[t])),$$

$$\Bigg(0.\,\sqrt{\left(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}}$$

$$\sqrt{\left(\frac{1. - 1.\,\text{Clip}[Po[t], \{0., 1.\}]}{1. + e^{0.115189\,(-413.235 - Y[t])}} + \left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)\text{Clip}[Po[t], \{0., 1.\}]\right)}\Bigg/$$

$$\left(\left(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\right) +$$

$$\frac{30\,000.\left(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}} - 1.\,Po[t]\right)}{\left(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}}\Big\},$$

$$\Big\{\{468.927, 0.\}, \Big\{\Bigg(0.\,\sqrt{\left(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}}$$

$$\sqrt{\left(\frac{1. - 1.\,\text{Clip}[Po[t], \{0., 1.\}]}{1. + e^{0.115189\,(-413.235 - Y[t])}} + \left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)\text{Clip}[Po[t], \{0., 1.\}]\right)}\Bigg/$$

$$\left(\left(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\right),$$

$$\Bigg(30.6186\,\sqrt{\left(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}}$$

$$\sqrt{\left(\frac{1. - 1.\,\text{Clip}[Po[t], \{0., 1.\}]}{1. + e^{0.115189\,(-413.235 - Y[t])}} + \left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)\text{Clip}[Po[t], \{0., 1.\}]\right)}\Bigg/$$

$$\left(\left(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\right)\Big\}\Big\},$$

$$\{Y[t], Po[t]\}\Big\}, \{\{Y, Po\}, \{-427, 0.17\}\}, \{t, 0\}\Bigg]$$

Out[ ]=

10.296575

In[ ]:=

```
(*Calculation of deflection power spectrum from simulated deflections*)

Yrestd1 = -421.68; (*resting position of IHB with channels blocked [nm]*)
(*Yrestd=-407;(*resting position with gating spring cut [nm]*)*)
```

```
ap0 = ap["PathComponent", 1] - Yrestd1;
(* subtract resting position from deflection time series*)


resampfactor = factot;(*resempling factor for time series*)

randtime100 = 0.005 * 10; (*starting time to ensure that values
 of time series are independent of initial value of simulation*)

(*round to multiple of time-step of simulation to avoid interpolation*)
randtimenm = Round[randtime100, deltatsima];
stpoint = IntegerPart[randtime100 / deltatsima];

deltatsampl = Round[deltatsima * resampfactor, deltatsima];
(*time-step of resampled time series*)
tsampl = 0.005; (* length of a single window of the splitted time series*)
endpoint = IntegerPart[Length[ap["PathComponent", 1]["Values"]] -
    tsampl / deltatsima - stpoint];(*end point of time series*)
tsamplnm = Round[tsampl, deltatsima];
(*round to fraction of delta of time series to avoid interpolation*)
(*determine length of sampled time series*)
b1 = Table[f[i], {i, randtimenm, tsamplnm + randtimenm, deltatsampl}];
lengthares = Length[b1];

ncopiesa = (totallength - randtimenm) / (tsamplnm + deltatsampl);
(*number of non-overlapping windows*)
ncopies = IntegerPart[ncopiesa];
(*round number of overlapping windows to nearest integer *)



ares0 = Table[0, {ncopies}];
(*table of non-overlapping blocks of deflection time series*)
sizea = lengthares;(*length of a single block*)
fousampled0 = Table[0, {ncopies}, {lengthares}];

(*derivation of power spectra and cross-power spectra*)
Do[

  ares0[[ll]] = ap0["Values"][[(ll - 1) * tsampl / deltatsima + stpoint
     ;; IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];
```

```
    fousampled0[[ll]] = ResourceFunction["WelchSpectralEstimate"][
     ares0[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
     "OverlapOffset" → None, "SegmentSize" → sizea];
       ,
{ll, ncopies}];

powspmeansampledt0 = Drop[Mean[fousampled0], -1];
(*mean of deflection power spectra of all blocks*)


lengthpsd = Length[powspmeansampledt0]
(*tables of values of simulated psd*)
meandataplot0 = Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt0[[i]][[2]]},
   {i, 1, lengthpsd}]; (*deflection psd*)
```

Out[ ]=

```
2501
```

```
(*Fig 4A plots of unstimulated deflection power spectrum vs
  frequency with channels blocked. Set D=32, fmult=0 in parameters.*)


plotstyleY = Lighter[■];


meandataplot0c = {100, meandataplot0[[1]][[2]]};

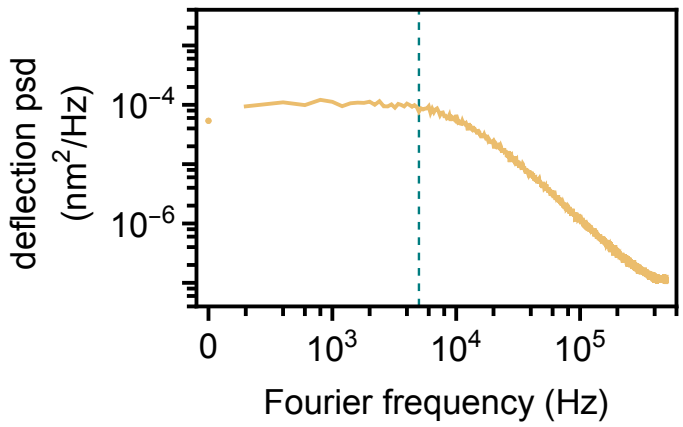


commondirgridlines = {Dashed, Thickness → 0.01 / 2};
yLineStyle = Directive[Darker[■], Dashed, Thickness → 0.01 / 2];

plotrangeY = {{80, 6 * 10^5}, {0.4 * 10^(-7), 4 * 10^(-3)}};

ListLogLogPlot[{meandataplot0},
  PlotStyle → plotstyleY, Joined → True, PlotRange → plotrangeY,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"Fourier frequency (Hz)", "deflection psd \n (nm^2/Hz)"},
  GridLines → {{5000}, {}},
  GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLogPlot[{meandataplot0c}, PlotStyle → plotstyleY]]
```

Out[ ]=



In[ ]:=

```
(*Fig 4B plots of stimulated deflection power spectrum vs
  frequency with channels blocked. Set D=32, fmult=180 in parameters.*)


plotstyleY = Lighter[■];


meandataplot0c = {100, meandataplot0[[1]][[2]]};

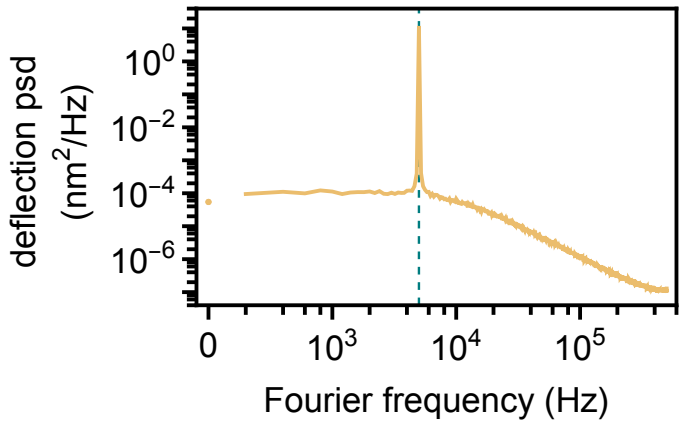


commondirgridlines = {Dashed, Thickness → 0.01 / 2};
yLineStyle = Directive[Darker[■], Dashed, Thickness → 0.01 / 2];

plotrangeY = {{80, 6 * 10^5}, {0.4 * 10^(-7), 4 * 10^(1)}};

ListLogLogPlot[{meandataplot0},
  PlotStyle → plotstyleY, Joined → True, PlotRange → plotrangeY,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"Fourier frequency (Hz)", "deflection psd \n (nm^2/Hz)"},
  GridLines → {{5000}, {}},
  GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLogPlot[{meandataplot0c}, PlotStyle → plotstyleY]]
```

Out[ ]=



In[ ]:=
```
(*Calculation of deflection psd with gating spring cut*)
SeedRandom[47741] (* set seed for reproducibility of results *)

gx = 2 * kbTd * lPd;
y0in = Yrestd;

t1 = AbsoluteTime[];

(*define Ito process for simulations of deflection.
  Fzd multiplied by 180 for stimulated psd,
0 for unstimulated psd. For channels blocked set kGSd term to zero*)
proc = ItoProcess[
  lPd * dY[t] == (-kPd) * (Y[t] - Yupd) * dt - 0 * kGSd * (Y[t] - Yugd - Dvd * Po[t]) * dt +
     Sqrt[gx] * dn1[t] + Fzd * fmult * Sin[2 * Pi * wstd * (t - tstartFd)] * dt &&
   taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd] * dPo[t] ==
    (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] - Po[t]) dt +
     Sqrt[taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd]]
      Sqrt[(Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] (1 - Clip[Po[t], {0.0, 1.0}]) + (1 - Pinf2[
              Yhd, kGSd, Dvd, kbTd, Ncd]) Clip[Po[t], {0.0, 1.0}]) / Ncd] dn2[t],
  {Y[t], Po[t]}, {{Y, Po}, {y0in, Pinfrest}}, t,
{n1 ≈ WienerProcess[], n2 ≈ WienerProcess[]}]

totallength = 1.014;(*length of simulation*)
factot = 4;
deltatsima = 0.000001 / factot;(*time-step of simulation*)
ap = RandomFunction[proc, {0, totallength, deltatsima}];
(*simulation of deflection and curent*)

t2 = AbsoluteTime[];
t2 - t1
```

Out[ ]=
```
RandomGeneratorState[ Method: ExtendedCA
                      State hash: -8833868631830390932 ]
```

Out[ ]=

$$\text{ItoProcess}\Bigg[\Bigg\{\Bigg\{0. - 50\,658.3\,(407. + Y[t]),\ \Bigg(0.\sqrt{\left(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}}\sqrt{\left(\frac{1. - 1.\,\text{Clip}[Po[t], \{0., 1.\}]}{1. + e^{0.115189\,(-413.235 - Y[t])}} + \left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)\text{Clip}[Po[t], \{0., 1.\}]\right)}\Bigg/\left(\left(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\right) + \frac{30\,000.\left(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}} - 1.\,Po[t]\right)}{\left(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}}\Bigg\},$$

$$\Bigg\{\{468.927, 0.\}, \Bigg\{\Bigg(0.\sqrt{\left(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}}\sqrt{\left(\frac{1. - 1.\,\text{Clip}[Po[t], \{0., 1.\}]}{1. + e^{0.115189\,(-413.235 - Y[t])}} + \left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)\text{Clip}[Po[t], \{0., 1.\}]\right)}\Bigg/\left(\left(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\right),$$

$$\Bigg(30.6186\sqrt{\left(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}}\sqrt{\left(\frac{1. - 1.\,\text{Clip}[Po[t], \{0., 1.\}]}{1. + e^{0.115189\,(-413.235 - Y[t])}} + \left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)\text{Clip}[Po[t], \{0., 1.\}]\right)}\Bigg/\left(\left(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\right)\Bigg\}\Bigg\},$$

$$\{Y[t], Po[t]\}\Bigg\}, \{\{Y, Po\}, \{-427, 0.17\}\}, \{t, 0\}\Bigg]$$

Out[ ]=

10.098858

In[ ]:=

```
(*Calculation of deflection power spectrum from simulated deflections*)

Yrestd1 = -407;(*resting position with gating spring cut [nm]*)

ap0 = ap["PathComponent", 1] - Yrestd1;
(* subtract resting position from deflection time series*)
```

```
resampfactor = factot;(*resempling factor for time series*)

randtime100 = 0.005 * 10; (*starting time to ensure that values
 of time series are independent of initial value of simulation*)

(*round to multiple of time-step of simulation to avoid interpolation*)
randtimenm = Round[randtime100, deltatsima];
stpoint = IntegerPart[randtime100 / deltatsima];

deltatsampl = Round[deltatsima * resampfactor, deltatsima];
(*time-step of resampled time series*)
tsampl = 0.005; (* length of a single window of the splitted time series*)
endpoint = IntegerPart[Length[ap["PathComponent", 1]["Values"]] -
    tsampl / deltatsima - stpoint];(*end point of time series*)
tsamplnm = Round[tsampl, deltatsima];
(*round to fraction of delta of time series to avoid interpolation*)
(*determine length of sampled time series*)
b1 = Table[f[i], {i, randtimenm, tsamplnm + randtimenm, deltatsampl}];
lengthares = Length[b1];

ncopiesa = (totallength - randtimenm) / (tsamplnm + deltatsampl);
(*number of non-overlapping windows*)
ncopies = IntegerPart[ncopiesa];
(*round number of overlapping windows to nearest integer *)



ares0 = Table[0, {ncopies}];
(*table of non-overlapping blocks of deflection time series*)
sizea = lengthares;(*length of a single block*)
fousampled0 = Table[0, {ncopies}, {lengthares}];

(*derivation of power spectra and cross-power spectra*)
Do[

  ares0[[ll]] = ap0["Values"][[(ll - 1) * tsampl / deltatsima + stpoint
     ;; IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];



   fousampled0[[ll]] = ResourceFunction["WelchSpectralEstimate"][
    ares0[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
```

```
    "OverlapOffset" → None, "SegmentSize" → sizea];
      ,
{ll, ncopies}];

powspmeansampledt0 = Drop[Mean[fousampled0], -1];
(*mean of deflection power spectra of all blocks*)


lengthpsd = Length[powspmeansampledt0]
(*tables of values of simulated psd*)
meandataplot0 = Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt0[[i]][[2]]},
   {i, 1, lengthpsd}]; (*deflection psd*)
```

Out[ ]=

```
2501
```

```
(*Fig 4A plots of unstimulated deflection power spectrum vs
  frequency with gating spring cut. Set D=32, fmult=180 in parameters.*)


plotstyleY = Lighter[■];


meandataplot0c = {100, meandataplot0[[1]][[2]]};

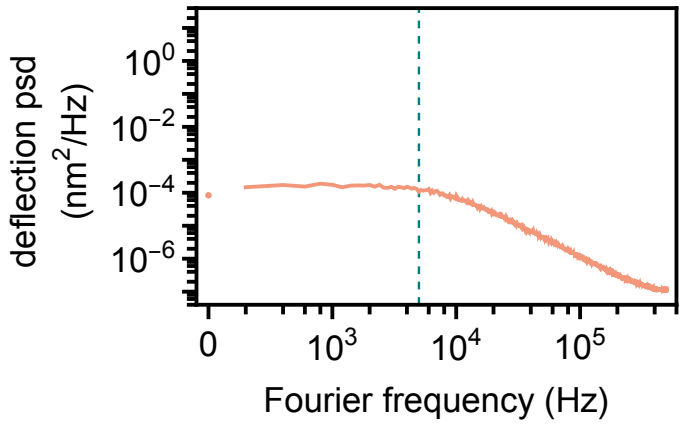


commondirgridlines = {Dashed, Thickness → 0.01 / 2};
yLineStyle = Directive[Darker[■], Dashed, Thickness → 0.01 / 2];

plotrangeY = {{80, 6 * 10^5}, {0.4 * 10^(-7), 4 * 10^(1)}};

ListLogLogPlot[{meandataplot0},
  PlotStyle → plotstyleY, Joined → True, PlotRange → plotrangeY,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"Fourier frequency (Hz)", "deflection psd \n (nm^2/Hz)"},
  GridLines → {{5000}, {}},
  GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLogPlot[{meandataplot0c}, PlotStyle → plotstyleY]]
```

Out[ ]=



```
(*Fig 4B plots of stimulated deflection power spectrum vs
  frequency with gating spring cut. Set D=32, fmult=180 in parameters.*)


plotstyleY = Lighter[■];


meandataplot0c = {100, meandataplot0[[1]][[2]]};

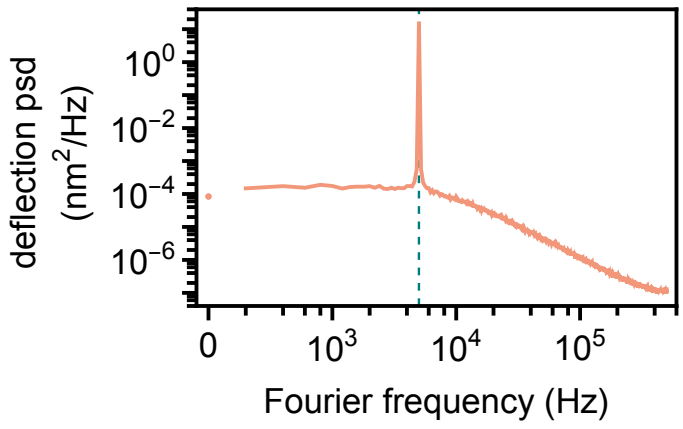


commondirgridlines = {Dashed, Thickness → 0.01 / 2};
yLineStyle = Directive[Darker[■], Dashed, Thickness → 0.01 / 2];

plotrangeY = {{80, 6 * 10^5}, {0.4 * 10^(-7), 4 * 10^(1)}};

ListLogLogPlot[{meandataplot0},
  PlotStyle → plotstyleY, Joined → True, PlotRange → plotrangeY,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicks, None}},
  FrameLabel → {"Fourier frequency (Hz)", "deflection psd \n (nm^2/Hz)"},
  GridLines → {{5000}, {}},
  GridLinesStyle → Directive[■, Dashed, Thickness → 0.01 / 2]];
Show[%, ListLogLogPlot[{meandataplot0c}, PlotStyle → plotstyleY]]
```

Out[ ]=



In[ ]:=

```
(*Calculation of ratio of DC and stimulus-
```

```
 frequency PSDs and ratio of first overtone and stimulus frequency vs stimulus-
 frequency amplitude in the case in which the channels are blocked*)
fmult1 = {0, 1, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110,
   120, 130, 140, 150, 160, 170, 172, 173, 175, 177, 178, 179, 180};

DC = Table[0, {Length[fmult1]}];
st = Table[0, {Length[fmult1]}];
o1 = Table[0, {Length[fmult1]}];

SetSharedVariable[DC];
SetSharedVariable[st];
SetSharedVariable[o1];
SetSharedVariable[fmult];


t1 = AbsoluteTime[];

ParallelDo[


  SeedRandom[47 741];

gx =  2 * kbTd * lPd;
  y0in = Yrestd;



  proc = ItoProcess[
    lPd * ⅆY[t] == (-kPd) * (Y[t] - Yupd) * ⅆt - kGSd * (Y[t] - Yugd - Dvd * 1) * ⅆt +
       Sqrt[gx] * ⅆn1[t] + Fzd * fmult1[[pp]] * Sin[2 * Pi * wstd * (t - tstartFd)] * ⅆt &&
     taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd] * ⅆPo[t] ==
      (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] - Po[t]) ⅆt +
       Sqrt[taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd]] Sqrt[
         (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] (1 - Clip[Po[t], {0.0, 1.0}]) + (1 - Pinf2[
                 Yhd, kGSd, Dvd, kbTd, Ncd]) Clip[Po[t], {0.0, 1.0}]) / Ncd] ⅆn2[t],
    {Y[t], Po[t]}, {{Y, Po}, {y0in, Pinfrest}}, t,
{n1 ≈ WienerProcess[], n2 ≈ WienerProcess[]}];


totallength = 2 * 0.507;
  factot = 4;
  deltatsima = 0.000001 / factot;
   ap = RandomFunction[proc, {0, totallength, deltatsima}];
```

```
  ap0 = ap["PathComponent", 1] - Yrestd;

  ap1 = ap["PathComponent", 2] - Pinfrest;


  resampfactor = 4;

  deltatsampl = Round[deltatsima * resampfactor, deltatsima];

  randtime100 = 0.005 * 10;

  stpoint = IntegerPart[randtime100 / deltatsima];

  tsampl = 0.005;

  randtimenm = Round[randtime100, deltatsima];
  tsamplnm = Round[tsampl, deltatsima];

  b1 = Table[f[i], {i, randtimenm, tsamplnm + randtimenm, deltatsampl}];
lengthares = Length[b1];

  sizea = lengthares;

  Yrestd1 = -421.681; (*channels blocked D=32 nm and Po=0.17*)

  ap0 = ap["PathComponent", 1] - Yrestd1; (*ONLY§ FOR NONLIN CASE*)
  resampfactor = factot;
randtime100 = 0.005 * 10;
randtimenm = Round[randtime100, deltatsima];
stpoint = IntegerPart[randtime100 / deltatsima];
deltatsampl = Round[deltatsima * resampfactor, deltatsima];
tsampl = 0.005;
endpoint = IntegerPart[
    Length[ap["PathComponent", 1]["Values"]] - tsampl / deltatsima - stpoint];
tsamplnm = Round[tsampl, deltatsima];
b1 = Table[f[i], {i, randtimenm, tsamplnm + randtimenm, deltatsampl}];
lengthares = Length[b1];
Clear[ares];
totallength2 = totallength;
ncopiesa = (totallength2 - randtimenm) / (tsamplnm + deltatsampl);
ncopies = IntegerPart[ncopiesa];
ares0 = Table[0, {ncopies}];
sizea = lengthares;
```

```
fousampled0 = Table[0, {ncopies}, {lengthares}];


  Do[

   ares0[[ll]] = ap0["Values"][[(ll - 1) * tsampl / deltatsima + stpoint ;;
       IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];



     fousampled0[[ll]] = ResourceFunction["WelchSpectralEstimate"][
     ares0[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
     "OverlapOffset" → None, "SegmentSize" → sizea];


      ,
{ll, ncopies}];
  powspmeansampledt0 = Drop[Mean[fousampled0], -1];



  lengthpsd = Length[powspmeansampledt0];


  meandataplot0 =
   Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt0[[i]][[2]]}, {i, 1, lengthpsd}];



  DC[[pp]] = meandataplot0[[1]][[2]];
  st[[pp]] = meandataplot0[[26]][[2]];
  o1[[pp]] = meandataplot0[[51]][[2]];
  ,
  {pp, 1, Length[fmult1]}];




t2 = AbsoluteTime[];
t2 - t1
```

Out[ ]=
```
80.950566
```

In[ ]:=
```
DCovst = Table[{fmult1[[i]], DC[[i]] / st[[i]]}, {i, 1, Length[fmult1]}];
o1ovst = Table[{fmult1[[i]], o1[[i]] / st[[i]]}, {i, 1, Length[fmult1]}];
```

In[ ]:=
```
(*Fig11 C,D DC/SF and O1/SF vs stimulus-force amplitude for channels blocked*)
plotrange1 = {Full, Full};
plotstyle1 = Lighter[■];
framelabels1 = {{"DC/SF", None}, {"stimulus-force amplitude (pN)", None}};

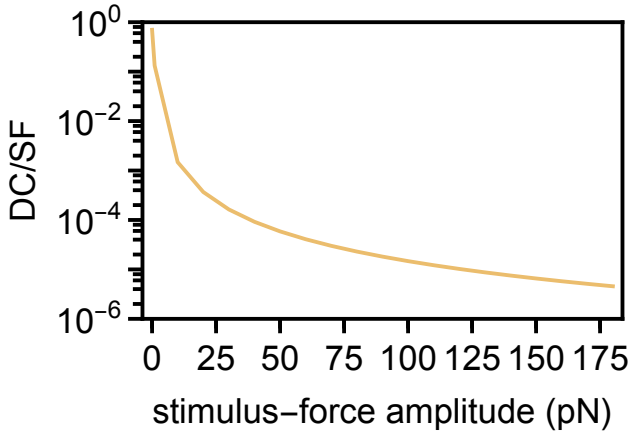

plotrange2 = {{10^-1, 11}, Full};
plotstyle2 = Lighter[■];
framelabels2 = {{"O1/SF", None}, {"stimulus-force amplitude (pN)", None}};


Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1000, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1000, 25}];
XTicks = Union[Xticks1, Xticks3];

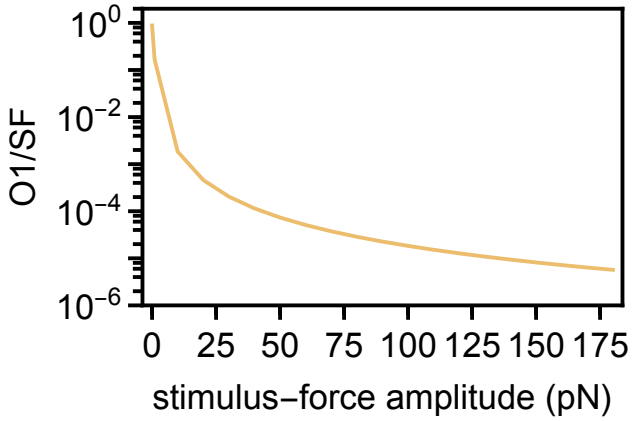

ListLogPlot[{DCovst}, Joined → True, PlotRange → {All, {10^-6, 1}},
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyle1, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabels1, FrameTicks → {{YTicks, None}, {XTicks, None}}]

ListLogPlot[{o1ovst}, PlotRange → {All, {10^-6, 2 * 1}}, Joined → True,
 PlotRange → plotrange2, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyle2,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabels2,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

Out[ ]=



Out[ ]=



In[ ]:=

```
(*Calculation of ratio of DC and stimulus-
 frequency PSDs and ratio of first overtone and stimulus frequency vs stimulus-
 frequency amplitude in the case in which the gating spring is cut*)
fmult1 = {0, 1, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110,
   120, 130, 140, 150, 160, 170, 172, 173, 175, 177, 178, 179, 180};

DC = Table[0, {Length[fmult1]}];
st = Table[0, {Length[fmult1]}];
o1 = Table[0, {Length[fmult1]}];

SetSharedVariable[DC];
SetSharedVariable[st];
SetSharedVariable[o1];
SetSharedVariable[fmult];
```

```
t1 = AbsoluteTime[];

ParallelDo[


  SeedRandom[47741];

gx =  2 * kbTd * lPd;
  y0in = Yrestd;



  proc = ItoProcess[
    lPd * dY[t] == (-kPd) * (Y[t] - Yupd) * dt - 0 * kGSd * (Y[t] - Yugd - Dvd * Po[t]) * dt +
       Sqrt[gx] * dn1[t] + Fzd * fmult1[[pp]] * Sin[2 * Pi * wstd * (t - tstartFd)] * dt &&
     taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd] * dPo[t] ==
      (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] - Po[t]) dt +
       Sqrt[taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd]] Sqrt[
         (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] (1 - Clip[Po[t], {0.0, 1.0}]) + (1 - Pinf2[
                 Yhd, kGSd, Dvd, kbTd, Ncd]) Clip[Po[t], {0.0, 1.0}]) / Ncd] dn2[t],
    {Y[t], Po[t]}, {{Y, Po}, {y0in, Pinfrest}}, t,
{n1 ≈ WienerProcess[], n2 ≈ WienerProcess[]}];


  totallength = 2 * 0.507;
  factot = 4;
  deltatsima = 0.000001 / factot;
   ap = RandomFunction[proc, {0, totallength, deltatsima}];


  ap0 = ap["PathComponent", 1] - Yrestd;

ap1 = ap["PathComponent", 2] - Pinfrest;


resampfactor = 4;

  deltatsampl = Round[deltatsima * resampfactor, deltatsima];

  randtime100 = 0.005 * 10;

  stpoint = IntegerPart[randtime100 / deltatsima];
```

```
  tsampl = 0.005;

  randtimenm = Round[randtime100, deltatsima];
  tsamplnm = Round[tsampl, deltatsima];

  b1 = Table[f[i], {i, randtimenm, tsamplnm + randtimenm, deltatsampl}];
lengthares = Length[b1];

  sizea = lengthares;

  (*Yrestd=-421.681;*)(*channels blocked D=32 nm and Po=0.17*)
  Yrestd1 = -407; (*gating spring cut D=32 nm and Po=0.17*)
  ap0 = ap["PathComponent", 1] - Yrestd1; (*ONLY§ FOR NONLIN CASE*)
  resampfactor = factot;
randtime100 = 0.005 * 10;
randtimenm = Round[randtime100, deltatsima];
stpoint = IntegerPart[randtime100 / deltatsima];
deltatsampl = Round[deltatsima * resampfactor, deltatsima];
tsampl = 0.005;
endpoint = IntegerPart[
    Length[ap["PathComponent", 1]["Values"]] - tsampl / deltatsima - stpoint];
tsamplnm = Round[tsampl, deltatsima];
b1 = Table[f[i], {i, randtimenm, tsamplnm + randtimenm, deltatsampl}];
lengthares = Length[b1];
Clear[ares];
totallength2 = totallength;
ncopiesa = (totallength2 - randtimenm) / (tsamplnm + deltatsampl);
ncopies = IntegerPart[ncopiesa];
ares0 = Table[0, {ncopies}];
sizea = lengthares;

fousampled0 = Table[0, {ncopies}, {lengthares}];


Do[

   ares0[[ll]] = ap0["Values"][[(ll - 1) * tsampl / deltatsima + stpoint ;;
      IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];



       fousampled0[[ll]] = ResourceFunction["WelchSpectralEstimate"][
     ares0[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
```

```
      "OverlapOffset" → None, "SegmentSize" → sizea];


       ,
{ll, ncopies}];
  powspmeansampledt0 = Drop[Mean[fousampled0], -1];



lengthpsd = Length[powspmeansampledt0];


  meandataplot0 =
   Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt0[[i]][[2]]}, {i, 1, lengthpsd}];



  DC[[pp]] = meandataplot0[[1]][[2]];
  st[[pp]] = meandataplot0[[26]][[2]];
  o1[[pp]] = meandataplot0[[51]][[2]];
  ,
  {pp, 1, Length[fmult1]}];




t2 = AbsoluteTime[];
t2 - t1
```

Out[ ]=

```
81.564760
```

In[ ]:=

```
DCovst = Table[{fmult1[[i]], DC[[i]] / st[[i]]}, {i, 1, Length[fmult1]}];
o1ovst = Table[{fmult1[[i]], o1[[i]] / st[[i]]}, {i, 1, Length[fmult1]}];
```

In[ ]:=

```
(*Fig11 C,D DC/SF and O1/SF vs stimulus-force amplitude for gating spring cut*)
plotrange1 = {Full, Full};
plotstyle1 = Lighter[■];
framelabels1 = {{"DC/SF", None}, {"stimulus-force amplitude (pN)", None}};

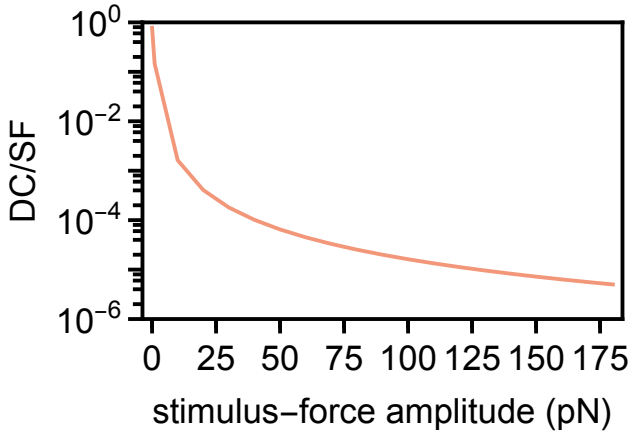

plotrange2 = {{10^-1, 11}, Full};
plotstyle2 = Lighter[■];
framelabels2 = {{"O1/SF", None}, {"stimulus-force amplitude (pN)", None}};


Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1000, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1000, 25}];
XTicks = Union[Xticks1, Xticks3];

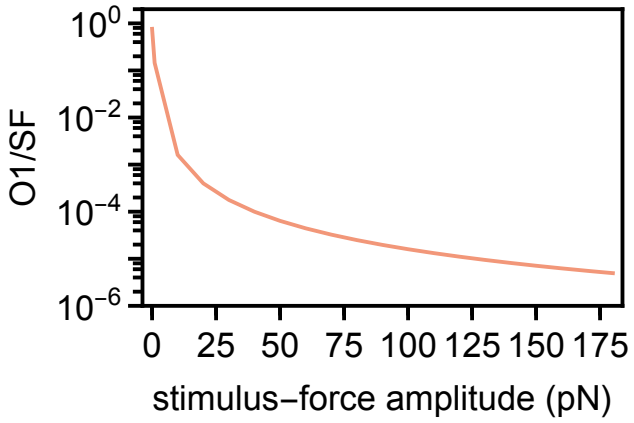

ListLogPlot[{DCovst}, Joined → True, PlotRange → {All, {10^-6, 1}},
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyle1, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabels1, FrameTicks → {{YTicks, None}, {XTicks, None}}]

ListLogPlot[{o1ovst}, PlotRange → {All, {10^-6, 2 * 1}}, Joined → True,
 PlotRange → plotrange2, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyle2,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabels2,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

Out[ ]=



Out[ ]=



In[ ]:=

```
(*Calculation of ratio of DC and stimulus-
 frequency PSDs and ratio of first overtone and stimulus frequency vs stimulus-
 frequency amplitude in the default case*)
fmult1 = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100,
   110, 120, 130, 140, 150, 160, 170, 172, 173, 175, 177, 178, 179, 180};

DC0 = Table[0, {Length[fmult1]}];
st0 = Table[0, {Length[fmult1]}];
o10 = Table[0, {Length[fmult1]}];

DC1 = Table[0, {Length[fmult1]}];
st1 = Table[0, {Length[fmult1]}];
o11 = Table[0, {Length[fmult1]}];

DCv = Table[0, {Length[fmult1]}];
stv = Table[0, {Length[fmult1]}];
o1v = Table[0, {Length[fmult1]}];

DCco = Table[0, {Length[fmult1]}];
stco = Table[0, {Length[fmult1]}];
o1co = Table[0, {Length[fmult1]}];

SetSharedVariable[DC0];
SetSharedVariable[st0];
SetSharedVariable[o10];
SetSharedVariable[DC1];
SetSharedVariable[st1];
SetSharedVariable[o11];
```

```
SetSharedVariable[DCco];
SetSharedVariable[stco];
SetSharedVariable[o1co];
SetSharedVariable[DCv];
SetSharedVariable[stv];
SetSharedVariable[o1v];
SetSharedVariable[fmult];


t1 = AbsoluteTime[];

ParallelDo[

  SeedRandom[47 741];

  gx =  2 * kbTd * lPd;
  y0in = Yrestd;



  proc = ItoProcess[
    lPd * ⅆY[t] == (-kPd) * (Y[t] - Yupd) * ⅆt - kGSd * (Y[t] - Yugd - Dvd * Po[t]) * ⅆt +
       Sqrt[gx] * ⅆn1[t] + Fzd * fmult1[[pp]] * Sin[2 * Pi * wstd * (t - tstartFd)] * ⅆt &&
     taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd] * ⅆPo[t] ==
      (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] - Po[t]) ⅆt +
       Sqrt[taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd]] Sqrt[
         (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] (1 - Clip[Po[t], {0.0, 1.0}]) + (1 - Pinf2[
                 Yhd, kGSd, Dvd, kbTd, Ncd]) Clip[Po[t], {0.0, 1.0}]) / Ncd] ⅆn2[t],
    {Y[t], Po[t]}, {{Y, Po}, {y0in, Pinfrest}}, t,
{n1 ≃ WienerProcess[], n2 ≃ WienerProcess[]}];


  totallength = 2 * 0.507;
  factot = 4;
  deltatsima = 0.000001 / factot;
   ap =  RandomFunction[proc, {0, totallength, deltatsima}];


  ap0 = ap["PathComponent", 1] - Yrestd;

  ap1 = ap["PathComponent", 2] - Pinfrest;


resampfactor = 4;
```

```
  deltatsampl = Round[deltatsima * resampfactor, deltatsima];

  randtime100 = 0.005 * 10;

  stpoint = IntegerPart[randtime100 / deltatsima];

  tsampl = 0.005;

  randtimenm = Round[randtime100, deltatsima];
  tsamplnm = Round[tsampl, deltatsima];

  b1 = Table[f[i], {i, randtimenm, tsamplnm + randtimenm, deltatsampl}];
lengthares = Length[b1];

  sizea = lengthares;


  ap0 = ap["PathComponent", 1] - Yrestd; (*ONLY§ FOR NONLIN CASE*)

    ap1 = ap["PathComponent", 2] - Pinfrest; resampfactor = factot;
randtime100 = 0.005 * 10;
randtimenm = Round[randtime100, deltatsima];
stpoint = IntegerPart[randtime100 / deltatsima];
deltatsampl = Round[deltatsima * resampfactor, deltatsima];
tsampl = 0.005;
endpoint = IntegerPart[
    Length[ap["PathComponent", 1]["Values"]] - tsampl / deltatsima - stpoint];
tsamplnm = Round[tsampl, deltatsima];
b1 = Table[f[i], {i, randtimenm, tsamplnm + randtimenm, deltatsampl}];
lengthares = Length[b1];
Clear[ares];
totallength2 = totallength;
ncopiesa = (totallength2 - randtimenm) / (tsamplnm + deltatsampl);
ncopies = IntegerPart[ncopiesa];
ares0 = Table[0, {ncopies}];
ares1 = Table[0, {ncopies}];
sizea = lengthares;

fousampled0 = Table[0, {ncopies}, {lengthares}];
fousampled1 = Table[0, {ncopies}, {lengthares}];

  cfousampled = Table[0, {ncopies}, {lengthares}];
```

```
Do[

    ares0[[ll]] = ap0["Values"][[(ll - 1) * tsampl / deltatsima + stpoint ;;
       IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];
     ares1[[ll]] = ap1["Values"][[(ll - 1) * tsampl / deltatsima + stpoint ;;
       IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];



        fousampled0[[ll]] = ResourceFunction["WelchSpectralEstimate"][
      ares0[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
      "OverlapOffset" → None, "SegmentSize" → sizea];

    cfousampled[[ll]] = ResourceFunction["WelchSpectralEstimate"][
      ares1[[ll]], ares0[[ll]], sizea / tsampl, "Window" → None,
      "Detrend" → None, "OverlapOffset" → None, "SegmentSize" → sizea];

     fousampled1[[ll]] = ResourceFunction["WelchSpectralEstimate"][
      ares1[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
      "OverlapOffset" → None, "SegmentSize" → sizea];


       ,
{ll, ncopies}];
  powspmeansampledt0 = Drop[Mean[fousampled0], -1];

powspmeansampledt1 = Drop[Mean[fousampled1], -1];
  cpowspmeansampledt = Drop[Re[Mean[cfousampled]], -1];

lengthpsd = Length[powspmeansampledt0];


  meandataplot0 =
   Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt0[[i]][[2]]}, {i, 1, lengthpsd}];
  meandataplot1 =
   Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt1[[i]][[2]]}, {i, 1, lengthpsd}];
  meandataplot1cpsd =
   Table[{cpowspmeansampledt[[i]][[1]], cpowspmeansampledt[[i]][[2]]}, {i, 1, lengthpsd}];
  coher01 = Table[{meandataplot0[[i]][[1]], meandataplot1cpsd[[i]][[2]] /
      Sqrt[(meandataplot0[[i]][[2]] * meandataplot1[[i]][[2]])]}, {i, 1, lengthpsd}];

  fc = 610;
  filtersq1 = Table[{meandataplot0[[i]][[1]],
```

```
    Abs[1 / (1 + I * meandataplot1[[i]][[1]] / fc)]^2}, {i, 1, Length[meandataplot0]}];
meandataplot1filt = Table[{meandataplot0[[i]][[1]],
    filtersq1[[i]][[2]] * meandataplot1[[i]][[2]]}, {i, 1, Length[filtersq1]}];

DC0[[pp]] = {fmult1[[pp]], meandataplot0[[1]][[2]]};
st0[[pp]] = {fmult1[[pp]], meandataplot0[[26]][[2]]};
o10[[pp]] = {fmult1[[pp]], meandataplot0[[51]][[2]]};

DC1[[pp]] = {fmult1[[pp]], meandataplot1[[1]][[2]]};
st1[[pp]] = {fmult1[[pp]], meandataplot1[[26]][[2]]};
o11[[pp]] = {fmult1[[pp]], meandataplot1[[51]][[2]]};

DCv[[pp]] = {fmult1[[pp]], meandataplot1filt[[1]][[2]]};
stv[[pp]] = {fmult1[[pp]], meandataplot1filt[[26]][[2]]};
o1v[[pp]] = {fmult1[[pp]], meandataplot1filt[[51]][[2]]};

DCco[[pp]] = {fmult1[[pp]], coher01[[1]][[2]]};
stco[[pp]] = {fmult1[[pp]], coher01[[26]][[2]]};
o1co[[pp]] = {fmult1[[pp]], coher01[[51]][[2]]};



,
{pp, 1, Length[fmult1]}];
```

```
t2 = AbsoluteTime[];
t2 - t1
```

Out[ ]=

```
118.857721
```

In[ ]:=

In[ ]:=

```
DCovst0 = Table[{fmult1[[i]], DC0[[i]][[2]] / st0[[i]][[2]]}, {i, 1, Length[fmult1]}];
o1ovst0 = Table[{fmult1[[i]], o10[[i]][[2]] / st0[[i]][[2]]}, {i, 1, Length[fmult1]}];
DCovst1 = Table[{fmult1[[i]], DC1[[i]][[2]] / st1[[i]][[2]]}, {i, 1, Length[fmult1]}];
o1ovst1 = Table[{fmult1[[i]], o11[[i]][[2]] / st1[[i]][[2]]}, {i, 1, Length[fmult1]}];
```

In[ ]:=

```
(*Fig11 C,D deflection DC/SF and O1/SF vs stimulus-
 force amplitude for default case*)
plotrange1 = {Full, Full};
plotstyle1 = Lighter[■];
framelabels1 = {{"DC/SF", None}, {"stimulus-force amplitude (pN)", None}};

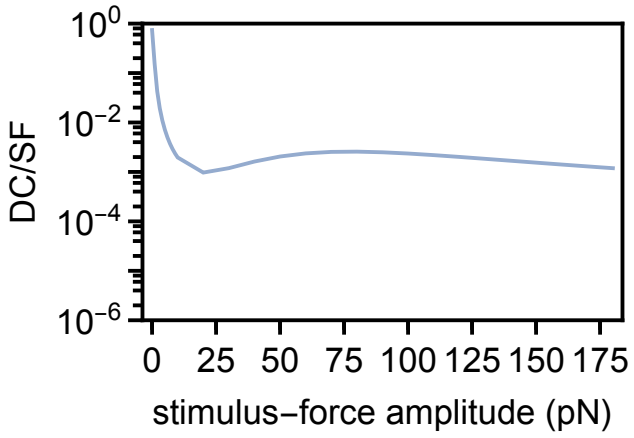

plotrange2 = {{10^-1, 11}, Full};
plotstyle2 = Lighter[■];
framelabels2 = {{"O1/SF", None}, {"stimulus-force amplitude (pN)", None}};


Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1000, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1000, 25}];
XTicks = Union[Xticks1, Xticks3];

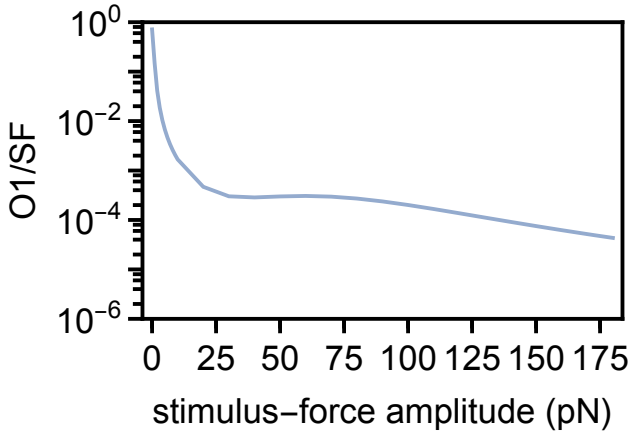

ListLogPlot[{DCovst0}, Joined → True, PlotRange → {All, {10^-6, 1}},
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyle1, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabels1, FrameTicks → {{YTicks, None}, {XTicks, None}}]

ListLogPlot[{o1ovst0}, PlotRange → {All, {10^-6, 1}}, Joined → True,
 PlotRange → plotrange2, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyle2,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabels2,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

Out[ ]=



Out[ ]=



In[ ]:=
```
(*Fig11 C,
D current DC/SF and O1/SF vs stimulus-force amplitude for default case*)
plotrange1 = {Full, Full};
plotstyle1 = Directive[{Dashed, Lighter[■]}];
framelabels1 = {{"DC/SF", None}, {"stimulus-force amplitude (pN)", None}};

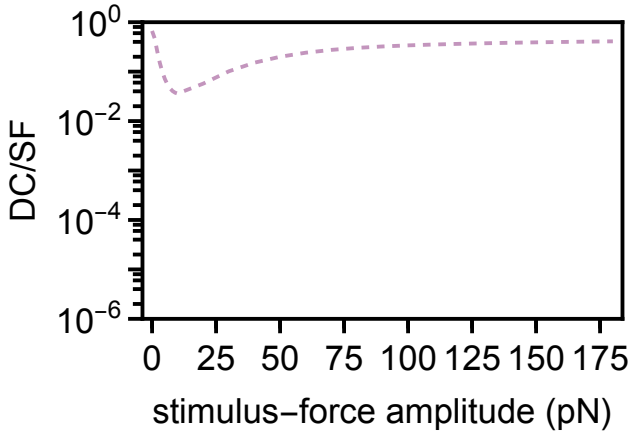

plotrange2 = {{10^-1, 11}, Full};
plotstyle2 = Directive[{Dashed, Lighter[■]}];
framelabels2 = {{"O1/SF", None}, {"stimulus-force amplitude (pN)", None}};


Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1000, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1000, 25}];
XTicks = Union[Xticks1, Xticks3];

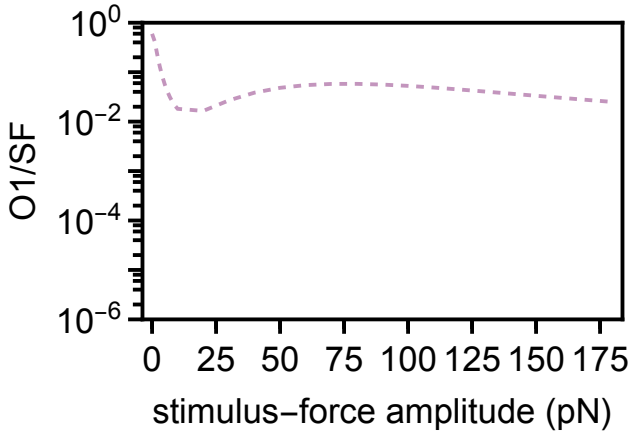

ListLogPlot[{DCovst1}, Joined → True, PlotRange → {All, {10^-6, 1}},
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyle1, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabels1, FrameTicks → {{YTicks, None}, {XTicks, None}}]

ListLogPlot[{o1ovst1}, PlotRange → {All, {10^-6, 1}}, Joined → True,
 PlotRange → plotrange2, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyle2,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabels2,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

Out[ ]=


DC/SF
10^{0}
10^{-2}
10^{-4}
10^{-6}
0 25 50 75 100 125 150 175
stimulus–force amplitude (pN)

Out[ ]=


O1/SF
10^{0}
10^{-2}
10^{-4}
10^{-6}
0 25 50 75 100 125 150 175
stimulus–force amplitude (pN)

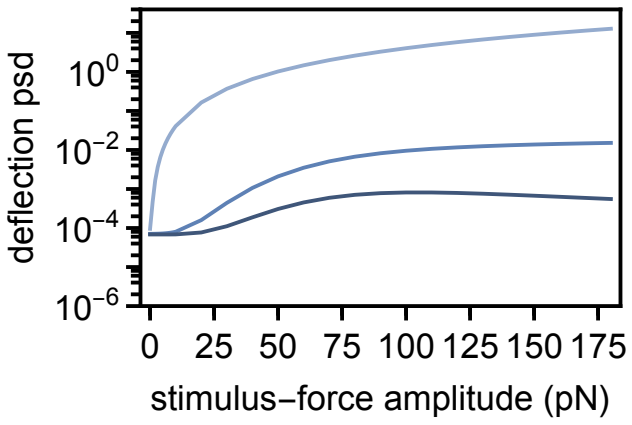
In[ ]:=

```
(*Fig5 A deflection DC,SF and O1 vs stimulus-force amplitude for default case*)
plotrange1 = {Full, Full};
plotstyle1 = {■, Lighter[■], Darker[■]};
framelabels1 = {{"deflection psd", None}, {"stimulus-force amplitude (pN)", None}};


Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1000, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1000, 25}];
XTicks = Union[Xticks1, Xticks3];


ListLogPlot[{DC0, st0, o10}, Joined → True, PlotRange → {All, {10^-6, 4 * 10^1}},
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyle1, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabels1, FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

Out[ ]=

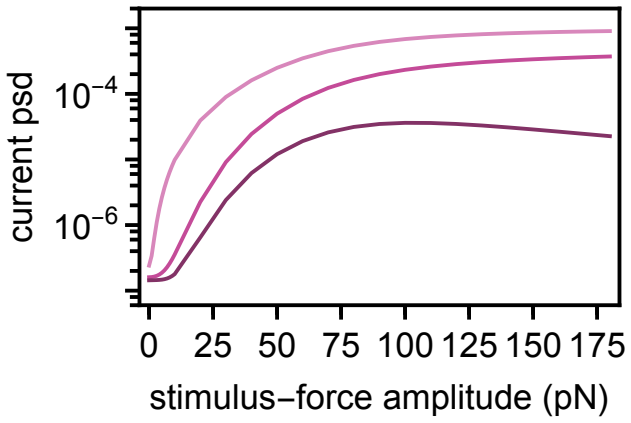


In[ ]:=
```
(*Fig5 B current DC,SF and O1 vs stimulus-force amplitude for default case*)
plotrange1 = {Full, Full};
plotstyle1 = {■, Lighter[■], Darker[■]};
framelabels1 = {{"current psd", None}, {"stimulus-force amplitude (pN)", None}};


Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1000, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1000, 25}];
XTicks = Union[Xticks1, Xticks3];


ListLogPlot[{DC1, st1, o11}, Joined → True, PlotRange → {All, {0.6 * 10^-7, 2 * 10^-3}},
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyle1, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabels1, FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

Out[ ]=



In[ ]:=

```
(*Fig5 C coherency DC,SF and O1 vs stimulus-force amplitude for default case*)

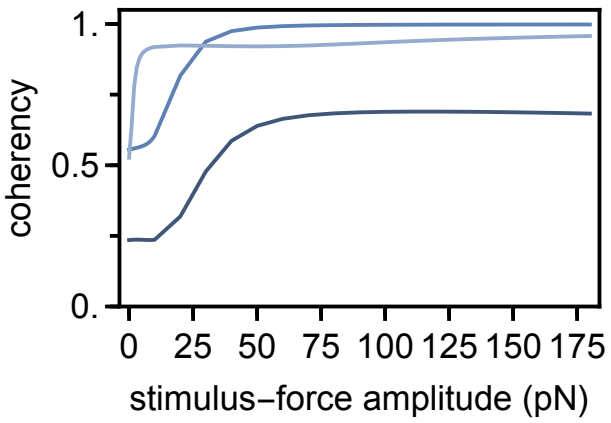
plotrange1 = {Full, Full};
plotstyle1 = {■, Lighter[■], Darker[■]};
framelabels1 = {{"coherency", None}, {"stimulus-force amplitude (pN)", None}};


Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1, 0.5}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1, 0.5 / 2}];
YTicksb = Union[Yticks1, Yticks3];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1000, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1000, 25}];
XTicks = Union[Xticks1, Xticks3];


ListPlot[{DCco, stco, o1co}, Joined → True, PlotRange → {0, 1.05},
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyle1, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabels1, FrameTicks → {{YTicksb, None}, {XTicks, None}}]
```

Out[ ]=



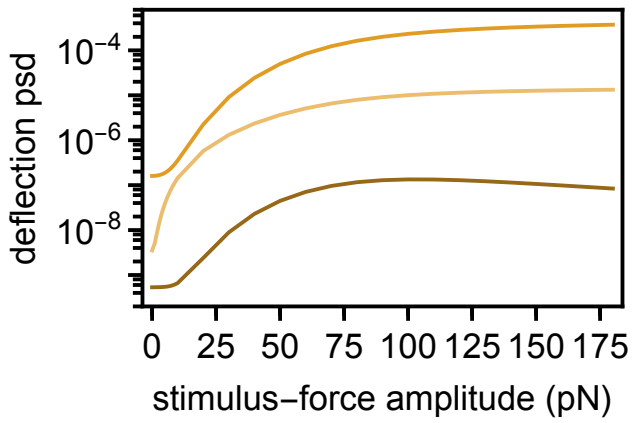
In[ ]:=

```
(*Fig5 B voltage DC,SF and O1 vs stimulus-force amplitude for default case*)
plotrange1 = {Full, Full};
plotstyle1 = {■, Lighter[■], Darker[■]};
framelabels1 = {{"deflection psd", None}, {"stimulus-force amplitude (pN)", None}};


Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1000, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1000, 25}];
XTicks = Union[Xticks1, Xticks3];


ListLogPlot[{DCv, stv, o1v}, Joined → True, PlotRange → {All, {2 * 10^-10, 8 * 10^-4}},
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyle1, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabels1, FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

Out[ ]=



In[ ]:=

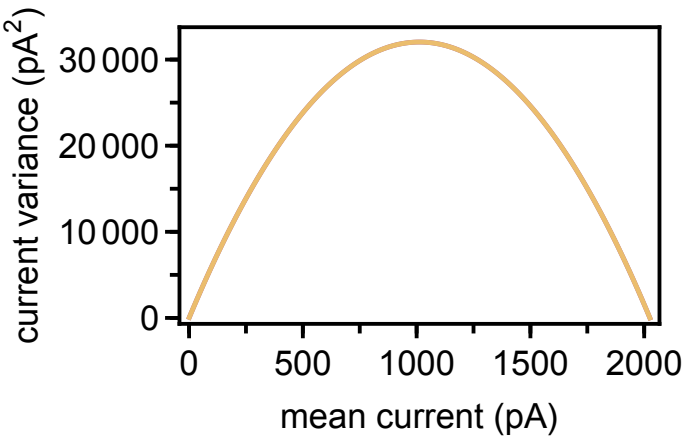
```
(*Calculation of current variance vs mean current*)

pady = 120;
padx = 70;
pad2 = 30;
fntsize = 18;
plotstylevar = {{Thickness → 0.01, ■}, {Thickness → 0.01, Lighter[■]}};
framelabelsvar = {{"current variance (\*SuperscriptBox[\(pA\), \(2\)])", None},
   {"mean current (pA)", None}};
Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 3000, 500}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 3000, 500 / 2}];
XTicks = Union[Xticks1, Xticks3];
Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 50 000, 10 000}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 50 000, 10 000 / 2}];
YTicks = Union[Yticks1, Yticks3];
```

```
(*calculation of current variance for a gating swing of 1.75 nm*)
Dvd = 4.4 * kbTd / (kGSd * Delta);
Imaxd = 2025 / Ncd;
kgd = Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2);
sigma2[Imax_, Imean_, Nc_, kGS_, kgd_, kPd_] :=
  (Imax * Imean - Imean^2 / Nc) (1 + kGS^2 / (kgd (kGS + kPd) - kGS^2));
sigma2val = Table[sigma2[Imaxd, Imean, Ncd, kGSd, kgd, kPd], {Imean, 1, 2025}];
(*table of values of current variance*)
sigmavalkgsz = Table[sigma2[Imaxd, Imean, Ncd, kGSd * 0, kgd, kPd], {Imean, 1, 2025}];
(*table of values of current variance with gating swing equal to zero*)

(*Fig2 current variance vs mean current*)
ListPlot[{sigma2val, sigmavalkgsz}, PlotRange → {All, All}, Joined → True,
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylevar, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsvar, FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

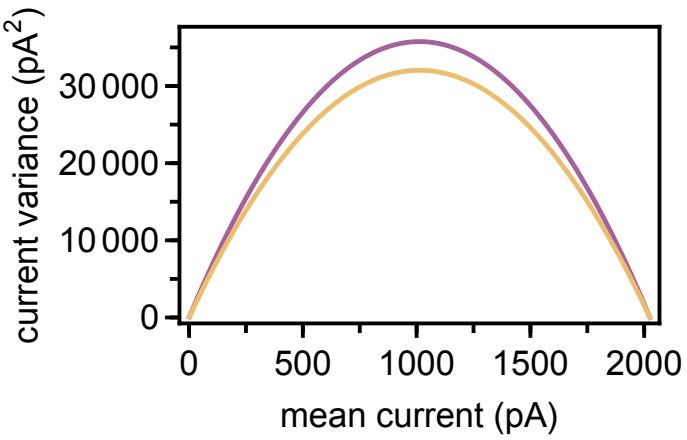
Out[ ]=



In[ ]:=
```
(*calculation of current variance for a gating swing of 32 nm*)
Dvd = 32;
Imaxd = 2025 / Ncd;
kgd = Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2);
sigma2[Imax_, Imean_, Nc_, kGS_, kgd_, kPd_] :=
  (Imax * Imean - Imean^2 / Nc) (1 + kGS^2 / (kgd (kGS + kPd) - kGS^2));
sigma2val = Table[sigma2[Imaxd, Imean, Ncd, kGSd, kgd, kPd], {Imean, 1, 2025}];
(*table of values of current variance*)
sigmavalkgsz = Table[sigma2[Imaxd, Imean, Ncd, kGSd * 0, kgd, kPd], {Imean, 1, 2025}];
(*table of values of current variance with gating swing equal to zero*)

(*Fig2 current variance vs mean current*)
ListPlot[{sigma2val, sigmavalkgsz}, PlotRange → {All, All}, Joined → True,
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylevar, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsvar, FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

Out[ ]=



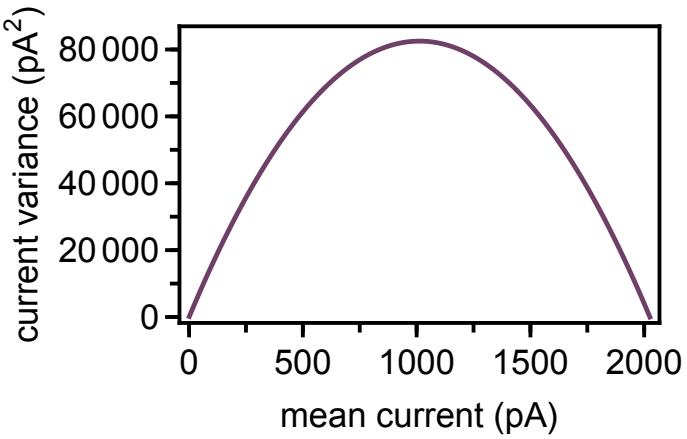
In[ ]:=
```
(*calculation of current variance for a gating swing of 77.6 nm*)
Dvd = 77.6;
Imaxd = 2025 / Ncd;
kgd = Ncd * kbTd / (Pinfrest (1 - Pinfrest) * Dvd^2);
sigma2[Imax_, Imean_, Nc_, kGS_, kgd_, kPd_] :=
  (Imax * Imean - Imean^2 / Nc) (1 + kGS^2 / (kgd (kGS + kPd) - kGS^2));
sigma2val = Table[sigma2[Imaxd, Imean, Ncd, kGSd, kgd, kPd], {Imean, 1, 2025}];
sigmavalkgsz = Table[sigma2[Imaxd, Imean, Ncd, kGSd * 0, kgd, kPd], {Imean, 1, 2025}];

Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 3000, 500}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 3000, 500 / 2}];
XTicks = Union[Xticks1, Xticks3];
Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 100 000, 20 000}];
Yticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 100 000, 20 000 / 2}];
YTicks = Union[Yticks1, Yticks3];

plotstylevar = {{Thickness → 0.01, Darker[■]}, {Thickness → 0.01, Lighter[■]}};

(*Fig2 current variance vs mean current*)
ListPlot[{sigma2val}, PlotRange → {All, All}, Joined → True,
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylevar, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsvar, FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

Out[ ]=



In[ ]:=
```
(*Calculation of SNRpeaks*)
(*Calculation of fluctuation power with and without fluctuation
 suppression. Gating swing can be set to 1.75 nm or 32 nm. *)
t1 = AbsoluteTime[];
fmult = {0, 1, 5, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150,
```

```
   160, 170, 172, 173, 175, 177, 178, 179, 180}; (*table of values of force-
 stimulus amplitude for calculation of fluctuation power*)
areaback0 = Table[0, {Length[fmult]}];
areatot0 = Table[0, {Length[fmult]}];
areaback1 = Table[0, {Length[fmult]}];
areatot1 = Table[0, {Length[fmult]}];
areaback1filt = Table[0, {Length[fmult]}];
areatot1filt = Table[0, {Length[fmult]}];
SetSharedVariable[areaback0];
SetSharedVariable[areatot0];
SetSharedVariable[areaback1];
SetSharedVariable[areatot1];
SetSharedVariable[areaback1filt];
SetSharedVariable[areatot1filt];
SetSharedVariable[fmult];



ParallelDo[
  SeedRandom[47 741];

gx =  2 * kbTd * lPd;
  y0in = Yrestd;

  proc = ItoProcess[
    lPd * dY[t] == (-kPd) * (Y[t] - Yupd) * dt - kGSd * (Y[t] - Yugd - Dvd * Po[t]) * dt +
       Sqrt[gx] * dn1[t] + fmult[[pp]] * Fzd * Sin[2 * Pi * wstd * (t - tstartFd)] * dt &&
     taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd] * dPo[t] ==
      (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] - Po[t]) dt +
       Sqrt[taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd]] Sqrt[
         (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] (1 - Clip[Po[t], {0.0, 1.0}]) + (1 - Pinf2[
                 Yhd, kGSd, Dvd, kbTd, Ncd]) Clip[Po[t], {0.0, 1.0}]) / Ncd] dn2[t],
    {Y[t], Po[t]}, {{Y, Po}, {y0in, Pinfrest}}, t,
{n1 ≈ WienerProcess[], n2 ≈ WienerProcess[]}];

  totallength = 2 * 0.507;
  factot = 4;
  deltatsima = 0.000001 / factot;
   ap = RandomFunction[proc, {0, totallength, deltatsima}];

  ap0 = ap["PathComponent", 1] - Yrestd;

  ap1 = ap["PathComponent", 2] - Pinfrest;
  resampfactor = factot;
```

```
randtime100 = 0.005 * 10;
randtimenm = Round[randtime100, deltatsima];
stpoint = IntegerPart[randtime100 / deltatsima];

deltatsampl = Round[deltatsima * resampfactor, deltatsima];
tsampl = 0.005;
endpoint = IntegerPart[
    Length[ap["PathComponent", 1]["Values"]] - tsampl / deltatsima - stpoint];
tsamplnm = Round[tsampl, deltatsima];
b1 = Table[f[i], {i, randtimenm, tsamplnm + randtimenm, deltatsampl}];
lengthares = Length[b1];

Clear[ares];
totallength2 = totallength;

ncopiesa = (totallength2 - randtimenm) / (tsamplnm + deltatsampl);
ncopies = IntegerPart[ncopiesa];
ares0 = Table[0, {ncopies}]; (*a resampled*)
ares1 = Table[0, {ncopies}];
sizea = lengthares;

fousampled0 = Table[0, {ncopies}, {lengthares}];
fousampled1 = Table[0, {ncopies}, {lengthares}];
cfousampled = Table[0, {ncopies}, {lengthares}];
Do[

   ares0[[ll]] = ap0["Values"][[(ll - 1) * tsampl / deltatsima + stpoint ;;
      IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];
    ares1[[ll]] = ap1["Values"][[(ll - 1) * tsampl / deltatsima + stpoint ;;
      IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];

       fousampled0[[ll]] = ResourceFunction["WelchSpectralEstimate"][
     ares0[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
     "OverlapOffset" → None, "SegmentSize" → sizea];
    cfousampled[[ll]] = ResourceFunction["WelchSpectralEstimate"][
     ares1[[ll]], ares0[[ll]], sizea / tsampl, "Window" → None,
     "Detrend" → None, "OverlapOffset" → None, "SegmentSize" → sizea];
    fousampled1[[ll]] = ResourceFunction["WelchSpectralEstimate"][
     ares1[[ll]], sizea / tsampl, "Window" → None, "Detrend" → None,
     "OverlapOffset" → None, "SegmentSize" → sizea];
   ,

{ll, ncopies}];
```

```
  powspmeansampledt0 = Drop[Mean[fousampled0], -1];

  powspmeansampledt1 = Drop[Mean[fousampled1], -1];

cpowspmeansampledt = Drop[Re[Mean[cfousampled]], -1];
  lengthpsd = Length[powspmeansampledt0];


  meandataplot0 = Table[{0, 0}, {i, 1, lengthpsd}];
  meandataplot1 = Table[{0, 0}, {i, 1, lengthpsd}];
meandataplot1cpsd = Table[{0, 0}, {i, 1, lengthpsd}];


  meandataplot0 =
   Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt0[[i]][[2]]}, {i, 1, lengthpsd}];
  meandataplot1 =
   Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt1[[i]][[2]]}, {i, 1, lengthpsd}];
meandataplot1cpsd =
   Table[{cpowspmeansampledt[[i]][[1]], cpowspmeansampledt[[i]][[2]]}, {i, 1, lengthpsd}];
  fc = 610;
  filtersq1 = Table[{meandataplot0[[i]][[1]],
     Abs[1 / (1 + I * meandataplot1[[i]][[1]] / fc)]^2}, {i, 1, Length[meandataplot0]}];
  meandataplot1filt = Table[{meandataplot0[[i]][[1]],
     filtersq1[[i]][[2]] * meandataplot1[[i]][[2]]}, {i, 1, Length[filtersq1]}];

  (*calculation of background area for deflection psd*)
  data0 = Table[meandataplot0[[nn]][[2]], {nn, 2, Length[meandataplot0]}];
  dataest0 = MedianFilter[data0, 15]; (*use 15 point median filter
   for estimate of background data from freq 2.8 kHz upward (point 15) *)
  dataestwfreq0 = Table[{meandataplot0[[nn + 1]][[1]], dataest0[[nn]]},
    {nn, 8 + 7, Length[meandataplot0] - 8 + 7}]; (*for frequencies lower than 2.8 kHz,
  the background psd is set equal to the actual psd*)
  data0part = Table[meandataplot0[[nn]][[2]], {nn, 2, 8 + 7}];
  dataest0part = data0part;
  dataestwfreq0part =
   Table[{meandataplot0[[nn + 1]][[1]], dataest0part[[nn]]}, {nn, 1, 8 + 7 - 1}];
  dataest0tot = Union[
    {{meandataplot0[[1]][[1]], meandataplot0[[2]][[2]]}}, dataestwfreq0part, dataestwfreq0];
  dataest0tot[[1]][[2]] = 0.00007091605149763008`;
  (*DC value is set equal to the corresponding value for the unstimulated psd*)
  areaback0[[pp]] = {fmult[[pp]],
    Integrate[Interpolation[dataest0tot, InterpolationOrder → 1][x], {x, 0, 5 * 10^5}]};
  (*area of the background deflection psd is obtained using
   a linear interpolation between points*)
```

```
  areatot0[[pp]] = {fmult[[pp]],
    Integrate[Interpolation[meandataplot0, InterpolationOrder → 1][x], {x, 0, 5 * 10^5}]};
  (*area of stimulated psd is obtained using a linear interpolation between points*)

  (*calculation of background area for current psd*)
  data1 = Table[meandataplot1[[nn]][[2]], {nn, 2, Length[meandataplot1]}];
  dataest1 = MedianFilter[data1, 15];
  dataestwfreq1 = Table[{meandataplot1[[nn + 1]][[1]], dataest1[[nn]]},
    {nn, 8 + 7, Length[meandataplot1] - 8 + 7}];
  data1part = Table[meandataplot1[[nn]][[2]], {nn, 2, 8 + 7}];
  dataest1part = data1part;
  dataestwfreq1part =
   Table[{meandataplot1[[nn + 1]][[1]], dataest1part[[nn]]}, {nn, 1, 8 + 7 - 1}];
  dataest1tot1 = Union[
    {{meandataplot1[[1]][[1]], meandataplot1[[2]][[2]]}}, dataestwfreq1part, dataestwfreq1];
  areaback1[[pp]] =
   {fmult[[pp]], Integrate[Interpolation[dataest1tot1, InterpolationOrder → 1][x],
      {x, 200, 5 * 10^5}] + meandataplot1[[2]][[2]] * meandataplot1[[2]][[1]] / 2};
  areatot1[[pp]] = {fmult[[pp]],
    Integrate[Interpolation[meandataplot1, InterpolationOrder → 1][x], {x, 0, 5 * 10^5}]};

  (*calculation of background area of voltage psd*)
  data1filt = Table[meandataplot1filt[[nn]][[2]], {nn, 2, Length[meandataplot1filt]}];
  dataest1filt = MedianFilter[data1filt, 15];
  dataestwfreq1filt = Table[{meandataplot1filt[[nn + 1]][[1]], dataest1filt[[nn]]},
    {nn, 8 + 7, Length[meandataplot1filt] - 8 + 7}];
  data1filtpart = Table[meandataplot1filt[[nn]][[2]], {nn, 2, 8 + 7}];
  dataest1filtpart = data1filtpart;
  dataestwfreq1filtpart =
   Table[{meandataplot1filt[[nn + 1]][[1]], dataest1filtpart[[nn]]}, {nn, 1, 8 + 7 - 1}];
  dataest1totfilt = Union[{{meandataplot1filt[[1]][[1]], meandataplot1filt[[2]][[2]]}},
    dataestwfreq1filtpart, dataestwfreq1filt];
  areaback1filt[[pp]] =
   {fmult[[pp]], Integrate[Interpolation[dataest1totfilt, InterpolationOrder → 1][x],
      {x, 200, 5 * 10^5}]} + meandataplot1filt[[2]][[2]] * meandataplot1filt[[2]][[1]] / 2;
  areatot1filt[[pp]] = {fmult[[pp]], Integrate[
     Interpolation[meandataplot1filt, InterpolationOrder → 1][x], {x, 0, 5 * 10^5}]};

,
  {pp, 1, Length[fmult]}];

t2 = AbsoluteTime[];
t2 - t1
```

Out[ ]=

```
94.785093
```

In[ ]:=

```
areapeacks0 = Table[{0}, {i, 1, Length[areatot0]}];
areapeacks1 = Table[{0}, {i, 1, Length[areatot1]}];

Do[areapeacks0[[pp]] = {areatot0[[pp]][[1]], areatot0[[pp]][[2]] - areaback0[[pp]][[2]]},
  {pp, 1, Length[areatot0]}];
Do[areapeacks1[[pp]] = {areatot1[[pp]][[1]], areatot1[[pp]][[2]] - areaback1[[pp]][[2]]},
  {pp, 1, Length[areatot1]}];

snrdefp = Table[{areapeacks0[[pp]][[1]], areapeacks0[[pp]][[2]] / 1.9630761547464095},
   {pp, 2, Length[areapeacks0]}] ;(*deflection SNR peaks is defined
 as the power of the peaks divided by the power of unstimulated psd*)
snrcurrp = Table[{areapeacks0[[pp]][[1]], areapeacks1[[pp]][[2]] / 0.004967582098481333},
   {pp, 2, Length[areapeacks0]}];(*current SNR peaks is defined as
 the power of the peaks divided by the power of unstimulated psd*)
```

In[ ]:=

```
(*plot settings*)

Remove[XTicks]
Remove[YTicks]
TickLength = 0.02;
TickLength1 = 0.03;
Yticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, -24, 10, 2}];
Yticks2 = Table[{10^k, Null, {0, TickLength1}}, {k, -23, 10, 2}];
TickLength2 = 0.02;
aa = Table[0, {38}, {4}];
For[k = 1, k ≤ 38, k++,
aa[[k]] = Table[{l * 10^(-k + 10), , {0, TickLength2}}, {l, 2, 8, 2}];
]
YTicks = Union[Yticks1, Yticks2, aa[[1]], aa[[2]], aa[[3]],
   aa[[4]], aa[[5]], aa[[6]], aa[[7]], aa[[8]], aa[[9]], aa[[10]], aa[[11]], aa[[12]],
   aa[[13]], aa[[14]], aa[[15]], aa[[16]], aa[[17]], aa[[18]], aa[[19]], aa[[20]],
   aa[[21]], aa[[22]], aa[[23]], aa[[24]], aa[[25]], aa[[26]], aa[[27]], aa[[28]]];

pady = 100;
padx = 70;
pad2 = 30;
fntsize = 18;
```

In[ ]:=
```
(*Fig S5 plots of SNRpeaks for a gating swing of 32 nm and
  resting current of 0.17. Set D=32 and Prest=0.17 in parameters. *)

plotrangeSNRY = {Full, Full};
plotstyleSNRY = {{Thickness → 0.01, Lighter[■]}, {Thickness → 0.01, Darker[■]}};
framelabelsSNRY =
  {{"deflection SNR", None}, {"stimulus-force amplitude (pN)", None}};

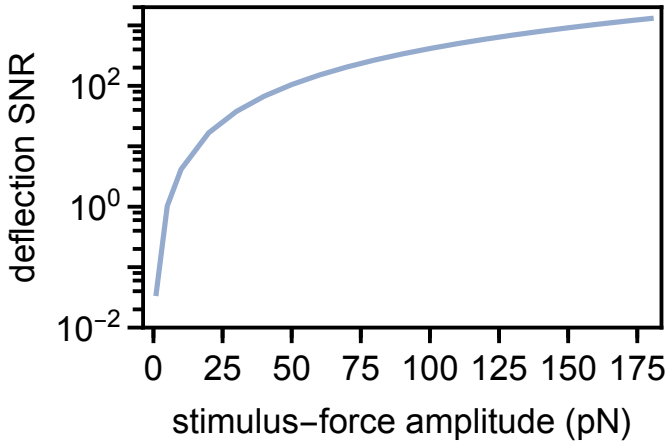

plotrangeSNRP = {{10^-1, 11}, Full};
plotstyleSNRP = {{Thickness → 0.01, Lighter[■]}, {Thickness → 0.01, Darker[■]}};
framelabelsSNRP =
  {{"current or voltage SNR", None}, {"stimulus-force amplitude (pN)", None}};


Xticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 1000, 25}];
Xticks3 = Table[{k, Null, {0, 0.02}}, {k, 0, 1000, 25}];
XTicks = Union[Xticks1, Xticks3];

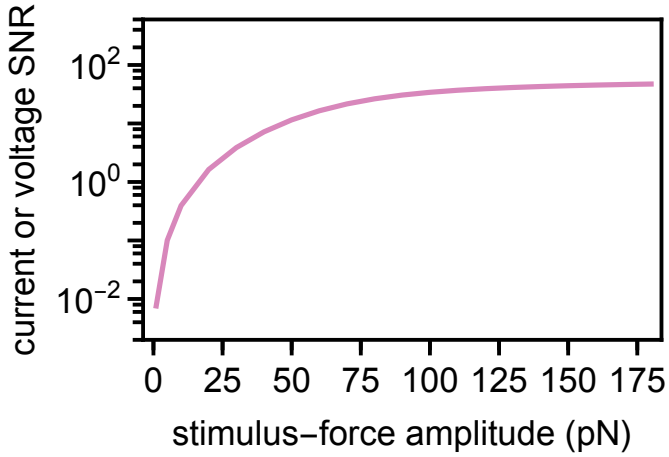

ListLogPlot[{snrdefp}, Joined → True, PlotRange → {All, {10^-2, 2 * 10^3}},
 ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
 LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleSNRY, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → framelabelsSNRY, FrameTicks → {{YTicks, None}, {XTicks, None}}]

ListLogPlot[{snrcurrp}, PlotRange → {All, {0.2 * 10^-2, 6 * 10^2}}, Joined → True,
 PlotRange → plotrangeSNRP, ImagePadding → {{pady, pad2}, {padx, pad2}},
 ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleSNRP,
 Background → White, Frame → {True, True, True, True},
 FrameStyle → Directive[Black, Thick], FrameLabel → framelabelsSNRP,
 FrameTicks → {{YTicks, None}, {XTicks, None}}]
```

Out[ ]=



Out[ ]=




```
(*Derivation of current and deflection
 for histeresis curves (noise terms set to zero)*)
SeedRandom[47741] (* set seed for reproducibility of results *)

gx =  2 * kbTd * lPd;
y0in = Yrestd;

t1 = AbsoluteTime[];

(*define Ito process for simulations of deflection and current
 Fzd multiplied by 180 for stimulated psd, 0 for unstimulated psd*)
```

```
proc = ItoProcess[
  lPd * ⅆY[t] == (-kPd) * (Y[t] - Yupd) * ⅆt - kGSd * (Y[t] - Yugd - Dvd * Po[t]) * ⅆt +
     0 * Sqrt[gx] * ⅆn1[t]  + Fzd * fmult * Sin[2 * Pi * wstd * (t - tstartFd)] * ⅆt &&
   taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd] * ⅆPo[t] ==
    (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] - Po[t]) ⅆt +
     0 * Sqrt[taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd]]
      Sqrt[(Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] (1 - Clip[Po[t], {0.0, 1.0}]) + (1 - Pinf2[
               Yhd, kGSd, Dvd, kbTd, Ncd]) Clip[Po[t], {0.0, 1.0}]) / Ncd] ⅆn2[t],
  {Y[t], Po[t]}, {{Y, Po}, {y0in, Pinfrest}}, t,
{n1 ≃ WienerProcess[], n2 ≃ WienerProcess[]}]

totallength = 1.014;(*length of simulation*)
factot = 4;
deltatsima = 0.000001 / factot;(*time-step of simulation*)
ap = RandomFunction[proc, {0, totallength, deltatsima}];
(*simulation of deflection and curent*)

t2 = AbsoluteTime[];
t2 - t1


(*Calculation of power spectra from simulated deflection and current*)

(*Yrestd=-412.681; (*resting position of IHB with channels blocked [nm]*)
Yrestd=-407;(*resting position with gating spring cut [nm]*)*)

ap0 = ap["PathComponent", 1] - Yrestd;
(* subtract resting position from deflection time series*)

ap1 = ap["PathComponent", 2] - Pinfrest;
(*subtract resting current from current time series*)

resampfactor = factot;(*resempling factor for time series*)

randtime100 = 0.005 * 10; (*starting time to ensure that values
 of time series are independent of initial value of simulation*)

(*round to multiple of time-step of simulation to avoid interpolation*)
randtimenm = Round[randtime100, deltatsima];
stpoint = IntegerPart[randtime100 / deltatsima];

deltatsampl = Round[deltatsima * resampfactor, deltatsima];
(*time-step of resampled time series*)
tsampl = 0.005; (* length of a single window of the splitted time series*)
```

```
endpoint = IntegerPart[Length[ap["PathComponent", 1]["Values"]] -
    tsampl / deltatsima - stpoint];(*end point of time series*)
tsamplnm = Round[tsampl, deltatsima];
(*round to fraction of delta of time series to avoid interpolation*)
(*determine length of sampled time series*)
b1 = Table[f[i], {i, randtimenm, tsamplnm + randtimenm, deltatsampl}];
lengthares = Length[b1];

ncopiesa = (totallength - randtimenm) / (tsamplnm + deltatsampl);
(*number of non-overlapping windows*)
ncopies = IntegerPart[ncopiesa];
(*round number of overlapping windows to nearest integer *)



ares0 = Table[0, {ncopies}];
(*table of non-overlapping blocks of deflection time series*)
ares1 = Table[0, {ncopies}];
(*table of non-overlapping blocks of deflection time series*)
sizea = lengthares;(*length of a single block*)
fousampled0 = Table[0, {ncopies}, {lengthares}];
(*table of deflection power spectrum of each block*)
fousampled1 = Table[0, {ncopies}, {lengthares}];
(*table of current power spectrum of each block*)
cfousampled = Table[0, {ncopies}, {lengthares}];
(*table of cross-power spectrum of each block*)

(*derivation of power spectra and cross-power spectra*)
Do[

  ares0[[ll]] = ap0["Values"][[(ll - 1) * tsampl / deltatsima + stpoint ;;
      IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];
    ares1[[ll]] = ap1["Values"][[(ll - 1) * tsampl / deltatsima + stpoint ;;
      IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];



   fousampled0[[ll]] = ResourceFunction["WelchSpectralEstimate"][ares0[[ll]],
    sizea / tsampl, "Window" → None, "Detrend" → None, "OverlapOffset" → None,
    "SegmentSize" → sizea]; (*deflection power spectrum of single block*)
    cfousampled[[ll]] = ResourceFunction["WelchSpectralEstimate"][
    ares1[[ll]], ares0[[ll]], sizea / tsampl, "Window" → None,
    "Detrend" → None, "OverlapOffset" → None, "SegmentSize" → sizea];
```

```
  (*cross-power spectrum of single block*)
    fousampled1[[ll]] =
   ResourceFunction["WelchSpectralEstimate"][ares1[[ll]], sizea / tsampl,
    "Window" → None, "Detrend" → None, "OverlapOffset" → None, "SegmentSize" → sizea];
  (*current power spectrum of single block*)
      ,
{ll, ncopies}];

powspmeansampledt0 = Drop[Mean[fousampled0], -1];
(*mean of deflection power spectra of all blocks*)

powspmeansampledt1 = Drop[Mean[fousampled1], -1];
(*mean of current power spectra of all blocks*)

cpowspmeansampledt = Drop[Re[Mean[cfousampled]], -1];
(*mean of cross-power spectra of all blocks*)

lengthpsd = Length[powspmeansampledt0]
(*tables of values of simulated psd*)
meandataplot0 = Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt0[[i]][[2]]},
   {i, 1, lengthpsd}]; (*deflection psd*)
meandataplot1 = Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt1[[i]][[2]]},
   {i, 1, lengthpsd}]; (*current psd*)
meandataplot1cpsd = Table[{cpowspmeansampledt[[i]][[1]], cpowspmeansampledt[[i]][[2]]},
   {i, 1, lengthpsd}]; (*cross-psd*)
coher01 = Table[{meandataplot0[[i]][[1]],
    meandataplot1cpsd[[i]][[2]] / Sqrt[(meandataplot0[[i]][[2]] * meandataplot1[[i]][[2]])]},
   {i, 1, lengthpsd}]; (*coherency*)

fc = 610; (*corner frequency of the hair-cell membrane [Hz]*)
filtersq1 =
  Table[{meandataplot0[[i]][[1]], Abs[1 / (1 + I * meandataplot1[[i]][[1]] / fc)]^2},
   {i, 1, Length[meandataplot0]}];(*table of values of the filter*)
meandataplot1filt =
  Table[{meandataplot0[[i]][[1]], filtersq1[[i]][[2]] * meandataplot1[[i]][[2]]},
   {i, 1, Length[filtersq1]}];(*table of values of the voltage psd*)




(*Calculation of histeresis curves for a gating swing of 32 nm. Set D=32 nm*)
fmult = {180};
```

```
ap1 = ap["PathComponent", 2];
ap0 = ap["PathComponent", 1];
Povals = ap1["Values"][[10^5 + 67 ;; 10^5 + 67 + 800 ;; 1]];
(* table of values of current in one cycle of sinusoidal stimulus force*)
Yvals = ap0["Values"][[10^5 + 67 ;; 10^5 + 67 + 800 ;; 1]];
(* table of values of deflections in one cycle of sinusoidal stimulus force*)
Fval = Table[(Fzd * fmult[[Length[fmult]]] * Sin[2 * Pi * wstd * (t - tstartFd)]) / 1000,
   {t, ap1["Times"][[10^5 + 67]], ap1["Times"][[10^5 + 67 + 800]], deltatsima}];
(* table of values of force in one cycle of sinusoidal stimulus force*)

(*plot settings*)
plotstyleP = {Lighter[■]};


pady = 100;
padx = 70;
pad2 = 30;
fntsize = 18;
plotstylehist = {{Thickness → 0.01, Lighter[■], Thickness → 0.01},
   {Thickness → 0.01, ■}, {Thickness → 0.01, Darker[■]}};
framelabelshist = {{"current", None}, {"stimulus force (pN)", None}};
Xticks1 = Table[{k, k, {0, 0.03}}, {k, -400, 400, 100}];
Xticks2 = Table[{k, Null, {0, 0.02}}, {k, -400, 400, 50}];
XTickshist = Union[Xticks1, Xticks2];
YTicks = {{0.0, "0.0", {0, 0.03`}}, {0.5, 0.5, {0, 0.03`}}, {-0.5, -0.5, {0, 0.03`}},
   {1.0, "1.0", {0, 0.03`}}, {-1.0, "-1.0", {0, 0.03`}}, {0.1, Null, {0, 0.02`}},
   {0.2, Null, {0, 0.02`}}, {0.3, Null, {0, 0.02`}}, {0.4, Null, {0, 0.02`}},
   {0.6, Null, {0, 0.02`}}, {0.7, Null, {0, 0.02`}}, {0.8, Null, {0, 0.02`}},
   {0.9, Null, {0, 0.02`}}, {-0.1, Null, {0, 0.02`}}, {-0.2, Null, {0, 0.02`}},
   {-0.3, Null, {0, 0.02`}}, {-0.4, Null, {0, 0.02`}}, {-0.6, Null, {0, 0.02`}},
   {-0.7, Null, {0, 0.02`}}, {-0.8, Null, {0, 0.02`}}, {-0.9, Null, {0, 0.02`}}};

HistFPo = Table[{Fval[[i]], Povals[[i]]}, {i, 1, Length[Fval]}];
(*table of values for histeresis curve curr vs force amplitude*)

(*Fig S1C plot of hist curve of current vs
  stimulus force for a gating swing of 32 nm. Set D=32 nm.*)
ListPlot[HistFPo, PlotRange → {{-200, 200}, {-0.1, 1.1}}, Joined → True,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleP,
  Background → White, Frame → {True, True, True, True},
  FrameStyle → Directive[Black, Thick], FrameLabel → framelabelshist,
  Axes → False, FrameTicks → {{YTicks, None}, {XTickshist, None}}];
```

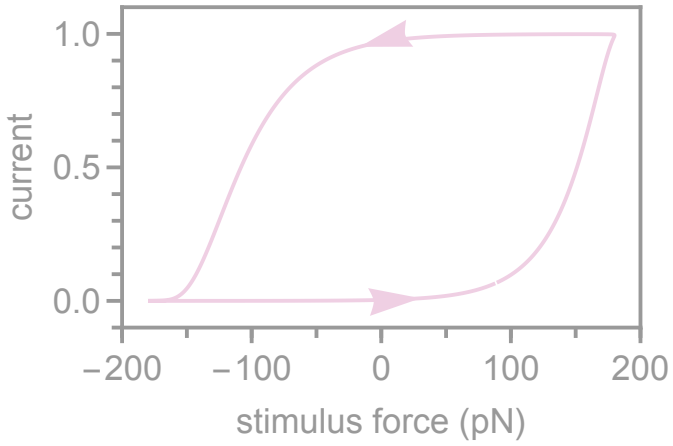
```
Show[{ % /. Line[x_] :> Sequence[Arrowheads[{{0.1, 0.4}, {0.1, 0.92}}], Arrow[x]] & /@
    {{1, 0 Degree, 90 Degree, Green}}}]


HistYPo = Table[{Yvals[[i]], Povals[[i]]}, {i, 1, Length[Fval]}];
(*table of values for histeresis curve curr vs force amplitude*)

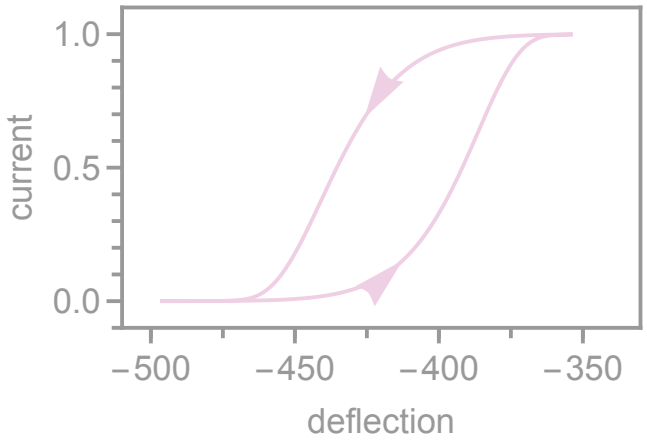
(*Fig S1E plot of hist curve of current vs
  deflection for a gating swing of 32 nm. Set D=32 nm.*)
plotstylehist = {{Thickness → 0.01, Lighter[■], Thickness → 0.01},
    {Thickness → 0.01, ■}, {Thickness → 0.01, Darker[■]}};
framelabelshist = {{"current", None}, {"deflection", None}};
Xticks1 = Table[{k, k, {0, 0.03}}, {k, -500, 405, 50}];
Xticks2 = Table[{k, Null, {0, 0.02}}, {k, -500, 405, 50 / 2}];
XTicks = Union[Xticks1, Xticks2];
YTicks = {{0.0, "0.0", {0, 0.03`}}, {0.5, 0.5, {0, 0.03`}}, {-0.5, -0.5, {0, 0.03`}},
    {1.0, "1.0", {0, 0.03`}}, {-1.0, "-1.0", {0, 0.03`}}, {0.1, Null, {0, 0.02`}},
    {0.2, Null, {0, 0.02`}}, {0.3, Null, {0, 0.02`}}, {0.4, Null, {0, 0.02`}},
    {0.6, Null, {0, 0.02`}}, {0.7, Null, {0, 0.02`}}, {0.8, Null, {0, 0.02`}},
    {0.9, Null, {0, 0.02`}}, {-0.1, Null, {0, 0.02`}}, {-0.2, Null, {0, 0.02`}},
    {-0.3, Null, {0, 0.02`}}, {-0.4, Null, {0, 0.02`}}, {-0.6, Null, {0, 0.02`}},
    {-0.7, Null, {0, 0.02`}}, {-0.8, Null, {0, 0.02`}}, {-0.9, Null, {0, 0.02`}}};


ListPlot[HistYPo, PlotRange → {{-510, -330}, {-0.1, 1.1}},
  Joined → True, ImagePadding → {{pady, pad2}, {padx, pad2}},
  ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleP,
  Background → White, Frame → {True, True, True, True},
  FrameStyle → Directive[Black, Thick], FrameLabel → framelabelshist,
  Axes → False, FrameTicks → {{YTicks, None}, {XTicks, None}}];
Show[{ % /. Line[x_] :> Sequence[Arrowheads[{{0.1, 0.5}, {0.1, 0.04}}], Arrow[x]] & /@
    {{1, 0 Degree, 90 Degree, Green}}}]
```

Out[1756]=



Out[1765]=




```
(*Calculation of sensitivity for a gating swing of 32 nm. Set D=32 nm.*)
(*Fig S1D plot of sensitivity vs stimulus
  force for a gating swing of 1.75 nm. Set D=1.75 nm.*)
fval = Table[n, {n, -180, 180, 0.5}];(* table of values of stimulus-force*)
nf = Length[fval];
PYval = Table[0, {nf}];
PFval = Table[0, {nf}];
FYval = Table[0, {nf}];

(*Solve equations of motion numerically to find current,
deflection and stimulus-force as a function of time*)
gx =  2 * kbTd * lPd;
y0in = Yrestd;
Do[sp = NDSolve[{lPd * Y'[t] ==
       (-kPd) * (Y[t] - Yupd) - kGSd * (Y[t] - Yugd - Dvd * Po[t]) + Fzd * fval[[ll]] ,
```

```
      taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd] * Po'[t] ==
       (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] - Po[t]),
      Y[0] == Yrestd, Po[0] == Pinfrest} , {Y, Po}, {t, 10}];
  PYval[[ll]] = {Y[9] /. First[sp], Po[9] /. First[sp]};
  (* deflection as a function of time*)
  PFval[[ll]] = {fval[[ll]], Po[9] /. First[sp]};
  (* current as a function of time*)
  FYval[[ll]] = {Y[9] /. First[sp], fval[[ll]]};(* force as a function of time*)
  ,
{ll, nf}];

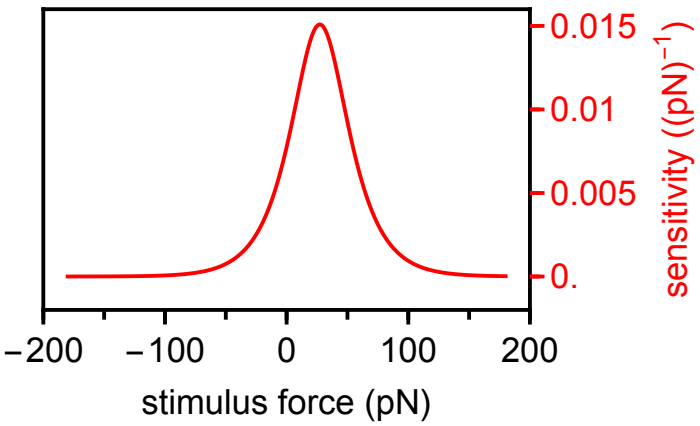
Xticks1 = Table[{k, k, {0, 0.03}}, {k, -400, 400, 100}];
Xticks2 = Table[{k, Null, {0, 0.02}}, {k, -400, 400, 50}];
XTickshist = Union[Xticks1, Xticks2];


fPfval = Interpolation[PFval];
dfPfval = Table[{x, fPfval'[x]}, {x, -180, 180}];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 0.02, 0.005}];
Yticks2 = Table[{k, Null, {0, 0.02}}, {k, 0, 0.02, 0.005}];
plotrange1 = {{-200, 200}, {-0.002, 0.016}};


YTickshist = Union[Yticks1, Yticks2];
ListPlot[dfPfval, Joined → True, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → {{None, Style["sensitivity ((pN)^-1)", Red, FontSize → 18]},
   {"stimulus force (pN)", None}}, PlotRange → plotrange1,
 FrameTicksStyle → {{Black, Red}, {Black, Black}}, LabelStyle → {FontSize → 18},
 Axes → {False, False}, PlotStyle → {Red},
 FrameTicks → {{ None, YTickshist}, {XTickshist, None}}]
```

Out[2004]=




```
(*Calculation of histeresis curves
  for a gating swing of 1.75 nm. Set D=1.75 nm*)
fmult = {2800};


ap1 = ap["PathComponent", 2];
ap0 = ap["PathComponent", 1];
Povals = ap1["Values"][[10^5 + 67 ;; 10^5 + 67 + 800 ;; 1]];
(* table of values of current in one cycle of sinusoidal stimulus force*)
Yvals = ap0["Values"][[10^5 + 67 ;; 10^5 + 67 + 800 ;; 1]];
(* table of values of deflections in one cycle of sinusoidal stimulus force*)
Fval = Table[(Fzd * fmult[[Length[fmult]]] * Sin[2 * Pi * wstd * (t - tstartFd)]) / 1000,
   {t, ap1["Times"][[10^5 + 67]], ap1["Times"][[10^5 + 67 + 800]], deltatsima}];
(* table of values of force in one cycle of sinusoidal stimulus force*)

(*plot settings*)
plotstyleP = {Lighter[■]};


pady = 100;
padx = 70;
pad2 = 30;
fntsize = 18;
plotstylehist = {{Thickness → 0.01, Lighter[■], Thickness → 0.01},
   {Thickness → 0.01, ■}, {Thickness → 0.01, Darker[■]}};
framelabelshist = {{"current", None}, {"stimulus force (pN)", None}};
Xticks1 = Table[{k, k, {0, 0.03}}, {k, -3000, 3000, 2000}];
Xticks2 = Table[{k, Null, {0, 0.02}}, {k, -3000, 3000, 1000}];
XTickshist = Union[Xticks1, Xticks2];
```

```
YTicks = {{0.0, "0.0", {0, 0.03`}}, {0.5, 0.5, {0, 0.03`}}, {-0.5, -0.5, {0, 0.03`}},
    {1.0, "1.0", {0, 0.03`}}, {-1.0, "-1.0", {0, 0.03`}}, {0.1, Null, {0, 0.02`}},
    {0.2, Null, {0, 0.02`}}, {0.3, Null, {0, 0.02`}}, {0.4, Null, {0, 0.02`}},
    {0.6, Null, {0, 0.02`}}, {0.7, Null, {0, 0.02`}}, {0.8, Null, {0, 0.02`}},
    {0.9, Null, {0, 0.02`}}, {-0.1, Null, {0, 0.02`}}, {-0.2, Null, {0, 0.02`}},
    {-0.3, Null, {0, 0.02`}}, {-0.4, Null, {0, 0.02`}}, {-0.6, Null, {0, 0.02`}},
    {-0.7, Null, {0, 0.02`}}, {-0.8, Null, {0, 0.02`}}, {-0.9, Null, {0, 0.02`}}};

HistFPo = Table[{Fval[[i]], Povals[[i]]}, {i, 1, Length[Fval]}];
(*table of values for histeresis curve curr vs force amplitude*)

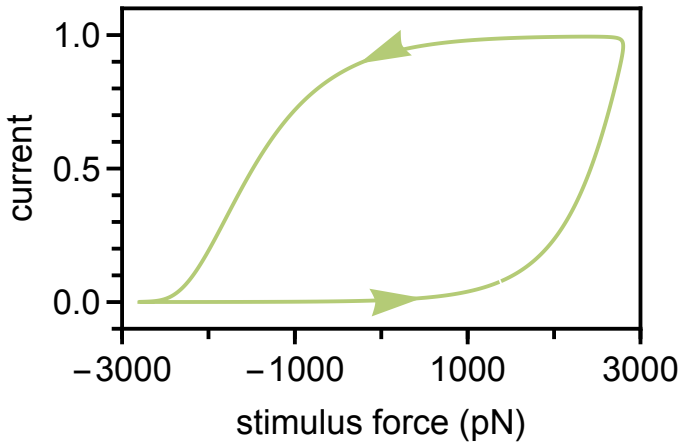
(*Fig S1C plot of hist curve of current vs stimulus
  force for a gating swing of 1.75 nm. Set D=1.75 nm.*)
ListPlot[HistFPo, PlotRange → {{-3000, 3000}, {-0.1, 1.1}},
  Joined → True, ImagePadding → {{pady, pad2}, {padx, pad2}},
  ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleP,
  Background → White, Frame → {True, True, True, True},
  FrameStyle → Directive[Black, Thick], FrameLabel → framelabelshist,
  Axes → False, FrameTicks → {{YTicks, None}, {XTickshist, None}}];
Show[{ % /. Line[x_] :> Sequence[Arrowheads[{{0.1, 0.4}, {0.1, 0.92}}], Arrow[x]] & /@
    {{1, 0 Degree, 90 Degree, Green}}}]
HistYPo = Table[{Yvals[[i]], Povals[[i]]}, {i, 1, Length[Fval]}];


HistYPo = Table[{Yvals[[i]], Povals[[i]]}, {i, 1, Length[Fval]}];
(*table of values for histeresis curve curr vs force amplitude*)

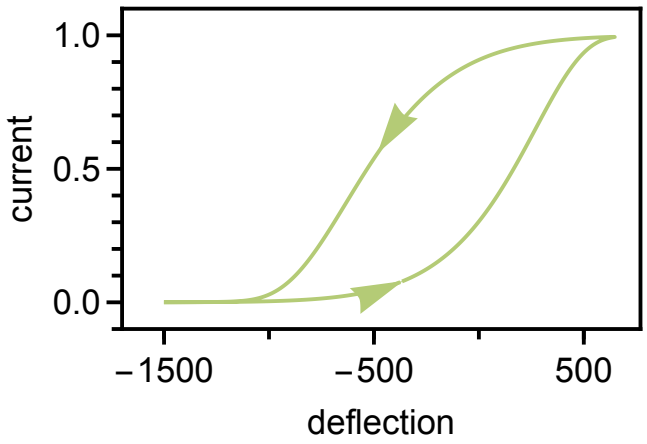
(*Fig S1E plot of hist curve of current vs
  deflection for a gating swing of 1.75 nm. Set D=1.75 nm.*)
plotstylehist = {{Thickness → 0.01, Lighter[■], Thickness → 0.01},
    {Thickness → 0.01, ■}, {Thickness → 0.01, Darker[■]}};
framelabelshist = {{"current", None}, {"deflection", None}};
Xticks1 = Table[{k, k, {0, 0.03}}, {k, -1500, 3000, 1000}];
Xticks2 = Table[{k, Null, {0, 0.02}}, {k, -1500, 3000, 1000 / 2}];
XTicks = Union[Xticks1, Xticks2];
YTicks = {{0.0, "0.0", {0, 0.03`}}, {0.5, 0.5, {0, 0.03`}}, {-0.5, -0.5, {0, 0.03`}},
    {1.0, "1.0", {0, 0.03`}}, {-1.0, "-1.0", {0, 0.03`}}, {0.1, Null, {0, 0.02`}},
    {0.2, Null, {0, 0.02`}}, {0.3, Null, {0, 0.02`}}, {0.4, Null, {0, 0.02`}},
    {0.6, Null, {0, 0.02`}}, {0.7, Null, {0, 0.02`}}, {0.8, Null, {0, 0.02`}},
    {0.9, Null, {0, 0.02`}}, {-0.1, Null, {0, 0.02`}}, {-0.2, Null, {0, 0.02`}},
    {-0.3, Null, {0, 0.02`}}, {-0.4, Null, {0, 0.02`}}, {-0.6, Null, {0, 0.02`}},
    {-0.7, Null, {0, 0.02`}}, {-0.8, Null, {0, 0.02`}}, {-0.9, Null, {0, 0.02`}}};
```

```
ListPlot[HistYPo, PlotRange → {{-1700, 770}, {-0.1, 1.1}},
  Joined → True, ImagePadding → {{pady, pad2}, {padx, pad2}},
  ImageSize → 400, LabelStyle → {FontSize → fntsize}, PlotStyle → plotstyleP,
  Background → White, Frame → {True, True, True, True},
  FrameStyle → Directive[Black, Thick], FrameLabel → framelabelshist,
  Axes → False, FrameTicks → {{YTicks, None}, {XTicks, None}}];
Show[{ % /. Line[x_] :→ Sequence[Arrowheads[{{0.1, 0.5}, {0.1, 0.001}}], Arrow[x]] & /@
   {{1, 0 Degree, 90 Degree, Green}}}]
```

Out[3976]=



Out[3986]=



```
(*Calculation of sensitivity for a gating swing of 1.75 nm. Set D=1.75 nm.*)
fval = Table[n, {n, -2800, 2800, 1}];(* table of values of stimulus-force*)
nf = Length[fval];
PYval = Table[0, {nf}];
PFval = Table[0, {nf}];
FYval = Table[0, {nf}];

(*Solve equations of motion numerically to find current,
deflection and stimulus-force as a function of time*)
gx =  2 * kbTd * lPd;
y0in = Yrestd;
Do[sp = NDSolve[{lPd * Y'[t] ==
       (-kPd) * (Y[t] - Yupd) - kGSd * (Y[t] - Yugd - Dvd * Po[t]) + Fzd * fval[[ll]] ,
      taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd] * Po'[t] ==
       (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] - Po[t]),
      Y[0] == Yrestd, Po[0] == Pinfrest} , {Y, Po}, {t, 10}];
  PYval[[ll]] = {Y[9] /. First[sp], Po[9] /. First[sp]};
  (* deflection as a function of time*)
  PFval[[ll]] = {fval[[ll]], Po[9] /. First[sp]};
  (* current as a function of time*)
  FYval[[ll]] = {Y[9] /. First[sp], fval[[ll]]};(* force as a function of time*)
  ,
{ll, nf}];
```

```
(*Fig S1D plot of sensitivity vs stimulus
  force for a gating swing of 1.75 nm. Set D=1.75 nm.*)

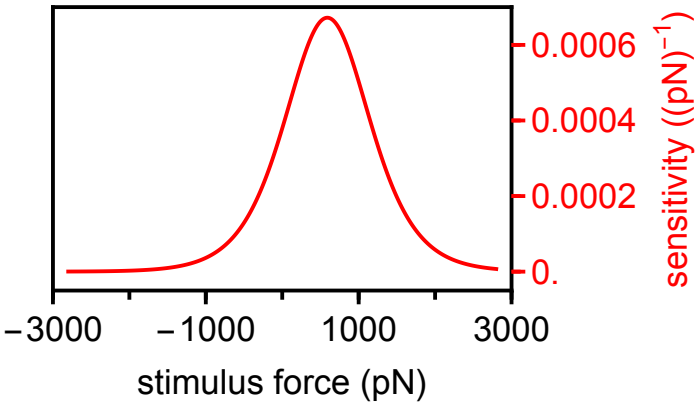
(*plot settings*)
Xticks1 = Table[{k, k, {0, 0.03}}, {k, -3000, 3000, 2000}];
Xticks2 = Table[{k, Null, {0, 0.02}}, {k, -3000, 3000, 1000}];
XTickshist = Union[Xticks1, Xticks2];


fPfval = Interpolation[PFval];
dfPfval = Table[{x, fPfval'[x]}, {x, -2800, 2800}];

Yticks1 = Table[{k, k, {0, 0.03}}, {k, 0, 0.0006, 0.0002}];
Yticks2 = Table[{k, Null, {0, 0.02}}, {k, 0, 0.0006, 0.0002}];
plotrange1 = {{-3000, 3000}, {-0.00005, 0.0007}};


YTickshist = Union[Yticks1, Yticks2];
ListPlot[dfPfval, Joined → True, Background → White,
 Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
 FrameLabel → {{None, Style["sensitivity ((pN)^-1)", Red, FontSize → 18]},
   {"stimulus force (pN)", None}}, PlotRange → plotrange1,
 FrameTicksStyle → {{Black, Red}, {Black, Black}}, LabelStyle → {FontSize → 18},
 Axes → {False, False}, PlotStyle → {Red},
 FrameTicks → {{None, YTickshist}, {XTickshist, None}}]
```

Out[4090]=



In[5231]:=

```
(*Calculation of CSNR badnwidths*)

(*Derivation of unstimulated power spectra*)
fmult = 0;
SeedRandom[47 741] (* set seed for reproducibility of results *)

gx =  2 * kbTd * lPd;
y0in = Yrestd;

t1 = AbsoluteTime[];

(*define Ito process for simulations of deflection and current
 Fzd multiplied by 180 for stimulated psd, 0 for unstimulated psd*)
proc = ItoProcess[
  lPd * dY[t] == (-kPd) * (Y[t] - Yupd) * dt - kGSd * (Y[t] - Yugd - Dvd * Po[t]) * dt +
     Sqrt[gx] * dn1[t]  + Fzd * fmult * Sin[2 * Pi * wstd * (t - tstartFd)] * dt &&
   taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd] * dPo[t] ==
    (Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] - Po[t]) dt +
     Sqrt[taofun[alphad, Yhd, kGSd, Dvd, kbTd, taorest, Pinfrest, Ncd]]
      Sqrt[(Pinf2[Yhd, kGSd, Dvd, kbTd, Ncd] (1 - Clip[Po[t], {0.0, 1.0}]) + (1 - Pinf2[
               Yhd, kGSd, Dvd, kbTd, Ncd]) Clip[Po[t], {0.0, 1.0}]) / Ncd] dn2[t],
  {Y[t], Po[t]}, {{Y, Po}, {y0in, Pinfrest}}, t,
{n1 ≈ WienerProcess[], n2 ≈ WienerProcess[]}]

totallength = 1.014;(*length of simulation*)
factot = 4;
deltatsima = 0.000001 / factot;(*time-step of simulation*)
ap = RandomFunction[proc, {0, totallength, deltatsima}];
(*simulation of deflection and curent*)

t2 = AbsoluteTime[];
t2 - t1
```

Out[5232]=

```
RandomGeneratorState[ Method: ExtendedCA
                      State hash: -8833868631830390932 ]
```

Out[5236]=

$$\text{ItoProcess}\Big[\Big\{\Big\{0. + 27.061\,(-1872.\,(407. + Y[t]) - 468.\,(512.44 - 32.\,Po[t] + Y[t])),$$
$$\Bigg(0.\,\sqrt{\left(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}}$$
$$\sqrt{\left(\frac{1. - 1.\,\text{Clip}[Po[t], \{0., 1.\}]}{1. + e^{0.115189\,(-413.235 - Y[t])}} + \left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)\text{Clip}[Po[t], \{0., 1.\}]\right)}\Bigg/$$
$$\left(\left(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\right) +$$
$$\frac{30\,000.\left(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}} - 1.\,Po[t]\right)}{\left(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}}\Big\},$$
$$\Big\{\{468.927, 0.\}, \Big\{\Bigg(0.\,\sqrt{\left(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}}$$
$$\sqrt{\left(\frac{1. - 1.\,\text{Clip}[Po[t], \{0., 1.\}]}{1. + e^{0.115189\,(-413.235 - Y[t])}} + \left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)\text{Clip}[Po[t], \{0., 1.\}]\right)}\Bigg/$$
$$\left(\left(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\right),$$
$$\Bigg(30.6186\,\sqrt{\left(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}}$$
$$\sqrt{\left(\frac{1. - 1.\,\text{Clip}[Po[t], \{0., 1.\}]}{1. + e^{0.115189\,(-413.235 - Y[t])}} + \left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)\text{Clip}[Po[t], \{0., 1.\}]\right)}\Bigg/$$
$$\left(\left(\frac{1}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\left(1. - \frac{1.}{1. + e^{0.115189\,(-413.235 - Y[t])}}\right)^{0.5}\right)\Big\}\Big\},$$
$$\{Y[t], Po[t]\}\Big\}, \{\{Y, Po\}, \{-427, 0.17\}\}, \{t,$$
$$0\}\Big]$$

Out[5242]=

10.448250

In[5540]:=

```
(*Calculation of power spectra from simulated deflection and current*)

(*Yrestd=-412.681; (*resting position of IHB with channels blocked [nm]*)
Yrestd=-407;(*resting position with gating spring cut [nm]*)*)
```

```
ap0 = ap["PathComponent", 1] - Yrestd;
(* subtract resting position from deflection time series*)

ap1 = ap["PathComponent", 2] - Pinfrest;
(*subtract resting current from current time series*)

resampfactor = factot;(*resempling factor for time series*)

randtime100 = 0.005 * 10; (*starting time to ensure that values
 of time series are independent of initial value of simulation*)

(*round to multiple of time-step of simulation to avoid interpolation*)
randtimenm = Round[randtime100, deltatsima];
stpoint = IntegerPart[randtime100 / deltatsima];

deltatsampl = Round[deltatsima * resampfactor, deltatsima];
(*time-step of resampled time series*)
tsampl = 0.005; (* length of a single window of the splitted time series*)
endpoint = IntegerPart[Length[ap["PathComponent", 1]["Values"]] -
    tsampl / deltatsima - stpoint];(*end point of time series*)
tsamplnm = Round[tsampl, deltatsima];
(*round to fraction of delta of time series to avoid interpolation*)
(*determine length of sampled time series*)
b1 = Table[f[i], {i, randtimenm, tsamplnm + randtimenm, deltatsampl}];
lengthares = Length[b1];

ncopiesa = (totallength - randtimenm) / (tsamplnm + deltatsampl);
(*number of non-overlapping windows*)
ncopies = IntegerPart[ncopiesa];
(*round number of overlapping windows to nearest integer *)



ares0 = Table[0, {ncopies}];
(*table of non-overlapping blocks of deflection time series*)
ares1 = Table[0, {ncopies}];
(*table of non-overlapping blocks of deflection time series*)
sizea = lengthares;(*length of a single block*)
fousampled0 = Table[0, {ncopies}, {lengthares}];
(*table of deflection power spectrum of each block*)
fousampled1 = Table[0, {ncopies}, {lengthares}];
(*table of current power spectrum of each block*)
cfousampled = Table[0, {ncopies}, {lengthares}];
(*table of cross-power spectrum of each block*)
```

```
(*derivation of power spectra and cross-power spectra*)
Do[

  ares0[[ll]] = ap0["Values"][[(ll - 1) * tsampl / deltatsima + stpoint
     ;; IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];
    ares1[[ll]] = ap1["Values"][[(ll - 1) * tsampl / deltatsima + stpoint
     ;; IntegerPart[ll * tsampl / deltatsima + stpoint] ;; resampfactor]];




    fousampled0[[ll]] = ResourceFunction["WelchSpectralEstimate"][ares0[[ll]],
    sizea / tsampl, "Window" → None, "Detrend" → None, "OverlapOffset" → None,
    "SegmentSize" → sizea]; (*deflection power spectrum of single block*)
    cfousampled[[ll]] = ResourceFunction["WelchSpectralEstimate"][
    ares1[[ll]], ares0[[ll]], sizea / tsampl, "Window" → None,
    "Detrend" → None, "OverlapOffset" → None, "SegmentSize" → sizea];
  (*cross-power spectrum of single block*)
    fousampled1[[ll]] =
   ResourceFunction["WelchSpectralEstimate"][ares1[[ll]], sizea / tsampl,
    "Window" → None, "Detrend" → None, "OverlapOffset" → None, "SegmentSize" → sizea];
  (*current power spectrum of single block*)
     ,
{ll, ncopies}];

powspmeansampledt0 = Drop[Mean[fousampled0], -1];
(*mean of deflection power spectra of all blocks*)

powspmeansampledt1 = Drop[Mean[fousampled1], -1];
(*mean of current power spectra of all blocks*)

cpowspmeansampledt = Drop[Re[Mean[cfousampled]], -1];
(*mean of cross-power spectra of all blocks*)

lengthpsd = Length[powspmeansampledt0]
(*tables of values of simulated psd*)
meandataplot0base = Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt0[[i]][[2]]},
   {i, 1, lengthpsd}]; (*deflection psd*)
meandataplot1base = Table[{powspmeansampledt0[[i]][[1]], powspmeansampledt1[[i]][[2]]},
   {i, 1, lengthpsd}]; (*current psd*)
meandataplot1cpsdbase = Table[{cpowspmeansampledt[[i]][[1]], cpowspmeansampledt[[i]][[2]]},
   {i, 1, lengthpsd}]; (*cross-psd*)
coher01base = Table[{meandataplot0[[i]][[1]],
```

```
    meandataplot1cpsd[[i]][[2]] / Sqrt[(meandataplot0[[i]][[2]] * meandataplot1[[i]][[2]])]},
   {i, 1, lengthpsd}]; (*coherency*)

fc = 610; (*corner frequency of the hair-cell membrane [Hz]*)
filtersq1 =
  Table[{meandataplot0[[i]][[1]], Abs[1 / (1 + I * meandataplot1base[[i]][[1]] / fc)]^2},
   {i, 1, Length[meandataplot0]}];(*table of values of the filter*)
meandataplot1filtbase =
  Table[{meandataplot0[[i]][[1]], filtersq1[[i]][[2]] * meandataplot1base[[i]][[2]]},
   {i, 1, Length[filtersq1]}];(*table of values of the voltage psd*)
```

Out[5564]=

```
2501
```

In[5275]:=

```
(*Bandwidths for deflection psd*)
n1 = 26; (*vector to 5 kHz position of psd*)
pp = 1;


fitpeakl = LinearModelFit[{meandataplot0[[n1 - 1]], meandataplot0[[n1]]}, x, x];
fitpeakr = LinearModelFit[{meandataplot0[[n1]], meandataplot0[[n1 + 1]]}, x, x];
eqfitpeak = fitpeakl[x];
fpeakinvl = InverseFunction[eqfitpeak /. x → # &];
peakvalat09 = meandataplot0[[n1]][[2]] * 0.1;

freqpeakl = fpeakinvl[peakvalat09];
freqpeakr = 2 * meandataplot0[[n1]][[1]] - freqpeakl;
deltafpeak = freqpeakr - freqpeakl;

indexfreqpakl = Table[meandataplot0[[n]][[1]], {n, n1, 1, -1}];
indexnoisesearch = n1 - FirstPosition[indexfreqpakl, _?(# < freqpeakl &)][[1]] + 1;

noisecheck1 = Table[{(meandataplot0[[n]][[2]] - meandataplot0[[n - 1]][[2]]) /
      (meandataplot0[[3]][[1]] - meandataplot0[[2]][[1]])}, {n, indexnoisesearch, 2, -1}];

noisecheck =
  Table[{noisecheck1[[n + 1]] / noisecheck1[[n]]}, {n, 1, Length[noisecheck1] - 1}];

noiser1 = indexnoisesearch - 14;
(*side bandwidths 14 frequency point after stimulus frequency*)

freqnoiser1 = meandataplot0[[noiser1]][[1]];
freqnoisel1 = freqnoiser1 - deltafpeak / 2;
freqnoisel2 = freqpeakl - meandataplot0[[noiser1]][[1]] + freqpeakr;
freqnoiser2 = freqnoisel2 + deltafpeak / 2;

peakl = {freqpeakl, 10^(-5)}
peakr = {freqpeakr, 10^(-5)}

noisel1 = {freqnoisel1, 10^(-5)}
noiser1 = {freqnoiser1, 10^(-5)}

noisel2 = {freqnoisel2, 10^(-5)}
noiser2 = {freqnoiser2, 10^(-5)}
```

Out[5294]=

$$\left\{4819.99,\ \frac{1}{100\,000}\right\}$$

Out[5295]=

$$\left\{5180.01, \frac{1}{100\,000}\right\}$$

Out[5296]=

$$\left\{1819.99, \frac{1}{100\,000}\right\}$$

Out[5297]=

$$\left\{2000., \frac{1}{100\,000}\right\}$$

Out[5298]=

$$\left\{8000., \frac{1}{100\,000}\right\}$$

Out[5299]=

$$\left\{8180.01, \frac{1}{100\,000}\right\}$$

In[5313]:=

```
(*plot settings*)
Remove[XTicks]
Remove[YTicks]
TickLength = 0.02;
Xmin = 0;
Xmax = 10 - 1;
Ymin = 0;
Ymax = 20;
deltaY = 2;
plotrangepsd = {{10^(2), 10^6}, {0.5 * 10^(-12), 10^(-2)}};
plotrangecoher = {{10^(2), 10^6}, {-1.05, 1.05}};
framelabels =
  {{"power spectrum (\!\(\*SuperscriptBox[\(nm\), \(2\)]\)/Hz)", None},
   {"noise frequency (Hz)", None}};
TickLength1 = 0.03;
Yticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, -24, 10, 2}];
Yticks2 = Table[{10^k, Null, {0, TickLength1}}, {k, -23, 10, 2}];
TickLength2 = 0.02;
aa = Table[0, {38}, {4}];
For[k = 1, k ≤ 38, k++,
aa[[k]] = Table[{l * 10^(-k + 10), , {0, TickLength2}}, {l, 2, 8, 2}];
]
YTicks = Union[Yticks1, Yticks2, aa[[1]], aa[[2]], aa[[3]],
   aa[[4]], aa[[5]], aa[[6]], aa[[7]], aa[[8]], aa[[9]], aa[[10]], aa[[11]], aa[[12]],
   aa[[13]], aa[[14]], aa[[15]], aa[[16]], aa[[17]], aa[[18]], aa[[19]], aa[[20]],
   aa[[21]], aa[[22]], aa[[23]], aa[[24]], aa[[25]], aa[[26]], aa[[27]], aa[[28]]];
```

```
Xticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, 3, 6, 1}];
aa2 = Table[0, {20}, {4}];
For[k = 1, k ≤ 10, k++,
aa2[[k]] = Table[{l * 10^k, , {0, TickLength2}}, {l, 2, 8, 2}];
]

Xticks2 = {{100, 0, {0, 0.03}}};

XTicks = Union[Xticks1, Xticks2, aa2[[2]], aa2[[3]], aa2[[4]], aa2[[5]], aa2[[6]],
   aa2[[7]], aa2[[8]], aa2[[9]], aa2[[10]], aa2[[11]], aa2[[12]], aa2[[13]], aa2[[14]],
   aa2[[15]], aa2[[16]], aa2[[17]], aa2[[18]], aa2[[19]], aa2[[20]], Xticks2];


pady = 120;
padx = 70;
pad2 = 30;
fntsize = 18;


plotstylecoh = {{Thickness → 0.01, RGBColor[0 * 251 / 255, 0 * 41 / 255, 0 * 67 / 255]},
   {Thickness → 0.01, RGBColor[106 / 255, 90 / 255, 205 / 255]},
   {Thickness → 0.01, RGBColor[225 / 255, 156 / 255, 36 / 255]}};
plotstylepsd = {{Thickness → 0.01, ■}, {Thickness → 0.01, ■},
   {Thickness → 0.01, ■}, {Thickness → 0.01, ■}, {Thickness → 0.01, ■}};

meandataplot0c = {100, meandataplot0[[1]][[2]]};
meandataplot1c = {100, meandataplot1[[1]][[2]]};
plotstylepeaklr = {Darker[■]};
plotstylenoiselr = {Darker[■]};
plotstyleY = {Darker[■]};

Xticks1 = Table[{10^k, Superscript[10, k], {0, TickLength1}}, {k, 3, 6, 1}];
aa2 = Table[0, {20}, {4}];
aa3 = Table[0, {20}, {4}];
For[k = 1, k ≤ 10, k++,
aa2[[k]] = Table[{l * 10^k, l * Superscript[10, k], {0, TickLength2}}, {l, 4, 8, 4}];
]

For[k = 1, k ≤ 10, k++,
aa3[[k]] = Table[{l * 10^k, Null, {0, TickLength2}}, {l, 2, 8, 2}];
]

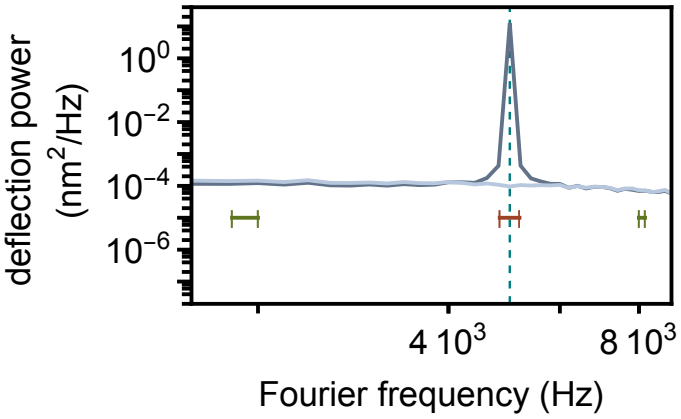
Xticks2 = {{100, 0, {0, 0.03}}};
```

```
XTicksbb = Union[Xticks1, Xticks2, aa2[[2]], aa2[[3]], aa2[[4]], aa2[[5]], aa2[[6]], aa2[[7]],
    aa2[[8]], aa2[[9]], aa2[[10]], aa2[[11]], aa2[[12]], aa2[[13]], aa2[[14]], aa2[[15]],
    aa2[[16]], aa2[[17]], aa2[[18]], aa2[[19]], aa2[[20]], aa3[[2]], aa3[[3]], aa3[[4]],
    aa3[[5]], aa3[[6]], aa3[[7]], aa3[[8]], aa3[[9]], aa3[[10]], aa3[[11]], aa3[[12]], aa3[[13]],
    aa3[[14]], aa3[[15]], aa3[[16]], aa3[[17]], aa3[[18]], aa3[[19]], aa3[[20]], Xticks2];
```

In[5406]:=

```
(*FigS6 B, defelction CSNR bandwidths*)
plotrange1 = {{1570, 9006}, {0.2 * 10^(-7), 0.4 * 10^(2)}};
ListLogLogPlot[{meandataplot0},
  PlotStyle → plotstyleY, Joined → True, PlotRange → plotrange1,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicksbb, None}},
  FrameLabel → {"Fourier frequency (Hz)",
    "deflection power \n (\!\(\*SuperscriptBox[\(nm\), \(2\)]\)/Hz)"},
  GridLines → {{5000}, {}}, GridLinesStyle →
   Directive[■, Dashed, Thickness → 0.01 / 2]];
psd00 = Show[%, ListLogLogPlot[{meandataplot0c}, PlotStyle → plotstyleY]];
Show[%,
  ListLogLogPlot[{peakl}, PlotStyle → plotstylepeaklr, PlotMarkers → {"|", 10}]];
Show[%,
  ListLogLogPlot[{peakr}, PlotStyle → plotstylepeaklr, PlotMarkers → {"|", 10}]];
Show[%, ListLogLogPlot[{peakl, peakr}, Joined → True, PlotStyle → plotstylepeaklr]];
Show[%,
  ListLogLogPlot[{noisel1}, PlotStyle → plotstylenoiselr, PlotMarkers → {"|", 10}]];
Show[%,
  ListLogLogPlot[{noiser1}, PlotStyle → plotstylenoiselr, PlotMarkers → {"|", 10}]];
Show[%,
  ListLogLogPlot[{noisel1, noiser1}, Joined → True, PlotStyle → plotstylenoiselr]];
Show[%,
  ListLogLogPlot[{noisel2}, PlotStyle → plotstylenoiselr, PlotMarkers → {"|", 10}]];
Show[%,
  ListLogLogPlot[{noiser2}, PlotStyle → plotstylenoiselr, PlotMarkers → {"|", 10}]];
Show[%,
  ListLogLogPlot[{noisel2, noiser2}, Joined → True, PlotStyle → plotstylenoiselr]];
Show[%,
 ListLogLogPlot[{meandataplot0base}, PlotStyle → {Lighter[■]}, Joined → True]]
```

Out[5418]=


deflection power
(nm^2/Hz)
10^0
10^{-2}
10^{-4}
10^{-6}
4 10^3
8 10^3
Fourier frequency (Hz)

In[5419]:=

```
(*Bandwidths for current psd*)
fitpeaklc = LinearModelFit[{meandataplot1[[n1 - 1]], meandataplot1[[n1]]}, x, x];
fitpeakrc = LinearModelFit[{meandataplot1[[n1]], meandataplot1[[n1 + 1]]}, x, x];

eqfitpeakc = fitpeaklc[x];
fpeakinvlc = InverseFunction[eqfitpeakc /. x → # &];
peakvalat09c = meandataplot1[[n1]][[2]] * 0.2;
freqpeaklc = fpeakinvlc[peakvalat09c];
freqpeakrc = 2 * meandataplot1[[n1]][[1]] - freqpeaklc;
(fitpeaklc[freqpeaklc] + fitpeakrc[meandataplot1[[n1]][[1]] + fitpeakrc[freqpeakrc]) *
  (meandataplot1[[n1]][[1]] - freqpeaklc);
deltafpeakc = freqpeakrc - freqpeaklc;

indexfreqpaklc = Table[meandataplot1[[n]][[1]], {n, n1, 1, -1}];
indexnoisesearchc = n1 - FirstPosition[indexfreqpaklc, _?(# < freqpeaklc &)][[1]] + 1;

noisecheckc1 = Table[{(meandataplot1[[n]][[2]] - meandataplot1[[n - 1]][[2]]) /
     (meandataplot1[[3]][[1]] - meandataplot1[[2]][[1]])}, {n, indexnoisesearchc, 2, -1}];

noisecheckc =
  Table[{noisecheckc1[[n + 1]] / noisecheckc1[[n]]}, {n, 1, Length[noisecheckc1] - 1}];

noiser1c = indexnoisesearchc - 14;
freqnoiser1c = meandataplot1[[noiser1c]][[1]];
freqnoisel1c = freqnoiser1c - deltafpeakc / 2;
freqnoisel2c = freqpeaklc - meandataplot1[[noiser1c]][[1]] + freqpeakrc;
freqnoiser2c = freqnoisel2c + deltafpeakc / 2;

peaklp = {freqpeaklc, 10^(-8)}
peakrp = {freqpeakrc, 10^(-8)}

noisel1p = {freqnoisel1c, 10^(-8)}
noiser1p = {freqnoiser1c, 10^(-8)}

noisel2p = {freqnoisel2c, 10^(-8)}
noiser2p = {freqnoiser2c, 10^(-8)}
```

Out[5437]=

$$\left\{4839.99, \frac{1}{100\,000\,000}\right\}$$

Out[5438]=

$$\left\{5160.01, \frac{1}{100\,000\,000}\right\}$$

Out[5439]=

$$\left\{1839.99,\ \frac{1}{100\,000\,000}\right\}$$

Out[5440]=

$$\left\{2000.,\ \frac{1}{100\,000\,000}\right\}$$

Out[5441]=

$$\left\{8000.,\ \frac{1}{100\,000\,000}\right\}$$

Out[5442]=

$$\left\{8160.01,\ \frac{1}{100\,000\,000}\right\}$$

In[5477]:=

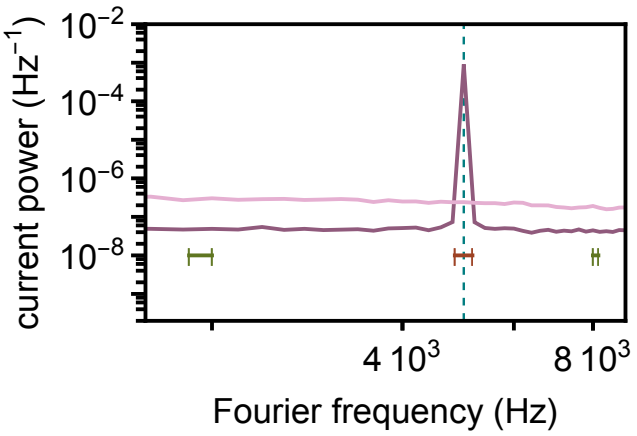
```
(*FigS6 D CSNR current bandwidths*)
meandataplot1c = {100, meandataplot1[[1]][[2]]};

plotstylepeaklr = {Darker[■]};
plotstylenoiselr = {Darker[■]};
plotstyleP = {Darker[■]}
plotrange3 = {{1570, 9006}, {0.2 * 10^(-9), 10^(-2)}};
ListLogLogPlot[{meandataplot1}, PlotStyle → plotstyleP,
  Joined → True, Frame → True, PlotRange → plotrange3,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicksbb, None}},
  FrameLabel → {"Fourier frequency (Hz)", "current power (Hz^-1)"},
  GridLines → {{5000}, {}}, GridLinesStyle →
   Directive[RGBColor[0, 125 / 255, 128 / 255], Dashed, Thickness → 0.01 / 2]];
psd11 = Show[%, ListLogLogPlot[{meandataplot1c}, PlotStyle → plotstyleP]];
Show[%,
  ListLogLogPlot[{peaklp}, PlotStyle → plotstylepeaklr, PlotMarkers → {"|", 10}]];
Show[%,
  ListLogLogPlot[{peakrp}, PlotStyle → plotstylepeaklr, PlotMarkers → {"|", 10}]];
Show[%, ListLogLogPlot[{peaklp, peakrp}, Joined → True, PlotStyle → plotstylepeaklr]];
Show[%, ListLogLogPlot[{noisel1p},
   PlotStyle → plotstylenoiselr, PlotMarkers → {"|", 10}]];
Show[%, ListLogLogPlot[{noiser1p},
   PlotStyle → plotstylenoiselr, PlotMarkers → {"|", 10}]];
Show[%,
  ListLogLogPlot[{noisel1p, noiser1p}, Joined → True, PlotStyle → plotstylenoiselr]];
Show[%, ListLogLogPlot[{noisel2p},
   PlotStyle → plotstylenoiselr, PlotMarkers → {"|", 10}]];
Show[%, ListLogLogPlot[{noiser2p},
   PlotStyle → plotstylenoiselr, PlotMarkers → {"|", 10}]];
Show[%,
  ListLogLogPlot[{noisel2p, noiser2p}, Joined → True, PlotStyle → plotstylenoiselr]];
Show[%, ListLogLogPlot[{meandataplot1base}, PlotStyle → {Lighter[■]}, Joined → True]]
```

Out[5480]=

{■}

Out[5493]=


current power (Hz^{-1})
10^{-2}
10^{-4}
10^{-6}
10^{-8}
$4\ 10^3$
$8\ 10^3$
Fourier frequency (Hz)

In[5620]:=

```
(*voltage CSNR bandwidths definition*)
fitpeaklc =
  LinearModelFit[{meandataplot1filt[[n1 - 1]], meandataplot1filt[[n1]]}, x, x];
fitpeakrc = LinearModelFit[{meandataplot1filt[[n1]], meandataplot1filt[[n1 + 1]]}, x, x];

eqfitpeakc = fitpeaklc[x];
fpeakinvlc = InverseFunction[eqfitpeakc /. x → # &];
peakvalat09c = meandataplot1filt[[n1]][[2]] * 0.2;
freqpeaklc = fpeakinvlc[peakvalat09c];
freqpeakrc = 2 * meandataplot1filt[[n1]][[1]] - freqpeaklc;
(fitpeaklc[freqpeaklc] + fitpeakrc[meandataplot1filt[[n1]][[1]]] +
    fitpeakrc[freqpeakrc]) * (meandataplot1filt[[n1]][[1]] - freqpeaklc);
deltafpeakc = freqpeakrc - freqpeaklc;

indexfreqpaklc = Table[meandataplot1filt[[n]][[1]], {n, n1, 1, -1}];
indexnoisesearchc = n1 - FirstPosition[indexfreqpaklc, _?(# < freqpeaklc &)][[1]] + 1;

noisecheckc1 = Table[{(meandataplot1filt[[n]][[2]] - meandataplot1filt[[n - 1]][[2]]) /
     (meandataplot1filt[[3]][[1]] - meandataplot1filt[[2]][[1]])},
   {n, indexnoisesearchc, 2, -1}];

noisecheckc =
  Table[{noisecheckc1[[n + 1]] / noisecheckc1[[n]]}, {n, 1, Length[noisecheckc1] - 1}];

noiser1c = indexnoisesearchc - 14;
freqnoiser1c = meandataplot1filt[[noiser1c]][[1]];
freqnoisel1c = freqnoiser1c - deltafpeakc / 2;
freqnoisel2c = freqpeaklc - meandataplot1filt[[noiser1c]][[1]] + freqpeakrc;
freqnoiser2c = freqnoisel2c + deltafpeakc / 2;

peaklp = {freqpeaklc, 10^(-10)}
peakrp = {freqpeakrc, 10^(-10)}

noisel1p = {freqnoisel1c, 10^(-10)}
noiser1p = {freqnoiser1c, 10^(-10)}

noisel2p = {freqnoisel2c, 10^(-10)}
noiser2p = {freqnoiser2c, 10^(-10)}
```

Out[5638]=

$$\left\{4839.99,\ \frac{1}{10\,000\,000\,000}\right\}$$

Out[5639]=

$$\left\{5160.01,\ \frac{1}{10\,000\,000\,000}\right\}$$

Out[5640]=

$$\left\{1839.99,\ \frac{1}{10\,000\,000\,000}\right\}$$

Out[5641]=

$$\left\{2000.,\ \frac{1}{10\,000\,000\,000}\right\}$$

Out[5642]=

$$\left\{8000.,\ \frac{1}{10\,000\,000\,000}\right\}$$

Out[5643]=

$$\left\{8160.01,\ \frac{1}{10\,000\,000\,000}\right\}$$

In[5651]:=

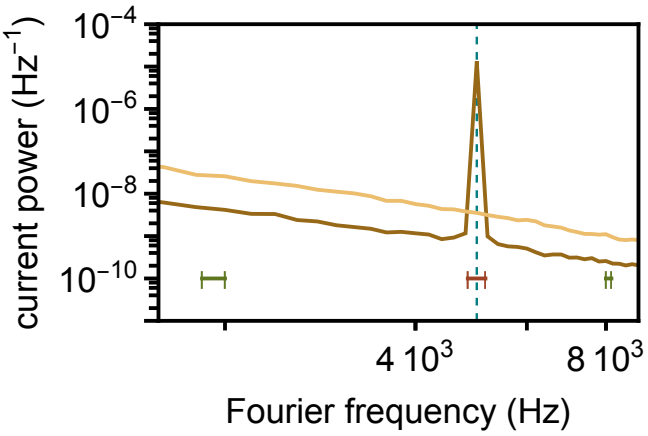
```
plotstylepeaklr = {Darker[■]};
plotstylenoiselr = {Darker[■]};
plotstyleP = {Darker[■]}
plotrange3 = {{1570, 9006}, {10^(-11), 10^(-4)}};

ListLogLogPlot[{meandataplot1filt}, PlotStyle → plotstyleP,
  Joined → True, Frame → True, PlotRange → plotrange3,
  ImagePadding → {{pady, pad2}, {padx, pad2}}, ImageSize → 400,
  LabelStyle → {FontSize → fntsize}, PlotStyle → plotstylepsd, Background → White,
  Frame → {True, True, True, True}, FrameStyle → Directive[Black, Thick],
  FrameTicks → {{YTicks, None}, {XTicksbb, None}}, FrameLabel →
   {"Fourier frequency (Hz)", "current power \!\(\*TemplateBox[{RowBox[{\"(\",
      \"Hz\"}], RowBox[{\"-\", \"1\"}]},\"Superscript\"]\))"},
  GridLines → {{5000}, {}}, GridLinesStyle → Directive[
    RGBColor[0, 125 / 255, 128 / 255], Dashed, Thickness → 0.01 / 2]];
psd11 = Show[%, ListLogLogPlot[{meandataplot1c}, PlotStyle → plotstyleP]];
Show[%,
  ListLogLogPlot[{peaklp}, PlotStyle → plotstylepeaklr, PlotMarkers → {"|", 10}]];
Show[%,
  ListLogLogPlot[{peakrp}, PlotStyle → plotstylepeaklr, PlotMarkers → {"|", 10}]];
Show[%, ListLogLogPlot[{peaklp, peakrp}, Joined → True, PlotStyle → plotstylepeaklr]];
Show[%, ListLogLogPlot[{noisel1p},
   PlotStyle → plotstylenoiselr, PlotMarkers → {"|", 10}]];
Show[%, ListLogLogPlot[{noiser1p},
   PlotStyle → plotstylenoiselr, PlotMarkers → {"|", 10}]];
Show[%,
  ListLogLogPlot[{noisel1p, noiser1p}, Joined → True, PlotStyle → plotstylenoiselr]];
Show[%, ListLogLogPlot[{noisel2p},
   PlotStyle → plotstylenoiselr, PlotMarkers → {"|", 10}]];
Show[%, ListLogLogPlot[{noiser2p},
   PlotStyle → plotstylenoiselr, PlotMarkers → {"|", 10}]];
Show[%,
  ListLogLogPlot[{noisel2p, noiser2p}, Joined → True, PlotStyle → plotstylenoiselr]];
Show[%,
 ListLogLogPlot[{meandataplot1filtbase}, PlotStyle → {Lighter[■]}, Joined → True]]
```

Out[5653]=

{■}

Out[5666]=


current power (Hz^{-1})
10^{-4}
10^{-6}
10^{-8}
10^{-10}
$4\ 10^3$
$8\ 10^3$
Fourier frequency (Hz)